THE CURVATURE SECTOR OF THREE-DIMENSIONAL ELECTRODYNAMICS: SUBSEQUENTIAL LIMITS AND NON-GAUSSIANITY

Abstract. We construct continuum and infinite-volume curvature limit points of noncompact lattice electrodynamics with Wilson fermions. Harmonic averaging gives reflection-positive laws with Gaussian bounds at every charge. At small charge, renormalization on two reference tori and chessboard estimates give non-Gaussianity in every limit point and almost every ergodic component, also with all flux sectors included.

The finite laws use four-component antiperiodic fermions, Wilson parameter one, $\eta = L^{-N}$, $R = hL^T$ and $h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}$. They average harmonic modes in zero flux (av) or over all fluxes (af). As defined in Section 2, $\mathcal{S}$ denotes Schwartz two-forms, H_{-4} the negative Hermite–Sobolev space, C_0 the free Maxwell covariance, κ_4 the fourth cumulant, and $b(f)$ the massive four-fermion loop coefficient.

Theorem A (Subsequential curvature construction)**.** *Fix $0 < \bar{m} \le 1$ and a sufficiently large odd L.*

(i) For every charge $e > 0$ and every volume-independent mass assignment $\mu_N = \bar{m}\eta + m_{0,N} > -1/2$, both families of physical curvature laws are tight on H_{-4}. Every sequence with $N, T \to \infty$ has a weakly convergent subsequence. Each limit μ is a law on tempered closed two-forms, is invariant under translations and the cubic group, is reflection positive at every coordinate plane, and obeys

$$\int e^{F(f)}\, d\mu(F) \le e^{C_0(f,f)/2} \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S} \text{ real}).$$

Its moments satisfy the OS linear-growth condition. All fixed moments and joint cumulants converge along the selected subsequence, and each coordinate direction has a reconstruction with a positive Hamiltonian.

(ii) For sufficiently small charge there are $\sigma > 0$ and one mass assignment $m_{0,N}(e)$, independent of the surrounding volume, with the following property. For every real compact smooth two-form β with $f = dd^*\beta$ and $b(f) > 0$, there are $e_\beta, C_\beta > 0$ such that, for $0 < e \le e_\beta$, every limit point satisfies

$$\left|\kappa_4^\mu(f) - e^4 b(f)\right| \le C_\beta e^{4+\sigma}, \qquad \kappa_4^\mu(f) \ge \tfrac{1}{2} e^4 b(f) > 0.$$

The same two conclusions hold in almost every translation-ergodic component of each limit point. A probe with $b(f) > 0$ exists.

Contents

1. Introduction 3
2. The model and the curvature observable 4
3. The source identity and the two reference volumes 7
4. Positive finite laws and Gaussian domination 9
5. Compactness and the continuum curvature law 10
6. Coordinate reconstruction and ergodic components 11
7. From products to conditional expectations 12
8. Comparison with pinned harmonic modes 13
9. A small-field ultraviolet flow 14
10. A compact probe 20
11. Two reference tori and normalized source jets 21
12. Transfer to every target volume 22
13. Cumulants in limits and in their components 23
14. Proof of the construction theorem 23
Appendix A. The finite Wilson and photon transfer calculations 25
Appendix B. Readout estimates, compactness and tensor regularity 33
Appendix C. Invariant projections, disintegration and reconstruction 37
Appendix D. Positive mass and the removal of harmonic pinning 42
Appendix E. Domains, absolute norms and the choice of constants 43
Appendix F. Fermion inverses, local gauges and complete recoveries 47
Appendix G. Photon minimizers, white roots and reconstruction 52
Appendix H. The growing collar and irrelevant contraction 56
Appendix I. The scalar determinant correction 58
Appendix J. The normalized one-step map 60
Appendix K. Reference geometry and one torus-independent pair 63
Appendix L. Terminal comparison and the pressure estimate 66
Appendix M. Compatible compact probes 70
Appendix N. The free loop and its lattice contacts 71
Appendix O. The large-mass sign 72
Appendix P. Normalized reference jets 79
Appendix Q. The normalized pressure defect 88
Appendix R. Chessboard transfer on the actual finite laws 90
Appendix S. Cumulants and invariant conditional statistics 95
References 97

1. Introduction

1.1. **The field and the construction.** The electromagnetic observable used here is an average of curvature against a smooth test. This avoids choosing a representative of the gauge potential and gives a random distribution rather than a pointwise field. The lattice measure contains the Maxwell weight and the four-component Wilson determinant. We integrate the fermions exactly and construct laws of this gauge-invariant observable.

The order of the argument is useful. Existence follows from a uniform bound on the Laplace transform of the actual finite laws. It does not require a comparison of two cutoffs or a perturbative expansion. Non-Gaussianity is proved afterwards, by quantitative estimates that remain valid in arbitrarily large volumes. Thus every cofinal sequence has limit points, and the same fourth-cumulant estimate applies to every one of them.

There are two finite regulators. Both average the constant part of the gauge potential over a period of the determinant. The first retains zero flux; the second sums all integer flux sectors. A constant potential has zero curvature but changes a charged fermion's holonomy around a torus. Averaging this potential therefore changes the curvature weight. It is part of the specified regulator, rather than an arbitrary choice of gauge.

1.2. **The two estimates behind the theorem.** Reflection positivity gives more than the eventual Hilbert-space construction. On a finite torus it allows an external curvature source to be reflected until it disappears, which proves Gaussian domination. After the physical rescaling, this yields uniformly bounded second moments in a negative Hermite–Sobolev space. A compact inclusion gives tightness. The same Laplace bound controls higher moments and allows the lattice symmetries and reflection inequalities to pass to a continuum, infinite-volume limit.

Gaussian domination alone permits a free limit. The second part of the proof distinguishes the interacting law. For a compact exact probe, the fourth cumulant has a positive free fermion coefficient at order e^4. A calculation on reference tori of sides $R^*(e)$ and $LR^*(e)$ controls the error. Their physical sizes depend on charge but not on the ultraviolet cutoff or the target volume. Reflection and chessboard inequalities then bound products in distinct cells of a much larger torus. The normalization of these inequalities is supplied by a separate pressure estimate on the same two reference tori.

Only the reference calculation uses the small-field renormalization group. The complement of its domain is estimated using Gaussian tails of the actual curvature law. No representation of every large-field history by one small terminal interaction is required. The local renormalization step nevertheless keeps its exact fermionic recovery, determinant factors and Gaussian normalization, since these enter the pressure estimate and the normalized source derivatives.

1.3. **Why products of cell observables are needed.** A positive fourth cumulant of a law does not imply that almost every phase is non-Gaussian. A mixture of Gaussian laws can have a positive fourth cumulant when its variances differ. We therefore estimate products of centered second and fourth Hermite statistics at an arbitrary number of distinct cells. On passing to a limit, spatial averages and all even moments bound the conditional statistics in its ergodic components. This yields the fourth-cumulant expansion in almost every component, with the same error bound as in the full law.

The spatial-averaging argument uses the invariant projection of a discrete subgroup and then conditions on the full translation-invariant sigma algebra. It makes no assumption that ergodicity under all translations implies ergodicity under that subgroup.

1.4. **Relation to the renormalization group.** The local operator construction is the block-averaging scheme developed by Bałaban and Bałaban–O'Carroll–Schor, in Dimock's form for QED in three dimensions [1, 2, 3, 4, 5]. The photon is averaged with an axial gauge on the fluctuations

and a Landau representative for the background. Fermionic activities carry ordinary and covariant-difference fields. Relevant scalar mass and vacuum terms are extracted, and the remaining activity contracts under change of scale.

Two aspects of the present use of that scheme deserve explanation. First, the relevant pair is chosen by a backward fixed point with vanishing terminal coordinates. The local constructions agree under flat periodization, so one pair is used on both reference tori. Second, the source calculation carries a finite jet algebra, with separate variables for the cells. Connected estimates retain their complete supports and give an error for each covered cell. This is the input needed by the all-volume and component arguments.

Reflection positivity, Gaussian domination and chessboard estimates follow the methods of [8, 7]; Wilson transfer positivity is related to [10, 12]. The probabilistic reconstruction and ergodic decomposition use the framework of [11, 6, 9]. We state the specific finite identities and estimates used below so that the normalization and the geometry of their applications are explicit.

1.5. **Scope and reading path.** The limiting laws need not be unique. Moment determinacy means that a given hierarchy determines its law; it does not say that different cutoff subsequences have the same hierarchy. Full rotational covariance and charged fermion correlation functions are not constructed. Coordinate reflection positivity yields positive Hamiltonians, while relativistic reconstruction requires additional Euclidean covariance. Higher-order clustering in translation-ergodic components is obtained in the mean and for diagonal separation.

The model and observable are defined next. The Gaussian argument then constructs the limit points. The reference calculation explains the counterterm selection, and the probe and transfer estimates establish non-Gaussianity. The final proof joins these results before the technical appendices. The main sections state the required estimates and explain their roles; the appendices give the finite transfer algebra, local operator construction, activity estimates, jet bounds and continuum coefficient calculation.

2. The model and the curvature observable

2.1. **Cochains, harmonic modes and flux.** We first work in lattice units on $\Lambda_n = \prod_{j=1}^{3} \mathbb{Z}/n_j\mathbb{Z}$, with $n_j \geq 2$. A k-cochain assigns a number to each positively oriented k-cell. The real spaces are denoted $C^k(n)$; their scalar product is the counting scalar product. The coboundary is the forward oriented difference. In particular, for a one-cochain A,

$$(dA)_{ij}(x) = A_i(x) + A_j(x + e_i) - A_i(x + e_j) - A_j(x), \quad i < j.$$

All differential operators in lattice units use this convention. Write d^* for the counting adjoint. The spaces

$$V_g = dC^0(n), \qquad V_h = \{A : A_j(x) \text{ is constant in } x\}, \qquad V_t = (V_g \oplus V_h)^{\perp}$$

give an orthogonal decomposition of $C^1(n)$. The kernel of d is $V_g \oplus V_h$. Consequently the Maxwell quadratic form is strictly positive on V_t.

For a closed integer two-cochain z, its flux $w_{ij}(z)$ is the sum of z_{ij} over a coordinate two-torus. Closedness makes this sum independent of the position of that torus. The three fluxes form $w(z) \in \mathbb{Z}^3$. Zero flux is equivalent to $z = dm$ for an integer one-cochain m. One way to see this is to integrate z successively along the three coordinate directions: closedness removes the dependence on the integration order, and the three zero periods are exactly the conditions for the resulting integer primitive to be periodic. This discrete cohomology fact is also used in the sector decomposition below.

2.2. **The Wilson determinant.** The spinor space consists of four-component functions on $\mathbb{Z}^3$ that are antiperiodic under each period $n_j e_j$. Its dimension is $4\,|\Lambda_n|$. With the Pauli matrices σ_j, set

$\gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j)$. For a complex one-cochain B, extended antisymmetrically to oriented bonds, define

$$(D[B]\psi)(x) = 3\psi(x) - \frac{1}{2}\sum_{|x'-x|_2=1} (1 - \gamma\cdot(x'-x))e^{iB(x,x')}\psi(x'). \tag{2.1}$$

The Wilson parameter is one. At charge $q > 0$ and real lattice mass μ, put

$$D_A = D[qA], \qquad \mathrm{Dt}(B) = \det(D[B] + \mu).$$

The two Pauli blocks are adjoints for real B. Thus $\mathrm{Dt}(B)$ is the squared modulus of a two-component determinant. It can vanish, but is nonnegative. Gauge conjugation and the periodicity of the link phases give

$$\mathrm{Dt}(B + d\lambda) = \mathrm{Dt}(B), \qquad \mathrm{Dt}(B + 2\pi m) = \mathrm{Dt}(B) \quad (m \in C^1(n;\mathbb{Z})). \tag{2.2}$$

Charge conjugation gives $\mathrm{Dt}(-B) = \mathrm{Dt}(B)$. The determinant is covariant under the lattice isometries with their spin lifts. These statements concern the full four-component determinant on the same antiperiodic space.

For the auxiliary spin covariance used in scalar extraction, we also use the matrix that interchanges the two Pauli blocks:

$$\gamma_4 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

It is a Hermitian involution and anticommutes with each γ_j, $j = 1, 2, 3$.

2.3. **The two finite laws.** The all-flux representation is convenient for reflection. Put $r_q = 2\pi/q$, let dA in the next integral denote Lebesgue measure on the link variables, and define

$$\omega_n(\Phi) = \int_{[0,r_q)^{\mathrm{links}}} dA \sum_{\substack{z\in C^2(n;\mathbb{Z}) \\ dz=0}} \Phi(dA + r_q z)e^{-\|dA+r_q z\|^2/2}\mathrm{Dt}(qA). \tag{2.3}$$

The curvature in lattice units is $F^u = dA + r_q z$. It is closed. Denote by $\omega_n^{(w)}$ the same expression restricted to flux $w(z) = w$, and set

$$Z_n = \omega_n(1), \quad Z_n^0 = \omega_n^{(0)}(1), \qquad E_n^{\mathrm{af}}\Phi = \frac{\omega_n(\Phi)}{Z_n}, \quad E_n^{\mathrm{av}}\Phi = \frac{\omega_n^{(0)}(\Phi)}{Z_n^0}. \tag{2.4}$$

All sums converge for observables of at most exponential growth. The determinant is bounded on the compact link domain and the integer-cochain sum has Gaussian decay. The two normalizations are strictly positive; a gauge field with positive determinant has a neighborhood of positive measure. The explicit transfer factorization below gives such a field in the mass range $\mu > -1/2$.

The zero-flux law has a noncompact transverse representation. For $\theta \in [-\pi, \pi]^3$ define the constant one-cochain $(H_\theta)_j = \theta_j/(qn_j)$. Unfolding the zero-flux integer sum and then removing the exact gauge variables gives

$$E_n^{\mathrm{av}}\Phi = \frac{\displaystyle\int_{V_t} d\sigma(A) \int_{[-\pi,\pi]^3} d\theta\, \Phi(dA)e^{-\|dA\|^2/2}\mathrm{Dt}(q(A + H_\theta))}{\displaystyle\int_{V_t} d\sigma(A) \int_{[-\pi,\pi]^3} d\theta\, e^{-\|dA\|^2/2}\mathrm{Dt}(q(A + H_\theta))}. \tag{2.5}$$

The constant Jacobian from the gauge quotient cancels between the two integrals. Exactness of zero-flux z, the change $(A, z) \mapsto (A + r_q m, z - dm)$, and (2.2) prove this identity. In particular, the average is weighted by the determinant before normalization. It is not an average of independently normalized twisted laws.

The same change of variables shows that every all-flux configuration has the ordinary real noncompact representative on a contractible region: there z can be removed by an integer primitive.

This explains why local estimates on real one-cochains apply to both regulators. Their global flux restrictions still enter the transfer and pressure arguments.

2.4. **Physical units and the readout.** Fix $0 < \bar{m} \le 1$, choose the large odd block factor L, and set

$$\eta = L^{-N}, \qquad h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad R = hL^T, \qquad n = R/\eta. \tag{2.6}$$

We consider $N, T \ge 0$ with $n \ge 2$; in particular $R \ge 1$. The lattice charge and mass are

$$q = e\sqrt{\eta}, \qquad \mu_N = \bar{m}\eta + m_{0,N} > -\tfrac{1}{2}. \tag{2.7}$$

A mass assignment is a real sequence $m_{0,N}$ satisfying this inequality and independent of T. For compactness it is otherwise arbitrary. Non-Gaussianity uses the particular assignment selected by the renormalization group.

The physical curvature is $F = \eta^{-3/2} F^u$. For a physical plaquette $\sigma = (x; ij)$, let

$$c(\sigma) = x + \tfrac{\eta}{2}(e_i + e_j), \qquad Q(\sigma) = c(\sigma) + [-\eta/2, \eta/2)^3.$$

For fixed ij these cubes partition the physical torus. A Schwartz two-form is first periodized and then averaged:

$$f_\eta(\sigma) = \eta^{-3} \int_{Q(\sigma)} \sum_{k \in \mathbb{Z}^3} f_{ij}(y + Rk)\, dy, \qquad X_{\eta,R}(f) = \eta^3 \sum_\sigma f_\eta(\sigma) F(\sigma). \tag{2.8}$$

The step field $F^\sharp$ equals $F(\sigma)$ on each component's averaging cube and is extended periodically to $\mathbb{R}^3$. Then $F^\sharp(f) = X_{\eta,R}(f)$. This convention retains the plaquette centers and the electric signs under reflection.

Let $P^{\mathrm{nat}}_{N,T}$, nat $\in \{\mathrm{av}, \mathrm{af}\}$, be the law of $F^\sharp$ under (2.4), with (2.7). A vacuum action factor $\exp(-\epsilon_{0,N} n^3)$ cancels in these laws. It must still be retained when partition functions on different reference volumes are compared.

2.5. **The topology of the limit.** Write $\mathcal{S} = \mathcal{S}(\mathbb{R}^3; \Lambda^2)$ and let $\mathcal{S}'$ denote its real tempered dual, complexified on tests. We use a Hilbert topology for the subsequence argument. For a multi-index $k \in \mathbb{N}_0^3$ and component $c \in \{12, 13, 23\}$, let $h_{k,c}$ be the normalized Hermite function in that component. It has eigenvalue $\lambda_k = 2\,|k|_1 + 4$ for $1 + |x|_2^2 - \Delta$. Define

$$\|F\|_s^2 = \sum_{k,c} \lambda_k^{2s} \left| F(h_{k,c}) \right|^2, \qquad H_s = \{F \in \mathcal{S}' : \|F\|_s < \infty\}. \tag{2.9}$$

The spaces are separable; H_{-3} embeds compactly in H_{-4}. Their Borel sigma algebras are the restrictions of the cylinder sigma algebra of $\mathcal{S}'$.

A limit point means weak convergence on H_{-4} along a sequence with $N \to \infty$ and $T \to \infty$. No relation between these two rates is imposed. Tightness will hold over all cutoffs, but continuum and infinite-volume symmetry conclusions concern these cofinal limits.

The free Maxwell covariance on two-forms is

$$C_0(f, g) = \langle f, \Pi_c g \rangle_{L^2}, \qquad \widehat{*\Pi_c g}(p) = \left(1 - \frac{pp^T}{|p|_2^2}\right) \widehat{*g}(p). \tag{2.10}$$

Here $*g = (g_{23}, -g_{13}, g_{12})$. A distribution is closed when $F(d^*\phi) = 0$ for every Schwartz three-form ϕ. The physical step interpolation need not be exactly closed as a continuum distribution at finite mesh; the discrete identity and convergence of the test differential imply closedness in the limit.

2.6. **Sources, moments and the compact probe.** For finitely many real tests f_j, the normalized source logarithm of a law ρ is

$$W_\rho(z) = \log \int \exp\left(\sum_j z_j F(f_j)\right) d\rho(F), \qquad W_\rho(0) = 0.$$

Its derivatives are joint cumulants; derivatives of its exponential are ordinary moments. Gaussian domination supplies the analytic neighborhood used for these derivatives. For a real centered readout, $\kappa_4^\rho(f) = \rho(F(f)^4) - 3\rho(F(f)^2)^2$.

The probes used to detect an interaction are real compact smooth two-forms β, with

$$a = d^*\beta, \qquad f = da = dd^*\beta, \qquad b(f) = -6\,\mathrm{Tr}\big[(S_{\bar m} i\gamma\cdot a)^4\big], \quad S_{\bar m} = (\bar m + \gamma\cdot\partial)^{-1}. \tag{2.11}$$

The trace is well defined through the Hilbert–Schmidt products proved below. Positivity is established for a particular probe, not assumed for every f. This exact form of the probe also allows its lattice version to be obtained by translating the connection in the fermion determinant, with the free photon quadratic contribution removed explicitly.

3. The source identity and the two reference volumes

The proof has a probabilistic part and a local analytic part. The probabilistic bound constructs laws without identifying the limits from different subsequences. The analytic part measures an interaction that all these laws retain. An elementary translation of the photon integral connects the two.

3.1. **A compact exact probe.** Choose a transverse real lattice one-cochain a_η whose physical curl is the exact discrete probe. Its compact construction from β is given below. Put

$$\widehat{a} = \sqrt{\eta}\, a_\eta, \qquad g = d\widehat{a}, \qquad X = \langle g, F^u\rangle, \qquad v_\eta = \|g\|^2.$$

Thus X is a physical curvature readout, with $g = \eta^{3/2} d_\eta a_\eta$. The equality of this discrete exact probe with the cell average of a smooth test tending to f is important: it allows the identity below to be used in the same laws whose continuum limits were constructed.

Write $Z(t)$ for the transverse, harmonic-averaged determinant integral in (2.5), with the connection $A + H_\theta$ replaced by $A + H_\theta + (t/e)\widehat{a}$. The mass and the normalized photon measure are unchanged. Define $\Phi(t) = Z(t)/Z(0)$. The external connection enters the Wilson phase as $t\eta a_\eta$, since $q = e\sqrt{\eta}$.

Lemma 3.1 (Translation of the photon source)**.** *For the harmonic-averaged law and every complex z,*

$$E^{\mathrm{av}} \exp(zX - z^2 v_\eta/2) = \Phi(ez). \tag{3.1}$$

More generally, let $\widehat{a}_c$ be mirror translates of this cochain with pairwise disjoint supports of $d\widehat{a}_c$, and put $X_c = \langle d\widehat{a}_c, F^u\rangle$. The multivariable determinant ratio then obeys

$$E^{\mathrm{av}} \exp\left(\sum_c z_c X_c - \tfrac{1}{2}\sum_c v_{\eta,c} z_c^2\right) = \Phi((ez_c)_c), \qquad v_{\eta,c} = \|d\widehat{a}_c\|^2. \tag{3.2}$$

Proof. For real z, complete the Maxwell square:

$$-\tfrac{1}{2}\|dA\|^2 + z\,\langle d\widehat{a}, dA\rangle - \tfrac{1}{2}z^2\|d\widehat{a}\|^2 = -\tfrac{1}{2}\|d(A - z\widehat{a})\|^2.$$

The transverse translation $A \mapsto A + z\widehat{a}$ preserves Lebesgue measure and moves the source into the determinant. It leaves the harmonic average and the mass unchanged. This proves (3.1) on the real axis. On compact complex sets, the finite-dimensional Gaussian dominates the exponential source and the finitely many Wilson phases, so both sides are entire. The identity theorem gives the complex statement. In several variables the same translation is made with $\sum_c z_c\widehat{a}_c$. Disjoint curvature supports eliminate its cross terms. Successive one-variable identity theorems prove (3.2). □

This identity is an exact property of the interacting law. Removing the determinant leaves a Gaussian curvature readout. The departure of Φ from one measures the interaction at positive charge.

3.2. **Removing the free photon contribution.** Let $H_m(x;v)$ be the Hermite polynomials defined by

$$\exp(zx - z^2v/2) = \sum_{m\geq 0} \frac{z^m}{m!} H_m(x;v).$$

In particular, $H_2(x;v) = x^2 - v$ and $H_4(x;v) = x^4 - 6vx^2 + 3v^2$. Differentiating the exact identity gives

$$EH_m(X;v_\eta) = e^m \Phi^{(m)}(0), \qquad \kappa_4(X) = e^4 \left(\Phi^{(4)}(0) - 3\Phi^{(2)}(0)^2\right). \tag{3.3}$$

The second equality uses charge symmetry, hence centering. It also follows directly from $EX^4 - 3(EX^2)^2$; the free variance terms cancel. This is why fourth derivatives of an unnormalized determinant alone would not suffice.

For several cells, each derivative acts only on its own source variable. If c_m is a reference derivative, the operator $(\partial_{t_c}^m - c_m)|_{t_c=0}$ corresponds to the centered statistic $e^{-m}H_m(X_c; v_{\eta,c}) - c_m$. An unperturbed factor is annihilated. In a connected expansion, a contribution survives a product of such operators only if its supports cover every marked cell. An error per covered cell consequently yields an estimate at every product order, rather than an estimate whose constant increases uncontrolled with the number of cells.

3.3. **Why two tori suffice.** The fourth derivatives are calculated on a reference torus of side $R^*(e)$. Its cell decomposition contains a fixed number of cells once the block factor is chosen. A second torus of side $LR^*(e)$ controls the normalization of the transfer to large volumes. The renormalization group uses the same selected bare pair on both tori.

For an unnormalized reflection-positive functional, the chessboard inequality bounds a product at selected cells by reflected full-cell products. Passing to normalized expectations introduces the partition function. The defect

$$D = L^3 \log Z_1 - \log Z_L,$$

where subscripts indicate these relative widths, controls this cost. The linear extensive vacuum factor cancels in D. The fermionic and Maxwell parts that are not exactly extensive remain and must be estimated. Convexity of pressure in each side length and a positive transfer operator make the estimate available for every larger aligned torus. For the harmonic-averaged law, positivity of flux Fourier sums bounds the zero-flux numerator, while convexity of its partition function controls the normalization. The remaining reference flux tail is bounded by Gaussian moments.

The all-volume result has the form

$$\left| E_k^{\mathrm{nat}} \prod_{c\in S} \left(e^{-m} H_m(X_c; v_\eta) - c_m\right) \right| \leq (\Gamma^{\mathrm{nat}} \Theta_m^{\mathrm{nat}})^{|S|}, \qquad m = 2, 4. \tag{3.4}$$

The constants on the right are formed on the reference tori. There is no dependence on the target width or on the ultraviolet cutoff. The error for $m = 2$ is retained when converting the $m = 4$ estimate into a cumulant. Its size and the pressure cost leave a positive power of charge beyond e^4.

Finally, take a subsequential limit at a fixed product order. Uniform Gaussian bounds justify moment convergence and removal of the exact probe's discretization error. Spatial averaging then gives an almost-sure bound on the corresponding invariant conditional statistic. Sending the product order to infinity is a separate probabilistic argument. This last step is what extends the positive fourth cumulant to almost every ergodic component of the limit.

4. Positive finite laws and Gaussian domination

The existence argument begins with the curvature measure itself. The fermions have already been integrated, and their four-component Wilson determinant is nonnegative. This ordinary positivity is insufficient for reflection positivity: the latter compares the two sides of a plane, with the sign of the electric curvature reversed. A slice factorization of the determinant supplies that additional structure. It also gives a Gaussian bound whose constants do not grow with the number of sites. This is the estimate that will allow both cutoffs to be removed without constructing a Cauchy sequence of renormalization group trajectories.

All norms in this section are native counting norms. Write d for the lattice coboundary and P_d for the orthogonal projection of two-cochains onto dC^1. The exact definitions of E^{av} and E^{af} are those of Section 2. Neither their determinant nor their normalization is replaced by a free-field value.

Theorem 4.1 (Finite Gaussian domination)**.** *Let all three sides of a torus be at least two, let $q > 0$, and let its native mass satisfy $\mu > -1/2$. For every real two-cochain g,*

$$E^{\mathrm{av}} \exp\langle g, F^u\rangle \le \exp\big(\tfrac{1}{2}\|P_d g\|^2\big), \tag{4.1}$$

$$E^{\mathrm{af}} \exp\langle g, F^u\rangle \le \exp\big(\tfrac{1}{2}\|g\|^2\big). \tag{4.2}$$

Both laws are invariant under $F^u \mapsto -F^u$ and lattice translations; on a cubic torus they are covariant under signed coordinate permutations. In particular their means vanish and their covariance forms are bounded by the right-hand quadratic forms in (4.1)–(4.2).

The two proofs use the same positive fermion slice, but treat the photon variables differently. For the averaged law we first put a Proca quadratic on the full real link space. Its layer kernel is a Fourier integral with a positive matrix-valued weight. Cauchy–Schwarz across a mirror removes the electric source on its fixed layer. A maximizing source with the fewest nonzero coordinates must therefore be zero. Removing the Proca mass produces the partition-weighted harmonic average, rather than the pinned law.

For the all-flux law, compact link coordinates and integer curvature coordinates form the slice space. The electric Gaussian is a positive heat operator followed by the projection onto the local Gauss-law subspace. A source translates this operator by a commuting unitary. Schatten Hölder first removes the electric source in one direction; using a second direction removes the magnetic source left by the first step. The proofs, including the scalar determinants and the boundary planes on odd tori, appear in Appendix A.

Theorem 4.2 (Finite reflection positivity)**.** *Fix a coordinate reflection r of a cubic torus. Let P be a bounded function of plaquette curvatures in one half of the torus, with all its plaquettes contained in that half and with magnetic plaquettes supported strictly between its two boundary slices. For either native law,*

$$E^{\mathrm{nat}}\big(\overline{P(r^*F^u)}P(F^u)\big) \ge 0, \qquad \mathrm{nat} \in \{\mathrm{av}, \mathrm{af}\}.$$

In particular this holds for finite linear combinations of curvature characters supported strictly between the two boundary planes. The reflection acts by tensor pullback: plaquettes containing the reflected direction acquire the electric sign.

For the averaged law the reflected integral is a square in the positive Proca transfer space, and remains so after the Proca limit. For the all-flux law it is the Hilbert–Schmidt square of a slab insertion. Bloch angles resolve the two electric fluxes; selecting zero spatial flux and integrating these angles gives the averaged law. Thus the same slab calculation proves positivity in the zero-flux sector. A boundary magnetic observable can be rewritten using $dF^u = 0$ before the insertion is formed; this point is made explicit in Lemma A.5. The statement above is sufficient for all continuum half-space tests, since their fixed compact supports lie strictly inside the admitted lattice half-space at fine mesh.

5. Compactness and the continuum curvature law

We now return to physical units. At (N,T), $\eta = L^{-N}$, $R = hL^T$, $q = e\sqrt{\eta}$, and $\mu_N = \bar{m}\eta + m_{0,N} > -1/2$. The law $P^{\mathrm{nat}}_{N,T}$ is the distribution on H_{-4} of the actual cell readout $F^\sharp$ from Section 2; in particular $F^\sharp(f) = X_{\eta,R}(f)$. The same letter F will denote the distributional curvature under a limiting law. This notation does not identify measures at different cutoffs.

The physical form of Theorem 4.1 is

$$E^{\mathrm{nat}} \exp X_{\eta,R}(f) \le \exp(\tfrac{1}{2}\|f_\eta\|^2_\eta) \qquad (f \text{ real}). \tag{5.1}$$

The cell averages have no loss for a test supported in one fundamental cube. For a Schwartz test, periodization is controlled by

$$\|f_\eta\|_\eta \le p_R(f) := \|f\|_2 + 26R^{-2}\||x|^2 f\|_{L^2(\mathbb{R}^3\setminus[-R/2,R/2)^3)}.$$

Here $p_R \le p_1$ and $p_R(f) \to \|f\|_2$. For the Hermite basis h_κ fixed in Section 2, $p_1(h_\kappa) \le 31\lambda_\kappa$. These bounds convert the finite-dimensional Gaussian estimate into a bound in one fixed space of distributions.

Theorem 5.1 (Cofinal limit points)**.** *For every charge $e > 0$, every admissible mass assignment, and either native law, with $R \ge 1$ and at least two native sites in every direction,*

$$\sup_{N,T} \int \|F\|^2_{-3}\, dP^{\mathrm{nat}}_{N,T}(F) \le 61. \tag{5.2}$$

The family is tight on H_{-4} and on $\mathcal{S}'$ for both its strong and weak-$*$ topologies. Every sequence of cutoffs has a weakly convergent subsequence on H_{-4}. If $N_i, T_i \to \infty$, every such limit ρ has the following properties.

(i) It is carried by closed two-form distributions, is invariant under $F \mapsto -F$ and translations, and is covariant under the cubic group. Its translation representation on $L^2(\rho)$ is strongly continuous.

(ii) For real $f \in \mathcal{S}$,

$$\rho(e^{F(f)}) \le e^{C_0(f,f)/2}, \qquad C_0(f,g) = \langle f, \Pi_c g\rangle_2,$$

where Π_c is the orthogonal projection onto the closed two-forms in L^2. The readout extends to a bounded map $X : L^2(\mathbb{R}^3;\Lambda^2) \to L^2(\rho)$, and the bound holds for all real L^2 tests with $X(f)$ in place of $F(f)$. For every $s > 3/2$, $\rho(H_{-s}) = 1$ and $\rho(\|F\|^2_{-s}) \le \sum_\kappa \lambda_\kappa^{-2s}$.

(iii) For any fixed finite family of Schwartz tests, the joint moment generating functions converge uniformly on compact subsets of the complex source space. All their moments and cumulants consequently converge. The limiting generating function is entire, and its logarithm is analytic on a fixed neighborhood of zero controlled only by the L^2 norms of the tests.

(iv) The moments S_n are symmetric tempered distributions. For real L^2 tests,

$$|S_n(f_1,\dots,f_n)| \le 2\sqrt{2^n n!}\prod_i \|f_i\|_2.$$

For the usual Schwartz seminorm $q_{6n}(\Psi) = \max_{|\alpha|\le 6n}\sup_x(1+|x|^2)^{3n}|\partial^\alpha\Psi(x)|$, including the component maximum, there is a fixed C such that

$$|S_n(\Psi)| \le 2C^n\sqrt{n!}\, q_{6n}(\Psi).$$

Thus the hierarchy has the OS linear-growth regularity. The moments determine ρ uniquely among probability laws on $\mathcal{S}'$.

(v) The law is reflection positive at every coordinate plane, including the corresponding inequality for finite sequences of positive-time Schwartz tensors.

Proof. The second moment of a Hermite readout is at most $961\lambda_\kappa^2$. Since $\sum_\kappa \lambda_\kappa^{-4} < 1/16$, summing with the H_{-3} weights gives (5.2). A closed H_{-3} ball is compact in H_{-4}, so Chebyshev and Prokhorov give tightness. This argument concerns the complete probability law, rather than separately chosen finite-dimensional marginals.

At a cofinal limit, truncated exponential functions and (5.1) give the L^2 Gaussian bound. Uniform exponential integrability upgrades weak convergence of each finite family of readouts to locally uniform convergence of its entire generating function. Discrete closedness passes to distributional closedness. A closed distribution annihilates the orthogonal complement of the closed L^2 two-forms, which sharpens the quadratic bound to C_0. Lattice translations approximate every physical translation; the readout error tends to zero in the same Gaussian estimate.

For fixed compact tests strictly in a positive half-space, the finite reflection inequality holds at sufficiently fine mesh and large volume. Its characteristic functions converge, so the inequality passes to the limit. The span of these characters is dense in the half-space L^2 space. This gives reflection positivity for arbitrary half-space observables, and then for polynomial tensors by their tempered extension. The detailed periodization, tensor, and changing- test estimates are in Appendix B. □

Moment determinacy is a statement about a fixed limiting hierarchy. It does not assert that two different cutoff subsequences have the same hierarchy. The theorem gives limit points along every cofinal sequence, without making a uniqueness or universality claim about those points. An iterated limit, when it exists, is also a cofinal limit point: in the bounded-Lipschitz metric on H_{-4} one chooses a sufficiently large second cutoff to approximate each intermediate limit within $1/i$. This elementary diagonal selection will also be used in the pinned comparison below.

6. Coordinate reconstruction and ergodic components

Reflection positivity supplies a positive Hamiltonian in each coordinate direction, but it need not remove all translation-invariant phases. We therefore keep the invariant sigma-algebra in the argument. This also prevents a cumulant of a mixture from being confused with the cumulants of its components.

Let $U(a)G = G \circ \tau_{-a}^*$ on $L^2(\rho)$. For $J \subset \{1,2,3\}$ write $\mathcal{I}_J$ for the invariant sigma-algebra of translations in the coordinate subspace spanned by e_j, $j \in J$, and $P_J = \mathbb{E}_\rho[\cdot \mid \mathcal{I}_J]$. For $J = \{j\}$ abbreviate $P_j = P_{\{j\}}$. The projections commute and $P_J = \prod_{j\in J} P_j$. Their ranges are the corresponding fixed-vector spaces.

Theorem 6.1 (Inheritance and coordinate reconstruction)**.** *Let ρ be a cofinal limit point from Theorem 5.1 and write its translation-ergodic decomposition as $\rho = \int \rho_\xi \, d\nu(\xi)$. Outside one ν-null set, each component is a law on closed two-form distributions, invariant and ergodic under all translations, invariant under the three coordinate reflections, and reflection positive at every coordinate plane. Moreover*

$$\rho_\xi(e^{X(f)}) \le e^{\|f\|_2^2/2} \qquad (f \in L^2 \text{ real}), \tag{6.1}$$

and $\rho_\xi(X(f)) = 0$. Each component has the same tensor regularity bounds and is carried by H_{-s} for every $s > 3/2$.

For ρ or any such component, reconstruction in direction e_j gives a Hilbert space $\mathcal{H}_j$, a unit vacuum vector Ω_j, a nonnegative self-adjoint H_j, and strongly continuous transverse translations $V_j(y)$ commuting with e^{-tH_j}. If $[\cdot]$ is the positive-half-space quotient map and P_j is formed under that law, then

$$P_{\ker H_j}[G] = [P_j G]. \tag{6.2}$$

On an ergodic component the vectors fixed by both e^{-tH_j} and all $V_j(y)$ are exactly the multiples of Ω_j, and $\dim\ker H_j$ is either one or infinite.

On each component all curvature moments cluster in the mean and when $\min_j |a_j| \to \infty$. *The two-point function tends to zero as* $|a| \to \infty$ *in any manner. Higher-order limits along a coordinate axis or plane are given by* P_J *and need not be products of means.*

There are two nonautomatic inheritance steps. First, an invariant weight belongs to each positive and negative half-space: translate compact approximants far into that half-space and use invariance. Reflection positivity then forces reflection to act as the identity on the invariant space. Multiplying a positive-half-space test by the square root of any nonnegative invariant weight preserves positivity. Disintegration therefore gives reflection positivity componentwise.

Second, an unconditional Gaussian bound is not simply a componentwise bound. For $G = e^{X(f)}$, average $U(a)G$ over a large cube. Its nth moment is bounded by the Gaussian estimate for the sum of n translated tests. The L^2 overlap of two translates tends to zero as their separation tends to infinity. For each fixed n, the region where two averaging points remain close has vanishing relative volume. The invariant average therefore has L^n norm at most $e^{\|f\|_2^2/2}$ for every n, hence has that pointwise bound. Countable determining families and continuity give a single exceptional set for all tests. Full proofs are in Appendix C.

For reconstruction, put

$$b_j(F,G) = \langle \mathcal{R}_j F, G\rangle_{L^2(\rho)}, \qquad F, G \in L^2(\Sigma_j^+, \rho),$$

where $(\mathcal{R}_j F)(F_0) = F(r_j^* F_0)$ is the linear pullback on functions of the configuration F_0. The quotient by the null space of b_j is completed to $\mathcal{H}_j$. Positive translations induce $[G] \mapsto [U(te_j)G]$. An iterated Cauchy–Schwarz estimate proves that these are self-adjoint contractions, and their semigroup is e^{-tH_j} with $H_j \geq 0$. The spectral projection at zero is the quotient of the invariant conditional expectation in (6.2).

This identification yields a useful mixing statement. For all $F, G \in L^2$,

$$\lim_{|s|\to\infty} \sup_{y \perp e_j} \big|\langle F, U(se_j + y)(1 - P_j)G\rangle\big| = 0. \tag{6.3}$$

For tests on opposite half-spaces the modulus is bounded by $\|[\mathcal{R}_j F]\| \|e^{-sH_j}(1 - P_{\ker H_j})[G]\|$, independently of y. Compact approximation extends this to all L^2 functions. Decomposing $1 - \prod_j P_j$ into the three terms containing $1 - P_j$ proves diagonal clustering even when the three coordinates diverge at different rates. Ergodicity turns the full invariant projection into the product of means. Applying the Gaussian inheritance argument to a single axis gives $P_j X(f) = 0$, which yields the stronger all-direction two-point decay. These conclusions make no claim of full rotational covariance or of relativistic reconstruction.

7. From products to conditional expectations

The non-Gaussian argument below will produce estimates for products of polynomial readouts in arbitrarily many distinct cells. The following lemma is the precise passage from those estimates to the invariant part. A bound for only one fixed product order would not suffice.

Lemma 7.1 (All-order product bound)**.** *Let* U *be a measure-preserving action of* $\mathbb{Z}^d$ *on a probability space, let* G *be real and belong to every finite* L^p*, and suppose that for every* $J \geq 1$ *and pairwise distinct* $y_1, \dots, y_J$*,*

$$\left| \mathbb{E} \prod_{i=1}^{J} U(y_i) G \right| \leq u^J. \tag{7.1}$$

If P_0 *is projection onto the* $\mathbb{Z}^d$*-invariant functions, then* $|P_0 G| \leq u$ *almost surely. If this action is a subaction of continuous translations, the same bound holds for* $\mathbb{E}[G \mid \mathcal{I}_{\{1,2,3\}}]$.

Proof. Let A_n average over $\{0,\dots,n-1\}^d$. The mean ergodic argument gives $A_nG \to P_0G$ in L^2. For even J, expansion of $(A_nG)^J$ has at most $\binom{J}{2}n^{d(J-1)}$ collision tuples. Hölder bounds each collision term by $\|G\|_J^J$; all the other terms are bounded by (7.1). Hence

$$\mathbb{E}|A_nG|^J \le u^J + \binom{J}{2} n^{-d}\|G\|_J^J.$$

Fatou along an almost surely convergent subsequence gives $\|P_0G\|_J \le u$ for every even J. Letting $J \to \infty$ gives the pointwise bound. The continuous invariant sigma-algebra is contained in the discrete one, so conditional expectation of P_0G gives the last assertion. Appendix C supplies the mean-ergodic argument and the measurable identifications. □

For a translation-ergodic decomposition, the final conditional expectation is $\xi \mapsto \rho_\xi(G)$. The lemma therefore controls the conditional second and fourth Hermite statistics once the all-volume, all-order product estimate has been passed to a limit. It does not infer non-Gaussian components from the cumulant of the mixture.

8. Comparison with pinned harmonic modes

The harmonic average is important to the finite positive structure. Returning to the zero-harmonic law requires an additional argument: flat connections have zero curvature but change the charged holonomy, and therefore change the fermion determinant. A positive native mass gives a simple uniform comparison at fixed mesh.

Proposition 8.1 (Positive-mass comparison)**.** *On a cubic native torus of side n, let $q \ge 0$ and $\mu > 0$. Put*

$$\tau_* = n\,\mathrm{arcsinh}(\mu/12), \qquad \delta_n = 110n^3e^{-\tau_*},$$

and suppose $\tau_* \ge 1$. For every real link field and every flat twist, the four-component determinants are positive and

$$|\log\Delta^\theta(qA) - \log\Delta^0(qA)| \le 2\delta_n.$$

The twisted and averaged normalized curvature laws have densities relative to the pinned law lying between $e^{-4\delta_n}$ and $e^{4\delta_n}$. Their expectations of a bounded observable of modulus at most one consequently differ by at most $e^{4\delta_n}-1$.

The proof in Appendix D uses only accretivity of the Wilson operator at positive mass. It obtains a zero-free complex twist strip of width proportional to $n\mu$, estimates the nonconstant Fourier coefficients of the logarithmic determinant, and then normalizes. It is uniform in the real photon field and does not invoke a photon large-field expansion.

Corollary 8.2 (Iterated pinned limits)**.** *Fix a charge and a mass assignment with $\mu_N > 0$ for all sufficiently large N. At each such fixed N, the infinite-volume weak limit points of the pinned laws on H_{-4} are exactly those of the averaged laws. Any continuum limit point obtained from these intermediate infinite-volume limits is a cofinal limit point of the averaged family. It therefore has the conclusions of Theorem 5.1; if the mass assignment and charge satisfy the non-Gaussian theorem below, it also has that theorem's cumulant and component conclusions.*

Proof. For fixed N, μ_N is fixed and positive, while $n = R/\eta \to \infty$. Thus $\delta_n \to 0$, and Proposition 8.1 makes the total-variation distance of the pushed-forward laws tend to zero. Their weak limit points coincide. The diagonal selection following Theorem 5.1 makes each iterated limit a cofinal averaged limit point. □

The order of limits in this corollary matters. No uniform lower bound on μ_N has been assumed as $N \to \infty$. In particular a bound $|m_{0,N}| \le C(e\sqrt{\eta})^\gamma$ with $0 < \gamma < 2$ does not imply $\bar{m}\eta + m_{0,N} > 0$. The unconditional existence theorem uses the averaged and all-flux regulators and the weaker admission $\mu_N > -1/2$; the pinned corollary uses its stated extra positivity.

9. A small-field ultraviolet flow

The ultraviolet analysis is used on two reference volumes. It has two purposes: to choose one bare mass and one vacuum constant on both volumes, and to compare their normalized partition functions. It is not necessary to construct an interacting flow on every large torus. Once the two reference partition functions have been controlled, reflection positivity will carry the resulting local estimate to arbitrary volumes.

The reference mass in the Gaussian fermion operator is positive. The relevant mass produced by the interaction is kept in the activity, and is selected by a backward boundary condition. This distinction permits a single analytic flow without presupposing the sign of the selected bare lattice mass. The second important distinction is between the activity and the scalar determinant correction. An external connection produces a scalar determinant term which is independent of the variables integrated at the current step. We retain it exactly instead of normalizing it away.

9.1. **Scales, reference tori and domains.** Fix an odd integer L, a number $b > 0$, and an exponent $0 < \varepsilon \le 1/200$. We choose $1 - \varepsilon < \alpha < 1$. The integers and constants used below are fixed in the order described in Appendix E; none depends on the cutoff or on a reference volume. Put

$$a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad v_j = -\log e_j, \quad r_j = \min\{L^i : i \in \mathbb{N}_0, L^i \ge v_j^r\}, \quad p_j = v_j^p, \quad p_{0,j} = v_j^{p_0}. \tag{9.1}$$

Here r, p_0, p are fixed exponents, with $r \ge 2$, $p_0 > 3r + 2$ and $p > 2p_0$; their further inequalities are specified in that appendix. The thresholds p_j are small-field thresholds, not thermodynamic pressures. The mass scale is

$$h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad h' = hL^{-m_0}, \qquad K = N + \log_L h',$$

where m_0 is fixed so that L^{-m_0} belongs to the uniform reference mass range. Thus $L^{-1} < \bar{m}h \le 1$ and $L^{-m_0-1} < \bar{m}h' \le L^{-m_0}$.

Let $M = L^m$ be a sufficiently large power of L. Define

$$R_{\mathrm{top}} = h'Mr_K, \qquad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{h'L^T : T \in \mathbb{N}_0, h'L^T \le R_{\mathrm{top}}\}. \tag{9.2}$$

Both R^* and LR^* belong to $\mathcal{T}(e)$. On a torus of physical side $R \in \mathcal{T}(e)$, level j has unit-lattice side $l_j = R/a_j$ and fine mesh $\delta_j = L^{-j}$. Its fine lattice therefore has the same number of sites as the original lattice. The stopping level is the first j for which

$$l_j = Mr_j. \tag{9.3}$$

For all sufficiently small $e_K = e\sqrt{h'}$ this level exists once $N \ge N_{\min}(R, e)$. The differences $K - j_*$ and the physical quantities $a_*, e_*, l_*, \bar{m}a_*$ are independent of N above this threshold. For $R = R^*$,

$$\begin{aligned} j_* &\in \{K-1, K-2\}, \qquad & a_* &\in \{h'/L, h'/L^2\}, \\ L^{-m_0-3} &< \bar{m}a_* \le L^{-m_0}, \qquad & e_K/L &\le e_* \le e_K/\sqrt{L}. \end{aligned} \tag{9.4}$$

The elementary proof, including the cutoff threshold, is given in Appendix K.

The analytic domains must control complete paths, not merely the values of the connection at their endpoints. At a fixed level write $g = e_j$, $\delta = \delta_j$ and $l = l_j$. For a bond function A define

$$\|A\|_0 = \sup_{x,\mu} |A_\mu(x)|, \quad \|A\|_1 = \sup_{x,\mu,\nu} |\partial_\nu A_\mu(x)|, \quad \|A\|_{1,\alpha} = \sup_{\substack{0<|x-x'|\le 1 \\ \mu,\nu}} \frac{|\partial_\nu A_\mu(x') - \partial_\nu A_\mu(x)|}{|x - x'|^\alpha},$$

where ∂_ν is the forward difference divided by δ. The restriction $0 < |x - x'| \le 1$ includes the initial mesh. We use the open domains

$$\mathcal{R}_j = \{A : \|A\|_0 < g^{-3/4+\varepsilon},\ \|A\|_1 < g^{-3/4+2\varepsilon},\ \|A\|_{1,\alpha} < g^{-3/4+3\varepsilon}\}, \tag{9.5}$$

$$\mathcal{G}_j = \{A : \|A\|_0 < g^{-1+\varepsilon}\}, \tag{9.6}$$

$$\mathcal{X}_j = \{C : \|C\|_0 < g^{-1+2\varepsilon},\ \|C\|_1 < g^{-1+3\varepsilon}\}. \tag{9.7}$$

The letter C denotes an external connection in these three displays; the photon covariance elsewhere always carries a scale subscript.

The gauge-saturated domain $c\widetilde{\mathcal{R}}_j$, $0 < c \le 1$, consists of $A = A_0 + A_1$, where $A_1 \in c\mathcal{R}_j$, A_0 is real, and on the enlargement of each M-cube by L layers it has a real gauge representative whose three norms are less than c times the thresholds in (9.5). The domain $\widetilde{\mathcal{G}}_j$ is defined in the same way with $A_1 \in \mathcal{G}_j$, a five-layer enlargement, and only the sup norm. The margin $\widetilde{\mathcal{R}}_j^-$ restricts the global complex part to $(1 - g^\varepsilon)\mathcal{R}_j$. A splitting or a local representative is required to exist; it is not part of the variable. These are open sets, since a sufficiently small global complex perturbation can be absorbed by A_1. They are invariant under real gauges and under addition of real constant connections. If e_0 is small enough,

$$\widetilde{\mathcal{R}}_j + \mathcal{X}_j \subset \widetilde{\mathcal{G}}_j, \qquad g\|\operatorname{Im}(A + C)\|_0 \le g^{1/4+\varepsilon} + g^{2\varepsilon}, \quad g\|d(A + C)\|_0 \le 4g^{1/4+2\varepsilon} + 2g^{3\varepsilon}. \tag{9.8}$$

Rescaling $A_L(x) = L^{-1/2}A(x/L)$ maps the next-level domains into $L^{-3/4}$ times the present domains, hence into the margin.

9.2. **The Gaussian operators and their local support.** We use the same covariant hierarchy at every step. In a unit block let $y_0 = x, y_1, \dots, y_j = y$ be the successive block centers. If $\tau A(u, v)$ is the average of the six coordinate-order line sums, define

$$\phi_{j,A}(y, x) = \exp\Big(ig\delta \sum_{i<j} \tau A(y_{i+1}, y_i)\Big), \qquad (Q_j(A)f)(y) = \delta^3 \sum_{x \in B_j(y)} \phi_{j,A}(y, x) f(x).$$

This is the phase of the average line sum, not the average of phases. The transpose is taken for the bilinear pairing $\delta^3 \sum$. In particular $Q_j(A)Q_j^T(-A) = 1$; at a real connection $Q_j^T(-A) = Q_j(A)^*$. All phases in a row of Q_j lie inside its unit block. The next-level average is the composition of the last L-block average and Q_j.

Set $\mathfrak{m}_j = \bar{m}a_j$ and $b_j = b(1 - L^{-1})/(1 - L^{-j})$ for $j \ge 1$. The Wilson operator on the fine lattice, in level coordinates, uses the spin projections $P_\mu^\pm = (1 \pm \gamma_\mu)/2$ and is

$$\begin{aligned}(D_A f)(x) = -\delta^{-1} \sum_{\mu=1}^{3} [&e^{ig\delta A(x, x+\delta e_\mu)} P_\mu^- f(x + \delta e_\mu) \\ &+ e^{ig\delta A(x, x-\delta e_\mu)} P_\mu^+ f(x - \delta e_\mu) - f(x)].\end{aligned}$$

With $K_j(A) = D_A + \mathfrak{m}_j$ put

$$S_j(A) = (K_j(A) + b_j Q_j^T(-A)Q_j(A))^{-1}, \tag{9.9}$$

$$H_j^F(A) = b_j S_j(A) Q_j^T(-A), \qquad \bar{H}_j^F(A) = \mathsf{C}^{-1} H_j^F(-A)\mathsf{C}, \tag{9.10}$$

$$D_j(A) = b_j - b_j^2 Q_j(A) S_j(A) Q_j^T(-A), \tag{9.11}$$

$$\Gamma_j(A) = (D_j(A) + (b/L)Q^T(-A)Q(A))^{-1}, \qquad T_j(A) = b_j^{-1} Q_j(A) S_j(A)^{-1}. \tag{9.12}$$

Here C is the charge-conjugation matrix, $\mathsf{C}^{-1}\gamma_\mu^T\mathsf{C} = -\gamma_\mu$. At $j = 0$, $Q_0 = H_0^F = T_0 = 1$, $D_0 = K_0$, and $\Gamma_0 = (K_0 + (b/L)Q^T(-A)Q(A))^{-1}$; no b_0 or S_0 is used. The photon minimizer is denoted by H_j^P, to distinguish it from the fermion map H_j^F. It minimizes $\|dA\|_2^2$ at fixed uncharged block

average and in Landau gauge. Its axial representative differs from it by an explicitly retained real gauge.

Theorem 9.1 (Uniform Gaussian primitives)**.** *There are $\gamma > 0$, $C < \infty$, $\alpha_0 > 0$ and a positive reference mass upper bound $\mathfrak{m}_0$, depending only on the fixed scheme, with the following properties. They hold at every level, on every torus of side an odd multiple of LM at least $13LM$, for $0 < \mathfrak{m}_j \le \mathfrak{m}_0$, $0 < e_j \le e_0$, and all $A \in \widetilde{\mathcal{G}}_j$.*

The operators (9.9)–(9.12) are defined and holomorphic on this domain. Their fine-to-coarse maps, and their covariant Hölder differences, have uniformly summable rows and transposed rows. The coarse covariance satisfies

$$|\Gamma_j(A)(y, y')| \le Ce^{-\gamma d(y,y')};$$

at a real connection the completed coarse matrix $D_j(A) + (b/L)Q^T(-A)Q(A)$ is invertible and its inverse has bounded Hilbert norm. No mass-uniform bound for the uncompleted $D_j(A)^{-1}$ is asserted. The fine inverse has the unit-block Hilbert bound

$$\|1_{\Delta_y} S_j(A) 1_{\Delta_{y'}}\|_{2\to 2} \le Ce^{-\gamma d(y,y')},$$

and, at the terminal positive reference mass, the pointwise bound $|S_j(A)(x, x')| \le Cd(x, x')^{-2}e^{-\gamma d(x,x')}$ away from the diagonal, with the usual $C\delta^{-2}$ diagonal bound. Constants in this last bound may depend on a fixed lower bound for the terminal reference mass.

The photon covariance on the axial fluctuation space has a real positive square root $R_{\mathrm{loc},j}$. In a local orthonormal basis $\{\Upsilon\}$,

$$|R_{\mathrm{loc},j}(\Upsilon, \Upsilon')| \le Ce^{-\gamma d(\Upsilon,\Upsilon')}, \qquad |\partial H_j^P(b, b')| + |\delta_\alpha \partial H_j^P((b, \tilde{b}), b')| \le Ce^{-\gamma d(b,b')}.$$

There are holomorphic weakened families on $|s_c| \le M^{\alpha_0}$ which recover these exact operators at $s = 1$. If the active parameter cubes split into link-components, rows and columns supported in one component read only that component, and are zero at the opposite indices outside it. Their connection supports include every path and interface bond meeting the component. Polymer resolvent pieces have both indices inside their polymer, depend only on bonds meeting it, and satisfy the preceding exponential bounds multiplied by $e^{-\kappa_w d_M(X)}$ for any fixed prescribed κ_w, after choosing M large enough.

The recovery is literal: $T_j H_j^F = 1$, with barred analogue. On a polymer it is the sum $T_j^{(1)}\psi + T_j^{(2)}\chi$ of an interior field dependence and a nearest-neighbor Hölder boundary pair. Both rows are uniformly bounded on $\widetilde{\mathcal{R}}_j + \mathcal{X}_j$.

The construction and proof are given in Appendices F and G. The fine reference mass is not replaced by a positive physical mass after tuning. Positivity enters only the Gaussian precision in (9.9); all relevant mass shifts remain in the interaction. The weakened photon square root is not required to be a square root at $s \ne 1$. Its purpose is locality. Its physical value $s = 1$ is the exact square root, so the integrated white variables have the original fluctuation covariance.

9.3. **Activities and relevant extraction.** An M-polymer is a face-connected union of M-cubes. Write $d_M(X)$ for the infimum, in M-cube units, of lengths of connected finite trees meeting every closed cube of X; the trees need not pass through cube centers. The rooted polymer count satisfies

$$\sup_{\square} \sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0$$

for absolute constants κ_0, K_0. We take $\kappa \ge 7\kappa_0 + 7$ and retain further reserved decay for periodization.

The activity $E_j(X; A, C)$ is an even balanced Grassmann polynomial in a fine field ψ and an independent Hölder field χ. On physical fields the latter becomes

$$(\delta_{\alpha,A+C}\psi)(x, x') = |x - x'|^{-\alpha}\big(e^{ie_j\delta_j\tau(A+C)(x,x')}\psi(x') - \psi(x)\big).$$

Every fine leg lies in X. A Hölder pair either has both ends and all six paths inside X, or is a nearest-neighbor pair with at least one end in X. The connection supports are exactly the bonds meeting X. These conventions allow a boundary recovery without hiding an enlargement in the activity norm.

For antisymmetric kernels of fine degree n and Hölder degree m use the counting integrals with weights δ_j^3 and δ_j^6, respectively, and set

$$\|E_j\|_j = \sup_{X,A,C} e^{\kappa d_M(X)} \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \|E_{nm}(X;A,C)\|_1, \quad h_1 = e_j^{-1/4}, \quad h_2 = e_j^{-1/4+\varepsilon}. \tag{9.13}$$

The supremum uses $(A,C) \in \widetilde{\mathcal{R}}_j \times \mathcal{X}_j$. Products obey the Banach algebra inequality. Substitution of a kernel into a field is bounded by its absolute row sum; integration against a unit white Berezin covariance is a contraction at field weight at least one. The fine measure factors in this norm are essential to its cutoff independence.

The auxiliary spin group Sp is generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_4$. For $S \in \mathrm{Sp}$ define the diagonal coordinate action by

$$S^{-1}\gamma_\nu S = r_\nu(S)\gamma_\nu \quad (1 \le \nu \le 3), \qquad r(S) = \mathrm{diag}(r_1(S), r_2(S), r_3(S)). \tag{9.14}$$

Together with translations $a \in M\mathbb{Z}^3$, this acts on coordinates by $gx = r(S)x + a$, on bond fields by pullback, and on spinors by $(gf)(x) = Sf(g^{-1}x)$. Barred spinors transform by the inverse spin matrix. In particular $S = \gamma_\mu$, $1 \le \mu \le 3$, fixes coordinate μ and reverses the other two, while $S = \gamma_4$ reverses all three. These are the specified auxiliary spin actions; no invariance under coordinate permutations is used. Activities are invariant under this group, real gauges in either connection slot, charge reversal, and the barred reality involution. The local quadratic extraction is a full 4×4 matrix before it is summed. Its sum commutes with the four specified spin generators. It is therefore scalar, and the reality involution makes it real. Denote the resulting mass and vacuum coordinates by $m(E)$ and $\epsilon(E)$, with the action sign convention: they are the negatives of the local quadratic and vacuum sums. Their norms satisfy

$$|\epsilon(E)| \le C\|E\|_j, \qquad |m(E)| \le Ce_j^{1/2}\|E\|_j. \tag{9.15}$$

No nonscalar mass channel is discarded.

After subtracting these coordinates, we transport the local quadratic coefficient to its mean by covariant line factors. Call the resulting normalization $\mathcal{R}E$. On physical fields this is the exact identity

$$\sum_X E(X) = -\epsilon(E)l_j^3 - m(E)\langle\bar\psi, \psi\rangle + \sum_X \mathcal{R}E(X).$$

The adjusted local vacuum and quadratic coefficients vanish *pointwise* at zero connections. This pointwise cancellation is what the subsequent Cauchy estimate uses. Local mass matrices need not be scalar before summation.

Proposition 9.2 (Irrelevant contraction)**.** *For the fixed choices above,*

$$\|\mathcal{L}\mathcal{R}E\|_{j+1} \le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon}\|E\|_j.$$

With $w_j^E = e_j^{1/4-7\varepsilon}$, L can be chosen so that

$$\frac{\|\mathcal{L}\mathcal{R}E\|_{j+1}}{w_{j+1}^E} \le \frac{1}{16}\frac{\|E\|_j}{w_j^E}.$$

The proof in Appendix H uses a growing collar for the affine connection jet. On a small torus it caps the collar at $l_j/5$ and uses the periodic mean-zero derivative estimate. The cap is paid within the stated domain, not by imposing an extra large-volume hypothesis. Vacuum coefficients gain two source powers and quadratic coefficients gain one; the remaining degrees contract by direct fermion scaling. The complex constant gauge cost is paid only where it occurs.

9.4. **The exact normalized blocking step.** The starting density is the original fermion determinant with reference mass $\bar{m}\eta$, relevant shift $m_{0,N}$ and a vacuum constant. At level j its small-field density is represented as

$$Y_j^x(A;C) = \frac{e^{-\epsilon_j l_j^3 + \sum_X P_j(X,C)}}{\det D_j(0)} \int D\Psi\, e^{-\langle\bar{\Psi}, D_j(H_j^P A + C)\Psi\rangle - m_j\langle\bar{\psi},\psi\rangle + \sum_X E_j(X; H_j^P A, C; \psi, \chi)}, \tag{9.16}$$

where $\psi = H_j^F(H_j^P A + C)\Psi$, with barred analogue, and $\chi = \delta_{\alpha, H_j^P A + C}\psi$. At level zero the denominator and the interpretation use the level-zero convention above. The scalar determinant correction is initialized by $P_0 = 0$ and always satisfies $P_j(X, 0) = 0$.

The photon fluctuation is $Z = H_j^P C_j R_{\mathrm{loc},j}\zeta$, where ζ has the standard Gaussian product law in its local orthonormal basis. The product law is restricted to $|\zeta(\Upsilon)| \le p_{0,j}$ and normalized. Its missing Gaussian probability is recorded in the vacuum coordinate. For fermions use the normalized unit white law of $\omega, \bar{\omega}$, with $\Psi_f = \Psi_c + \Gamma_j\omega$ and $\bar{\Psi}_f = \bar{\Psi}_c + \bar{\omega}$. This asymmetric substitution has covariance Γ_j exactly; no factorization of its native matrix entries is assumed.

Theorem 9.3 (Normalized one-step map)**.** *Suppose both successive sides are admitted, the reference mass at the next level is at most $\mathfrak{m}_0$, and*

$$|m_j| \le w_j^m = e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le 4w_j^E.$$

There is a canonical normalized map, holomorphic in the two connection slots and in the input data on the corresponding complex windows, whose physical value is the exact Gaussian/Berezin blocking of (9.16). With $m_j' = m_j + m(E_j)$ it is

$$m_{j+1} = Lm_j', \qquad \epsilon_{j+1} = L^3[\epsilon_j + \epsilon(E_j) + g_j], \tag{9.17}$$

$$E_{j+1} = \mathcal{L}\mathcal{R}E_j + \mathcal{L}\delta E_j^{\det} + E_j^*, \tag{9.18}$$

$$P_{j+1}(Y, C) = \sum_{X:\bar{X}^L = LY} [P_j(X, C_L) + E_j^{\det}(X, C_L)]. \tag{9.19}$$

Here g_j is the exact negative logarithm per unit volume of the white Gaussian box probability, with $0 \le g_j \le e_j^7$. The determinant pieces are analytic and local, even under charge reversal, and

$$E_j^{\det}(X, A) = E_j^d(X, A) - E_j^d(X, 0), \qquad \delta E_j^{\det}(X; A, C) = E_j^d(X, A + C) - E_j^d(X, C).$$

The resolvent representation gives

$$|E_j^d(X, A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}, \quad \|\mathcal{L}\delta E_j^{\det}\|_{j+1} \le w_{j+1}^E, \quad \|E_j^*\|_{j+1} \le e_{j+1}^{1/4-6\varepsilon}.$$

The last term is Lipschitz on the unit input window:

$$\|\Delta E_j^*\|_{j+1} \le e_{j+1}^{1/4-6\varepsilon}\left(\frac{|\Delta m_j|}{w_j^m} + \frac{\|\Delta E_j\|_j}{w_j^E}\right).$$

The zero-external-connection map is obtained by setting $C = 0$; the relevant coordinates are the same for both maps.

The complete proof is in Appendices I and J. The map is built from five differences: the bare kinetic difference under the new photon fluctuation; the old normalized activity under the new photon fluctuation; its relevant mass term; the old activity under the new fermion fluctuation; and its relevant mass term. The recovery $T^{(1)} + T^{(2)}$ is inserted before the second localization. The independent white coordinates then give an ordinary hard-core polymer expansion. A connected cluster sum is taken before reblocking, so the scalar, relevant and irrelevant parts all belong to the same normalized map.

The new scalar determinant correction is independent of the active variables at this step. Consequently it enters additively after taking the connected logarithm. The old scalar determinant

correction remains outside the same integral and is carried by (9.19). At its physical value the determinant identities telescope with the exact free Gaussian constants to the denominator $\det D_{j+1}(0)$. This is why no additional normalization hypothesis is required.

9.5. **One pair on the reference volumes.** The auxiliary Gaussian operators on a smaller reference torus are defined by periodizing those on R_{top}, including the antiperiodic character. This choice gives exact local agreement of the normalized maps. It does not replace the physical operators: at $s = 1$, exact Gaussian operators intertwine the covering map, so their periodization is their actual value on the smaller torus.

Theorem 9.4 (Agreement on the full local support)**.** *Let $R, R' \in \mathcal{T}(e)$, $R \le R'$, and use the prescribed projected primitives. On every polymer whose lifted bounding box and a $6(2L+1)M$ collar inject into the smaller torus, the two normalized maps agree after lifting all their field indices, connections, parameter cubes and white coordinates. This agreement includes the scalar determinant correction, both recovery pieces, every contour, and every term of the connected cluster expansion. In particular all small-polymer relevant extractions agree. The initial data $E_0 = P_0 = 0$ therefore give the same mass and vacuum coordinates until the smaller stopping level.*

Appendix K.3 proves this statement. Row and column component support eliminate all nonzero covering classes before the kernel is compared. The scalar determinant traces and the Grassmann monomials are consequently compared on their whole supports. Absolute convergence permits termwise comparison of the connected sum.

Theorem 9.5 (Tuned pair and two-reference pressure)**.** *For the fixed scheme there is $e_1 > 0$ such that, if $e\sqrt{h'} \le e_1$, then for every sufficiently large N there is a real pair $P_N(e) = (m_{0,N}, \epsilon_{0,N})$, independent of the reference torus and of its harmonic twist. Its trajectory is defined on every $R \in \mathcal{T}(e)$ once $N \ge N_{\min}(R, e)$, and satisfies*

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|E_j^x\|_j \le 4 e_j^{\varepsilon} w_j^E, \qquad |\epsilon_*| \le 5 c_{\mathrm{fl}} e_*^{\varepsilon} w_*^E. \tag{9.20}$$

The terminal mass and vacuum coordinates vanish on R_{top}. The external-connection activity and scalar determinant correction are analytic on their declared domains. Every terminal external connection whose sum of the three norms is less than $e_*^{-1+3\varepsilon}$ rescales into $\mathcal{X}_j$ at all earlier levels.

Let ρ_n be the determinant expectation in the averaged law before normalizing it to a probability, with the selected native mass $\mu_N = \bar{m}\eta + m_{0,N}$ and $n = R/\eta$. Put

$$p_F = p^0(\bar{m}\eta) + \epsilon_{0,N}, \qquad p^0(u) = \frac{2}{(2\pi)^3} \int_{[-\pi,\pi]^3} \log \left[(u + \sum_\nu (1 - \cos p_\nu))^2 + \sum_\nu \sin^2 p_\nu \right] dp.$$

With the explicit decreasing function $\varepsilon_5(x) = 312 \log[(1 - \exp(-x/4))^{-1}]$, exponentially small as $x \to \infty$, one has

$$|\log \rho_n - p_F n^3| \le \varepsilon_5(\bar{m}R) + 3 e_*^{1/4-9\varepsilon}. \tag{9.21}$$

Consequently, on the two reference tori,

$$\left| L^3 \log \rho_{R^*/\eta} - \log \rho_{LR^*/\eta} \right| \le (L^3 + 1) \left[\varepsilon_5(\bar{m}R^*) + 3 e_K^{1/4-9\varepsilon} \right]. \tag{9.22}$$

Proof. On R_{top} impose $m_{j_{\mathrm{top}}} = 0$ and solve the mass recursion backwards while solving the activity recursion forwards. In the normalized product window the fixed-point map has differences

$$b_j' \le \tfrac{1}{4} b_{j+1} + c_{\mathrm{fl}} e_j^{\varepsilon} a_j, \qquad a_j' \le \tfrac{1}{16} a_{j-1} + e_{j-1}^{\varepsilon} (a_{j-1} + b_{j-1}).$$

For e_1 small these give a contraction $5/16$. The determinant forcing cancels in the difference, since it is independent of the input trajectory. This produces the unique trajectory in the chosen product window. The vacuum recursion is a finite scalar telescope with terminal value zero. The stronger bounds (9.20) follow by inserting the fixed point in the same estimates. Theorem 9.4 carries it to the smaller reference volumes with the identical pair.

At the stop level fix a real gauge of the mean-zero block field. The bounded terminal curvature makes its fine minimizer small on the larger potential domain. A finite-volume Berezin perturbation estimate compares the terminal integrand with the free constant-twist coarse determinant, with error $O(e_*^{1/4-9\varepsilon})$. The Gaussian domination of Theorem 4.1 bounds every excluded white coordinate and terminal curvature by its exact one-dimensional Gaussian tail. The union of these tails is summable over the ultraviolet levels, since the thresholds grow as powers of v_j. The retained scalar normalizers turn this comparison into a bound for ρ_n.

The constant-twist free determinant is a product of its Fourier eigenvalues. Periodization of the exponentially decaying Fourier coefficients of its logarithm gives

$$|\log \rho_n^{\mathrm{free}} - p^0(\bar{m}\eta)n^3| \le \varepsilon_5(\bar{m}R).$$

The complete estimates, including harmonic averaging and all scalar terms, are proved in Appendix L. They yield (9.21). The extensive term $p_F n^3$ cancels exactly in the difference between R^* and LR^*, which proves (9.22). □

The theorem selects a torus-independent bare pair; it does not assert $\mu_N > 0$. Indeed its bound on $m_{0,N}$ is larger than $\bar{m}\eta$ at sufficiently fine mesh. The positivity needed for the averaged/all-flux laws is supplied by their four-spin paired determinant and by the finite-model argument, not by a positive selected mass assumption. Nothing here identifies the pinned zero-harmonic law with the averaged law.

9.6. **The terminal interface for external probes.** The next section uses only the following consequences of the flow. At the reference stop level the positive reference mass belongs to the fixed interval in (9.4). All constants for fine fermion kernels and complete external legs may therefore be chosen independently of e, N and the placement of a compact probe. For $A \in \widetilde{\mathcal{G}}_*$ the inverse of $D_A + \bar{m}a_* + b_* Q_*^T(-A)Q_*(A)$ is analytic and has the fine pointwise and unit-block bounds of Theorem 9.1. Its coarse Gaussian covariance and the two physical legs have uniformly summable exponential kernels. The activity with an external slot satisfies (9.20); its source-free relevant coordinates are the selected m_*, ϵ_*.

The scalar determinant correction obeys (9.19) literally and is the sum of local even determinant differences. Its physical value, multiplied by the terminal coarse determinant and the exact free normalization constants, is the original fine determinant ratio. This identity holds on an open complex external-connection domain. It therefore also holds coefficientwise in a ring of finite jets, with one independent jet variable for each source cell. Taking jets introduces no new ultraviolet normalizer and does not alter the selected pair. Source dependence of the terminal Berezin integral is retained rather than evaluated against the source-free measure.

Finally, the photon hierarchical coordinates are orthogonal coordinates of the original curvature Gaussian space. A new fluctuation white coordinate is a norm-one linear curvature functional; a terminal plaquette curvature is a functional of norm at most one. The count at level j is at most $2l_*^3 L^{3(j_*-j)}$ white coordinates and $3l_*^3$ terminal plaquettes. These facts, proved in Appendix G.6, are the only Gaussian coordinate inputs needed for the later large-field jet estimate. The common reference partition defect is (9.22); the later volume comparison will use it with the fixed number $N^* = 8L^3$ of reflected cells.

10. A compact probe

Gaussian domination gives a compact family of curvature laws, but does not distinguish an interacting limit from the free Maxwell law. We use a compact exact test to make that distinction. The coefficient is a massive fermion loop. Its sign is not fixed for an arbitrary test; a dilation makes its local constant-field contribution dominant and produces a test with positive coefficient.

For a real compact two-form β, put

$$a = d^*\beta, \qquad f = da = dd^*\beta, \qquad S_m = (m + \gamma\cdot\partial)^{-1}, \qquad b(f) = -6\,\mathrm{Tr}(S_{\bar m} i\gamma\cdot a)^4.$$

The definition is independent of the choice of β giving f: a compact closed and coclosed one-form is harmonic and hence vanishes. The product in the trace is trace class. We use four spin components, as in the finite model.

Theorem 10.1 (The free coefficient)**.** *For real $a \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,*

$$-\frac{1}{4}\,\mathrm{Tr}(S_m i\gamma\cdot a)^4 = \frac{1}{240\pi m^5}\int |\operatorname{curl} a|^4 + O_a(m^{-7}) \quad (m \ge 1).$$

Consequently, for every $0 < \bar m \le 1$ there is a real compact two-form β with $b(dd^*\beta) > 0$. If a_η is the compatible lattice probe constructed in Appendix M, and

$$b_{\eta,R} = \partial_t^4 \log \frac{\det(D[t\eta a_\eta] + \bar m\eta)}{\det(D[0] + \bar m\eta)}\Big|_{t=0},$$

then, for $R \ge D_\beta + 1$,

$$|b_{\eta,R} - b_R(f)| \le C_\beta \eta^{1/2}, \qquad |b_R(f) - b(f)| \le C_\beta e^{-\bar m(R-D_\beta)}.$$

Here b_R uses the antiperiodic continuum inverse on the torus; the constants are independent of η and R.

The proof, including the Wilson contact terms, is given in Appendices N–O. For a concrete probe let $\rho(x) = \exp[-(1-|x|^2)^{-1}]$ in the unit ball and zero outside, and take $a = (\partial_2\rho, -\partial_1\rho, 0)$. If $\beta_{12}^{(l)}(x) = l^2\rho(x/l)$, with other components zero, then $d^*\beta^{(l)} = la(x/l)$ and

$$b(dd^*\beta^{(l)}) = \frac{l^3}{10\pi\bar m^5}\int |\operatorname{curl} a|^4 + O_a(l\bar m^{-7}).$$

The integral is positive, so a sufficiently large fixed l works. The probe is fixed before the charge is made small.

11. Two reference tori and normalized source jets

The selected mass is obtained from the RG construction, rather than adjusted separately in each surrounding volume. Theorem 9.5, together with Theorem 9.4, gives one pair at each fine cutoff and two reference tori of sides $R^*(e)$ and $LR^*(e)$. Their relative pressure defect is bounded. The reference side grows only as a power of $\log e^{-1}$. We estimate sources on the smaller torus and use positivity to transfer those estimates to every larger target torus.

Divide the smaller reference torus into $n_c = 8L^3$ mirror cells of physical side $R^*/(2L)$. Place the fixed probe in the central three quarters of one cell and make its reflected copies in the other cells. Let X_c be the corresponding curvature readouts, and let v_η be their common free variance. Their supporting plaquettes are disjoint. With

$$H_2(x;v) = x^2 - v, \qquad H_4(x;v) = x^4 - 6vx^2 + 3v^2,$$

write

$$N_0(t) = \frac{\det(D[t\eta a_\eta] + \bar m\eta)}{\det(D[0] + \bar m\eta)}, \qquad c_m = N_0^{(m)}(0) \quad (m = 2, 4).$$

Thus $c_4 - 3c_2^2 = b_{\eta,R^*}$. The reference comparison is between normalized source ratios, not between unnormalized determinants.

Theorem 11.1 (Reference-cell jets)**.** *Fix $s \in (0,1)$ and sufficiently small $\epsilon > 0$ as in the RG parameter choice, and put*

$$\alpha_{\rm jet} = s(1/4 - 9\epsilon) - 12\epsilon > 0.$$

For sufficiently small charge, independently of the fine cutoff, the selected mass satisfies

$$\Theta_4^0 \le C_\beta e^{\alpha_{\rm jet}}, \qquad (1+|c_2|)\Theta_2^0 \le C_\beta e^{\alpha_{\rm jet}}, \qquad |c_2| + |c_4|^{1/2} \le C_\beta e^{-6\epsilon}.$$

Here

$$(\Theta_m^0)^{n_c} = E^{\rm av}_{\eta,R^*} \prod_c \{e^{-m} H_m(X_c; v_\eta) - c_m\} \ge 0.$$

The constants depend on the scheme, $\bar m$ and the fixed probe, but not on the cutoff.

Appendix P proves this theorem. Its main mechanism is simple to state. The source ratio is expanded in the finite algebra $\mathbb{C}[t_c : c]/(t_c^5 : c)$. Its free part is a product of one-cell ratios. The interacting corrections are organized by the cells their full connection supports meet. A correction meeting just one cell is small; a correction meeting two separated cells is exponentially small in their separation. Applying $\partial_{t_c}^m - c_m$ kills an uncovered free factor. Only families covering all cells remain. The covering sum is therefore small to a power per cell.

The construction keeps the actual terminal Grassmann integral and its vacuum normalizer. On the small-field set the terminal integral is close to one in an absolute coefficient norm. The harmonic twist enters through a periodic analytic tube; its oscillation is exponentially small in the terminal side. The complement is controlled by Gaussian domination of the actual curvature law. A finite-dimensional two-constants estimate transfers that real bound to the small complex source polydisc used for the jets. None of these steps replaces the interacting photon law by a Gaussian law.

12. Transfer to every target volume

The all-flux transfer representation has a positive operator independent of the time length; this is the content of A.3. It gives separate convexity of the logarithm of the partition function in the three integer side multipliers. A second consequence is that a full mirror product does not increase under lengthening one side, apart from its natural power. Normalized expectations require both facts: the partition-function defect must be paid.

Let Z_k and $\bar Z_k$ be the all-flux and zero-flux partition functions on side kR^*, at the same native mass and charge, and set

$$D_L = L^3 \log Z_1 - \log Z_L, \quad \bar D_L = L^3 \log Z_1 - \log \bar Z_L, \quad 0 \le D_L \le \bar D_L.$$

The matter part of this defect is bounded by the two-reference RG theorem. The photon and flux terms are evaluated explicitly in Appendix Q; in particular

$$\frac{\bar D_L}{(L-1)^3 n_c} \le \frac{a_L}{n_c} \log \frac{2\pi}{e^2 R^*} + C_L, \qquad a_L = \frac{L^3-1}{(L-1)^3}.$$

Theorem 12.1 (Normalized volume transfer)**.** *Let $k = 1$ or an integer $k \ge L$. For a set J of distinct mirror cells on the target torus and $m = 2, 4$, put $O_{m,c} = e^{-m} H_m(X_c; v_\eta) - c_m$. Then*

$$\left| E^{\rm nat}_{\eta,kR^*} \prod_{c \in J} O_{m,c} \right| \le (\Gamma_* \Theta_m)^{|J|}, \qquad {\rm nat} \in \{{\rm av}, {\rm af}\},$$

where a common choice for the two laws is

$$\Gamma_* = \exp\left\{ \frac{(L^3-1)\log(2\pi/(e^2R^*)) + K_L + D_L^{\rm mat} + 1}{(L-1)^3 n_c} \right\}, \qquad \Theta_m = \Theta_m^0 + 2e^6 \dot M_m.$$

Here $D_L^{\mathrm{mat}} = L^3 \log \rho_n - \log \rho_{Ln}$ *is the normalized matter defect,* K_L *is the finite Laplacian constant in Lemma Q.2, and*

$$\dot{M}_4 = 600 n_c^2 v_\eta^2 e^{-4} + |c_4|, \qquad \dot{M}_2 = 12 n_c v_\eta e^{-2} + |c_2|.$$

It is enough to impose $6 n_c e^2 R^* \log e^{-1} \le \pi^2$.

The proof is given in Appendix R. All-flux transfer uses the full positive trace directly. For the averaged law, three successive Fourier positivity reductions show that its unnormalized mirror product is at most the full mirror product. Separate convexity of the zero-sector partition then pays its own normalizer through $\bar{D}_L$. The reference flux tail is paid by actual high moments in both cases. The constant Γ_* grows at most as Ce^{-2a_L/n_c}, a small fixed loss because $n_c = 8L^3$.

13. Cumulants in limits and in their components

Choose the parameter window so that

$$\sigma = \alpha_{\mathrm{jet}} - \frac{2a_L}{n_c} > 0, \qquad \alpha_{\mathrm{jet}} = s(1/4 - 9\epsilon) - 12\epsilon.$$

For example, $s = 1/2$, $0 < \epsilon \le 10^{-4}$, and odd $L \ge 5$ already leave a positive margin; increasing L and decreasing ϵ as required by the RG estimates preserves it. The threshold e is chosen after all these constants.

Theorem 13.1 (Uniform probe cumulants)**.** *For the selected, target-volume-independent mass and a fixed compact probe, there are* $C_\beta < \infty$, $e_\beta > 0$ *and* $\sigma > 0$ *such that, for* $0 < e \le e_\beta$, *all sufficiently fine cutoffs and all sufficiently large target volumes,*

$$|\kappa_4(X(f)) - e^4 b(f)| \le Ce^{4+\sigma} + C(\eta^2 + e^4 \eta^{1/2}), \qquad |\kappa_2(X(f)) - \|f\|_2^2| \le Ce^{2-6\epsilon} + C\eta.$$

Every cofinal limit point μ *consequently satisfies*

$$|\kappa_4^\mu(f) - e^4 b(f)| \le Ce^{4+\sigma}.$$

For almost every translation-ergodic component μ_ξ *of each such limit point, the same estimate holds with* μ_ξ *in place of* μ. *If* $b(f) > 0$, *reducing* e_β *gives* $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$, *and therefore these components, and the limit law itself, are non-Gaussian.*

For completeness, the component step uses the product assertion in Theorem 12.1, not just its one-cell case. At each fixed order it passes to the limit by 5.1. The bounds hold for every collection of distinct translates in the reference cell lattice. Lemma 7.1 then bounds the invariant conditional Hermite statistics almost surely. Conditional second and fourth statistics give the component cumulant; no ergodicity of this lattice subgroup is needed. Appendix S gives the normalization and subsequence details. The mass assignment is independent of the chosen probe; only the small-charge ceiling used to detect its cumulant depends on the probe.

14. Proof of the construction theorem

The estimates now give the theorem on the first page. We spell out the order of choices because a volume-dependent mass or a probe-dependent trajectory would give a different assertion.

Proof of Theorem A. For part (i), fix the charge and the volume-independent native masses with $\mu_N > -1/2$. Theorems 4.1 and 4.2 apply at every finite cutoff. Neither their Gaussian bound nor their reflection-positive form contains a constant depending on N, T, the charge, or the chosen admissible mass. The physical test map identifies each law with a Borel probability on H_{-4}. The uniform H_{-3} second-moment bound in Theorem 5.1 and the compact inclusion $H_{-3} \hookrightarrow H_{-4}$ make these probabilities tight. Prokhorov's theorem therefore gives a convergent subsequence of each cofinal sequence.

Discrete closedness passes to distributional closedness. Lattice translations approximating a prescribed continuum translation, and the exact cubic symmetries, pass to the same limit. Real Laplace transforms are uniformly bounded by the Gaussian estimate, so the finite-dimensional generating functions converge locally uniformly in their complex variables. This proves convergence of every fixed joint moment. The generating function is nonzero in a sufficiently small neighborhood of the origin; derivatives of its logarithm give convergence of joint cumulants. The resulting law has the Maxwell Laplace bound, and its Hermite expansion gives the tensor-distribution growth estimate stated in Theorem 5.1. Positive-half tests with support strictly inside the half-space transfer the finite reflection inequalities to every coordinate plane. Density then gives the full continuum positive-half algebra. The reconstruction in Theorem 6.1 supplies each positive coordinate Hamiltonian.

For part (ii), choose the RG scheme once, before the probe or target volume: an odd L, the block sizes and collars, and the exponents s, ϵ so that

$$\alpha_{\mathrm{jet}} = s(1/4 - 9\epsilon) - 12\epsilon > \frac{2a_L}{n_c}, \qquad n_c = 8L^3, \qquad a_L = \frac{L^3 - 1}{(L-1)^3}.$$

The positive number $\sigma = \alpha_{\mathrm{jet}} - 2a_L/n_c$ is fixed by this choice. Shrink the common charge ceiling to satisfy the hypotheses of Theorem 9.5. The backward mass equation and forward activity equation give the pair $P_N(e) = (m_{0,N}(e), \epsilon_{0,N}(e))$ on the largest reference torus. Theorem 9.4 gives the identical pair on both tori used below. The mass coordinate, rather than a new solution of that equation in each target volume, defines the mass assignment of the finite laws. At the finitely many meshes preceding its definition, set $m_{0,N} = 0$. For the later meshes, the bound in (9.20) gives $\mu_N > -1/2$. Thus part (i) applies to this assignment. The vacuum coordinate is retained in the reference partition functions, although it cancels from each normalized probability.

Fix a compact real β, put $a = d^*\beta$, $f = da$, and fit its discretization into one reference cell. The side $R^*(e)$ tends to infinity as $e \downarrow 0$, so this requires only a probe-dependent reduction of the charge ceiling. It does not change $P_N(e)$. Apply Theorem 11.1 to the second and fourth centered Hermite statistics. Its error is $O(e^{\alpha_{\mathrm{jet}}})$ after taking the n_cth root. The reference partition defect is controlled by (9.22), with the same pair and the same normalizers.

Theorem 12.1 carries these statistics to every larger target torus of the specified cutoff family, in both regulators. Its pressure cost is at most Ce^{-2a_L/n_c}. The reference flux-tail terms have higher powers of e in the chosen exponent window. Consequently the transferred fourth statistic, and the second statistic multiplied by $1 + |c_2|$, are bounded by Ce^{σ} at every product order and every collection of distinct cells. The one-cell consequence and the algebraic Hermite cancellation (S.4), valid for both regulators, give

$$\kappa_4(X(f)) = e^4(c_4 - 3c_2^2) + O_\beta(e^{4+\sigma}) + O_\beta(\eta^2).$$

The discrete test correction and the Wilson comparison in Theorem 10.1 replace $c_4 - 3c_2^2$ by $b(f)$, with the additional error $O_\beta(\eta^{1/2})$. These are exactly the uniform estimates of Theorem 13.1.

For a fixed charge in this window, both reference sides are fixed. Every cofinal sequence eventually has sufficiently fine mesh and a target side exceeding the larger reference side. Part (i) passes the fixed fourth cumulant to every selected limit point and removes the mesh errors. This proves the first assertion of part (ii).

For the component assertion, first pass the distinct-cell product estimate to the limit at each fixed order. The moment bounds in Theorem 5.1 justify this passage for its polynomial statistics. Lemma 7.1, applied separately to the second and fourth centered Hermite statistics, bounds their conditional expectations on the translation-invariant sigma-algebra. Theorem 6.1 makes the component laws centered and supplies their moments. Converting their conditional Hermite statistics to the fourth cumulant proves the same error estimate almost surely, as detailed in Appendix S.

Finally Theorem 10.1 supplies a compact exact probe with $b(f) > 0$. For any such probe shrink e_β until $C_\beta e_\beta^\sigma \le b(f)/2$. The fourth cumulant is then positive in the limit law and in almost every

component. A Gaussian law has vanishing fourth cumulants, so both non-Gaussian conclusions follow. □

14.1. **What the theorem leaves open.** The theorem constructs curvature laws with a prescribed lattice regulator and proves a uniform non-Gaussian cumulant estimate. Compactness gives limit points for every cofinal sequence; it does not identify those limit points with one another. The moment determinacy of each individual law does not supply that identification.

The coordinate reconstruction uses reflection positivity and translations in a chosen coordinate direction. Full rotational covariance would require an additional comparison of directions beyond the cubic symmetries retained by the regulator. Likewise the ergodic component conclusions proved here do not give arbitrary-direction clustering of every higher correlation. No charged fermion hierarchy is constructed by the curvature compactness argument.

The pinned-harmonic comparison of Corollary 8.2 has its own positive-mass hypothesis and its stated iterated order of limits. The selected counterterm is not shown to satisfy that extra hypothesis. The averaged and all-flux laws in Theorem A therefore remain the regulators of the unconditional construction.

14.2. **Notation used across the proof.**

Symbol	Meaning
$\eta = L^{-N}$, $R = hL^T$	physical mesh and torus side
$q = e\sqrt{\eta}$, $\mu_N = \bar{m}\eta + m_{0,N}$	native charge and mass
F^u, $\eta^{-3/2}F^u$, $F^\sharp$	native, physical, and embedded curvature
E^{av}, E^{af}	normalized averaged and all-flux laws
C_0, H_s	Maxwell covariance and Hermite Hilbert scale
$a = d^*\beta$, $f = da$, $b(f)$	potential probe, curvature test, free quartic coefficient
a_j, e_j, $\mathfrak{m}_j$	scale mesh, charge, and reference mass
Q_j, H_j^P, H_j^F	block average, photon and fermion minimizers
S_j, Γ_j	fine and coarse fermion inverses
E_j, m_j, ϵ_j	activity and relevant mass and vacuum coordinates
$P_N(e)$, R^*, n_c	selected bare pair, reference side, reflected-cell count

Appendix A. The finite Wilson and photon transfer calculations

A.1. **The Wilson determinant on a cycle.** We work at native spacing one and distinguish direction 0 as time. The cross-section Σ has $v = n_1 n_2$ sites and $l = 2v$ spatial links. Spatial fermion characters are fixed throughout this calculation. Let $\mathcal{V}$ be the two-spinor space on Σ, of dimension $r = 2v$, and let $\mathcal{K} = \mathcal{V} \oplus \mathcal{V}$. After a constant spin change of basis, the time gamma matrix is $\mathrm{diag}(1, -1)$. Write A_t for the spatial links in slice t and a_t for its forward temporal links. The spatial unitary shifts are $\tau_i(A)$ and the temporal phase is $u(a)\phi(x) = e^{iqa(x)}\phi(x)$. The two spatial parts of the Wilson operator are

$$B(A) = (3+\mu) - \tfrac{1}{2}\sum_{i=1}^{2}(\tau_i + \tau_i^*), \qquad C(A) = \tfrac{1}{2}\sum_{i=1}^{2} c_i(\tau_i - \tau_i^*),$$

where the fixed spin matrices c_i are the off-diagonal blocks of the spatial gamma matrices. The opposite block is $-C^*$. The unitary shifts give

$$1 + \mu \le B \le 5 + \mu, \qquad \|C\| \le 2.$$

For $\mu > -1$ define

$$S(A) = \begin{pmatrix} B^{-1} & -B^{-1}C \\ -C^*B^{-1} & B + C^*B^{-1}C \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -C^* & 1 \end{pmatrix} \begin{pmatrix} B^{-1} & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & -C \\ 0 & 1 \end{pmatrix} > 0. \tag{A.1}$$

Both S and S^{-1} are bounded uniformly over the spatial real fields on this fixed cross-section. They are continuous and depend only on the link phases. Positivity here uses $\mu > -1$, not positivity of the native mass.

Lemma A.1 (Exact cyclic factorization)**.** *Let the time character be χ_0, with $|\chi_0| = 1$. Put $g_t = (u(a_t)^* \oplus u(a_t)^*)S(A_t)$. Then the actual four-component determinant is*

$$\Delta^\chi(qA) = (-\chi_0)^r \prod_{t=0}^{n_0-1} \det B(A_t) \det u(a_t) \det_{\mathcal{K}}(1 - \overline{\chi_0} g_{n_0-1} \cdots g_0). \tag{A.2}$$

The scalar determinants in the product are on $\mathcal{V}$.

Proof. On $\mathcal{V}^{n_0}$ let $\mathbb{B}, \mathbb{C}$ be the diagonal slice operators and let $(\mathbb{T}\psi)_t = u(a_t)\psi_{t+1}$, with $\psi_{n_0} = \chi_0\psi_0$. The Wilson matrix is

$$\mathbb{D} = \begin{pmatrix} \mathbb{B} - \mathbb{T}^* & \mathbb{C} \\ -\mathbb{C}^* & \mathbb{B} - \mathbb{T} \end{pmatrix}.$$

Write $\widehat{\mathbb{T}} = \mathbb{T} \oplus \mathbb{T}$, $\mathbb{S} = \mathrm{diag}(S(A_t))$, and $\mathbb{N} = \left(\begin{smallmatrix} \mathbb{B} & 0 \\ -\mathbb{C}^* & -1 \end{smallmatrix}\right)$. Block multiplication gives

$$\mathbb{D}(\mathbb{T} \oplus 1) = \mathbb{N}\widehat{\mathbb{T}}(1 - \widehat{\mathbb{T}}^*\mathbb{S}).$$

Thus

$$\det \mathbb{D} = (-1)^{n_0 r} \det \mathbb{B} \det \mathbb{T} \det(1 - \widehat{\mathbb{T}}^*\mathbb{S}).$$

The determinant of the cyclic shift is $\det \mathbb{T} = ((-1)^{n_0-1}\chi_0)^r \prod_t \det u(a_t)$. For a backward cyclic operator W with step matrices g_t and one seam factor $\overline{\chi_0}$, $\det(1 - W) = \det(1 - \overline{\chi_0} g_{n_0-1} \cdots g_0)$. One proof is to expand $\log\det(1 - zW)$ near zero: traces vanish unless the power is a multiple of n_0, and the remaining traces are n_0 times the trace of the corresponding monodromy power. The resulting determinant identity extends to all z because both sides are polynomials. Substitution gives (A.2), including its sign and all scalar factors. □

Let $\mathcal{F} = \bigoplus_{p=0}^{2r} \bigwedge^p \mathcal{K}$. For an operator L on $\mathcal{K}$, write $\mathfrak{G}(L) = \bigoplus_p \bigwedge^p L$. The identities

$$\mathfrak{G}(L_1 L_2) = \mathfrak{G}(L_1)\mathfrak{G}(L_2), \quad \mathfrak{G}(L^*) = \mathfrak{G}(L)^*, \quad \mathrm{Tr}\,\mathfrak{G}(L) = \det(1 + L)$$

follow respectively from functoriality, the induced exterior inner product, and expansion of the determinant by principal minors. In particular $L \geq 0$ implies $\mathfrak{G}(L) \geq 0$. At each site let N_x count its four Fock occupations and put $Q_x^{\mathrm{ch}} = 2 - N_x$. These self-adjoint operators commute and have integer spectra. Set

$$\Gamma(a) = \exp(iq \sum_x a(x) Q_x^{\mathrm{ch}}), \qquad Q_{\mathrm{tot}} = r - N_{\mathrm{tot}}.$$

The scalar $\det u(a) = e^{2iq\sum_x a(x)}$ is precisely what changes the occupation phase $-N_x$ into the charge $2 - N_x$. It must not be discarded when converting a determinant into a Fock trace.

Writing $\chi_0 = -e^{i\theta}$, Lemma A.1 therefore gives

$$\Delta^\chi(qA) = \mathrm{Tr}_{\mathcal{F}} \left[e^{i\theta Q_{\mathrm{tot}}} \prod_{t=n_0-1}^{0} \Gamma(a_t) \det B(A_t) \mathfrak{G}(S(A_t)) \right]. \tag{A.3}$$

The ordered product has later times to the left. The global number operator commutes with all its factors, so cyclic relabelling is legitimate. With antiperiodic time character the first factor is one. For that case define the positive slice data

$$s(A) = \det B(A)^{1/2}, \qquad R_s(A) = \mathfrak{G}(S(A))^{1/2}. \tag{A.4}$$

Cyclicity of the trace splits each positive slice factor into its two square roots:

$$\Delta(qA) = \mathrm{Tr}_{\mathcal{F}} \prod_{t=n_0-1}^{0} s(A_{t+1}) R_s(A_{t+1}) \Gamma(a_t) R_s(A_t) s(A_t). \tag{A.5}$$

The spaces and all data in this formula depend on the spatial cross-section, q, and μ, and not on the number of time slices.

For completeness, ordinary determinant positivity is independent of this transfer calculation. In the paired representation $\gamma_i = \mathrm{diag}(\sigma_i, -\sigma_i)$ the lower Pauli Wilson block is the adjoint of the upper one at every real field and every unitary fermion character. Thus $\Delta = |\det D_+|^2 \geq 0$. At a time-independent spatial field and zero temporal field, (A.2) with antiperiodic time gives $\Delta = (\det B)^{n_0} \det(1 + S^{n_0}) > 0$. Continuity makes the finite partition functions strictly positive on an open set; it is not necessary to assert that the determinant never vanishes.

The finite invariances can also be checked before integration. Translations conjugate the Wilson matrix by the corresponding fermion shift; their seam characters are preserved by this unitary conjugation. For a signed coordinate permutation r, the four-component gamma representation admits a unitary U_r with $U_r \gamma_i U_r^{-1} = \sum_j r_{ji} \gamma_j$. Proper permutations use the Pauli spin rotations in each block; a coordinate reflection is obtained by exchanging the two blocks and conjugating by its Pauli matrix. Pulling back sites and links and applying U_r therefore preserves the determinant, with the tensor action on curvature. All these maps preserve the Maxwell quadratic and the real or compact link measure, and biject closed integer cochains.

For charge conjugation use $\sigma_2 \overline{\sigma_i} \sigma_2 = -\sigma_i$ in the Pauli blocks. With real antiperiodic seam characters this gives $\sigma_2 \overline{D_+(qA)} \sigma_2 = D_-(-qA) = D_+(-qA)^*$, and hence $\Delta(-qA) = \Delta(qA)$. With general flat twists the same identity also reverses those twists; their Haar integral is unchanged. The changes $A \mapsto -A$ and, in the compact description, reduction of $-A$ modulo r_q with the compensating integer curvature shift, preserve the actual laws. These checks prove all the invariances used in Theorem 4.1.

A.2. **The real-link Proca parent and its mirrors.** For $\varepsilon > 0$ introduce, on the full real link space,

$$Z_\varepsilon(h) = \int dA \, e^{-\|dA+h\|^2/2 - \varepsilon \|A\|^2/2} \Delta(qA), \qquad h \in C^2.$$

The determinant is bounded because its entries depend on unit-modulus link phases on this finite lattice. These integrals are finite, continuous in h, nonnegative, and $Z_\varepsilon(0) > 0$. Split h into magnetic slice values h_t^m and electric layer values h_t^e. With the Fock data above put

$$W_t(A) = e^{-\|d_s A + h_t^m\|^2/2 - \varepsilon \|A\|^2/2} \det B(A) \mathfrak{G}(S(A)) \geq 0,$$
$$k_e(y) = \int_{\mathbb{R}^v} da \, e^{-\|y - d_s a + e\|^2/2 - \varepsilon \|a\|^2/2} \Gamma(a).$$

The symbol e in k_e is a layer source, not the physical charge. Using a different letter h^e when both occur avoids any ambiguity. The determinant formula gives the exact chain integral

$$Z_\varepsilon(h) = \int \prod_t dA_t \ \mathrm{Tr} \overleftarrow{\prod_t} k_{h_t^e}(A_{t+1} - A_t) W_t(A_t). \tag{A.6}$$

There is an integrable bound independent of all phases: each temporal integration is bounded by $(2\pi/\varepsilon)^{v/2}$, each slice matrix by a fixed multiple of $e^{-\varepsilon \|A_t\|^2/2}$. Consequently every interchange of the finitely many matrix operations and Gaussian integrations below is justified by absolute convergence.

Fourier transformation of the electric Gaussian gives

$$k_e(y) = \int_{\mathbb{R}^l} dp \, e^{i\langle p, y \rangle} e^{i \langle p, e \rangle} \varpi(p), \tag{A.7}$$

where

$$\varpi(p) = (2\pi)^{-l/2} (2\pi/\varepsilon)^{v/2} \exp\left[-\tfrac{1}{2} \|p\|^2 - \tfrac{1}{2\varepsilon} \sum_x (q Q_x^{\mathrm{ch}} - (d_s^* p)_x)^2 \right] > 0.$$

Indeed, in a common charge eigenbasis the remaining a integral is the Fourier transform of $e^{-\varepsilon\|a\|^2/2}$ at $qQ^{\mathrm{ch}} - d_s^* p$. This also proves $k_e(y)^* = k_{-e}(-y)$ and $\|\varpi(p)\| \le c_\varepsilon e^{-\|p\|^2/2}$. Only the factor $e^{i\langle p,e\rangle}$ contains the electric source, and it is unitary.

We describe the mirror inequality explicitly, since the opposite boundary on an odd torus has a different type from the first one. For $n_0 = 2m+1$ label slices $-m,\dots,m$, reflect $t \mapsto -t$, and cut the fixed electric layer between m and $-m$ by (A.7). The positive and negative half-paths are

$$\Phi_+(p, A_0) = \int \prod_{t=1}^{m} dA_t\, e^{-ip\cdot A_m} W_m(A_m) k_{h^e_{m-1}}(A_m - A_{m-1}) \cdots W_1(A_1) k_{h^e_0}(A_1 - A_0),$$

$$\Phi_-(A_0, p) = \int \prod_{t=1}^{m} dA_{-t}\, e^{ip\cdot A_{-m}} k_{h^e_{-1}}(A_0 - A_{-1}) W_{-1}(A_{-1}) \cdots k_{h^e_{-m}}(A_{-m+1} - A_{-m}) W_{-m}(A_{-m}).$$

There is one fixed site weight $W_0(A_0)$ and one fixed-layer phase. The chain becomes

$$Z_\varepsilon(h) = \int dp\, dA_0\ \mathrm{Tr}\big[e^{ip\cdot h^e_m} \varpi(p) \Phi_+ W_0 \Phi_-\big].$$

Put $X = \varpi^{1/2}\Phi_+ W_0^{1/2}$ and $Y = W_0^{1/2}\Phi_-\varpi^{1/2} e^{ip\cdot h^e_m}$. Hilbert–Schmidt Cauchy–Schwarz in $L^2(dp, dA_0)$ gives

$$Z_\varepsilon(h)^2 \le Z_\varepsilon(h^+) Z_\varepsilon(h^-). \tag{A.8}$$

Here h^+ keeps the positive half-source, reflects it to the negative half with the electric sign reversed, retains h^m_0, and sets $h^e_m = 0$; h^- is obtained from the negative half. To verify the identification, take the adjoint of Φ_+ and rename A_t as A_{-t}. The order of its factors reverses, the W factors remain self-adjoint, and $k_e(y)^* = k_{-e}(-y)$ produces exactly the reflected negative path. The two integrals of $\mathrm{Tr}\, XX^*$ and $\mathrm{Tr}\, Y^*Y$ are therefore the two mirrored chain integrals. The fixed-layer phase disappears from both squares.

For $n_0 = 2m$, reflect $t \mapsto 1 - t$. The two fixed electric layers are 0 and m, and there is no fixed site. Cutting both layers gives half-path matrices

$$\Phi_+(p', p) = \int \prod_{t=1}^{m} dA_t\, e^{-ip'\cdot A_m} W_m k_{m-1} \cdots k_1 W_1 e^{ip\cdot A_1},$$

$$\Phi_-(p, p') = \int \prod_{t=1-m}^{0} dA_t\, e^{-ip\cdot A_0} W_0 k_{-1} \cdots k_{1-m} W_{1-m} e^{ip'\cdot A_{1-m}}.$$

The abbreviations $k_t = k_{h^e_t}(A_{t+1} - A_t)$ and $W_t = W_t(A_t)$ apply only in these two formulas. The full chain is

$$\int dp\, dp'\ \mathrm{Tr}\big[e^{ip'\cdot h^e_m} \varpi(p') \Phi_+ e^{ip\cdot h^e_0} \varpi(p) \Phi_-\big].$$

Cauchy–Schwarz is applied to $\varpi(p')^{1/2}\Phi_+\varpi(p)^{1/2}$ and $\varpi(p)^{1/2}\Phi_-\varpi(p')^{1/2}$ with the two unitary phases attached to the latter at its ends. Taking adjoints and renaming $A_t \leftrightarrow A_{1-t}$ identifies the two squares with mirrored chains. Both fixed-layer sources are set to zero. This proves (A.8) also on an even torus. If a nontrivial temporal fermion character is retained, its factor $e^{i\theta Q_{\mathrm{tot}}}$ can be attached to an end; it is unitary and commutes with ϖ. The squared chains have the antiperiodic temporal character. The domination argument needs only the antiperiodic character in the reflected direction.

Lemma A.2 (Removal of the Proca source)**.** *For the antiperiodic four-component determinant, $Z_\varepsilon(h) \le Z_\varepsilon(0)$ for every real h.*

Proof. Let $C_\varepsilon = (d^*d + \varepsilon)^{-1}$. Completing the square in the integral with the determinant replaced by its supremum gives

$$Z_\varepsilon(h) \le C \exp\big[-\tfrac{1}{2}\langle h, (1 - dC_\varepsilon d^*)h\rangle\big] \le C e^{-\varepsilon\|h\|^2/(2(16+\varepsilon))}.$$

The last inequality follows from $1 - d(d^*d + \varepsilon)^{-1}d^* = \varepsilon(dd^* + \varepsilon)^{-1}$ and $\|d\|^2 \le 16$. The bound on d follows by applying Cauchy–Schwarz to the four links of each plaquette and using that each link belongs to four plaquettes. Dominated convergence gives continuity in h. Thus the maximum $M > 0$ is attained. Among maximizing sources choose one with the least number of nonzero plaquette coordinates.

If a coordinate is nonzero, choose as time one of its two directions and put that plaquette on a fixed layer. The two mirrored sources have values between zero and M, while (A.8) forces both values to equal M. Their numbers of nonzero coordinates have sum equal to twice the original number minus twice the number of nonzero fixed-layer coordinates. On an odd torus the fixed magnetic slice is retained in both copies and causes no change in this count. On an even torus both fixed electric layers are deleted. The chosen coordinate is deleted in either case, so one mirrored source has fewer nonzero coordinates. This contradicts minimality. The maximizing source is zero. □

The same calculations prove reflection positivity of the Proca curvature law. Insert a bounded positive-half curvature function P in Φ_+ before carrying out its integrations. Its reflected complex conjugate is exactly the adjoint path, since temporal links and electric curvatures change sign under reflection. On an odd torus the result is

$$\int dp\, dA_0\ \mathrm{Tr}\big[\varpi^{1/2}\Phi_+[P]W_0\Phi_+[P]^*\varpi^{1/2}\big] \ge 0;$$

on an even torus it is $\int dp\, dp' \|\varpi(p')^{1/2}\Phi_+[P]\varpi(p)^{1/2}\|_{\mathrm{HS}}^2$. Here $\|\cdot\|_{\mathrm{HS}}$ denotes the Hilbert–Schmidt norm. The support condition excludes dependence on the fixed electric layers; their Gaussian integrations are performed before the half-path insertion. Boundedness of P and the preceding Gaussian majorants justify these operations. Approximation extends the inequality to all square-integrable half-space functions for which the reflected product is defined.

A.3. **Removing the Proca mass.** The orthogonal Hodge decomposition is $C^1 = V_g \oplus V_h \oplus V_t$. Write $A = A_g + A_h + A_t$. Gauge invariance removes A_g from the determinant and the curvature, so its Gaussian integral cancels in normalized expectations. A harmonic field is specified by constants c_μ; its Gaussian norm is $|\Lambda| \sum_\mu c_\mu^2$. The determinant at $A_t + c$ is periodic in $\theta_\mu = q n_\mu c_\mu$, with period 2π, because adding one such period is a periodic lattice gauge transformation. Its Fourier series is finite: the determinant is a polynomial in the link phases. The normalized harmonic Gaussian has Fourier coefficient

$$\exp\left[-\frac{q^2}{2\varepsilon|\Lambda|}\sum_\mu n_\mu^2 k_\mu^2\right].$$

As $\varepsilon \downarrow 0$ it tends to one for $k = 0$ and zero otherwise. The same conclusion holds for each fixed spatial field, and the determinant's uniform bound permits dominated convergence under the transverse Maxwell Gaussian. Curvature insertions of growth $Ce^{C\|dA_t\|}$ are also dominated, since d is injective on the finite-dimensional space V_t.

Consequently the normalized Proca curvature expectations converge to

$$\frac{\int_{[-\pi,\pi]^3} d\theta \int_{V_t} dA_t\, e^{-\|dA_t\|^2/2}\Delta(q(A_t + H_\theta))G(dA_t)}{\int_{[-\pi,\pi]^3} d\theta \int_{V_t} dA_t\, e^{-\|dA_t\|^2/2}\Delta(q(A_t + H_\theta))}, \qquad (H_\theta)_\mu = \frac{\theta_\mu}{q n_\mu}.$$

This is $E^{\mathrm{av}}G$. Equivalently it is an average of normalized twisted laws with their positive partition functions as weights. The denominator is positive because the determinant is positive on an open set for at least one twist. Lemma A.2, applied to $G(F) = e^{-\langle h,F\rangle - \|h\|^2/2}$, passes to the limit and gives $E^{\mathrm{av}}e^{\langle g,F^u\rangle} \le e^{\|g\|^2/2}$. Since $F^u = dA_t$, replace g by $P_d g$ to obtain (4.1). The reflected squares pass to the same limit for bounded insertions; this proves the averaged part of Theorem 4.2 for odd site mirrors and even link mirrors. Site mirrors on even tori follow below from the zero-flux slab

trace. Together these arguments cover the coordinate mirrors needed here on rectangular tori of either parity.

A.4. **The all-flux slice space and its Bloch kernels.** Put $r_q = 2\pi/q$. A spatial slice consists of $x = (A, m) \in \mathcal{X} := [0, r_q)^l \times \mathbb{Z}^{\mathrm{plaq}(\Sigma)}$. Its magnetic curvature is $b(x) = d_s A + r_q m$. For consecutive slices, a temporal link field $a \in [0, r_q)^v$ and an integer electric field $c \in \mathbb{Z}^l$ give

$$E(x', x; a, c) = A' - A - d_s a + r_q c, \qquad m' = m + d_s c.$$

The latter constraint is exactly the closedness equation for the integer two-cochain on one layer.

The group $\mathbb{Z}^l$ acts on the extended slice space by $x \mapsto (A + r_q c, m - d_s c)$. For a Bloch angle $\vartheta \in \mathbb{R}^2$, the Hilbert space $\mathcal{H}_\vartheta$ consists of $\mathcal{F}$-valued functions satisfying

$$\Psi(A + r_q c, m - d_s c) = e^{-2\pi i \vartheta \cdot w(c)} \Psi(A, m), \qquad \|\Psi\|^2 = \int_{[0,r_q)^l} dA \sum_m \|\Psi(A, m)\|^2. \tag{A.9}$$

Here $w_i(c)$ is the sum of c along one fixed closed row in spatial direction i. This row sum is defined for every c, not just for closed c. Restriction to the indicated fundamental domain identifies $\mathcal{H}_\vartheta$ unitarily with $L^2([0, r_q)^l \times \mathbb{Z}^{\mathrm{plaq}}) \otimes \mathcal{F}$.

Translations $\tau_y \Psi(A, m) = \Psi(A - y, m)$ form a strongly continuous unitary representation of $\mathbb{R}^l$. The translated fundamental domain gives the same norm because the modulus in (A.9) is invariant. Let $E_b = -i\partial_{A_b}$ be its commuting self-adjoint generators, and put $\mathsf{H} = \exp(-\frac{1}{2}\sum_b E_b^2) \geq 0$. Fourier transformation of a Gaussian gives

$$\mathsf{H}\tau_z = \int_{\mathbb{R}^l} (2\pi)^{-l/2} e^{-\|y-z\|^2/2} \tau_y \, dy.$$

Unfolding the fundamental domain yields the kernel

$$\sum_{c \in \mathbb{Z}^l} \mathbf{1}[m' = m + d_s c] e^{2\pi i \vartheta \cdot w(c)} (2\pi)^{-l/2} e^{-\|A' - A + r_q c - z\|^2/2}.$$

The operators $U(a) = \Gamma(a)\tau_{d_s a}$ commute with these translations and form a unitary representation of $(\mathbb{R}/r_q\mathbb{Z})^v$. Indeed $\Gamma(r_q k) = 1$ because the charges are integers, and $w(d_s k) = 0$ for every periodic integer site function k. Normalized Haar averaging gives the orthogonal projection Π_G onto its invariant vectors:

$$\int_{[0,r_q)^v} U(a), da = r_q^v \Pi_G.$$

The integral is self-adjoint, idempotent after division by r_q^v, and acts as the identity exactly on those vectors. It commutes with H. Therefore

$$K_\vartheta = r_q^v \mathsf{H} \Pi_G \geq 0. \tag{A.10}$$

Its kernel is the preceding unfolded heat kernel with $z = d_s a$, integrated against $\Gamma(a)\, da$. A shift $E \mapsto E + h$ changes it to $K_\vartheta u_h$, where $u_h = \tau_{-h}$ is a unitary commuting with K_ϑ.

Define multiplication by

$$M(x) = s(A) e^{-\|b(x)\|^2/4} R_s(A), \qquad T_\vartheta = M^* K_\vartheta M, \qquad X_\vartheta = K_\vartheta^{1/2} M.$$

The factors s, R_s are bounded and periodic in the link phases; $b(A + r_q c, m - d_s c) = b(A, m)$. Thus M is well defined on $\mathcal{H}_\vartheta$. The one-step kernel is

$$\mathsf{T}_\vartheta(x', x; a, c) = \mathbf{1}[m' = m + d_s c] e^{2\pi i \vartheta \cdot w(c)} (2\pi)^{-l/2} e^{-\|E\|^2/2} M(x')^* \Gamma(a) M(x). \tag{A.11}$$

Its integral over a and sum over c is the kernel of T_ϑ.

For a bounded slab function P of $(E_0, \ldots, E_{s-1}; b_1, \ldots, b_{s-1})$, let $C_{s,\vartheta}[P]$ be the kernel obtained by multiplying s successive kernels (A.11), inserting P, integrating the $s-1$ interior slice variables

and the s temporal variables, and summing the s electric integer variables. The integer constraints are retained at every step. Its mirror is

$$P^\theta(E_0,\dots,E_{s-1};b_1,\dots,b_{s-1}) = \overline{P(-E_{s-1},\dots,-E_0;b_{s-1},\dots,b_1)}.$$

Proposition A.3 (Positive Bloch slab transfer)**.** *Every $C_{s,\vartheta}[P]$ is Hilbert–Schmidt and depends linearly on P. Moreover*

$$C_{s,\vartheta}[P^\theta] = C_{s,\vartheta}[P]^*, \qquad T_\vartheta = X_\vartheta^* X_\vartheta \ge 0, \qquad C_{s,\vartheta}[1] = T_\vartheta^s,$$

where X_ϑ is Hilbert–Schmidt and T_ϑ is trace class. Let $s_1+\cdots+s_b = n_0$ with $b \ge 2$, and read P_i on the consecutive slabs of these thicknesses. With $\kappa = (2\pi)^{l/2}$,

$$\kappa^{n_0}\operatorname{Tr}(C_{s_b,\vartheta}[P_b]\cdots C_{s_1,\vartheta}[P_1]) = \sum_w e^{2\pi i(\vartheta_1 w_{01}+\vartheta_2 w_{02})}\omega_n^{(w)}(\prod_i P_i). \tag{A.12}$$

The series is absolutely convergent. The spaces split orthogonally according to $w_{12} = \sum_p m_p$, each slab operator preserves those spaces, and (A.12) remains valid with a trace restricted to one spatial-flux summand and the same restriction on its series. All slice objects and κ are independent of n_0.

Proof. There is a fixed C such that the norm of the integrated one-step kernel is bounded by $Cu(x')u(x)$, where $u(x) = e^{-\|b(x)\|^2/4}$. To see this, drop the constraint and bound the electric sum by

$$\sup_y \sum_{c\in\mathbb{Z}^l} (2\pi)^{-l/2} e^{-\|y+r_q c\|^2/2} < \infty.$$

The a domain is compact and s, R_s are bounded. Furthermore $u \in L^2(\mathcal{X})$: $d_s A$ is bounded on the compact link box, and the sum over integer m is dominated by a translated finite-dimensional Gaussian. Iterating the bound gives $\|C_{s,\vartheta}[P](x',x)\| \le C_s\|P\|_\infty u(x')u(x)$, hence the Hilbert–Schmidt property and absolute convergence of all interior integrations.

Reversing the order of steps, exchanging endpoints, replacing a_j, c_j by their negatives, and relabelling interior variables reverses E_j and the Bloch phase. Since $\Gamma(a)^* = \Gamma(-a)$, the adjoint kernel is the kernel with P^θ. This proves the mirror identity without requiring positivity of individual matrix entries. Here the negative temporal variable is reduced modulo r_q back to its compact box. The resulting integral integer gauge shift is absorbed into c_j; its spatial gradient has zero row sum and leaves both the Bloch phase and the electric curvature unchanged. Thus the substitution is a measure-preserving bijection of the compact and integer integration variables.

Positivity of K_ϑ gives $T_\vartheta = X_\vartheta^* X_\vartheta$. For its trace-class property, $K_\vartheta \le r_q^v \mathsf{H}$. The half-heat kernel has Gaussian density $\pi^{-l/2}e^{-\|y\|^2}$. Unfolding its square gives a finite, constant heat diagonal

$$h_\vartheta(x,x) = (2\pi)^{-l/2}\sum_{d_s c=0} e^{2\pi i\vartheta\cdot w(c)} e^{-r_q^2\|c\|^2/2}.$$

Equivalently this is $\int |\mathsf{H}^{1/2}(x',x)|^2 dx'$, so it is nonnegative. Thus

$$\|X_\vartheta\|_{\mathrm{HS}}^2 \le r_q^v \|\mathsf{H}^{1/2}M\|_{\mathrm{HS}}^2 = r_q^v \int h_\vartheta(x,x)\operatorname{Tr}(M(x)^*M(x))\,dx < \infty.$$

The equality follows directly from the half-heat kernel and Tonelli; the last integrand is bounded by a constant times $u(x)^2$. For $P = 1$, composition of the one-step kernel gives $C_{s,\vartheta}[1] = T_\vartheta^s$.

The product in (A.12) has at least two Hilbert–Schmidt factors, so is trace class. Its trace is the integral of the cyclic product of their kernels. Expand each slab kernel into its individual steps. The Gaussian majorant just proved makes the resulting cyclic integral absolutely convergent: its scalar absolute value is bounded by a constant times $\prod_t u(x_t)^2$. Each step contributes $(2\pi)^{-l/2}$, giving κ^{-n_0}. Each slice's magnetic half-weight occurs twice, giving $e^{-\sum_t \|b_t\|^2/2}$. The Fock factors form exactly (A.5). The constraints say $dm = 0$ for the full integer two-cochain whose magnetic pieces

are m_t and electric pieces are c_t. The compact link variables range once over the full fundamental box, and each closed integer two-cochain is attained once. Their real curvature is $F^u = dA + r_q m$.

Finally $\sum_t w_i(c_t)$ is the integer electric flux through the coordinate two-torus in directions $0, i$. Hence the accumulated phase is the phase on the right of (A.12). Grouping the closed integer cochains by their three fluxes proves that identity. Absolute convergence also follows from positivity of each flux-sector measure and $\sum_w |\omega_n^{(w)}(\Phi)| \le \omega_n(|\Phi|)$. The sum of $d_s c$ over all spatial plaquettes is zero. Consequently each kernel preserves $\sum_p m_p$, and restricting the trace inserts the indicator of one spatial flux in this same cyclic integral. This proves the restricted identity. □

A.5. **All-flux domination and the zero-flux identification.** Let $Z(h) = \omega_n(e^{-\langle h, F^u\rangle - \|h\|^2/2})$. In the transfer construction use $M_t = s e^{-\|b+h_t^m\|^2/4} R_s$ and $X_t = K_0^{1/2} M_t$. The electric source changes K_0 to $K_0 u_{h_t^e}$, with a commuting unitary $u_{h_t^e}$. Cyclicity and (A.12) give

$$Z(h) = \kappa^{n_0} \operatorname{Tr} \overleftarrow{\prod_t} X_{t+1}^* u_{h_t^e} X_t = \kappa^{n_0} \operatorname{Tr} \overleftarrow{\prod_t} u_{h_t^e} X_t X_t^*.$$

Each X_t is Hilbert–Schmidt. Schatten Hölder with exponent $n_0 \ge 2$ gives

$$|Z(h)| \le \prod_t \left(\kappa^{n_0} \operatorname{Tr}(X_t X_t^*)^{n_0}\right)^{1/n_0}.$$

The factor indexed by t is Z with zero electric source and with the magnetic source h_t^m repeated in every time slice. On that source choose direction 1 as time. Its only nonzero component is in the pair $1, 2$, which is now electric; its new magnetic component is zero. Applying the same Hölder inequality in this direction bounds it by $Z(0)$. Hence $Z(h) \le Z(0)$ and (4.2) follows. This proof pays the exact factor κ^{n_0} in every term and applies to either parity of every side.

Corollary A.4 (Flux suppression)**.** *If h_w is the constant curvature with flux $r_q w$, then*

$$\frac{\omega_n^{(w)}(1)}{Z_n} \le e^{-\|h_w\|^2/2}, \qquad (h_w)_{ij} = \frac{r_q w_{ij}}{n_i n_j}.$$

Proof. In sector w, the pairing $\langle h_w, F^u\rangle$ is $\|h_w\|^2$, since the exact part is orthogonal to constants. The Gaussian bound at h_w and positivity of all other terms imply $e^{\|h_w\|^2} \omega_n^{(w)}(1)/Z_n \le e^{\|h_w\|^2/2}$. □

Zero flux is the averaged law, not the zero-harmonic law. Here is the exact identification. A closed integer two-cochain with zero flux is dm for an integer one-cochain; two such m give the same two-cochain precisely when their difference is closed. Translating the compact link box by $r_q m$ therefore unfolds the zero-flux integral to C^1/Γ, where $\Gamma = r_q\{m \in C^1(\mathbb{Z}) : dm = 0\}$. This quotient is compact in its gauge and harmonic directions and noncompact in its transverse direction. The determinant and curvature are invariant in the gauge direction, so its volume is an insertion-independent factor. The harmonic fundamental periods are $2\pi/(qn_i)$; a periodic integer cochain with unit period in direction i and its transverse sheets gives that generator. Orthogonal Hodge decomposition then gives another insertion-independent volume factor times

$$\int_{[-\pi,\pi]^3} d\theta \int_{V_t} dA\, e^{-\|dA\|^2/2} \Delta(q(A + H_\theta)) G(dA).$$

All factors cancel in the ratio. Thus $E_{n,0} = E_n^{\mathrm{av}}$. This also explains why the harmonic average can be recovered by the spatial-flux projection and the electric Bloch integral.

The integer assertion just used has no torsion or real-coefficient qualification. It follows by the cubical contraction in each coordinate circle: its cochain complex retracts, by integer successive differences along a chosen cut, onto one vertex and one edge, with zero differential. The discarded pairs are an edge and a vertex difference with coefficient one, so this elimination and its homotopy have integer entries. Tensoring the three contractions retracts the torus complex onto the exterior

algebra on its three circle edges. The degree-two coordinates of the retraction are exactly the sums over the three coordinate two-tori. The identity $1-\iota p = dK + Kd$ from the contraction therefore sends a closed degree-two cochain m with zero flux to dKm, with integer Km. In degree one it similarly identifies the three integral periods of closed one-cochains. This proves both the unfolding assertion and the claimed harmonic periods, without replacing the flux sum by a formal real Hodge decomposition.

A.6. **Boundary slices and reflected slab squares.**

Lemma A.5 (Removing end magnetic variables)**.** *Let $s \geq 2$. A bounded function of $(E_0, \dots, E_{s-1}; b_0, \dots, b_s)$ restricted to closed curvature is a slab function without the two end magnetic variables: substitute*

$$b_0 = b_1 - d_s E_0, \qquad b_s = b_{s-1} + d_s E_{s-1}.$$

This substitution commutes with the mirror operation and preserves boundedness and measurability. Consequently its slab insertion has the adjoint identity in Proposition A.3.

Proof. The lattice closedness equation is $b_{t+1} - b_t = d_s E_t$. This follows by writing the three signed differences in $(dF)_{012} = 0$. The two displayed identities are its first and last instances. Under reflection the electric variables are reversed and negated and the magnetic variables are reversed. Substitution of the two identities before or after this operation therefore gives the same function. The substitution is linear, so boundedness and Borel measurability are unchanged. □

Consider first a site reflection on an odd torus $n_0 = 2m+1$, with positive slices $0, \dots, m$. A bounded curvature function P in this half contains no magnetic observable at the fixed slice 0. It may contain magnetic curvature at m; if $m \geq 2$, use Lemma A.5 to express it as an insertion in $C_{m,\vartheta}[P]$. The reflected half gives the adjoint operator, and the unoccupied layer between m and $-m$ contributes $T_\vartheta = X_\vartheta^* X_\vartheta$. Therefore, in the full Bloch space or in one spatial-flux summand,

$$\omega_n^{\vartheta,\Pi}(\overline{P \circ r^*}P) = \kappa^{n_0} \| X_\vartheta \Pi C_{m,\vartheta}[P] \|_{\mathrm{HS}}^2 \geq 0. \tag{A.13}$$

Here Π is the identity or the orthogonal spatial-flux projection. It commutes with all the displayed operators. The small case $m = 1$ has no interior magnetic slice; the strict-half-space support used in Theorem 4.2 excludes end magnetic variables, so the insertion is already $C_{1,\vartheta}[P]$ and the same square applies. For an even torus the two boundaries are site slices and the support restriction removes both magnetic end variables. There is no unoccupied layer, and the square is

$$\omega_n^{\vartheta,\Pi}(\overline{P \circ r^*}P) = \kappa^{n_0} \| \Pi C_{n_0/2,\vartheta}[P] \|_{\mathrm{HS}}^2.$$

These are cyclic trace identities, not assertions about a positive pointwise determinant kernel.

At $\vartheta = 0$ and $\Pi = 1$ the functional is ω_n. At fixed spatial flux zero, integration over $[0,1]^2$ selects zero electric flux, by the absolutely convergent series (A.12). Its integral has total volume one. Both squares remain nonnegative under this integral. Division by the strictly positive corresponding partition function proves Theorem 4.2. The tensor electric sign in these formulas is the sign from $E_t \mapsto -E_{s-1-t}$; it is not an additional convention imposed on the curvature law.

Appendix B. Readout estimates, compactness and tensor regularity

B.1. **Cell averages and periodization.**

The native bound applies to the actual source readout because, in counting units, its source is $g = \eta^{3/2} f_\eta$. Hence $\|g\|^2 = \eta^3 \sum |f_\eta|^2 = \|f_\eta\|_\eta^2$, with no volume factor left over. We verify the bounds on this last norm. For a fundamental cube $Q_R = [-R/2, R/2)^3$, let $f_R(x) = \sum_{k \in \mathbb{Z}^3} f(x + Rk)$. The cubes centered at the plaquette centers tile the torus separately for each component. Jensen's inequality on each cube gives

$$\|f_\eta\|_\eta \leq \|f_R\|_{L^2(Q_R)}.$$

If the support of f fits in one fundamental cube, the terms of the periodization have disjoint interiors. Consequently the right side is $\|f\|_2$. For a general Schwartz test, the triangle inequality and Cauchy–Schwarz give

$$\begin{aligned}\|f_R\|_{L^2(Q_R)} &\le \|f\|_{L^2(Q_R)} + \sum_{k\neq 0} \|f\|_{L^2(Q_R+Rk)} \\ &\le \|f\|_2 + 4R^{-2}\left(\sum_{k\neq 0} |k|_\infty^{-4}\right)^{1/2} \||x|^2 f\|_{L^2(Q_R^c)}.\end{aligned}$$

We used $|x| \ge R|k|_\infty/2$ on $Q_R + Rk$. There are $24j^2 + 2$ points with $|k|_\infty = j$, so the sum is $24\zeta(2) + 2\zeta(4) = 4\pi^2 + \pi^4/45$; its square root times four is less than 26. This proves the stated p_R bound, also for vector tests. Its tail term tends to zero as $R \to \infty$. For $R \ge 1$ it is bounded by p_1.

Here and below the Hermite operator is $\mathsf{A} = 1 + |x|^2 - \Delta$, so its eigenvalues are $\lambda_\kappa = 2|k| + 4$. With $M = 1 + |x|^2$ and $P_i = -i\partial_i$, integration by parts yields

$$\|\mathsf{A}u\|_2^2 = \|Mu\|_2^2 + \|\Delta u\|_2^2 + 2\sum_i \langle P_i u, MP_i u\rangle - 6\|u\|_2^2.$$

Since $\mathsf{A} \ge 4$, $\|Mu\|_2^2 \le (11/8)\|\mathsf{A}u\|_2^2$. Applied to a normalized Hermite function, this gives $p_1(h_\kappa) \le 1 + 26\sqrt{11/8}\lambda_\kappa \le 31\lambda_\kappa$. The multiplicity of total Hermite degree j is $3(j+1)(j+2)/2$, including the three two-form components. Thus

$$Z_4 := \sum_\kappa \lambda_\kappa^{-4} = \frac{3}{32}\sum_{j=0}^{\infty} \frac{j+1}{(j+2)^3} = \frac{3}{32}(\zeta(2) - \zeta(3)) < \frac{1}{16}. \tag{B.1}$$

More generally $Z_{2s} < \infty$ precisely when $s > 3/2$.

B.2. **A common compact space.** The variance bound in Theorem 4.1, obtained by twice differentiating at zero, gives $E|F^\sharp(h_\kappa)|^2 \le 961\lambda_\kappa^2$. Summing its nonnegative terms with weights λ_κ^{-6} gives $E\|F^\sharp\|_{-3}^2 \le 961 Z_4 < 61$. For a deterministic lattice field the same computation uses $|X(f)| \le \|f_\eta\|_\eta \|F\|_\eta$ and gives $\|F^\sharp\|_{-3} \le 8\|F\|_\eta$. In particular the finite readout map is a continuous finite-dimensional map into H_{-3}; its pushforward is a Borel probability law.

We give the compactness argument to specify the topology of the limit. In a ball $K_b = \{\|F\|_{-3} \le b\}$ every Hermite coordinate is bounded. A diagonal subsequence makes each coordinate converge to a sequence t_κ. Fatou gives $\sum \lambda_\kappa^{-6}|t_\kappa|^2 \le b^2$ and this sequence defines a tempered distribution by its Hermite series. The H_{-4} tail of its difference from any member of K_b is, after restricting to $\lambda_\kappa > M$, at most $4b^2M^{-2}$. The remaining coordinates are finite in number. The subsequence therefore converges in H_{-4}, and K_b is compact there. Chebyshev gives $P_{N,T}(K_b^c) \le 61b^{-2}$. The Hilbert space H_{-4} is separable and complete, so Prokhorov's theorem gives every asserted subsequential weak limit.

For a bounded subset B of Schwartz space,

$$\sup_{f\in B} |F(f)| \le \|F\|_{-4} \sup_{f\in B} \|f\|_4.$$

The last supremum is finite. This proves continuity of the inclusion in the strong distribution topology, and hence in the weak topology. It proves tightness and convergence in both topologies without applying a compactness theorem to a nonmetrizable space. Portmanteau on K_b also gives $\rho(H_{-3}) = 1$.

B.3. **Limits of exponentials, moments and changing tests.** Let $P_k \Rightarrow \rho$ on H_{-4} along a cofinal sequence. For real Schwartz f, truncating $e^{F(f)}$ at a positive constant and using $p_{R_k}(f) \to \|f\|_2$ gives

$$\rho(e^{F(f)}) \le e^{\|f\|_2^2/2}.$$

Applying this inequality to tf shows that the mean is zero (differentiate at $t = 0$ from both sides) and that $\rho(F(f)^2) \le \|f\|_2^2$. The derivative is justified, for example, by $\rho(e^{|F(f)|}) \le \rho(e^{F(f)}) + \rho(e^{-F(f)}) < \infty$. The variance bound extends $f \mapsto F(f)$ uniquely to $X : L^2 \to L^2(\rho)$. A subsequence of L^2 approximants converges almost surely; Fatou gives the same exponential bound for that extension.

For complex Schwartz tests f_i, put $G_k(z) = E_k \exp(\sum z_i X_k(f_i))$. Its absolute square is $\exp(2X_k(g_z))$, where $g_z = \operatorname{Re} \sum z_i f_i$. The finite bound gives

$$\sup_k E_k \left| \exp\left(\sum z_i X_k(f_i) \right) \right|^2 \le \exp(2p_1(g_z)^2). \tag{B.2}$$

It is locally bounded in z. Continuous finite-dimensional readouts converge in law, and this bound gives uniform integrability; their exponential expectations therefore converge pointwise. The same bound justifies differentiating their exponential series on compact sets, so the G_k and G are entire. Local boundedness and the Cauchy formula, or normal-family compactness followed by pointwise identification, now give locally uniform convergence on $\mathbb{C}^s$. The Cauchy formula gives convergence of every derivative, hence of all mixed moments.

We record a zero-free neighborhood rather than assuming that an entire generating function has an entire logarithm. If a real linear readout satisfies $Ee^{tY} \le e^{t^2\sigma^2/2}$, Chernoff gives $\Pr(|Y| > a) \le 2e^{-a^2/(2\sigma^2)}$ and hence

$$E|Y|^p \le p\, 2^{p/2} \Gamma(p/2) \sigma^p \quad (p > 0). \tag{B.3}$$

For an even integer $p = 2j$ this is $2^{j+1} j! \sigma^{2j}$. For a complex readout $Y = \sum z_i X(f_i)$ the L^{2j} triangle inequality gives this same bound with σ replaced by a finite seminorm $\sum |z_i| (\| \operatorname{Re} f_i \|_2 + \| \operatorname{Im} f_i \|_2)$. For real tests use simply $\sum |z_i| \|f_i\|_2$. Symmetry under $F \mapsto -F$ cancels the odd powers. Thus, when $t = \sum |z_i| \|f_i\|_2$ for real tests,

$$|G(z) - 1| \le 2 \sum_{j \ge 1} \frac{2^j j! t^{2j}}{(2j)!} < \frac{1}{2} \quad (t \le 2/5).$$

Indeed the ratio of successive summands is at most t^2, and the first summand is $2t^2$, giving the bound $2t^2/(1 - t^2) < 1/2$ at $t = 2/5$. The finite functions have the same estimate with p_{R_k} in place of $\| \cdot \|_2$. On every compact subset of the open limiting domain, $p_{R_k}(f_i) \to \|f_i\|_2$ makes the estimate uniform for large k. Their principal logarithms therefore converge locally uniformly there and all cumulants converge. Complex tests can be split into their real and imaginary parts; this supplies the claimed neighborhood for any fixed complex family.

There is a useful estimate for a test that varies with the cutoff:

$$E_k |X_k(f_k) - X_k(f)|^2 \le p_1(f_k - f)^2 \longrightarrow 0 \quad \text{if } f_k \to f \text{ in } \mathcal{S}. \tag{B.4}$$

The exponential moment bounds and Hölder also give convergence of any fixed polynomial involving such changing tests. For characters the simpler inequality $|e^{iu} - e^{iv}| \le |u - v|$ suffices.

The sharper support statement follows after the limit. Each normalized Hermite coordinate now has variance at most one, so

$$\rho(\|F\|_{-s}^2) \le Z_{2s}, \qquad s > 3/2.$$

For completeness it also has an exponential square bound. With weights $w_\kappa = \lambda_\kappa^{-2s}/Z_{2s}$, Jensen gives $e^{a \sum \lambda_\kappa^{-2s} X(h_\kappa)^2} \le \sum w_\kappa e^{a Z_{2s} X(h_\kappa)^2}$. The one-dimensional Gaussian integration identity and the readout bound give $\rho(e^{bX(h_\kappa)^2}) \le (1 - 2b)^{-1/2}$ for $0 \le b < 1/2$. Applying finite partial sums and then monotone convergence proves

$$\rho(e^{a\|F\|_{-s}^2}) \le (1 - 2aZ_{2s})^{-1/2} \quad (2aZ_{2s} < 1).$$

B.4. **Closedness and continuum symmetries.** We spell out the only approximation needed to pass $dF = 0$. Let g be a compact smooth three-form. Average its coefficient on cubes centered at the three-cell centers, and apply the physical lattice coderivative d^*_η. This gives a two-cochain v_η with $\langle v_\eta, F\rangle_\eta = 0$ for every closed lattice F. In component ij, the two three-cell centers incident to the plaquette center are displaced by $\pm\eta e_k/2$, where $\{i,j,k\} = \{1,2,3\}$. Their difference divided by η is the cell average of the central difference of g. Taylor's formula therefore gives

$$\|v_\eta - (d^*g)_\eta\|_\eta \le C_g\eta^2$$

once the support fits in the fundamental cube. The finite variance bound implies $E_k|X_k(d^*g)|^2 \le C_g^2\eta_k^4$. Moment convergence then gives $\rho(F(d^*g)^2) = 0$. A countable Schwartz-dense family of compact three-form tests, followed by continuity of distributions, gives $dF = 0$ on one set of full measure.

The L^2 orthogonal complement of the closed two-forms is the closure of the smooth coderivatives of three-forms. To check this without a boundary assumption, apply the Hodge star and Fourier transform. The complement becomes the longitudinal vectors $p\varphi(p)$, with $\int |p|^2|\varphi(p)|^2dp < \infty$. Smooth functions supported away from $p = 0$ are dense in this weighted space; symmetrizing by $\varphi(p) \mapsto (\varphi(p) + \overline{\varphi(-p)})/2$ preserves the real condition. Their inverse transforms are Schwartz gradients. The continuity $X : L^2 \to L^2(\rho)$ and the preceding annihilation show $X(f) = X(\Pi_c f)$. Applying the exponential bound to $\Pi_c f$ gives the C_0 bound.

Let $a \in \mathbb{R}^3$ and choose $a_k \in \eta_k\mathbb{Z}^3$ with $|a_k - a| \le \sqrt{3}\eta_k/2$. Lattice covariance gives $E_k e^{iX_k(\tau_{a_k}f)} = E_k e^{iX_k(f)}$. Since $\tau_{a_k}f \to \tau_a f$ in Schwartz space, (B.4) passes this identity to ρ. Applying it to every real linear combination of finitely many tests proves equality of all cylindrical laws, hence translation invariance. Signed coordinate permutations act exactly on the plaquette centers, their averaging cubes, and the tensor indices; their covariance requires no approximation. Curvature sign symmetry likewise passes through the characteristic function.

The span of characters $e^{iF(f)}$ is dense in $L^2(\rho)$. For clarity, if an L^2 function is orthogonal to every character, its conditional expectation on the first s readouts has zero Fourier transform as a finite signed measure, and therefore is zero. Taking a countable determining family and letting s increase proves density. On these characters,

$$\|U(a)e^{iF(f)} - e^{iF(f)}\|_2 \le \|\tau_{-a}f - f\|_2 \longrightarrow 0.$$

Unitary boundedness extends strong continuity to all of L^2.

B.5. **Tempered moments and tensor positivity.** The even bound with $p = 2n$ and Cauchy–Schwarz give $E|X(f)|^n \le (2^{n+1}n!)^{1/2}\|f\|_2^n$. Hölder therefore bounds the n-point multilinear form by $2\sqrt{2^n n!}\prod\|f_i\|_2$. This harmless enlargement of the numerical constant is convenient for every n.

Here is an explicit tensor extension, including its order of growth. Expand a Schwartz tensor Ψ on $(\mathbb{R}^3)^n$ in products of the Hermite basis, with coefficients $c_{\boldsymbol{\kappa}}$. The multilinear bound and Cauchy–Schwarz give

$$\begin{aligned}|S_n(\Psi)| &\le 2\sqrt{2^n n!}\sum_{\boldsymbol{\kappa}}|c_{\boldsymbol{\kappa}}|\\ &\le 2\sqrt{2^n n!}\, Z_4^{n/2}\|\mathsf{A}_1^2\cdots\mathsf{A}_n^2\Psi\|_2.\end{aligned}$$

Every A_i^2 has finitely many fixed-coefficient terms, of derivative order at most four and polynomial degree at most four in its own variable. Multiplication of these finite sums has total coefficient sum at most C^n, with a fixed C. Consequently

$$|\mathsf{A}_1^2\cdots\mathsf{A}_n^2\Psi(x)| \le C^n\prod_i(1+|x_i|^2)^2\max_{|\alpha|\le 4n}|\partial^\alpha\Psi(x)| \le C^n q_{6n}(\Psi)(1+|x|^2)^{-n}.$$

Its squared integral in dimension $3n$ is $\pi^{3n/2}\Gamma(n/2)/\Gamma(2n)$, bounded by C^{2n}; the component count can be absorbed in the same constant. This proves the tensor bound of Theorem 5.1. Finite Hermite

sums converge in Schwartz space, so the extension is a unique tempered distribution. Its restriction to product tests is the original moment, and permutation symmetry follows there. The derivative and weight order is linear in n.

Reflection positivity first passes on the characters of compact smooth tests strictly in a positive half-space. At sufficiently fine mesh and large volume their averaging cubes are contained in that half, so Theorem 4.2 applies, with the exact tensor electric sign. Weak convergence of the characters gives the limiting positive matrix. The character-density argument just given, restricted to the countable half-space readouts, extends positivity to $L^2(\Sigma_j^+, \rho)$ because reflection preserves the law and the associated form is bounded by the product of L^2 norms. Every L^2 test vanishing on the negative half-space can be approximated in L^2 by compact smooth tests strictly in the positive half. Its readout is therefore half-space measurable; its polynomials lie in L^2 by the moment bounds. Translation invariance moves this result to every coordinate plane.

For the tensor formulation, finite sums of separated smooth tests strictly in the half-space give polynomial observables, and their OS quadratic form is nonnegative. These sums are dense among Schwartz tensors supported on the closed positive-time domain. Indeed a smooth tensor supported there is flat at its boundary; cutting off successively within a distance δ of the boundary converges in every Schwartz seminorm as $\delta \downarrow 0$. Spatial cutoffs then converge in the same topology. On a compact set away from the boundary, the ordinary product Hermite expansion, or a product partition of unity and Fourier expansion on boxes, gives convergence of separated test sums with supports in that half-space. The established tensor bounds make each entry of a finite OS quadratic form continuous in this approximation. This proves the tensor reflection inequality.

Finally suppose another probability law has the same compact-test moments. Its expectation of $\cosh(tF(f))$ is, by monotone convergence, the same even-moment series, finite for every real t. Hölder gives integrability of $\exp(\sum |t_i F(f_i)|)$ for a finite family. The multivariate exponential series can therefore be integrated term by term under both laws. Their characteristic functions coincide, hence all finite-dimensional readout laws coincide. Compact tests generate the distributional sigma-algebra: a Schwartz test is their Schwartz limit. The two probability laws are equal. This is moment determinacy of the given hierarchy, with no assertion that different cutoff limit points have that hierarchy.

Appendix C. Invariant projections, disintegration and reconstruction

C.1. **Invariant averages on the distribution space.** All sigma-algebras in this section are completed for the law at hand. For disintegration take the field space $\mathcal{S}'$ with its cylindrical sigma-algebra. It is standard Borel: its Hermite coordinates identify it with the Borel subset $\bigcup_{m\ge1}\{(t_\kappa) : \sum \lambda_\kappa^{-2m}|t_\kappa|^2 < \infty\}$ of $\mathbb{R}^{\mathbb{N}}$. A tempered distribution belongs to this union because a fixed Schwartz seminorm of h_κ grows at most polynomially in λ_κ; this follows by iterating the Hermite raising and lowering identities. Conversely each such sequence defines a distribution by its convergent pairing with Schwartz Hermite coefficients. On every H_{-m} the coordinate sigma-algebra is its Borel sigma-algebra, since the norm is the weighted coordinate square sum. Translations act jointly measurably: each coordinate $F(\tau_a f)$ is a measurable pointwise limit of its Hermite series, and its dependence on a is continuous for fixed F. Compact smooth tests also generate this sigma-algebra by Schwartz density. A countable dense rational family of such tests, stable under rational translations and the coordinate reflections, will be fixed throughout.

For a coordinate subspace $\Gamma_J \simeq \mathbb{R}^d$, $d \ge 1$, put

$$A_T^J G = T^{-d} \int_{[0,T]^d} U(a) G \, da.$$

The integral is a Bochner integral on $L^2(\rho)$, and $\|A_T^J\| \le 1$. We recall a short mean-ergodic proof suitable for every subgroup used here. The fixed-vector space is the orthogonal complement of the closed span of $U(b)G - G$, $b \in \Gamma_J$. An average fixes every fixed vector. On $U(b)G - G$ its norm is

at most $\|G\|$ times the relative volume of the symmetric difference of the cube and its translate, which tends to zero. Contractions and density prove $A_T^J \to P_J$ strongly on L^2. The same proof holds on L^p, $1 \le p < \infty$, after approximating by bounded functions. For discrete cube averages in $\mathbb{Z}^d$, volume is replaced by cardinality and the identical proof applies.

Fixed vectors are precisely the functions measurable for the invariant sigma-algebra. If $U(a)G = G$, every sublevel set of its real and imaginary parts is invariant modulo null sets; the converse follows first for indicators and then by approximation. Thus P_J is the conditional expectation onto $\mathcal{I}_J$. Every $U(a)$ commutes with P_J since translations commute. Projections for different coordinate directions commute: each is a strong limit of its coordinate averages, which commute with the averages of every other coordinate. Their product projects onto the intersection of the fixed spaces, giving $P_J = \prod_{j\in J} P_j$.

C.2. **A reflection fixes its axis-invariant space.** Fix j, and let $\mathcal{R}_j$ be the linear reflection pullback on $L^2(\rho)$. It is a self-adjoint unitary involution, because the law is reflection invariant. A function fixed by translations along e_j belongs to every positive and every negative j-half-space sigma-algebra. Indeed approximate it in L^2 by finite linear combinations of compact-test characters. Translate each approximant far into the desired half-space; the target is unchanged and the approximation error is unchanged by unitarity. Taking a subsequence if necessary proves the claimed half-space measurability.

Reflection preserves this invariant subspace, because $\mathcal{R}_j U(te_j) = U(-te_j)\mathcal{R}_j$. Its restricted involution has eigenvalues 1 and -1. Reflection positivity applies to every vector of this subspace, by the preceding half-space measurability. A negative eigenvector would have reflection form $-\|G\|_2^2$, so none exists. Consequently

$$\mathcal{R}_j G = G \quad \text{for every } G \in L^2(\mathcal{I}_j, \rho). \tag{C.1}$$

In particular this holds for all full-translation-invariant weights. If w is a bounded, nonnegative full-invariant function and $G \in L^2(\Sigma_j^+, \rho)$, then $\sqrt{w}G$ belongs to that half-space and

$$\rho(w\,\overline{\mathcal{R}_j G}\,G) = \rho(\overline{\mathcal{R}_j(\sqrt{w}G)}\sqrt{w}G) \ge 0. \tag{C.2}$$

Also reflection invariance and (C.1) give $\rho(w\,G \circ r_j^*) = \rho(wG)$ for every integrable G. These are the identities needed for componentwise inheritance.

C.3. **Constructing the component kernel.** We give the disintegration argument to keep track of one common exceptional set. Since $L^2(\rho)$ is separable, $L^2(\mathcal{I}_{\{1,2,3\}}, \rho)$ is separable. Choose a countable bounded family h_n generating this sigma-algebra modulo null sets: start with a dense family of bounded invariant functions and include their rational real sublevel sets. The map $H(F) = (h_n(F))_n$ takes values in the standard Borel space $\mathbb{R}^{\mathbb{N}}$. Let ν be its image law. The conditional distribution on the fibers of H exists on the standard Borel field space; write it ρ_ξ. It satisfies

$$\rho(G) = \int \rho_\xi(G)\, d\nu(\xi), \qquad P_{\{1,2,3\}}G(F) = \rho_{H(F)}(G) \tag{C.3}$$

for every integrable G. The elementary conditional-distribution theorem being used here is the regular conditional probability theorem for a Borel map between standard Borel spaces; it carries no dynamical hypothesis.

The conditional laws are stationary. For a rational translation a and a bounded cylindrical test G, invariance of every weight $w(H)$ gives $\rho(w(H)(U(a)G - G)) = 0$. The defining conditional identity implies $\rho_\xi(U(a)G) = \rho_\xi(G)$ almost everywhere. Impose these equalities for the countable rational translation group and a countable determining algebra of bounded characters. Outside their union of null sets they imply equality of the cylindrical laws. For arbitrary real a, approximate by rational a_n. The pointwise distributional continuity $F(\tau_{a_n} f) \to F(\tau_a f)$ and bounded convergence pass stationarity to every real translation.

To see ergodicity directly, fix a bounded member G of the countable determining algebra. The mean-ergodic theorem gives $A_T G \to PG$, with $P = P_{\{1,2,3\}}$, in $L^2(\rho)$. Choose a sequence $T_k \to \infty$ along which these squared errors are summable, for all countably many G by diagonal selection. Disintegration and Fubini imply that, for almost every ξ, the averages converge in $L^2(\rho_\xi)$ to the constant $\rho_\xi(G)$. On the other hand the stationary law ρ_ξ has its own strongly continuous translation representation: on bounded compact-test characters this follows from pointwise distributional continuity and bounded convergence, then from density on L^2. Its own mean-ergodic theorem says these averages converge to its invariant projection. Thus that projection sends every member of the determining algebra to its mean. The same character-density proof works under any Borel probability law, so the algebra is dense under ρ_ξ. The invariant space consists of constants. This is translation ergodicity, on the same full-measure set. The proof also applies verbatim to any nonzero coordinate subgroup if its invariant kernel is needed.

The set of closed distributions has original measure one, so conditional measure one almost everywhere. Apply (C.2) to a countable dense rational linear family of positive-half-space characters, and apply the weighted reflection-invariance identity to a countable determining family. The conditional identities give reflection positivity and reflection invariance under ρ_ξ simultaneously for the three coordinate reflections. Character density extends the inequality to the complete half-space $L^2(\rho_\xi)$ space. Stationarity moves it to every coordinate plane. Arbitrary rotations or coordinate permutations are not required to preserve an individual component; the coordinate reflections are inherited because they fix full-invariant weights.

C.4. **The Gaussian bound under invariant conditioning.** Fix a nonzero coordinate subgroup Γ_J and a real L^2 test f. Let $\sigma = \|f\|_2$ and $G = e^{X(f)}$. The unconditional bound gives $G \in L^p(\rho)$ for every finite p. Translation covariance extends to L^2 tests by continuity. Writing $f_a = \tau_a f$, the scalar overlap $c(a) = \langle f, f_a\rangle_2$ tends to zero as $|a| \to \infty$. For example approximate both functions in L^2 by compact functions, whose overlaps vanish once their supports separate. For fixed integer n, positivity and Fubini give

$$\begin{aligned}\rho((A_T^J G)^n) &= T^{-dn} \int_{[0,T]^{dn}} \rho\left(\exp X\left(\sum_{i=1}^n f_{a_i}\right)\right) \prod_i da_i \\ &\le T^{-dn} \int \exp\left(\frac{n}{2}\sigma^2 + \sum_{i<k} c(a_i - a_k)\right) \prod_i da_i.\end{aligned}$$

The integrand is at most $e^{n^2\sigma^2/2}$. Given $\delta > 0$, choose M so that $|c(a)| \le \delta$ when $|a| > M$. The region where at least one pair satisfies $|a_i - a_k| \le M$ has relative volume at most $C_{n,d}(M/T)^d$. On its complement the integrand is bounded by $\exp(n\sigma^2/2 + n(n-1)\delta/2)$. Letting $T \to \infty$ and then $\delta \downarrow 0$ proves $\limsup_T \rho((A_T^J G)^n) \le e^{n\sigma^2/2}$. The L^2 mean-ergodic convergence has an almost surely convergent subsequence. Fatou gives $\rho((P_J G)^n) \le e^{n\sigma^2/2}$ for every n. Since $P_J G \ge 0$, sending $n \to \infty$ yields the pointwise bound

$$P_J(e^{X(f)}) \le e^{\|f\|_2^2/2}. \tag{C.4}$$

This derivation is also valid with the closed-test quadratic C_0 in place of the L^2 norm: Π_c commutes with translations, and its translated overlaps likewise tend to zero.

Use (C.4) for a countable Schwartz-dense real linear family of tests and for its rational scalar multiples. By (C.3) the component laws satisfy the Gaussian bound on that family, on one common full-measure set. For a general Schwartz test choose approximants in Schwartz topology. Evaluations converge pointwise on every distribution, and Fatou extends the exponential inequality. Each component therefore has a bounded readout $X_\xi : L^2 \to L^2(\rho_\xi)$ and the exponential bound for all L^2 tests by the same extension argument as before. Differentiation at zero gives zero means. These are the component readouts; when necessary their equality with the unconditional readout is

understood under the disintegration identity, rather than as a simultaneously chosen pointwise map on all uncountably many L^2 tests. Closedness gives the C_0 sharpening on each component as well. Its tensor regularity follows from the identical Hermite proof.

The subgroup version of (C.4), at both signs and every rational multiple of f, implies

$$P_J X(f) = 0 \qquad (J \neq \varnothing). \tag{C.5}$$

Indeed the convex differentiable conditional generating function is bounded above by $e^{t^2\|f\|^2/2}$ and equals one at zero; the two one-sided derivatives must be zero. Conditional exponential integrability justifies differentiation. For generality this argument can also be applied to the subgroup kernel constructed above.

C.5. **The coordinate Hilbert space.** We now work with either ρ or one of the component laws just obtained, and fix j. On $\mathcal{E}_+ = L^2(\Sigma_j^+)$ define $b(F,G) = \langle \mathcal{R}_j F, G\rangle_2$. The form is Hermitian because $\mathcal{R}_j$ is a self-adjoint involution, positive by reflection positivity, and bounded by $\|F\|_2\|G\|_2$. The Cauchy–Schwarz inequality for this positive form shows that its null vectors form a linear subspace orthogonal for b to all of $\mathcal{E}_+$. The quotient and completion define $\mathcal{H}_j$ and a contraction $G \mapsto [G]$ with dense range. The vector $\Omega_j = [1]$ has norm one and $\langle \Omega_j, [G]\rangle = \rho(G)$.

Positive translations preserve the positive half-space. For $U_t = U(te_j)$, reflection gives $b(F, U_t G) = b(U_t F, G)$. Put $|G|_b = b(G,G)^{1/2}$. The positive-form Cauchy–Schwarz inequality gives

$$|U_t G|_b^2 = b(G, U_{2t} G) \leq |G|_b |U_{2t} G|_b.$$

Iteration and ordinary unitary boundedness give

$$|U_t G|_b \leq |G|_b^{1-2^{-n}} \|G\|_2^{2^{-n}} \longrightarrow |G|_b.$$

When $|G|_b = 0$ the first displayed inequality already forces $|U_t G|_b = 0$. Thus $[G] \mapsto [U_t G]$ is well defined and extends uniquely to a self-adjoint contraction $T(t)$. The semigroup law holds on a dense subspace, hence everywhere. Strong continuity at zero follows from $\|[U_t G - G]\| \leq \|U_t G - G\|_2$; contractions and density give it everywhere.

Every $T(t)$ is nonnegative because $T(t) = T(t/2)^2$. If $T(t)\psi = 0$, then $T(t/2)\psi = 0$, and iterating and using continuity at zero gives $\psi = 0$. In particular $T(1)$ is injective. Functional calculus therefore defines the nonnegative self-adjoint operator $H_j = -\log T(1)$. The unique nonnegative dyadic roots of $T(1)$ are $T(2^{-n})$, so $T(t) = e^{-tH_j}$ at every nonnegative dyadic rational. Strong continuity extends the identity to all $t \geq 0$. This also proves uniqueness of the generator. Its zero spectral projection is the projection onto the common fixed vectors of the semigroup, and it fixes Ω_j.

Transverse translations preserve the half-space and commute with $\mathcal{R}_j$. They therefore preserve b in both arguments and induce unitary $V_j(y)$, with inverse $V_j(-y)$. Their strong continuity follows from that on L^2 and the contractive quotient map. They commute with e^{-tH_j}.

Average the identity $\langle [F], T(t)[G]\rangle = \langle \mathcal{R}_j F, U_t G\rangle_2$ over $0 \leq t \leq S$. On the Hilbert-space side the spectral theorem sends the average to $P_{\ker H_j}$; on the probability-space side the mean-ergodic theorem sends it to P_j. Hence $P_{\ker H_j}[G] = [P_j G]$. By (C.1), the quotient norm equals the L^2 norm on $L^2(\mathcal{I}_j)$. This closed subspace maps isometrically into $\ker H_j$, and its image contains the dense set $P_{\ker H_j}[\mathcal{E}_+]$. It maps onto the entire kernel. This proves the projection identity and its precise probabilistic interpretation.

C.6. **Uniform transverse control and clustering.** Let $F = \mathcal{R}_j F'$ with $F', G \in \mathcal{E}_+$. By the projection identity, for $s \geq 0$ and $y \perp e_j$,

$$\begin{aligned} |\langle F, U(se_j + y)(1 - P_j) G\rangle_2| &= |\langle [F'], V_j(y) e^{-sH_j}(1 - P_{\ker H_j})[G]\rangle| \\ &\leq \|[F']\|\, \|e^{-sH_j}(1 - P_{\ker H_j})[G]\|. \end{aligned}$$

The last norm tends to zero by spectral dominated convergence, uniformly over every transverse y. For general $F, G \in L^2$, approximate by compact-test characters. For sufficiently large fixed r,

translate the first approximant by $-re_j$ into the negative half and the second by re_j into the positive half. The correlation is then the previous one with $s-2r$ in place of s. The approximation error is bounded uniformly in both s and y, since all translations are unitary and $\|1-P_j\|\le 1$. Letting the approximation error tend to zero gives

$$\lim_{|s|\to\infty}\sup_{y\perp e_j}|\langle F,U(se_j+y)(1-P_j)G\rangle_2|=0. \tag{C.6}$$

Negative s follows by taking the adjoint and replacing y by $-y$. This proves the mixing statement in the main text.

More generally let $a_j^{(k)}$ tend in absolute value to infinity for $j\in J$, and converge to $\bar a_j$ for $j\notin J$. Strong continuity removes the convergent coordinates. For the others use the commuting-projection decomposition

$$1=P_J+\sum_{\ell=1}^{|J|}P_{j_1}\cdots P_{j_{\ell-1}}(1-P_{j_\ell}).$$

The fixed term is P_J; each remaining correlation tends to zero by (C.6), which permits all the other translating coordinates to vary arbitrarily. Thus the weak limit is $P_JU(\bar a)$. On a full-translation-ergodic component $P_{\{1,2,3\}}G=\rho_\xi(G)1$. Taking $J=\{1,2,3\}$ proves clustering when every coordinate diverges, and the mean-ergodic theorem proves clustering averaged over cubes. All polynomial fields belong to L^2 by the Gaussian bound, so these conclusions apply to every split of a curvature moment.

For a two-point function an arbitrary sequence $|a^{(k)}|\to\infty$ has, from every subsequence, a further subsequence on which each coordinate either converges or diverges in absolute value, with at least one divergent coordinate. Its weak limit contains some P_j, and (C.5) annihilates the single-field readout. The two-point correlation therefore tends to zero along that subsequence, hence along the original sequence. This reasoning does not replace P_j by the mean for a higher polynomial.

The remaining statements about the ground space are elementary but useful in interpreting this distinction. A ground vector fixed by all transverse translations corresponds under the kernel isometry to a function fixed by the axis and every transverse direction. Full ergodicity makes it constant, so the vector is a multiple of Ω_j. If $\dim L^2(\mathcal{I}_j)<\infty$, this sigma-algebra has finitely many positive-measure atoms: otherwise arbitrarily large families of disjoint indicators would be linearly independent. Every translation permutes those atoms. Strong continuity makes this finite permutation trivial near zero (distinct indicators have a positive minimum distance); every translation is a finite sum of translations near zero, so it is trivial everywhere. Each atom is consequently full invariant and, by ergodicity, has measure zero or one. There is only one atom and the dimension is one. Thus $\dim\ker H_j$ is one or infinite.

C.7. **Separated products and invariant conditional bounds.** For completeness consider a discrete subgroup $\Gamma_0\simeq\mathbb{Z}^d$ and the real variable G in Theorem 7.1. Let A_M be its cube average. Mean-ergodic convergence gives $A_MG\to P_0G$ in L^2. Suppose the hypothesis controls the absolute expectation of every product of r distinct translates by u^r, for every even r, and $G\in L^r$ for every finite r. Expanding $E(A_MG)^r$, the fraction of ordered r-tuples with a repeated site is at most $r(r-1)/(2M^d)$. Each repeated-site product has absolute expectation at most $\|G\|_r^r$ by Hölder and stationarity. Every other product has absolute expectation at most u^r. Thus

$$\limsup_M E|A_MG|^r\le u^r.$$

An almost surely convergent subsequence and Fatou give $E|P_0G|^r\le u^r$. Sending even $r\to\infty$ proves $|P_0G|\le u$ almost surely. This argument uses control at all orders, not just a second- or fourth-moment estimate. If $\mathcal{I}$ is the invariant sigma-algebra of the full continuous translations, then $\mathcal{I}\subset\mathcal{I}_0$ and the tower identity gives $P_{\mathcal{I}}G=E[P_0G\mid\mathcal{I}]$. The same pointwise bound follows. No identification of continuous and discrete ergodic components is asserted.

Appendix D. Positive mass and the removal of harmonic pinning

D.1. **Accretivity and a complex twist strip.** Here n is the native side length and the real photon field is fixed. Let S_i denote the unitary forward hopping shift, including the photon phase, seam character and uniform flat phase $e^{i\theta_i/n}$. It commutes with γ_i. The native Wilson matrix, including its mass, is

$$D^\theta + \mu = \mu + \frac{1}{2}\sum_{i=1}^{3}\gamma_i(S_i - S_i^*) + \frac{1}{2}\sum_{i=1}^{3}(1-S_i)^*(1-S_i).$$

The gamma term is anti-Hermitian. Thus $\mathrm{Re}\langle v, (D^\theta+\mu)v\rangle \geq \mu\|v\|^2$, and finite dimensionality gives $\|(D^\theta+\mu)^{-1}\| \leq \mu^{-1}$ when $\mu > 0$. The paired determinant is nonnegative and now nonzero, hence strictly positive at every real twist.

For $\vartheta = \theta + i\tau$, the forward and backward hops are multiplied by $e^{-\tau_i/n}$ and $e^{\tau_i/n}$. Since $(1\pm\gamma_i)/2$ are projections,

$$\|\mathsf{E}_\tau\| := \|D^{\theta+i\tau} - D^\theta\| \leq 2\sum_i \sinh(|\tau_i|/n).$$

In the closed tube $|\tau_i| \leq \tau_* = n\,\mathrm{arcsinh}(\mu/12)$ this is at most $\mu/2$. The matrix $K_{\theta,\tau} = (D^\theta+\mu)^{-1}\mathsf{E}_\tau$ therefore has norm at most $1/2$. The same calculation gives invertibility in the larger convex open tube $|\tau_i| < n\,\mathrm{arcsinh}(\mu/6)$. The determinant is entire in the twist variables and periodic with period 2π in each: adding 2π to a real twist is a large gauge transformation, and the identity extends to complex twists by successive one-variable identity theorems.

On the convex zero-free tube choose its holomorphic logarithm $\ell(\vartheta)$, normalized to the real logarithm at zero. It is the real logarithm on all of $\mathbb{R}^3$, since their difference is a continuous function into $2\pi i\mathbb{Z}$. It is periodic as well: its increment under $2\pi e_i$ is constant in $2\pi i\mathbb{Z}$ on the connected tube and zero on its real part. Following the segment $\theta + it\tau$ and using the convergent matrix logarithm series gives the exact identity

$$\ell(\theta+i\tau) - \ell(\theta) = \mathrm{Tr}\log(1+K_{\theta,\tau}).$$

Both exponentials agree, and their continuous difference in $2\pi i\mathbb{Z}$ is zero at $t=0$. The matrix dimension is $4n^3$, so

$$|\ell(\theta+i\tau) - \ell(\theta)| \leq 4n^3\sum_{j\geq 1}\frac{2^{-j}}{j} = 4n^3\log 2. \tag{D.1}$$

The bound is uniform in the fixed real photon field.

D.2. **The nonconstant Fourier coefficients.** Let $c_k = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\ell(\theta)e^{-ik\cdot\theta}\,d\theta$. For $k \neq 0$ shift the contour in each variable to $\theta_i - i\tau_*\,\mathrm{sgn}\,k_i$. Periodicity cancels the vertical faces and holomorphy justifies the shift, entirely inside the larger tube. By (D.1),

$$c_k = e^{-\tau_*|k|_1}(c_k + \rho_k), \qquad |\rho_k| \leq 4n^3\log 2.$$

Consequently $|c_k| \leq 4n^3\log 2\, e^{-\tau_*|k|_1}/(1-e^{-\tau_*|k|_1})$. When $\tau_* \geq 1$ set $x = e^{-\tau_*} \leq e^{-1}$. The elementary geometric sum gives

$$\sum_{k\neq 0} x^{|k|_1} = \left(\frac{1+x}{1-x}\right)^3 - 1 \leq 24.9x, \qquad (1-x)^{-1} \leq 1.582.$$

The first inequality follows because its quotient by x has a positive-coefficient power series and its value at e^{-1} is less than 24.83. Thus $\sum_{k\neq 0}|c_k| \leq 110n^3e^{-\tau_*} = \delta_n$. The Fourier series is absolutely uniformly convergent and equals the smooth periodic function ℓ. It follows that $|\ell(\theta) - c_0| \leq \delta_n$ and hence $|\ell(\theta) - \ell(0)| \leq 2\delta_n$. This proves the determinant comparison, including its uniformity in the photon field.

D.3. **Normalization and the order of limits.** The raw twisted weight divided by the raw pinned weight is between $e^{-2\delta_n}$ and $e^{2\delta_n}$. Its partition function has the same comparison. Dividing the two weights by their actual partition functions gives the density bounds $e^{-4\delta_n}$ and $e^{4\delta_n}$. For the harmonic average, integrate the raw weight in θ with normalized Haar measure first; the identical raw comparison remains valid and yields the same normalized density bounds. Thus every measurable curvature function $|\Phi| \le 1$ satisfies the difference bound $e^{4\delta_n} - 1$. These comparisons pass to any measurable pushforward, in particular the actual $F^\sharp$ readout.

At a fixed sufficiently fine cutoff N with $\mu_N > 0$, $n = R/\eta$ tends to infinity with T. Then $\delta_n \to 0$ and the total-variation distance of the two readout laws tends to zero. Tightness of the averaged laws therefore gives tightness of the pinned laws at that fixed N, and their infinite-volume weak limit points coincide. Given a sequence of such intermediate limit points $\rho_N \Rightarrow \rho$, choose, for each member of the sequence, a volume $T_N \ge N$ so large that its averaged law is within $1/N$ of ρ_N in the bounded-Lipschitz metric on H_{-4}. This metric metrizes weak convergence on the separable Hilbert space. The triangle inequality makes $P^{\mathrm{av}}_{N,T_N} \Rightarrow \rho$, with both cutoffs divergent. The conclusions proved for cofinal averaged limits therefore apply to the iterated pinned limit. This is exactly the scope of Corollary 8.2.

The argument has no uniform conclusion when N and T are allowed to vary arbitrarily with $\mu_N \downarrow 0$. The factor $n^3 e^{-n \operatorname{arcsinh}(\mu_N/12)}$ must vanish for that stronger comparison. In particular an upper bound on the magnitude of a tuned mass counterterm does not imply $\mu_N = \bar{m}\eta + m_{0,N} > 0$. The existence and non-Gaussian results for the averaged and all-flux laws do not use this additional positive-mass condition.

Appendix E. Domains, absolute norms and the choice of constants

The proofs below use level coordinates. A level j has fine mesh $\delta = L^{-j}$, unit lattice $\mathbb{Z}^3/l\mathbb{Z}^3$, running charge $g = e_j$ and positive reference mass $\mathfrak{m} = \bar{m}a_j$. The bilinear pairing of fine spinors is $\delta^3 \sum$, and that of unit spinors is $\sum$. All spinors are antiperiodic in each direction. Connections and scalar gauge functions are periodic. A bond value is a component of the connection, so a line integral contains a factor δ for each fine bond. This convention is used in every phase, Stokes identity and fine measure below.

E.1. **Geometry and rooted counts.** The cubes are centered on $M\mathbb{Z}^3$; because L and M are odd, their faces contain no fine lattice sites. Fine sites, unit blocks, L-blocks and M-cubes are nested. A bond meeting a union of M-cubes has both ends in its one-layer enlargement. For a site x, let $\square(x)$ be its M-cube; for a bond or a local photon basis vector, let $\square(x)$ be the smallest union of M-cubes containing its support. The basis supports used below are either a single bond or a single L-block, and thus have diameter at most L.

Use throughout the same meeting-net metric as in the main text: $Md_M(X)$ is the infimum of the lengths of connected finite trees meeting every closed cube of X. The tree need not meet cube centers. It gives $d_M(\square) = 0$, $d_M(X) \le |X|_M - 1$ and $\operatorname{diam}(X) \le M(d_M(X) + 2)$. Traverse a net of length ℓ twice and cut the traversal into $\max(1, \lceil 2\ell/M \rceil)$ arcs of length at most M. Each arc has coordinate ranges of length at most M, and hence meets at most 27 closed cubes. Taking $\ell \downarrow Md_M(X)$ gives $|X|_M \le c_X(d_M(X) + 1)$, with $c_X = 54$. A face-connected union with n cubes has a spanning cube tree; traversing that tree twice, from a prescribed root, encodes the union by a walk of $2(n-1)$ nearest-neighbor steps. There are at most 36^{n-1} such unions. Hence

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le e^{\kappa_0} \sum_{n \ge 1} 36^{n-1} e^{-\kappa_0 n/54} \le K_0,$$

with $\kappa_0 = 54 \log 72 + \log 2$ and $K_0 = 2 \cdot 72^{54}$,

$$\sup_{\square} \sum_{X \supset \square} e^{-\kappa d_M(X)} \le K_0 \quad (\kappa \ge \kappa_0), \qquad \sum_{X \supset \square} (d_M(X) + 3)^2 e^{-(\kappa+2) d_M(X)} \le 9K_0. \tag{E.1}$$

The second inequality follows from $(t+3)^2e^{-2t}\le 9$. It is harmless to increase these explicit absolute constants. What matters is that they precede L, M and the charge in the order of choice.

For a connected union X let $\bar X^L$ be the smallest union of LM-cubes containing it. Every LM-cube contains L^3 M-cubes. Every net meeting all closed fine cubes of X also meets every closed coarse cube of $\bar X^L$. Scaling this same net gives

$$d_M(X)\ge Ld_M(L^{-1}\bar X^L).$$

For $Y=L^{-1}\bar X^L$ choose a fine root in LY and use (E.1). There are $L^3|Y|_M$ possible roots, and $|Y|_M\le 54[d_M(Y)+1]$. Splitting off κ_0 decay, then using $d_M(Y)+1\le e^{d_M(Y)}$, proves

$$\sum_{X:\bar X^L=LY} e^{-\kappa' d_M(X)}\le c_{\mathrm{rb}}L^3e^{-[L(\kappa'-\kappa_0)-1]d_M(Y)},\qquad c_{\mathrm{rb}}=54K_0.$$

Thus reblocking costs $c_{\mathrm{rb}}L^3$ and has the displayed reserved decay. A union of mutually compatible polymers has distinct components. A connected cluster of polymers has a connected union and satisfies the tree inequality $d_M(\bigcup_i X_i)\le\sum_i d_M(X_i)+3(n-1)$ for n polymers with connected incompatibility graph. To prove it, choose a spanning tree of that graph, meet a parent and its child at points of their common or adjacent cubes, and add a segment of length at most $3M$; the existing nets can then be joined in this order. These inequalities will account for every reblocking and cluster sum; volume factors enter only when the root is subsequently summed.

E.2. **Consistent choices of smallness parameters.** Choose first b, $\varepsilon\le 1/200$ and $\alpha\in(1-\varepsilon,1)$, then $\kappa\ge 7\kappa_0+7$. Choose an odd L so large that

$$L^{1/8-3\varepsilon/2}\ge 2400,\quad c_{\mathfrak{fl}}L^{-3/8+6\varepsilon}\le 1/16,\quad L(\kappa-6\kappa_0-4)-1\ge\kappa. \tag{E.2}$$

Here $c_{\mathfrak{fl}}$ is the absolute extraction/reblocking constant obtained from the norm estimates below; it can be taken to be $61\cdot 54K_0$ by enlarging the extraction constant. Primitive constants are then obtained for the fixed L, b, α and reference mass range. Let $\gamma_w>0$ be the smaller photon/fermion decay rate, and choose $\alpha_0>0$ below both permitted interpolation exponents. Choose a power $M=L^m$ satisfying

$$M\ge L^2,\quad M>4L,\quad M\ge L^{m_0+2},\quad \gamma_w M\ge 16,$$

and, for the finite numbers n_P, n_F of cubes read by the photon paths and fermion walk links, the inequalities

$$\alpha_0 n_P\frac{\log M}{M}\le\gamma_w/4,\qquad C_FM^{-1+2\alpha_0}e^{\kappa_w+\kappa_0+2}\le 1/2. \tag{E.3}$$

Also require $M^{\alpha_0}-1\ge 6c_X\exp(N_c\kappa)$, where $c_X=54$ and $N_c=(2L+1)^3$ bounds the number of ordinary cubes met by one parameter cube. This is possible after the fixed L,κ,α_0 have been chosen. The walk exponent may be reduced before this choice, so that $2\alpha_0<1$. All these conditions are possible as $M\to\infty$. The complete interpolation constants C_w are fixed after M. Let c_γ dominate all unit-bond, unit-site and local-basis exponential sums at rate γ_w; let c_{row} dominate all exact row and unit-lattice activity constants, and C_B the local photon regularity constant below. They are finite numbers depending on the fixed scheme alone.

Choose $r\ge 2$, then $p_0>3r+2$ and $p>2p_0$, so that the finite Gaussian tail bounds in L tend to zero faster than every required power of e_K. Choose $e_0<1$ after all these constants, so that for $0<g\le e_0$

$$3Mg^{\varepsilon}\le 1,\qquad g^{1/4}+g^{\varepsilon}\le 1,\qquad 4g^{1/4+2\varepsilon}+2g^{3\varepsilon}\le 1, \tag{E.4}$$

$$p_0(g)\le g^{-\varepsilon/4},\quad 2C_w^2c_\gamma^2c_Cg^{\varepsilon/4}\le 1,\quad 8C_B(-\log g)^p\le g^{-3/4+3\varepsilon}, \tag{E.5}$$

$$8g^{3/4-7\varepsilon/2}\le 1,\quad c_{\mathfrak{fl}}g^{\varepsilon}\le 1/16,\quad 2g^{\varepsilon}\le 1/4. \tag{E.6}$$

The extra finite inequalities used in the terminal proof have the form $C(-\log g)^Ag^B\le c$, with $B>0$, or $C(-\log g)^A\exp[-c(-\log g)^B]\le g^D$, with $B>1$. They can be imposed at the same stage.

The level-zero bare contour requires only finitely many inequalities of the first type. Finally decrease $e_1 \le e_0$ so that, for $v \ge -\log e_1$,

$$(1+3\log\sqrt{L}/v)^r < L, \quad r_* > 31L+16, \quad c_{\mathrm{row}}(LMv^r)^A e^{-Bv} \le c \tag{E.7}$$

for the finitely many A, B, c appearing below. Every positive exponent used here is displayed in the relevant estimate. The resulting e_1 is independent of $\bar{m}$. Conditions involving a fixed probe or $\log h'$ are imposed later by decreasing its physical charge threshold; they do not alter this order of construction.

E.3. **Elementary domain properties.** For $A \in \widetilde{\mathcal{R}}_j$ choose its defining splitting $A = A_0 + A_1$. On the enlarged cube write $A_0 = A' + d\lambda$. Since a five-layer cube lies inside the L-layer cube,

$$\|A_1 + C\|_0 < g^{-3/4+\varepsilon} + g^{-1+2\varepsilon} \le g^{-1+\varepsilon}, \qquad \|A'\|_0 < g^{-3/4+\varepsilon} \le g^{-1+\varepsilon}.$$

This proves the inclusion in (9.8) without changing the local real gauge. Imaginary parts arise only from $A_1 + C$, giving its first bound. Every plaquette lies in the one-layer enlargement of one M-cube, where $dA_0 = dA'$. The bound $\|dB\|_0 \le 2\|B\|_1$ gives its second bound. The sets are open by keeping A_0 fixed and perturbing A_1. Real gauges preserve the local bounds; a real constant has a linear gauge on every nonwrapping enlarged cube, and so can be added to A_0. Negation, conjugation and lattice symmetries preserve the definitions.

Rescaling multiplies the three norms by $L^{-1/2}$, $L^{-3/2}$ and $L^{-3/2-\alpha}$. Since $e_{j+1} = \sqrt{L}e_j$, a field in the next-level R domain rescales into $L^{-7/8+\varepsilon/2}R_j$: dividing the three scaled thresholds by the corresponding old thresholds gives $L^{-7/8+\varepsilon/2}$, $L^{-15/8+\varepsilon}$ and $L^{-15/8-\alpha+3\varepsilon/2}$. The first is the largest. For an external field the analogous ratios are $L^{-1+\varepsilon}$ and $L^{-2+3\varepsilon/2}$, both below one. Thus a terminal external field in its strict window remains in every earlier $\mathcal{X}_j$. Periodic extension to an odd covering torus preserves each local representative and each short path. It therefore preserves all these domains, including their margins.

E.4. **Absolute Grassmann estimates.** Expand a polynomial in ordered independent generators. Its absolute coefficient norm at weights h_i is submultiplicative: the coefficient of a product is a sum of signed products, and the weight of a surviving monomial is the product of its factors' weights. Antisymmetrising kernels gives the same norm as the ordered-monomial expansion with the factorials in (9.13). The chosen fine weights and the pair weights make the integral form invariant under the choice of fundamental domain; moving a generator through an antiperiodic seam changes only a sign.

If $\psi(\xi) = \sum_u K(\xi,u)\Psi(u)$, then substitution into a degree-n kernel gives

$$\|F(K\Psi)\|_a \le \|F\|_h \quad \text{provided} \quad a \sup_\xi \sum_u |K(\xi,u)| \le h.$$

Indeed sum the u indices after taking absolute values, then integrate the old kernel indices; this gives one row bound per leg. The same proof works with Hölder pair legs and with the two different barred/unbarred kernels. It does not require a column estimate unless the old norm itself uses that transpose. Fine pair legs retain δ^6, and ordinary fine legs retain δ^3 throughout this substitution.

For a unit white Berezin field use

$$\int d\mu_I(\omega)\, \omega_{i_1}\bar{\omega}_{i_1} \cdots \omega_{i_s}\bar{\omega}_{i_s} = 1$$

at distinct ordered indices, and zero on a monomial lacking a complete pairing. Integrating deletes paired generators and has absolute norm at most one when their weights are at least one. Arbitrary permutations are handled by the corresponding Grassmann sign. An even balanced polynomial remains even and balanced. Product Gaussian integration over local photon white coordinates is also a contraction for a supremum norm on its box. These facts apply to the whole polynomial

before extracting any scalar or quadratic coefficient, so they preserve every contact needed in the normalized map.

For kernels analytic on a connection domain, Cauchy's estimate holds in these absolute norms. It follows either by integrating each coefficient or by using the coefficient vector in a finite dimensional Banach space. Uniform bounds permit coefficientwise limits. A family analytic in the finite-dimensional input data and uniformly bounded in the activity norm has the same estimate in that Banach norm: approximate its contour integral by Riemann sums and use the norm bound. No positive-measure interpretation is applied to an intermediate complex contour.

E.5. **Full-matrix relevant extraction and its adjustment.** For a small polymer X, $d_M(X) \le L$, the vacuum extraction is $E_{00}(X;0,0)$. Let E_{20} be the quadratic kernel written as $\int dxdx'\,\bar\psi(x)E_{20}(x,x')\psi(x')$. Its mean matrix is

$$N(X) = |X|^{-1}\int dxdx'\,E_{20}(X;0,0;x,x').$$

Distribute its constant quadratic form uniformly over the fine sites of X. Translation invariance implies that summing this distribution over polymers produces one common matrix at every fine site. For each auxiliary symmetry (S,a) of (9.14), covariance conjugates the quadratic matrix by S and permutes the small polymers and their site distributions. Summing at zero connection therefore makes the common matrix commute with S. In particular it commutes with each of $\gamma_1,\gamma_2,\gamma_3,\gamma_4$, since the auxiliary group explicitly includes these four generators. Their products span the full matrix algebra on $\mathbb{C}^4$, so its commutant is scalar. Thus define the action mass coordinate by making this sum $-m(E)I$. Charge conjugation and the barred reality involution show $m(E)\in\mathbb{R}$ for a real activity. The action vacuum coordinate likewise makes the local vacuum sum $-\epsilon(E)l^3$. More explicitly, at any fixed ordinary cube $\square$, these translation-independent coordinates are

$$\epsilon(E) = -\sum_{X\supset\square,\ X\text{ small}} |X|^{-1}E_{00}(X;0,0), \qquad m(E)I = -\sum_{X\supset\square,\ X\text{ small}} N(X).$$

The local matrices $N(X)$ themselves need not be scalar.

The kernel norm gives

$$|E_{00}(X;0,0)| \le e^{-\kappa d_M(X)}\|E\|, \quad \int dxdx'|E_{20}(X;0,0;x,x')| \le 2h_1^{-2}e^{-\kappa d_M(X)}\|E\|.$$

The rooted count and $|X|\ge M^3$ imply

$$|\epsilon(E)| \le K_0M^{-3}\|E\|, \qquad |m(E)| \le CK_0M^{-3}g^{1/2}\|E\|.$$

The unimportant finite spin/factorial constant is absorbed in c_{fl}. Subtracting these local means gives an operator of norm at most $4\|E\|$, and its physical sum is exactly the irrelevant activity plus the two relevant terms.

For the quadratic adjustment retain the actual finite chain inside X. For $y,y_0\in X$, average over all shortest face-connected chains $\square_0,\dots,\square_n$ from the cube of y to that of y_0. Put $p_0=y$, p_{i+1} equal to the center of $\square_i$, and $p_{n+2}=y_0$. Every nontrivial pair (p_i,p_{i+1}) has distance at most M and all six rectilinear paths inside X: at the two ends both points lie in one cube, and between two successive centers the path is a straight coordinate segment in the adjacent cubes. Let $U_i=\prod_{a<i}u(p_a,p_{a+1})$, where $u(p,p')=\exp(ig\delta\tau(A+C)(p,p'))$, and use inverse transports for barred fields. Since the chain never revisits a cube,

$$\sum_i |p_i-p_{i+1}|_1 \le c'LM, \qquad \sum_i |p_i-p_{i+1}|^\alpha \le c'LM.$$

Choose e_0 so that $c'LMg^{\varepsilon} \le 1/30$. The imaginary-part bound in the domains gives $|U_i^{\pm 1}| \le \exp(1)$. Writing Mn for the chain average, define

$$T(y,y_0) = \mathrm{Mn}\, U_{n+2},$$
$$R(y,y_0)\chi = \mathrm{Mn} \sum_i U_i |p_i - p_{i+1}|^{\alpha} \chi(p_i, p_{i+1}),$$
$$\Theta\psi(y) = |X|^{-1} \int_X dy_0 [T(y,y_0)\psi(y_0) - R(y,y_0)\chi].$$

The barred rule uses U_i^{-1}. These images are odd linear forms, so they define an algebra homomorphism. Iterating $\psi(p) = u(p,p')\psi(p') - |p-p'|^{\alpha}\chi(p,p')$ shows that Θ fixes the evaluated physical field. Replace only the $(2,0)$ quadratic part by its image under Θ. At $A = C = 0$, its $(2,0)$ kernel is the constant matrix $|X|^{-2} \int dxdy\, E_{20}(x,y)$ and is zero after extraction. Every newly created Hölder pair is of the interior kind and all transport supports lie in X. Expanding the two arms gives

$$\|E_{20}^{\mathrm{nat}}\| \le \exp(2)\|E_{20}\|, \quad \|E_{11}^{\mathrm{nat}} - E_{11}\| \le \exp(2) c'LM \|E_{20}\|, \quad \|E_{02}^{\mathrm{nat}} - E_{02}\| \le \exp(2)(c'LM)^2 \|E_{20}\|.$$

In the weighted norm the cost is $\exp(2)(1 + c'LMg^{\varepsilon})^2 \le 10$. Thus the complete extraction and adjustment costs at most 40. Chain averaging is invariant under the required block translations and coordinate reflections, inverse transport implements charge reversal, and the endpoint gauge phases cancel against the two arms. The adjusted family has all the original symmetries, locality and analyticity. Restriction to $C = 0$ commutes with the construction.

Appendix F. Fermion inverses, local gauges and complete recoveries

The reference mass in this appendix is positive; the selected relevant shift is never included in a primitive inverse. Write $Q^{\sharp}(A) = Q^T(-A)$ and $Q_j^{\sharp}(A) = Q_j^T(-A)$. The fourth Hermitian Clifford matrix γ_4 anticommutes with the three spatial matrices; in particular $D_A^* = \gamma_4 D_A \gamma_4$ for real A. The inverse estimates must cover the entire gauge-saturated domain, including arbitrary harmonic backgrounds. We first prove them for a small global potential, then localize and keep all interface phases.

F.1. **The free completed operator.** Let $F_0 = D_0 + \mathfrak{m} + \mathrm{Pen}$ on a fine lattice of mesh δ. The penalty is a positive combination of projections onto averages over nested blocks of side at most L. Its coefficient at the largest block is at least a fixed positive number divided by L; the smaller coefficients lie in a fixed compact positive interval. The Wilson real form is

$$\mathrm{Re}\langle u, D_0 u\rangle = \frac{\delta}{2} \sum_{\mu=1}^{3} \|\partial_\mu u\|_2^2.$$

A coordinate-line telescoping sum in a block B of side ℓ gives

$$\|u - \mathrm{av}_B\, u\|_{2,B}^2 \le C\ell^2 \sum_{\mu} \|\partial_\mu u\|_{2,B}^2. \tag{F.1}$$

Indeed expand $2\#B \sum_x |u(x) - \mathrm{av}_B\, u|^2$ as $\sum_{x,y} |u(x) - u(y)|^2$, use the three coordinate segments, and count the pairs whose segment contains a given bond. This proves the inequality with an absolute constant, also for mirror blocks. The block average and the Wilson form control complementary modes.

We use a weighted estimate before converting it to a sup estimate. For a block decomposition with side function $\ell(x)$, choose a Lipschitz minorant λ satisfying $\ell/L \le \lambda \le \ell$, $\lambda \ge \delta$, and $|\lambda(x + \delta e_\mu) - \lambda(x)| \le \epsilon_0 \delta$, with $\epsilon_0 \le 1/(6L)$. One may take the infimum of $\ell(y) + \epsilon_0 d(x,y)$, then

truncate below by δ; the separated block sizes ensure the first lower bound. At a boundary of a restricted problem $\lambda \le (L+1)\delta$. Let c_1, C_1 be the fixed penalty constants such that

$$c_1 \sum_B \ell_B^{-1} \|P_B u\|^2 \le \mathrm{Re}\langle u, \mathrm{Pen}\, u\rangle, \qquad \|\lambda^{1/2}\,\mathrm{Pen}\, u\|^2 \le C_1\,\mathrm{Re}\langle u, \mathrm{Pen}\, u\rangle.$$

Put $N = \|\lambda^{-1/2}u\|$, $F = \|\lambda^{1/2}F_0 u\|$ and $X = \mathrm{Re}\langle u, F_0 u\rangle \le NF$. The three nonnegative terms of the real form separately give $\delta\|\nabla^+ u\|^2/2 \le X$, $\mathfrak{m}\|u\|^2 \le X$, and $\sum_B \ell_B^{-1}\|P_B u\|^2 \le X/c_1$. Moreover, $\mathfrak{m}\sup \ell \le L$ in the reference window, so $\|\mathfrak{m}\lambda^{1/2}u\| \le \sqrt{LX}$.

The essential additional estimate uses the first-order part of the Wilson operator, not just its real form. The Fourier symbol gives $\|\nabla^+ v\| \le \|D_0 v\|$ for every v; the Wilson square adds all the cross terms required for this inequality. Take $v = \lambda^{1/2}u$. The nearest-neighbor commutator satisfies

$$\|[D_0, \lambda^{1/2}]u\|^2 \le 18\epsilon_0^2 N^2,$$

since $|\sqrt{\lambda(x')} - \sqrt{\lambda(x)}| \le \epsilon_0\delta\lambda(x')^{-1/2}$ and each site has six neighbors. For a restricted inverse, the boundary term has norm at most $\sqrt{6(L+1)X}$: an exiting bond contributes $\delta^2|u(x')|^2/2$ to the real form and its weight is at most $(L+1)\delta$. Therefore, with $K_1 = \sqrt{L} + \sqrt{C_1} + \sqrt{6(L+1)}$,

$$\|\lambda^{1/2}\nabla^+ u\| \le F + K_1\sqrt{NF} + 6\epsilon_0 N.$$

Poincaré on each block and $\ell/L \le \lambda \le \ell$ now give

$$N^2 \le \frac{L}{c_1}NF + \frac{L^2}{8}\|\lambda^{1/2}\nabla^+ u\|^2 \le \frac{L}{c_1}NF + \frac{3L^2}{8}(F^2 + K_1^2 NF) + \frac{27}{2}L^2\epsilon_0^2 N^2.$$

The last coefficient is at most $27/72 < 1/2$; after absorption, $N \le C_I F$ with $C_I = 2L/c_1 + 3L^2K_1^2/4 + L$. Applying the same argument to the adjoint proves invertibility, including the infinite-volume local problem by closed and dense range. Conjugating by an exponential weight with sufficiently small logarithmic slope perturbs the completed operator by less than $1/(2C_I)$ in this weighted norm. Its Neumann series proves exponential decay with a fixed rate. The block penalty, rather than a lower reference mass bound, controls constant modes throughout.

For the sup and Hölder bounds use the bare Fourier inverse. Its symbol is

$$\widehat{K_0}(p) = \mathfrak{m} + \delta^{-1}\sum_\mu (1 - \cos\delta p_\mu) + i\delta^{-1}\sum_\mu \gamma_\mu \sin\delta p_\mu.$$

The inverse is the adjoint numerator divided by its scalar squared modulus. Near zero a derivative of order a is at most $C_a(|p| + \mathfrak{m})^{-1-a}$. Away from zero the Wilson term separates the denominator from zero, including the other Brillouin corners. A dyadic partition and three integrations by parts give

$$|K_0^{-1}(x,x')| \le C(d(x,x') + \delta)^{-2}, \qquad |\partial K_0^{-1}(x,x')| \le C(d(x,x') + \delta)^{-3}. \tag{F.2}$$

Taking a fractional difference in the same dyadic sum gives every exponent $\alpha < 1$. Apply this kernel to an interior cutoff of a solution of $F_0 u = f$. The cutoff commutator is a positive distance from the interior unit block, has uniformly bounded row sum, and is controlled by the weighted Hilbert estimate. The penalty term is a block constant, bounded by the Hilbert estimate divided by the square root of the block volume. Its convolution with (F.2) is integrable in three dimensions. If a unit block has fewer than L^2 mesh spacings, direct Hilbert-to-sup conversion costs only a fixed power of L; otherwise the interior estimate applies. Hence

$$\|1_{\Delta_y} F_0^{-1} 1_{\Delta_{y'}}\|_{\infty\to\infty} + \|1_{\Delta_y}\delta_\alpha F_0^{-1} 1_{\Delta_{y'}}\|_{\infty\to\infty} \le C_I e^{-\gamma_I d(y,y')}. \tag{F.3}$$

Repeat the interior argument for the adjoint to obtain the transposed bound. At a terminal mass in a fixed positive interval the fine kernel also satisfies (F.2) with exponential decay.

F.2. **The Schur family.** The inverse transport at $-A$ and the unit fine volume of a block give $Q_j Q_j^\sharp = 1$ at complex as well as real A. Composing averages coalesces their hierarchical line sums literally. Put $\beta = b/L$, $P = Q^\sharp Q$, and, for $j \geq 1$ and real y,

$$\begin{aligned} B_y &= (b_j + i\gamma_4 y + \beta P)^{-1}, \\ \mathrm{Pen}_y &= b_j Q_j^\sharp Q_j - b_j^2 Q_j^\sharp B_y Q_j, \\ F_y &= K_j + \mathrm{Pen}_y, \qquad S_y = F_y^{-1}. \end{aligned}$$

At $y = \infty$ take $\mathrm{Pen}_\infty = b_j Q_j^\sharp Q_j$. Multiplication on both sides proves

$$(D_j + \beta P + i\gamma_4 y)^{-1} = B_y + b_j^2 B_y Q_j S_y Q_j^\sharp B_y. \tag{F.4}$$

The cancellation uses $S_y - S_\infty = S_y b_j^2 Q_j^\sharp B_y Q_j S_\infty$. At level zero use $F_y = K_0 + \beta P + i\gamma_4 y$ directly, without b_0 or a fictitious level-zero Schur factor.

Since $P^2 = P$ and it commutes with constant spin matrices,

$$B_y = (b_j + i\gamma_4 y)^{-1}(1 - P) + (b_j + \beta + i\gamma_4 y)^{-1} P.$$

At real A the real part of Pen_y is nonnegative, with the next-block term controlling its constant subspace. The imaginary spin coefficients cause no loss in the free estimate. On a γ_4 eigenspace and either nested projection subspace, the coefficient has the form

$$z = \frac{b_j(a + i\epsilon y)}{b_j + a + i\epsilon y}, \qquad a \in \{0, \beta\}, \quad \epsilon \in \{-1, 1\}.$$

Direct multiplication gives

$$|z|^2 \leq b_j \Re z, \qquad \Re z\big|_{a=\beta} \geq \frac{b_j \beta}{b_j + \beta}.$$

Thus the two penalty inequalities used in the weighted proof hold uniformly in y, although Pen_y itself need not be self-adjoint. At level zero the Fourier identity for $D_0 + i\gamma_4 y$ still controls the ordinary gradient: anticommutation adds y^2 to the squared spin symbol. The large-y inverse there is $O((1 + |y|)^{-1})$ because its native mesh is one and the remaining operator is bounded. Set $w_j(y) = 1$ for $j \geq 1$, $w_0(y) = (1 + |y|)^{-1}$. The bound for B_y is

$$|B_y(y_1, y_2)| \leq \frac{C}{\sqrt{b_j^2 + y^2}} [1_{y_1 = y_2} + L^{-3} 1_{y_1, y_2 \text{ in one } L\text{-block}}]. \tag{F.5}$$

At complex A it is the same idempotent formula with a bounded projection kernel; orthogonality is not asserted there.

F.3. **Small potentials and local inverses.** Define the submultiplicative unit-block norm

$$N_\gamma(O) = \sup_y \sum_{y'} e^{\gamma d(y, y')} \| 1_{\Delta_y} O 1_{\Delta_{y'}} \|_{\infty \to \infty}.$$

We use its transpose and Hilbert variants as well. If $g\|A\|_\infty \leq \delta_A$ and $3L\delta_A \leq 1$, then

$$N_\gamma(F_y(A) - F_y(0)) \leq C_V \delta_A. \tag{F.6}$$

For a Wilson bond the estimate $|e^{ig\delta A} - 1| \leq g\delta |A| e^{g\delta |A|}$ cancels the δ^{-1} in the operator. Hierarchical paths inside a unit block have length at most $3/2$, and those in the next block at most $3L/2$. Projection differences therefore satisfy the same bound; their kernels read only the entire unit block or L-block. The resolvent identity for B_y retains its two local bounded factors. Choose $C_I C_V \delta_A \leq 1/2$. The Neumann series

$$S_y(A) = S_y(0) \sum_{n \geq 0} [-\{F_y(A) - F_y(0)\} S_y(0)]^n$$

converges in all three norms, is analytic, and is the exact inverse. The product rule

$$\delta_{\alpha,A}(hf)(x,x') = h(x)\delta_{\alpha,A}f(x,x') + |x-x'|^{-\alpha}(h(x')-h(x))U_A(x,x')f(x')$$

gives its covariant Hölder estimate. A long pair uses two sup bounds and keeps its complete phase; it is never treated as a short uncharged difference.

For $A \in \widetilde{\mathcal{G}}_j$, fix the defining real gauge on $\square^{\sim 5}$. Its representative plus the global complex part has norm below $2g^{-1+\varepsilon}$. Fold the cube of side $3M$ into a mirror torus of side $6M$, using reflection orientation on bonds and zero normal values on mirror seams. Let $A^\square$ be that extension. Its norm obeys $g\|A^\square\| \le 2g^\varepsilon$. With reflection extension $e_\square$ and restriction $r_\square$, define $S_y^\square = r_\square F_y (A^\square)^{-1} e_\square$, then restore the two real gauge phases. In the interior, all Wilson bonds and complete block paths lie inside the mirror chamber. Consequently $F_y S_y^\square f = f$ on its interior and $S_y^\square F_y f = f$ for an interior-supported field. The latter identity supplies a separate transposed parametrix. A change of local gauge rotates each inverse by its endpoint phases; no interface phase is discarded when local inverses are composed.

F.4. **Walks, weakening and two-ended locality.** Choose $\sum_\square h_\square^2 = 1$, with each cutoff supported in the cube of side $3M$ and derivative at most π/M. Set $S_y^* = \sum h_\square S_y^\square h_\square$. Then

$$F_y S_y^* = 1 - R_y, \qquad R_y = -\sum_\square [F_y, h_\square] S_y^\square h_\square.$$

The commutator couples only nearest neighbors or one complete block. Its norm is bounded by $C_V \pi/M$: a fine neighbor difference cancels the Wilson inverse mesh, and a block difference costs at most its diameter $3L$. At most 27 cutoffs overlap. Thus $N_\gamma(R_y) \le q_M = 54\pi C_V C_I/M < 1/2$, and

$$S_y = S_y^* \sum_{n\ge 0} R_y^n. \tag{F.7}$$

The reversed local-inverse identity gives a reversed series with the same bound, proving columns as well as rows.

A term reads its cutoff cubes, the complete averaging paths and the interface bonds. Its connected support $X(\omega)$ contains at most a fixed number $n_F(L)$ of additional cubes per link. For the weakening multiply each constituent by the product of the parameters on its full support. On $|s_c| \le M^{\alpha_0}$ the additional cost per link is at most $M^{2\alpha_0 n_F}$. Choose $2\alpha_0 n_F < 1$ before M and impose $q_M M^{2\alpha_0 n_F} e^{\kappa_w n_F} < 1/2$. Group the walk terms with the same support. Uniform convergence gives

$$\|1_{\Delta_y} S_y(X,s,A) 1_{\Delta_{y'}}\| \le C_w w_j(y) e^{-\gamma_w d(y,y') - \kappa_w d_M(X)}. \tag{F.8}$$

This holds for the row, transpose, Hilbert and covariant Hölder norms. Both indices lie in X; its connection support consists of the bonds meeting X. At $s = 1$ the series is the exact inverse. If nonzero parameter cubes split into components, a walk crossing components contains a zero parameter. Rows in a component therefore vanish at columns outside it and read only that component; reversed walks prove the column statement. Intermediate gauge phases cancel at matched indices and external phases remain, proving covariance on the full connection support.

F.5. **The terminal fine Green kernel.** At the two reference stops, m^{ref} lies in the fixed positive interval in (9.4). We spell out why the unit-block bound then also has the fine singularity required for the later continuum fermion coefficients. Choose a smaller fixed exponential rate. The dyadic proof of (F.2) gives the bare kernel $|K_0^{-1}(x,x')| \le C_m (d(x,x')+\delta)^{-2} e^{-\gamma_m d(x,x')}$. On a local small-potential representative the Wilson perturbation is a bounded six-neighbor kernel with row sum at most $Cg\|A\|_\infty$. Weighted convolutions of two copies of the above singular kernel are at most $C_m(d+\delta)^{-2} e^{-\gamma_m d/2}$: split the integration into the regions within $d/2$ of either endpoint and their complement; the radial integral of r^{-2} is finite in three dimensions, and exponential decay pays the remaining region. The same argument works for shifted nearest neighbors. The resulting pointwise

Neumann series converges when $C_m g\|A\|_\infty < 1/2$, a permitted fixed small-charge condition, and gives that bound for K_A^{-1}.

For the completed inverse use the exact Woodbury formula

$$S = K_A^{-1} - K_A^{-1} Q_j^\sharp (b_j^{-1} + Q_j K_A^{-1} Q_j^\sharp)^{-1} Q_j K_A^{-1}.$$

Each end factor averages an integrable fine singularity over a unit block, so its kernel and transposed kernel are bounded. The middle coarse inverse has a fixed Hilbert bound and exponential decay, by the completed block estimate and its exponential conjugation argument. The correction is thus bounded by $C_m e^{-\gamma_m d/2}$. The singularity of the first term dominates it at short distance. On gauge-saturated domains use the local gauges and the walk expansion; matched endpoint phases cancel, their real gauge parts have modulus one, and the reserved walk decay absorbs the preceding convolutions. This proves

$$|S_j(A)(x,x')| \le C_m (d(x,x')+\delta)^{-2} e^{-\gamma'_m d(x,x')},$$

including $C_m \delta^{-2}$ at coincident sites. Its constants depend on the fixed terminal reference-mass interval, not on the selected relevant mass, the fine mesh or the torus volume.

F.6. **Exact recovery without an inverse-mesh loss.** $T_j H_j^F = 1$ follows by multiplication, but its naive row contains δ^{-1}. Average Wilson neighbors over a unit block and transport them to its center. Interior bonds give an ordinary field dependence; an exiting bond gives an admitted nearest-neighbor Hölder pair. This defines exact kernels

$$T_j \psi = T_j^{(1)} \psi + T_j^{(2)} \chi, \qquad \chi = \delta_{\alpha,A} \psi. \tag{F.9}$$

More explicitly, let Y be the polymer under consideration, $x^{\nu,\sigma} = x + \sigma\delta e_\nu$, let $\phi_{A,y}(x)$ be its hierarchical phase, and put $P_{\nu,\sigma} = (1-\sigma\gamma_\nu)/2$ and $u_A(x,x') = \exp(ig\delta A(x,x'))$. Then

$$(T_j^{(1)}\psi)(y) = \delta^3 \sum_{x\in B_j(y)} \phi_{A,y}(x) \left[(1+\mathfrak{m}_j/b_j)\psi(x) - \frac{1}{b_j\delta} \sum_{x^{\nu,\sigma}\in Y} P_{\nu,\sigma}\{u_A(x,x^{\nu,\sigma})\psi(x^{\nu,\sigma}) - \psi(x)\} \right],$$

$$(T_j^{(2)}\chi)(y) = -b_j^{-1}\delta^{2+\alpha} \sum_{x\in B_j(y)} \phi_{A,y}(x) \sum_{x^{\nu,\sigma}\notin Y} P_{\nu,\sigma}\chi(x,x^{\nu,\sigma}).$$

The sums over ν,σ range over $\nu = 1,2,3$ and $\sigma = \pm 1$. On a nearest-neighbor pair, $\delta^\alpha \chi(x,x') = u_A(x,x')\psi(x') - \psi(x)$, which proves (F.9) with its exact powers of the fine mesh. $B_j(y) \subset Y$, so the first row has only interior ordinary indices and the second has exactly the permitted exiting pairs. The phase paths remain within $B_j(y)$, and the exiting connection bond has an endpoint there. Both rows are entire functions of precisely these connection values. To check the interior row, compare the hierarchical neighbor path with the path taking the nearest-neighbor bond first. If their first common block is at depth i, their closed loop fills with at most $6L^{2i}$ fine plaquettes. Its phase difference is bounded by $6L^{2(i-j)} g\|dA\| e^{6g\|dA\|}$. At most $C\delta^{-3}L^{1-i}$ sites have that first-coalescence depth. Multiplying by the averaging measure and by the Wilson inverse mesh gives a geometric sum bounded by $CLg\|dA\|e^{6g\|dA\|}$, uniformly in j. Boundary faces contain $O(\delta^{-2})$ sites; the field difference is represented in the independent pair kernel with its δ^6 measure. Its weighted row is bounded by $6b_-^{-1} e^{3g\|\operatorname{Im} A\|/2}$. Hence both exact rows are bounded on the stated domain, and every boundary pair and interface bond is retained. The barred rule is the charge-conjugate transpose. No fine-column assertion is needed for this recovery.

Composing (F.8) with $Q_j^\sharp$ gives the physical fermion legs and their transposed rows: the unit-block fine measure is one, and the phase modulus is at most $e^{3g\|\operatorname{Im} A\|/2}$. The Schur formula at $y = 0$ gives the coarse covariance. At a real connection the completed real form is strictly positive; eliminating

the fine variables proves invertibility of its coarse Schur complement. Finally $b_{j+1}^{-1} = L^{-1}b_j^{-1} + b^{-1}$ coalesces two Gaussian penalties after rescaling and gives

$$(H_{j+1}^F(A')\Psi')_L = H_j^F(A'_L)\Psi^c, \qquad \Psi^c = T_j(A'_L)(H_{j+1}^F(A')\Psi')_L.$$

These identities are checked by multiplication of the completed fine precision and supply the exact classical recovery in the step.

Appendix G. Photon minimizers, white roots and reconstruction

The Gaussian photon construction includes its gauge quotient. At mesh δ let d be the oriented coboundary with its inverse-mesh factor, and use $\delta^3 \sum$ in every inner product. The uncharged bond average is the average over the block and over the straight segment joining it to the next block, with the segment divided by its length. It commutes with d when the scalar and two-form averages use the corresponding cell averages. Denote this uncharged average by Q_j^P to distinguish it from a charged fermion average.

G.1. **The constrained minimum and its gauge.** On the affine space $Q_j^P\mathcal{A} = A$, minimize $\|d\mathcal{A}\|^2$. This finite-dimensional least-squares problem has a minimizer: split the space into the kernel and orthogonal complement of d and minimize on the complement. Every minimizer obeys $\langle d\mathcal{A}, dB\rangle = 0$ for $Q_j^P B = 0$. Two minimizers differ by a closed field in $\ker Q_j^P$. A closed periodic one-form is a constant plus an exact form. The average of the constant is itself, so a closed field in this kernel is $d\lambda$ with $Q_j^P\lambda = 0$, after subtracting its scalar mean. Thus the minimizers constitute exactly one restricted gauge orbit.

Let $\mathcal{N}_j$ be the scalar functions with $Q_j^P\lambda = 0$. The Landau condition is orthogonality of the divergence to $\Delta\mathcal{N}_j$. The axial condition is that each average of the six coordinate-order paths from a block center to an interior site vanishes at every earlier scale. Both choose one representative of the restricted orbit. For Landau gauge this follows from the positive scalar form $\|d\lambda\|^2$ on $\mathcal{N}_j$; for axial gauge define λ by those center-to-site path averages, successively from large blocks to small ones. Each average is zero at its center and has zero block mean after subtracting the mean, so this recursion preserves the restricted scalar condition. The resulting maps $H_j^P, H_j^{P,x}$ are real linear right inverses of Q_j^P and

$$H_j^{P,x} = H_j^P + dO_j, \qquad dH_j^{P,x} = dH_j^P. \tag{G.1}$$

They preserve constants, means and exact fields. In particular $H_j^P c = c$ for an arbitrary constant background. For any $B \in \ker Q_j^P$,

$$\|d(H_j^P A + B)\|^2 = \|dH_j^P A\|^2 + \|dB\|^2. \tag{G.2}$$

This proves the exact Gaussian split used in blocking and its independence from a choice of representative.

For an explicit operator formula let Π_j^{sc} be the scalar orthogonal projection onto $\Delta\mathcal{N}_j$, and put $G_j^P = (d^*d + d\Pi_j^{\mathrm{sc}}d^* + aQ_j^{P,*}Q_j^P)^{-1}$ with any fixed $a > 0$. Its real form is strictly positive: a vector in its kernel is closed, annihilated by Q_j^P, and Landau, hence zero. The same argument proves positivity of $Q_j^P G_j^P Q_j^{P,*}$. The normal equations therefore give

$$H_j^P = G_j^P Q_j^{P,*}(Q_j^P G_j^P Q_j^{P,*})^{-1}. \tag{G.3}$$

Changing the gauge convention changes only O_j in (G.1); all curvature Gaussian quantities are fixed.

G.2. **A mesh-independent analytic symbol.** We give details because a scalar zero mode must be removed before claiming an analytic strip. Put $n_\delta = \delta^{-1}$, an odd integer, and define

$$s(t) = 2n_\delta \sin(t/(2n_\delta)), \qquad u(t) = n_\delta^{-1} \sum_{a=-(n_\delta-1)/2}^{(n_\delta-1)/2} e^{ita/n_\delta}.$$

The telescoping geometric sum proves $u(t)s(t) = 2\sin(t/2)$. For $q = p + 2\pi m$, $p \in [-\pi, \pi]^3$, let

$$d_\mu(q) = \delta^{-1}(e^{i\delta q_\mu} - 1), \quad D(q) = \sum_\mu s(q_\mu)^2, \quad U(q) = \prod_\mu u(q_\mu), \quad V_\mu(q) = e^{i(1-\delta)q_\mu/2} u(q_\mu).$$

The star in the following formulas means evaluation at $-q$, not complex conjugation away from real momenta. Then $V_\mu(q)d_\mu(q) = \rho_\mu(p) = e^{ip_\mu} - 1$. The bond-average symbol is $U(q)V_\mu(q)$; its transpose is $U(q)V_\mu^*(q)$. Sum over the n_δ^3 representatives $m \in \{-(n_\delta-1)/2, \dots, (n_\delta-1)/2\}^3$. Set $D_0 = D(p)$, $U_0 = U(p)$, $s_\mu = s(p_\mu)$, and

$$\begin{aligned} X_\mu &= \sum_{m\neq 0} U(q)^2 u(q_\mu)^2/D(q), \qquad Y = \sum_{m\neq 0} U(q)^2/D(q)^2, \\ N_\mu &= U_0^2 u(p_\mu)^2 + D_0 X_\mu, \qquad N_G = U_0^2 + D_0^2 Y, \\ \mathcal{N} &= N_1 N_2 N_3, \qquad \Pi = \mathcal{N} - \sum_\mu s_\mu^2 X_\mu \prod_{\nu\neq\mu} N_\nu. \end{aligned}$$

Define

$$\begin{aligned} \Lambda_{\mu\nu} &= \delta_{\mu\nu} D_0/N_\mu - \rho_\mu \rho_\nu^* U_0^2 \mathcal{N}/(N_\mu N_\nu \Pi), \\ \omega_\nu &= \rho_\nu^* D_0 U_0^2 \mathcal{N}/(N_\nu N_G \Pi), \\ h_{\mu\nu}(p,m) &= U(q)V_\mu^*(q)\Lambda_{\mu\nu}/D(q) + d_\mu(q)U(q)\omega_\nu/D(q)^2 \quad (m \neq 0), \\ h_{\mu\nu}(p,0) &= \delta_{\mu\nu} U_0 V_\mu^*(p)/N_\mu \\ &\quad + d_\mu(p)\rho_\nu^* U_0^3 (X_\mu - u(p_\mu)^2 D_0 Y)\mathcal{N}/(N_\mu N_\nu N_G \Pi). \end{aligned}$$

These are the symbol of (G.3). To check this, multiply by the block-average symbol and sum aliases; the definitions of X_μ, Y give the identity matrix. The residual Euler equation is perpendicular to $\ker Q_j^P$, while contraction with d^* satisfies the restricted Landau condition. The uniqueness already proved identifies the solution with H_j^P. The second line for $m = 0$ is the algebraically canceled expression at $p = 0$; it is not a formula involving an uncanceled D_0^{-1}.

Here are quantitative strip bounds. On the real cube, $u(p_\mu) \geq 2/\pi$, $U_0 \geq (2/\pi)^3$, and $|D(q)| \geq c|m|^2$ for $m \neq 0$. The sums X_μ, Y are nonnegative and uniformly bounded, because $|u(q_\mu)| \leq C\langle m_\mu\rangle^{-1}$. The identity $\Pi/\mathcal{N} = U_0^2 \sum_\mu s_\mu^2 u(p_\mu)^2/(D_0 N_\mu)$ holds when $D_0 \neq 0$, and has its continuous value at zero. Since $\sum_\mu s_\mu^2/D_0 = 1$ on real momenta, it is a positive weighted average with uniformly positive summands. This gives a fixed positive lower bound for N_μ, N_G, Π on the real cube. The alias estimates and their first complex derivatives are summable uniformly in n_δ. Choose a strip width $c_s > 0$ small enough that their changes from the real cube are below half those lower bounds. Then every denominator remains separated from zero for $|\operatorname{Im} p| < c_s$. In that strip,

$$\begin{aligned} |h_{\mu\nu}(p,m)| &\leq C\langle m_\mu\rangle^{-1} \prod_\sigma \langle m_\sigma\rangle^{-1} |m|^{-2} \\ &\quad + C\langle m_\mu\rangle \prod_\sigma \langle m_\sigma\rangle^{-1} |m|^{-4} \quad (m \neq 0). \end{aligned}$$

Multiplication by one derivative and a Hölder difference costs $C|m|^{1+\alpha}$. The first term is summed by assigning the excess decay to the two coordinates other than μ; the resulting powers are 2 in coordinate μ and $1 + (1-\alpha)/2$ in each other coordinate. For the second term use equal powers

$1+(1-\alpha)/3$ in all three coordinates. Both are summable for $\alpha<1$, uniformly in the number of aliases.

For a fine offset z inside a unit cube, $K(p;z)=\sum_m e^{i(p+2\pi m)\cdot z}h(p,m)$ is periodic in p, analytic in the fixed strip, and uniformly bounded together with its first and Hölder derivatives. Shift each of its three Fourier contours towards the sign of a unit-block displacement. The shift gives $e^{-c_s|y-y'|_1}$. Periodizing over $l\mathbb{Z}^3$ gives the identical estimates with torus distance. This proves the photon minimizer bounds, including their mesh independence and their validity on a torus of any admitted side.

G.3. **The finite-dimensional axial fluctuation space.** The coordinates must preserve the rectangular cutoff as well as the Gaussian measure. At one unit-lattice blocking step, call a bond interior if its ends are in one L-block. The other bonds cross a face between two blocks. Among the L^2 bonds of each face, its central bond lies on the line through the two block centers. Let V be the real bond functions v that vanish on the central bonds and satisfy $\tau v=0$ inside each block, where τ averages the six coordinate-order paths from its center to a site. Only the interior bonds occur in this constraint; every noncentral face bond is an independent coordinate.

On the interior bonds of $B(0)$ the three coordinate reflections $\mathfrak{r}_1,\mathfrak{r}_2,\mathfrak{r}_3$ are commuting orthogonal involutions. They preserve $\ker\tau$, since reflection permutes the six paths. Their restrictions can therefore be diagonalized simultaneously over the reals: successively restrict to the $+1$ and -1 eigenspaces of each involution and choose an orthonormal basis in their common eigenspaces. Fix this basis $\{e_i^0\}$ once for the chosen L and translate it to e_i^y in every block. Complete it by the bond delta functions δ_b for positively oriented noncentral face bonds. Thus the orthonormal basis is

$$\{\Upsilon\}=\{e_i^y\}\cup\{\delta_b : b \text{ is a noncentral face bond}\}.$$

The map τ from the $3L^2(L-1)$ interior bond variables onto the L^3-1 site values away from the center is onto: for a prescribed site function f with $f(0)=0$, the interior gradient df has $\tau(df)(0,x)=f(x)$. Hence the interior kernel has dimension $2L^3-3L^2+1$. The three outgoing faces supply $3(L^2-1)$ further coordinates per block. Consequently

$$\dim V=(2L^3-2)(l/L)^3=2(1-L^{-3})l^3.$$

Each basis support is one block or one bond; its norm and its construction are independent of the torus and the level.

The map C_j restores the central bonds. It equals v off these bonds and, for a face $c=\langle y,y+Le_\mu\rangle$, is defined by

$$(C_jv)(b_c)=-\sum_{\substack{b \text{ crosses } c\\ b\neq b_c}} v(b)-\sum_{\substack{b \text{ interior in } B(y) \text{ or } B(y+Le_\mu)\\ b\parallel e_\mu}}\frac{n_c(b)}{L}v(b). \tag{G.4}$$

Here $n_c(b)$ counts the straight paths of length L from sites of $B(y)$ that traverse b; on an interior bond it lies between 1 and $L-1$. Each face bond is traversed by L paths, so

$$L^4(Q^PZ)(c)=L\sum_{b \text{ crosses } c} Z(b)+\sum_{b \text{ interior}} n_c(b)Z(b).$$

Thus (G.4) is exactly the equation $Q^PC_jv=0$. The axial constraint uses no central bonds, so $\tau C_jv=0$. Conversely, a field satisfying these two equations is recovered uniquely by setting its central bonds to zero and applying C_j. This proves that C_j is a bijection from V to $\ker Q^P\cap\ker\tau$. The formula reads only two neighboring blocks and their common face. A unit bound on all basis coordinates gives $|v(b)|\le(2L^3)^{1/2}$ on interior bonds and $|v(b)|\le1$ on the other noncentral bonds. There are at most $2L^3$ summands in a central-bond formula, all with coefficient of modulus at most

one; hence $\|C_j v\|_\infty \le 3L^{9/2} \sup_\Upsilon |v(\Upsilon)|$. Each interior bond occurs in at most two such formulas and each face bond in only one. Cauchy–Schwarz therefore gives

$$\|v\|_2^2 \le \|C_j v\|_2^2 \le (1 + 4L^3)\|v\|_2^2.$$

Translations by $M\mathbb{Z}^3$ and coordinate reflections act on this basis by signed permutations. Indeed M is a multiple of L, and for $gx = rx + a$, with r diagonal with entries ± 1, $ge_i^y = \pm e_i^{a+ry}$ by the joint eigenvector construction; also $g\delta_b = \pm\delta_{b'}$ with b' the positively oriented image bond. Central bonds are mapped to central bonds. These transformations preserve Q^P, τ and the unique restoring formula, so $gC_j = C_j g$. The minimizer commutes with them by its normal equations. The one-step precision is

$$\widetilde{\Delta}_j = C_j^*(dH_j^P)^* dH_j^P C_j.$$

The axial Stokes estimate supplies a lower bound uniformly in j. For an interior bond the six path constraints express its value as an average of strips, each containing at most $L - 1$ plaquettes. For a face bond, subtract its face average; the remaining difference is expressed by two interior strips and one face strip. Its norm is bounded by CL^2 times the curvature norm in the neighboring blocks. Since $Q^P B = 0$, the face average is also controlled by these strips. Summing the finite number of strips through a plaquette gives

$$c_P\|v\|^2 \le \langle v, \widetilde{\Delta}_j v\rangle \le C_P\|v\|^2.$$

The upper bound follows from the minimizer kernel just proved. Both constants depend on L, not j, l. The precision kernel decays exponentially in the distances of its local basis supports. Conjugate it by $e^{cd(\Upsilon,\Upsilon_0)}$. Its change has norm $O(c)$ by the Schur test, using the two-ended exponential kernel. For c small it changes every $(\widetilde{\Delta}_j + t^2)^{-1}$ by at most half its inverse norm. Thus

$$|(\widetilde{\Delta}_j + t^2)^{-1}(\Upsilon, \Upsilon')| \le C(c_P + t^2)^{-1} e^{-\gamma_P d(\Upsilon,\Upsilon')}.$$

The positive square root of the covariance is

$$R_{\mathrm{loc},j} = \widetilde{\Delta}_j^{-1/2} = \frac{2}{\pi}\int_0^\infty (\widetilde{\Delta}_j + t^2)^{-1}\, dt. \tag{G.5}$$

The integral converges in norm and gives a real positive operator, with the asserted exponential two-ended kernel. Functional calculus also gives $gR_{\mathrm{loc},j} = R_{\mathrm{loc},j} g$. The coordinate box $|\zeta(\Upsilon)| \le p_{0,j}$ is invariant under the signed permutations just proved and under $\zeta \mapsto -\zeta$; the standard Gaussian and its normalized restriction to the box are therefore invariant. No invariance under arbitrary orthogonal changes of this basis, or under coordinate permutations, is needed or asserted. These symmetries are the block translations and coordinate reflections used in extraction. This proves both the exact covariance and the local rows used below, with their actual cutoff measure.

G.4. **Weakening photon rows and columns.** Fix the translated M-cube partition and assign to each pair of indices one of the shortest coordinate-order cube paths, averaging over ties. Multiply its physical kernel by the product of parameters on the complete path and on the endpoint support cubes. A partition of unity on the starting endpoint gives the same recipe for the minimizer, including derivatives of its cutoff. At $s = 1$ the partition sums to one and every parameter product is one, so the physical kernel is recovered exactly. At $s \ne 1$ the weakened root is a local analytic kernel, not asserted to square to a covariance.

If a kernel joins supports at distance d, its parameter degree is at most $n_P(1 + d/M)$. Hence on $|s_c| \le M^{\alpha_0}$ its absolute value is at most the physical decay times $M^{\alpha_0 n_P} \exp(\alpha_0 n_P (\log M) d/M)$. Choose M so that the latter rate is below $\gamma_P/4$. Differentiating the endpoint partition gives bounds $3/M$ and $42/M^2$; they can be absorbed by the remaining decay. The same recipe reversed gives columns and their complete paths. A parameter component cannot be crossed by a nonzero row or column, because every connecting path then contains a zero parameter. This establishes the two-ended support condition used in comparison of reference tori. Only a finite number of endpoint

cubes is added to the support, including the whole basis-vector support; there is no implicit infinite convex hull enlargement.

G.5. **Reconstruction on the actual small-field domain.** For an arbitrary coarse real field A', the exact one-step Gaussian split is $A = H^{P,x}_{\mathrm{step}} A' + C_j R_{\mathrm{loc},j}\zeta$. Applying the fine minimizer gives

$$H^P_j A = (H^P_{j+1} A')_L - d\omega + Z, \qquad Z = H^P_j C_j R_{\mathrm{loc},j}\zeta. \tag{G.6}$$

The gauge ω is real; it is retained in both fermion arms and in the outgoing coarse field. This equality follows by the normal equations and (G.2): the classical part minimizes on the combined fiber, while Z is orthogonal to that part.

If $|\zeta(\Upsilon)| \le p_{0,j}$, the two exponential row bounds give all three Z norms bounded by $C_Z p_{0,j}$. Since $p_{0,j}$ is a power of v_j, impose $C_Z p_{0,j} \le \frac{1}{2} g^{-\varepsilon/2}$ by the final charge choice. This places the new fluctuation field inside the margin of $\widetilde{\mathcal{R}}^-_j$ and, after multiplying a white contour by its permitted radius, inside the declared complex domain.

The coarse curvature condition also gives a local representative of the classical field. Subtract its harmonic constant and take a coordinate-order tree gauge rooted near a chosen M-cube. Interior bonds at distance r from that root are bounded by $(2r + 4 + s_d)\|dA'\|$, where $s_d = (2L+1)M$. Seam bonds are bounded by $3l^2\|dA'\|/2$ but have distance at least $(l - 1 - s_d)/2$. Since $l \ge 13LM$, $s_d \le 3l/13$. Convolving with the exponential minimizer row bounds the seam contribution by $Cl^2 e^{-5\gamma_P l/26}\|dA'\|$, uniformly in l. The identity $\sup_{x>0} x^2 e^{-cx} < \infty$ pays this cost. On the chosen enlarged cube the constant is an exact local gauge; the local representative therefore has the required three small-field bounds whenever $8C_B p_j \le g^{-3/4+3\varepsilon}$. The rescaled classical field and the new fluctuation field consequently lie in the actual domain throughout the whole normalized integration, with arbitrary harmonic part. This proves the domain join rather than only a zero-harmonic version of it.

G.6. **Orthogonal coordinates of the curvature Gaussian.** Iterate (G.2). Every new fluctuation fiber is orthogonal, in the Maxwell form, to all later fibers and to the terminal curvature space. The map from a mean-zero exact curvature to its new fluctuation coordinates ζ_j and terminal curvature is therefore an isometry onto a direct orthogonal sum. Its inverse is the repeated reconstruction (G.6), with the gauges removed after taking d. The coordinates $\zeta_j(\Upsilon)$ are norm-one Gaussian linear functionals. A terminal plaquette value is the orthogonal projection of its original plaquette functional onto the terminal subspace and has norm at most one. Harmonic averaging adds an independent compact constant and does not change any of these curvature statements. At level j there are $2(1 - L^{-3})l_j^3 \le 2l_j^3$ new fluctuation coordinates, and $3l_*^3$ terminal plaquettes. These are exactly the coordinate counts used in the large-field union bound.

Appendix H. The growing collar and irrelevant contraction

We prove the contraction on a torus, including the small-torus cap. All Taylor expansions below are expansions of a bond component at its bond barycenter. A derivative component is based at the barycenter of the corresponding two-bond cell. This convention supplies the half-mesh correction in an exact discrete gradient and avoids a mesh-dependent quadratic remainder.

H.1. **Cutoffs with their finite differences.** Choose a one-dimensional C^1 function f equal to one on $[-1, 1]$, zero outside $[-2, 2]$, with $|f'| \le 2$ and Lipschitz constant of f' at most four. For a center x_0 let the bond cutoff χ_r be the product of $f((x_\nu - x_{0,\nu})/r)$ evaluated at its barycenter. If $4r + 11 < l$, it is a periodic cutoff supported in the embedded cube of radius $2r + 1$. The fundamental theorem of calculus on one fine bond gives

$$|\partial_\nu \chi_r| \le 2/r, \qquad |\partial_\nu \chi_r(x') - \partial_\nu \chi_r(x)| \le 12|x' - x|/r^2.$$

For a bond function with norms B_0, B_1, B_α on the containing cube, the product difference formula consequently gives

$$\|\chi_r b\|_0 \le B_0, \quad \|\chi_r b\|_1 \le B_1 + 2B_0/r, \quad \|\chi_r b\|_{1,\alpha} \le B_\alpha + 12B_1/r + 12B_0/r^2. \tag{H.1}$$

The last estimate uses $|x - x'|^{1-\alpha} \le 1$ for short pairs; long pairs use twice the derivative sup norm. The cutoff sees every bond of every path in the polymer's full support and connection dependence, since it is one on their entire containing cube.

The periodic mean-zero slope bound will pay the cap. A periodic first difference has mean zero. For a component $a = \partial_\nu b_\mu$ and its Hölder norm $[a]_\alpha$, sum $a(x) - a(y)$ over y after splitting a coordinate line into segments of length at most one. This gives

$$|a(x)| \le Cl[a]_\alpha. \tag{H.2}$$

The harmless absolute constant is retained in the collar estimates. No condition on the mean of b itself is needed.

H.2. **Removing the affine gauge jet.** Let X be a small polymer and choose $x_0 \in X$. Its full support and connection dependence lies in a cube of radius $\rho = 2LM$. A next-level connection, rescaled to the current level and put in a defining local real gauge, has bounds

$$B_0 \le L^{-7/8+\varepsilon/2} t_1, \quad B_1 \le L^{-15/8+\varepsilon} t_2, \quad B_\alpha \le L^{-15/8+3\varepsilon/2} t_3, \qquad t_i = g^{-3/4+i\varepsilon}.$$

Write $w = L^{-15/8+3\varepsilon/2} t_3$ and $v = L^{-15/8+\varepsilon} t_2$. For the real representative b, the exact discrete quadratic scalar

$$\lambda(x) = \sum_\mu b_\mu(x_0)(x_\mu - x_{0,\mu}) + \frac{1}{2}\sum_{\mu,\nu} s_{\mu\nu}(x_\mu - x_{0,\mu})(x_\nu - x_{0,\nu}) - \frac{\delta}{2}\sum_\mu s_{\mu\mu}(x_\mu - x_{0,\mu})$$

has gradient equal to the constant plus symmetric affine derivative jet, where $s_{\mu\nu}$ is the symmetric part of the derivative at x_0. Subtract this exact real gauge. The remaining real affine part is the antisymmetric curvature jet, and its coefficient is at most $2\min(v, Clw)$ by (H.2). The Taylor remainder has, on a cube of radius r, bounds $Cr^2 w, Crw, Cw$. Apply the same decomposition to the global complex part, except that its complex constant is kept for the holomorphic gauge identity below.

Use χ_ρ on the Taylor remainder and

$$\rho' = \min(g^{-\varepsilon}, l/5) \tag{H.3}$$

on the affine remainder. After reducing e_0, $\rho' \ge \rho$; every admitted side has $l \ge 13LM$, so $4\rho' + 11 < l$. The crucial affine sup estimate is

$$\rho' \min(v, Clw) \le Cg^{-\varepsilon} v \quad \text{if } \rho' = g^{-\varepsilon}, \qquad \rho' \min(v, Clw) \le Cl^2 w \quad \text{if } \rho' = l/5.$$

In the capped case $l < 5g^{-\varepsilon}$, so the right side is at most $25Cg^{-2\varepsilon} w$. Dividing by t_1 or t_2 uses $t_3/t_2 = g^{\varepsilon}$ and $t_3/t_1 = g^{2\varepsilon}$. Thus in both cases the cutoff affine term satisfies the same three domain bounds as the uncapped construction. The cutoff remainder uses (H.1) with $r = \rho$. A direct product estimate gives, after enlarging the absolute constant,

$$\|\widehat{b}\|_\infty \le 1382 L^{-15/8+3\varepsilon/2} t_1, \quad \|\partial\widehat{b}\|_\infty \le 1382 L^{-15/8+3\varepsilon/2} t_2, \quad \|\delta_\alpha \partial\widehat{b}\|_\infty \le 1382 L^{-15/8+3\varepsilon/2} t_3,$$

The choice $L^{1/8-3\varepsilon/2} \ge 2400$ implies that $t\widehat{b} \in \mathcal{R}_j$ for $|t| \le L^{7/4}$. The field equals the gauge-reduced original field on the full support and connection dependence of X, not just at its Grassmann indices.

The complex constant has a distinct cost. A constant complex connection on an embedded local support is $d\lambda_z$ with a linear complex scalar. At first take z real; local gauge invariance gives the physical activity with endpoint factors $e^{ig\lambda_z}$. Both sides are entire in z, so the identity theorem extends the same formula to the permitted complex z. A degree-n monomial costs at most $\exp(2n)$ after the charge and smallness conditions; a vacuum monomial has no cost. This is a finite endpoint

cost, not a gauge invariance assertion on a torus for an arbitrary complex constant. The winding and scalar determinant correction remain separately retained.

H.3. **The degree-by-degree bound.** For an adjusted small-polymer activity, its vacuum and $(2,0)$ quadratic kernels vanish pointwise at zero connections. Charge reversal makes the vacuum kernel even. The preceding collar produces an analytic one-variable function on $|t| \le L^{7/4}$ through the connection on the full support and connection dependence. Cauchy's formula gives a factor $L^{-7/2}$ for its vacuum value and $L^{-7/4}$ for its quadratic value. This application is to the independent Grassmann kernels; the physical Hölder evaluation is made afterwards.

Direct spinor scaling contributes L^{-n} to n ordinary legs, and $L^{-(1+\alpha)m}$ to m Hölder legs. Comparing the next weights to the old weights multiplies these by $L^{-n/8}$ and $L^{-m/8+\varepsilon m/2}$, respectively. The only degrees requiring the collar gains are $(0,0)$ and $(2,0)$. For $(1,1)$ or $(0,2)$ the additional Hölder scaling gives the needed gain because $\alpha > 1-\varepsilon$. For total degree at least four, direct scaling suffices even after the complex-constant endpoint factor $\exp(2(n+m))$; choose L so that each extra pair pays that factor. Small polymers thus obey the required $CL^{-1/4+2\varepsilon}$ bound after the finite extraction cost. For large polymers reserve one extra decay unit and use their $d_M(X) > L$ to pay the same factor, without Taylor subtraction. The rooted count and reblocking inequality then give

$$\|\mathcal{L}\mathcal{R}E\|_{j+1} \le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon} \|E\|_j.$$

Since $w^E_{j+1}/w^E_j = L^{1/8-7\varepsilon/2}$, its normalized coefficient is $c_{\mathfrak{fl}} L^{-3/8+11\varepsilon/2} \le 1/16$. All cutoffs were constructed inside the actual torus; the same proof therefore covers the smallest admitted reference volume.

Appendix I. The scalar determinant correction

At a step the Gaussian fluctuation determinant is $\delta Z_j(A) = \det(D_j(A) + \beta P(A))$, with $\beta = b/L$. It acts on unit-lattice spinors. At $j = 0$ it is $\det(K_0(A) + \beta P(A))$ on the original unit lattice. This is a coarse-rank determinant at every level. Taking the trace of a fine inverse without the block factors would introduce an unnecessary ultraviolet volume loss.

I.1. **An exact resolvent logarithm.** For real A, set $T = (D_j + \beta P)\gamma_4$; it is Hermitian and invertible, and $\det T = \delta Z_j$ because $\det \gamma_4 = 1$ on each four-spin site. For a real nonzero eigenvalue x use the branch $\operatorname{Log} x = \log|x| + i \arg x$, $0 \le \arg x \le \pi$. Direct integration gives

$$\operatorname{Log} x = \lim_{R\to\infty} [-i \int_0^R (x+iy)^{-1} dy + \log R + i\pi/2].$$

Insert (F.4) in the trace of $(T+iy)^{-1}$. The term B_y is explicitly integrable, because P has rank $4l^3/L^3$. It gives

$$c_j^d = 4(1-L^{-3}) \log b_j + 4L^{-3} \log(b_j + \beta).$$

The logarithmic divergence cancels $4l^3 \log R$, and the remaining $2\pi i l^3$ has exponential one. Hence

$$\delta Z_j(A) = \exp\left\{ c_j^d l^3 - i b_j^2 \int_0^\infty \operatorname{Tr}[\gamma_4 B_y Q_j S_y(A) Q_j^\sharp B_y] dy \right\}. \tag{I.1}$$

For $j = 0$ use the convergent scalar identity

$$\operatorname{Log} x - i\pi/2 = \int_1^\infty \frac{x}{y(x+iy)} dy - i \int_0^1 \frac{dy}{x+iy}.$$

It gives the logarithm $\int_1^\infty y^{-1} \operatorname{Tr}(F_0 S_y) dy - i \int_0^1 \operatorname{Tr}(\gamma_4 S_y) dy$; here $F_0 = K_0 + \beta P$. The large-y inverse is $O((1+y)^{-1})$, so the first integral converges. This is the first completed native Gaussian, with its whole normalizer retained.

Replace S_y by its complete polymer pieces in these formulas. Call the resulting local logarithms $\acute{E}_j(X, A)$. For $j \ge 1$, cyclicity puts $B_y \gamma_4 B_y$ next to $\mathcal{N}_X = Q_j S_y(X, A) Q_j^\sharp$. Each entry of $\mathcal{N}_X$ is

at most $C_{\mathrm{w}}\exp(3g\|\operatorname{Im}A\|)e^{-\kappa_{\mathrm{w}}d_M(X)}$, by the unit-block inverse bound and the block volume one. Further

$$|(B_y\gamma_4 B_y)(y_2,y_1)| \le (b_j^2+y^2)^{-1}[1_{y_1=y_2}+2\exp(1)L^{-3}1_{y_1,y_2\text{ in one block}}].$$

Summing y_2 costs at most $1+2\exp(1)$; summing y_1 costs only $M^3|X|_M$. Since $|X|_M \le c_X(d_M(X)+1)$ and $\int_0^\infty (b_j^2+y^2)^{-1}dy = \pi/(2b_j)$,

$$|\acute{E}_j(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}. \tag{I.2}$$

The level-zero formula gives the same bound because its lattice is unit and both integrals have integrable weights. Its F_0 factor couples only a neighbor or one L-block and has fixed norm there. Every trace index and complete average path belongs to X, so the scalar reads only bonds meeting X. Uniform convergence proves analyticity on the full open domain.

For real A the determinant is even under charge reversal and real under the barred involution. Average each local logarithm with its charge-reversed and barred-conjugate partners:

$$E_j^d(X,A) = \tfrac{1}{4}[\acute{E}_j(X,A)+\acute{E}_j(X,-A)+\overline{\acute{E}_j(X,\overline{A})}+\overline{\acute{E}_j(X,-\overline{A})}].$$

Exponentiating their physical sum gives the same determinant. One checks this first on the real domain using the operator involutions, and then extends along each complex line by the identity theorem. The gauge-saturated domain is star-shaped and open, so successive such lines cover it. Consequently

$$\frac{\delta Z_j(A)}{\delta Z_j(0)} = \exp\sum_X [E_j^d(X,A)-E_j^d(X,0)]. \tag{I.3}$$

All scalar constants are kept; the equality concerns the determinant itself and is unaffected by a choice of logarithm modulo $2\pi i$.

I.2. **The determinant difference on the field domain.** For $(A,C)\in\widetilde{\mathcal{R}}_j\times\mathcal{X}_j$ define

$$E_j^{\det}(X,A) = E_j^d(X,A)-E_j^d(X,0), \qquad \delta E_j^{\det}(X;A,C) = E_j^d(X,A+C)-E_j^d(X,C).$$

For small X, remove its defining real gauge on the full support and connection dependence, and cut the representative to a larger embedded cube. The sup bound of the cut representative is at most $2g^{-3/4+\varepsilon}$. Hence $C+tA_{\mathrm{cut}}\in\mathcal{G}_j$ on $|t|\le\frac{1}{4}g^{-1/4}$, by the domain inclusion calculation. For a polymer whose diameter is too large for that embedded cube, retain the direct bound (I.2) and reserve its extra decay. More explicitly split at $d_M(X)=v_j/4$; the second part gains $e^{-v_j/4}=g^{1/4}$, while in the first part the cube radius is at most $2M(v_j/4+4)$ and is embedded since $l\ge Mr_j$ and $r_j\gg v_j$. Its cutoff norm is paid by $8M(v_j/4+4)g^{\varepsilon}\le 1$. Cauchy's difference formula on the large circle and the two direct endpoint bounds give, for every X,

$$|\delta E_j^{\det}(X;A,C)| \le 8C_d M^3 g^{1/4} e^{-\kappa d_M(X)}.$$

Thus the scalar difference is local and even, vanishes at $A=0$, and has the claimed rooted norm after reblocking. The fixed factor $8C_dM^3L^3K_0$ is absorbed into $g^{-2\varepsilon}$ by the final small-charge condition. This yields $\|\mathcal{L}\delta E_j^{\det}\|\le e_{j+1}^{1/4-2\varepsilon}$.

I.3. **Free constants telescope exactly.** Coalesce two Grassmann block Gaussians by completing their square. For a finite-dimensional precision K and coisometry Q, integrating $\exp[-\beta\langle\bar\Psi'-Q\bar\Psi,\Psi'-Q\Psi\rangle-\langle\bar\Psi,K\Psi\rangle]$ produces the fluctuation determinant and the Schur precision $\beta-\beta^2Q(K+\beta Q^\sharp Q)^{-1}Q^\sharp$. The normalized block kernel has a scalar prefactor chosen so that its integral over Ψ' is one. Eliminating the fine variable in two orders gives the same result, so the scalar prefactors obey Let $n_{j+1}=(l_j/L)^3$. Completing the square, including the field and Berezin rescaling, gives

$$\det D_{j+1}(0) = \frac{c_j'\det D_j(0)}{\delta Z_j(0)}, \qquad c_j' = b^{4n_{j+1}}, \qquad Z_j^{\mathrm{block}}(0) := c_j'/\delta Z_j(0). \tag{I.4}$$

Thus the scalar fluctuation prefactor multiplying $1/\det D_j(0)$ is $\delta Z_j(0)/c'_j$, the reciprocal of $Z_j^{\mathrm{block}}(0)$, and it gives $1/\det D_{j+1}(0)$. This identifies the direction of the telescope as well as its normalization. This is a finite matrix identity, valid at complex connections where the inverses exist. At a source connection, split (I.3) as

$$E_j^{\det}(X, A+C) = \delta E_j^{\det}(X; A, C) + E_j^{\det}(X, C).$$

The first term is added to the active activity; the second is added to the scalar determinant correction. Formula (I.4) supplies the next free denominator. No fine determinant is silently counted twice, and no charged scalar is omitted from the later source derivatives.

Appendix J. The normalized one-step map

We construct the map before estimating it. Every intermediate algebra is finite-dimensional. Photon integrals are against the normalized product Gaussian on its box, fermion integrals against a unit white Berezin law. Absolute convergence below justifies their interchange with all polymer and contour sums. Put $B = A + C$, $m' = m + m(E)$ and $E^r = \mathcal{R}E$. The fine physical field and its independent Hölder counterpart are written together as $\psi^{\#}$.

J.1. **Two localization operations.** There are three independent kinds of parameter cubes: the translated M-cubes for fermion walks, the minimizer cubes of radius M, and the root cubes of radius LM. Cubes of different kinds carry different parameters even if their centers agree. A cube of any kind meets at most $(2L+1)^3$ ordinary M-cubes. Cubes of one kind are linked if they meet in a lattice site. Distinct components have disjoint sites.

For a seed polymer X, let $P_0(X)$ be the parameter cubes contained in X. For P disjoint from $P_0(X)$, let $Y(X,P)$ be the union of the M-cubes meeting X or any cube in $P_0(X) \cup P$. Define

$$\operatorname{Loc} G(Y) = \sum_{X,P:Y(X,P)=Y} \int_{[0,1]^P} ds_P\, \partial_P G(s,X)\big|_{s=1 \text{ on } P_0(X),\ s=0 \text{ elsewhere outside } P}. \tag{J.1}$$

Repeated fundamental theorems of calculus show $\sum_Y \operatorname{Loc} G(Y) = \sum_X G(1,X)$. If $Y(X,P)$ is disconnected, take a component not attached to X. The two-ended primitive support implies that G is independent of every parameter of that component. One of these parameters occurs in ∂_P, so the term vanishes. Thus only connected Y occur. At a nonzero term the active primitive row stays in its attached component, reads all its interface bonds, and has no columns outside it; the transposed statement handles barred arms. These are the exact locality facts behind (J.1).

On a weakening circle of radius $\rho = M^{\alpha_0}$, Cauchy gives $\|\partial_P G\| \le (\rho-1)^{-|P|} \sup \|G\|$. Ordinary cubes partition lattice sites, so every new ordinary cube of $Y \setminus X$ meets a cube of P, and $0 \le d_M(Y) - d_M(X) \le N_c|P|$. If $\|G(s,X)\| \le B_0 e^{-\kappa' d_M(X)}$, with $\kappa_0 \le \kappa' \le \kappa$, the choice of M yields

$$\|G_P(X)\| \le B_0 e^{-(\kappa'-\kappa_0)d_M(Y)} e^{-\kappa_0 d_M(X)} (6c_X)^{-|P|}.$$

There are at most $3|Y|_M$ parameter cubes contained in Y. Summing their subsets costs $(1 + (6c_X)^{-1})^{3|Y|_M} \le e^{(d_M(Y)+1)/2}$. The seed sum costs $K_0|Y|_M \le 2c_X K_0 e^{d_M(Y)/2}$. Consequently one localization is bounded by $4c_X K_0 B_0 e^{-(\kappa'-\kappa_0-1)d_M(Y)}$. This gives exactly $\kappa_1 = \kappa - \kappa_0 - 1$ and $\kappa_2 = \kappa - 2\kappa_0 - 2$. Both sums are on the full field and connection support, and introduce no mesh or volume loss.

J.2. **The five differences.** For an analytic algebra-valued function define

$$\mathcal{C}_\rho f = \frac{1}{2\pi i} \int_{|t|=\rho} \frac{f(t)}{t(t-1)} dt = f(1) - f(0), \qquad \rho > 1.$$

Its norm is at most $2 \sup_{|t|=\rho} \|f(t)\|/(\rho-1)$. Let $Z(s') = H_j^P(s') C_j R_{\mathrm{loc},j}(s')\zeta$, $J^{\#} = H_j^{\#}(s'', B+Z) - H_j^{\#}(s'', B)$, and $W^{\#} = H_j^{\#}(s'', B)\widetilde{\Gamma}_j(s'', B)\omega$, where $\widetilde{\Gamma}\omega = (\Gamma\omega, \bar{\omega})$ on the two arms. Let

$M_j = b_j - D_j$ and use its complete polymer pieces. The first three differences are

$$G_1(X) = \mathcal{C}_{\rho_1}\langle\bar\Psi, M_j(X, B+tZ)\Psi\rangle,$$
$$G_2(X) = \mathcal{C}_{\rho_2}E^r(X; A+tZ, C; \psi^\# + tJ^\#\Psi),$$
$$G_3(\square) = \mathcal{C}_{\rho_2}[-m'\langle\bar\psi + t\bar J\bar\Psi, \psi + tJ\Psi\rangle_\square].$$

At level zero the last two vanish, since $E_0 = 0$, and G_1 is the bare Wilson link difference under Z, including its two projectors and its inverse-mesh factor. Apply (J.1) to the first three functions and then recover each coarse Ψ using the literal local pieces $T^{(1)}\psi + T^{(2)}\chi$ of (F.9) on its resulting polymer. This recovery introduces only the admitted boundary pairs. Call their sum $E^{\le 3}$. The second localization acts on

$$G_{\le 3}(X) = E^{\le 3}(X; A, C; \psi^\# + W^\#),$$
$$G_4(X) = \mathcal{C}_{\rho_4}E^r(X; A, C; \psi^\# + tW^\#),$$
$$G_5(\square) = \mathcal{C}_{\rho_4}[-m'\langle\bar\psi + t\bar W, \psi + tW\rangle_\square].$$

Denote the complete resulting local exponent by $V(X)$. These are definitions at nonphysical weakening parameters. Their physical sum is exactly the difference of the original exponent under the new photon fluctuation and new fermion fluctuation substitutions, as follows from the two contour identities and the two localization identities. The kinetic sign is positive because $-D_j = -b_j + M_j$. The relevant mass signs are negative and are retained in both classes.

J.3. **All contour domains and the quantitative bounds.** Use $\rho_1 = \frac12 g^{-3/4+9\varepsilon/2}$, $\rho_2 = g^{-3/4+7\varepsilon}$, and $\rho_4 = g^{-1/4+3\varepsilon}$, decreasing e_0 so they exceed two. The weakened photon rows and the box give $\|Z(s')\|_i \le \frac12 g^{-\varepsilon/2}$ in all three norms. Multiplying by ρ_1 is at most half the margin $g^\varepsilon t_i$ after the finite row constant and logarithmic box thresholds are absorbed. Thus $A + tZ(s') \in \widetilde{\mathcal{R}}_j$ and $B + tZ(s') \in \widetilde{\mathcal{G}}_j$ on both photon circles. The difference $J^\#$ gains the first photon Cauchy factor. The uniform complete rows ensure that, at weights $h/2$ for old fields and $C_T h_1$ for recovered coarse fields, every substituted leg stays inside the original field weights. For ρ_2, the row difference $J^\#$ is bounded by $Cg^{3/4-9\varepsilon/2}$, so its multiplied radius is $Cg^{5\varepsilon/2}$, below the field margin. For ρ_4 the new white row has weight one, and $\rho_4/h_1 = g^{3\varepsilon}$; its Hölder row is similarly below h_2 after paying one factor g^ε. Thus all substituted fields remain inside their declared weights. There is no external Hölder constraint on C; its two norms and complete line phases suffice.

The kinetic class uses $h_1^2 = g^{-1/2}$ and $1/(\rho_1 - 1) \le 4g^{3/4-9\varepsilon/2}$, giving $Cg^{1/4-9\varepsilon/2}$. For $j = 0$ the direct link inequality $|e^{igZ(b)} - 1| \le 2g|Z(b)|$ gives the same bound with its two ordinary or boundary field weights. The two old-activity classes are bounded respectively by $Cg^{1-14\varepsilon}$ and $Cg^{1/2-10\varepsilon}$. The two mass classes use $|m'|h_1^2 \le 8g^{1/4-8\varepsilon}$, and their contours give bounds $Cg^{1-15\varepsilon}$ and $Cg^{1/2-11\varepsilon}$. Each is at most $g^{1/4-5\varepsilon}$ after choosing e_0, because every difference of its exponent from $1/4 - 5\varepsilon$ is strictly positive for $\varepsilon \le 1/200$. After the two finite localization costs,

$$\|V(X)\|_{(h/4;1)} \le H_0 e^{-\kappa_2 d_M(X)}, \qquad H_0 = 36c_X^2 K_0^2 g^{1/4-5\varepsilon}. \tag{J.2}$$

Any fixed extra constants in these inequalities are absorbed by $g^{-\varepsilon/2}$ in the final small-charge choice. Each bound is affine in (m, E) and continuous on the closed white box; the complex dependence is on the open connection and input domains.

J.4. **A connected logarithm with its normalizer.** Let $\widehat\mu_B$ be the product Gaussian conditioned on the white box and normalized to probability. For every union U of ordinary cubes put

$$K(U) = \sum_{\mathcal{X}:\,\cup_{X\in\mathcal{X}}X = U}\ \prod_{X\in\mathcal{X}}(e^{V(X)} - 1), \qquad H(U) = \int d\widehat\mu_B\, d\mu_I(\omega) K(U),$$

where $\mathcal{X}$ is a finite set of distinct polymers, and $K(\emptyset) = 1$. These are sums of whole products before integration, so overlapping fluctuation contacts are retained. Expanding the finite product $\prod_X [1 + (e^{V(X)} - 1)]$ gives $e^{\sum_X V(X)} = \sum_U K(U)$. The family defining $K(U)$ splits uniquely over the face-connected components of U. Compatible components have disjoint white coordinates and Berezin indices; their normalized integrals therefore factor. Thus the exact normalized integral is the hard-core polymer partition polynomial $\sum_{\mathcal{Y}\ \text{compatible}} \prod_{Y\in\mathcal{Y}} H(Y)$. The compatibility is absence of both overlap and face adjacency. This uses the independent white variables, rather than factorizing the nonlocal physical covariance.

We bound the two sums separately. Put $\beta_1 = \kappa_2 - \kappa_0$, $\kappa_4 = \beta_1 - 1$, and $H_1 = 7c_X K_0 e^{3\beta_1} H_0$. For any connected family with union Y, adjoining a child polymer by one overlap or adjacent cube gives $d_M(Y) \le \sum_i d_M(X_i) + 3(n-1)$. Hence the n-factor product bound, followed by the seed sum inside Y, gives

$$\|K(Y)\| \le e^{-\beta_1 d_M(Y)} x e^x, \qquad x = 2H_0 K_0 e^{3\beta_1} |Y|_M.$$

Impose $4H_0K_0c_Xe^{3\beta_1} \le 1$. Then $x \le (d_M(Y)+1)/2$, while $x \le 4H_0K_0c_Xe^{3\beta_1}e^{d_M(Y)/2}$. It follows that $\|K(Y)\| \le H_1 e^{-\kappa_4 d_M(Y)}$. Both normalized integrations are norm contractions, so the same bound holds for $H(Y)$. Its logarithm is

$$E^{\#}(Y) = \sum_{n\ge 1} \frac{1}{n!} \sum_{X_1\cup\cdots\cup X_n = Y} \varphi^T(X_1,\dots,X_n) \prod_i H(X_i), \tag{J.3}$$

where φ^T is the sum of connected graphs with edge weight -1 for incompatible polymers and zero otherwise. The graph identity follows by expanding the exponential formula for finite labeled sets; its constant term is the same normalized white integral, so no separate vacuum measure has been used.

For the cluster bound put $\beta_2 = \kappa_4 - 2\kappa_0 > 0$. The connected-graph tree inequality replaces $|\varphi^T|$ by a sum over labeled trees of incompatibility edges. One proof orders the vertices and groups each connected graph by its rooted breadth-first tree. Each group is an interval of edge sets with that tree as its least element; the alternating sum over the optional edges is zero or the tree sign. This proves the stated absolute bound. Root a tree at X_1. A child of a polymer can be rooted in one of at most $7|X_i|_M$ cubes. If it has c children, use

$$e^{-2\kappa_0 d_M(X)} |X|_M^c \le c! e^{\kappa_0} e^{-\kappa_0 d_M(X)}.$$

Sum leaves toward the root using (E.1). The Prüfer formula $(n-2)!/[(c_1-1)!\prod_{i>1} c_i!]$ sums the degree families. After division by $n!$, their sum is bounded by 4^{n-1}. The union tree inequality supplies $e^{3\beta_2(n-1)}$. Thus the resulting geometric ratio is bounded by $z_S = 28H_1e^{\kappa_0}K_0e^{3\beta_2}$. Impose $2z_S \le 1$. Absorbing the root factor $|Y|_M \le 2c_Xe^{d_M(Y)/2}$ with the final decay unit yields

$$\|E^{\#}(Y)\| \le c_S' H_0 e^{-\kappa_3 d_M(Y)}, \quad c_S' = 14c_X^2K_0^2e^{3\kappa}, \quad \kappa_3 = \beta_2 - 1 = \kappa - 5\kappa_0 - 4.$$

The tail with $n > n_0$ is at most 2^{-n_0} times the same bound. Both smallness requirements are consequences, for example, of $10^5 c_X^5 K_0^5 e^{6\kappa} g^{1/4-5\varepsilon} \le 1$, imposed after the scheme constants by choosing e_0. The same tree majorant applies in the finite Banach algebra and uniformly on compact complex domains. Hence the logarithm is analytic, its exponential is the exact normalized integral, and that integral is zero-free in the declared windows.

The scalar box probability is separate. For one coordinate let $c_B = \mathbb{P}(|Z| \le p_{0,j}) < 1$. There are $2(1-L^{-3})l_j^3$ coordinates, so restoring their total box probability gives the exact action vacuum increment

$$g_j = -2(1-L^{-3}) \log c_B \ge 0.$$

Gaussian tails imply $0 \le g_j \le Ce^{-p_{0,j}^2/2} \le g^7$. Unnormalizing the box integral restores precisely this factor. It is this restoration that gives the vacuum recursion in the main theorem, including its sign.

J.5. **The physical identity and the output window.** Complete the fermion square at the classical background B. The substitution is $\Psi_f = \Psi_c + \Gamma_j\omega$, $\bar\Psi_f = \bar\Psi_c + \bar\omega$. Its covariance is exactly Γ_j. The classical fine field is recovered by the two-level identity at the end of Appendix F. The physical five-packet sum is precisely the exponent difference between this field and the field with both new fluctuation variables. The normalized integral is $\exp\sum_Y E^{\#}(Y)$ by (J.3). Add the active determinant difference from Appendix I; its source-only part enters the scalar determinant correction outside the current integral. Coalesce the free constants by (I.4) and restore c_B. This proves the physical density identity with all three recursions of Theorem 9.3, for arbitrary harmonic background.

Reblocking $E^{\#}$ and putting the classical fine fields into it costs the fixed row constant and $c_{\mathrm{rb}}L^3$. The reserved decay and $L(\kappa_3-\kappa_0)-1\ge\kappa$ give the output polymer norm. The finite constants are paid by $g^{-\varepsilon}$, yielding $\|E^*\|\le g^{1/4-6\varepsilon}$, which is stronger than the next-charge version in the main theorem. Gauge covariance, charge reversal and the barred involution are preserved at every complete packet, its normalized integral and the connected sum. Setting $C=0$ commutes with the construction, so the two maps have identical source-free relevant coordinates.

Finally work on the input ball of radius four. The output bound holds uniformly there and the dependence on the input is analytic in its interior. A line through two points of the unit ball has a remaining radius at least two after its direction is normalized by $|\Delta m|/w^m + \|\Delta E\|/w^E$. Cauchy's derivative estimate and integration along the segment prove the Lipschitz bound in the main theorem. This is an estimate on the same normalized map, including scalar and quadratic contacts, rather than an estimate on a separately normalized numerator.

Appendix K. Reference geometry and one torus-independent pair

K.1. **The stop scale is stable under removing the cutoff.** Write $\widehat{j}=K-j$. Then $a_j=h'L^{-\widehat{j}}$, $v_j=v_K+\widehat{j}\log\sqrt{L}$, and the rounded threshold

$$\widehat{r}_{\widehat{j}}=\min\{L^i : L^i\ge(v_K+\widehat{j}\log\sqrt{L})^r\}$$

does not depend on N. For $R=h'L^T$ put $n_{\widehat{j}}=\log_L(R/(h'L^{-\widehat{j}}M\widehat{r}_{\widehat{j}}))$. These are integers. If $(1+3\log\sqrt{L}/v_K)^r<L$, the rounded threshold increases by at most one power of L at a step, so $n_{\widehat{j}+1}-n_{\widehat{j}}\in\{0,1\}$. At zero, $n_0\le 0$ because $R\le R_{\mathrm{top}}$; while $n_{\widehat{j}}\to\infty$ since its linear term dominates $r\log_L(v_K+\widehat{j}\log\sqrt{L})$. There is therefore a last zero $\widehat{j}_*=\max\{i\ge 0 : n_i=0\}$. It is independent of N. The stopping level is $j_*=K-\widehat{j}_*$. Set

$$N_{\min}(R,e)=\max(0,\widehat{j}_*+1-\log_L h').$$

For N above this threshold $j_*\ge 1$, and before that level $l_j>Mr_j$. Every side is an odd multiple of LM; by taking v_K large enough, $r_j>31L+16$ throughout the retained range. Thus both successive levels of each step and the terminal level are admitted. The reference mass never exceeds $\bar m h'\le L^{-m_0}$, and all charges are at most e_K. This proves the consistent parameter choices of every step.

For $i=1,2,3$ the three-step threshold condition gives $\widehat{r}_i\in\{\widehat{r}_0,L\widehat{r}_0\}$. For $R=R_{\mathrm{top}}$, $n_i=i-\log_L(\widehat{r}_i/\widehat{r}_0)$, so its last zero is 0 or 1. For $R=R^*$, $n_i=i-1-\log_L(\widehat{r}_i/\widehat{r}_0)$; $n_1\in\{-1,0\}$, $n_2\in\{0,1\}$ and $n_3\in\{1,2\}$, so its last zero is 1 or 2. Monotonicity rules out a later zero. This proves (9.4); more directly, substitute those two values of $\widehat{j}_*$ into a_j,e_j and $L^{-m_0-1}<\bar m h'\le L^{-m_0}$. The cutoff threshold and the physical stop data are consequently stable even though the number of ultraviolet steps grows.

K.2. **Projected primitives and their physical value.** Construct the primitive families first on the largest reference torus, with polymer decay $\kappa_w+\kappa_0$ reserved. Let the smaller side be l_1 and the larger side $l_2=Jl_1$, with J an odd power of L. For a periodic scalar kernel sum its covering

columns; for a spinor kernel use the antiperiodic character $(-1)^{n_1+n_2+n_3}$ on a translation by $l_1 n$. For a fine-to-coarse kernel retain the corresponding character at both its fine and coarse spinor indices. Pull back its connections and parameter families to the large torus before taking this sum. The construction is independent of the representatives chosen for the small-torus indices, by reindexing the covering sum.

Exponential bounds survive projection. Choose a shortest lift of the small-torus distance; the sum over other lifts is at most $64e^{-\gamma d_1}$ after halving the decay rate. Projecting a complete polymer support can identify some cubes, but a fixed small-torus polymer has at most the rooted count of its lifts. Reserve κ_0 and apply (E.1); this costs at most K_0. Thus all projected row, column and polymer bounds hold with $64K_0$ times the original constant. These projected families are the prescribed primitives on the reference bank.

At $s = 1$ they are the actual operators. For a physical inverse, pullback of a small-torus spinor intertwines the Wilson precision and the block projections, including the complete charged paths. The unique inverse therefore intertwines too. The projected photon minimizer satisfies the same normal equations and gauge constraints; uniqueness identifies it. The projected precision intertwines its resolvent and then its square root by (G.5). These arguments include the arbitrary constant background. Hence projection changes only the auxiliary nonphysical interpolation, not the physical Gaussian or the finite-model law.

K.3. **Agreement on full local support.** Put $d_U = 2L + 1$. For a finite lifted union $\widetilde{X}$ of M-cubes let $N(\widetilde{X})$ be its bounding box enlarged by $6d_U$ layers. Call it admissible when its largest box side is at most $l_1 - 14d_U M$. Then every side of $N(\widetilde{X})$ is at most $l_1 - 2d_U M$. Its distinct $l_1\mathbb{Z}^3$ translates are disjoint, so it embeds in both tori. Furthermore two of its sites at torus distance less than $2d_U M$ have the same shortest displacement in the lift: a different displacement would differ by $l_i n$ and have norm at least $l_i - |u - u'| \geq 2d_U M$. Thus all short pairs, paths, parameter links and cutoff supports used in a complete packet have identical geometry in the lift.

Set weakening parameters to zero outside its active lifted components. Consider a primitive row in one such component. Its two-ended locality eliminates every column in another component. In the projection sum, all nonzero covering classes belong to other translates of the lifted component and therefore vanish. The sole remaining class is its actual lifted column. For a transposed row the same argument eliminates nonzero covering rows. Its connection support is the whole component, including the interface bonds. Thus the projected and unprojected local primitives agree entry by entry, for both arms and every source connection in their declared domains.

Apply this argument in order to a localized packet. Its new fluctuation photon row reads its root and minimizer components and their basis supports. A fine fermion row at one of its seed indices reads its attached walk component. A Hölder row also reads the six paths of the pair; these paths lie inside the same bounding box. A coarse covariance row issued from the fine row remains in that same component. Finally the recovery reads either the same unit block or its explicit nearest-neighbor boundary pair. All these supports lie in $N(\widetilde{X})$; their possible endpoints use less than its reserved collar. They therefore agree before either contour is performed.

In (J.1), a term contributing to a flat output Y has its seed $X \subset Y$ and every parameter cube contained in Y. The same derivatives and parameter integrals consequently occur on both tori. A disconnected component gives zero on both. The first recovery is then identical, the second localization identical, and their white coordinate sets identical. Each local photon basis packet on the common lift has the same normalized box law; translations and coordinate reflections act on its coordinates by the same signed permutation proved in Appendix G.3. The ordered Berezin indices have the same order in the lift. Changing the torus fundamental domain multiplies each generator and its kernel by the same antiperiodic sign, which cancels in the physical monomial. Thus all local integrals agree with their normalizers, not merely their covariances.

The determinant trace is treated in the same way at both indices of its complete polymer piece, and its zero-field value is subtracted identically. In a connected cluster with flat union Y, every constituent polymer and the full support of its fluctuation variables is inside the same lift. Every term of (J.3) therefore agrees; its absolute tree majorant permits termwise comparison. Reblocking uses only the same nested cubes and field scaling, so the output activities and scalar determinant sums agree. This proves Theorem 9.4 for a pair of input families agreeing on flat polymers.

Every small polymer has lifted diameter at most $(L+2)M$. The retained condition $l_1 > (31L+16)M$ makes all such polymers admissible. Relevant extraction at zero sets every declared connection variable on the full support to zero and integrates the full quadratic matrix in this one lift. Consequently all local extractions, and hence their translation-invariant sums, agree. Begin with $E_0 = P_0 = 0$, then apply the preceding packet argument inductively. It proves that the relevant coordinates selected on the largest torus are the coordinates on every smaller torus until its stop. No agreement of separately weakened vertices is substituted for this complete normalized comparison.

K.4. **A fixed point with one backward coordinate.** Let j_t be the top-torus stop and take the finite product of the real Banach activity spaces with $E_0 = 0$. Its closed window is $|m_j| \le w_j^m$, $\|E_j\| \le w_j^E$. Use the maximum of the corresponding normalized norms. Define

$$\begin{aligned} m'_j &= L^{-1} m_{j+1} - m(E_j) \quad (j < j_t), \qquad m'_{j_t} = 0, \\ E'_j &= \mathcal{L}\mathcal{R} E_{j-1} + \mathcal{L} E^{\det}_{j-1} + E^*_{j-1}(m_{j-1}, E_{j-1}) \quad (j \ge 1), \qquad E'_0 = 0. \end{aligned}$$

The ratio $L^{-1} w^m_{j+1}/w^m_j$ is at most $1/4$ for the fixed L. Also $g^{1/2} w^E_j / w^m_j = g^{\varepsilon}$. The forward norm estimates give $\|E'_j\|/w^E_j \le 1/16 + 2e^{\varepsilon}_j \le 1$. The backward coordinate satisfies $|m'_j|/w^m_j \le 1/4 + c_{\mathfrak{fl}} e^{\varepsilon}_j \le 1$. Thus the map preserves the closed window.

For two inputs let a_j, b_j be their normalized differences. The linear extraction and the input Lipschitz bound give

$$b'_j \le \tfrac{1}{4} b_{j+1} + c_{\mathfrak{fl}} e^{\varepsilon}_j a_j, \qquad a'_j \le \tfrac{1}{16} a_{j-1} + e^{\varepsilon}_j (a_{j-1} + b_{j-1}).$$

The determinant forcing cancels exactly: it is independent of the input activity and relevant mass. Impose $c_{\mathfrak{fl}} e^{\varepsilon}_K \le 1/16$ and $2e^{\varepsilon}_K \le 1/4$. The map is a contraction of factor $5/16$ on a complete space. Its unique fixed point is real by the closed reality conditions. The physical recursions and induction show that this sequence is the actual trajectory of its first mass coordinate, rather than a sequence solving an auxiliary linearized map.

The improved activity bound follows from its recurrence with $A_0 = 0$: $A_j \le A_{j-1}/16 + 2e^{\varepsilon}_j$, hence $A_j \le 4e^{\varepsilon}_j$. The backward bound with terminal zero then gives $B_j \le (1/16)\sum_{i \ge 0} 4^{-i} = 1/12 < 1/8$. Define the vacuum coordinate by the exact finite telescope

$$\epsilon_{0,N} = -\sum_{i=0}^{j_t - 1} L^{-3i} [\epsilon(E_i) + g_i].$$

It makes the top terminal vacuum zero. At level j the same telescope starts at j. The activity part is bounded by $4.2 c_{\mathfrak{fl}} e_j^{1/4 - 6\varepsilon}$ because the geometric ratio is $L^{-23/8 - 3\varepsilon}$. The box part is at most $e_j^6 e_* \le 0.8 c_{\mathfrak{fl}} e_j^{1/4 - 6\varepsilon}$ after reducing e_1. Thus $|\epsilon_j| \le 5 c_{\mathfrak{fl}} e^{\varepsilon}_j w^E_j$.

Take this pair on every smaller reference torus. Flat agreement shows its local activity and relevant extractions are the same until the smaller stop. Their stated bounds consequently hold there, and the normalized map induction constructs the entire external-slot trajectory. Harmonic constants are local real gauges in every flat extraction, so the selected pair is independent of the harmonic twist as well as the reference torus. This proves the selection and agreement clauses of Theorem 9.5.

Appendix L. Terminal comparison and the pressure estimate

The comparison in this appendix is made in the actual averaged measure. Its positive density is the paired four-spin determinant at the selected real mass. The positive mass in the free operators below is the reference mass $\bar{m}\eta$; it is not the selected bare mass. Put $g = e_*$, $l = l_*$ and $m^{\mathrm{ref}} = \bar{m}a_*$. Then $0 < m^{\mathrm{ref}} \le 1$, $m^{\mathrm{ref}} l = \bar{m}R$, and

$$Mr_* \le l < LMr_*, \qquad (m^{\mathrm{ref}})^{-1} \le L^{m_0+1} l, \qquad e^2 R = g^2 l. \tag{L.1}$$

Thus every polynomial loss in l and $(m^{\mathrm{ref}})^{-1}$ in this appendix is a fixed power of $v_* = -\log g$.

L.1. **The free determinant at every real harmonic twist.** For native mass $0 < u \le 1$ write

$$w_u(p) = \left(u + \sum_{\nu=1}^{3} (1 - \cos p_\nu) \right)^2 + \sum_{\nu=1}^{3} \sin^2 p_\nu .$$

Antiperiodic plane waves diagonalize the constant-twist operator, with momenta $p_\nu = (2\pi a_\nu + \pi + \theta_\nu)/n$. The four-spin matrix $a + i\gamma \cdot s$ has eigenvalues $a \pm i|s|$, each twice. Consequently

$$\det(D_{H_\theta} + u) = \prod_{p \in P_{n,\theta}} w_u(p)^2 > 0. \tag{L.2}$$

This formula also applies on the very small tori when oriented links, rather than merely their endpoints, distinguish the two opposite hops.

Here is the quantitative Fourier estimate needed for its pressure. For fixed p_2, p_3 put $W = \sum_{\nu=2}^{3} (1 - \cos p_\nu)$, $S = \sum_{\nu=2}^{3} \sin^2 p_\nu$ and $M = 1 + u + W$. Then

$$w_u(p) = M(2\cosh E - 2\cos p_1), \qquad 4\sinh^2(E/2) = \frac{(u+W)^2 + S}{1 + u + W}.$$

The numerator is at least $u^2 + 4\pi^{-2}(p_2^2 + p_3^2)$, and the denominator is at most 6. Concavity of arcsinh on $[0, 3/(2\sqrt{6})]$ gives

$$E \ge \frac{u}{4} + \frac{(p_2^2 + p_3^2)^{1/2}}{6}.$$

The nonzero t-th Fourier coefficient in p_1 of $\log(2\cosh E - 2\cos p_1)$ is $-e^{-|t|E}/|t|$. Integrating its absolute value over the other two momenta, and permuting a largest component of x into the first position, therefore proves

$$|\widehat{\log w_u}(x)| \le 6|x|^{-3} e^{-u|x|/4} \quad (x \ne 0).$$

The sum over the twisted momentum grid retains just frequencies $x = nw$, with a phase of modulus one. There are at most $26t^2$ integer vectors with $|w| = t$, so

$$\left| \log\det(D_{H_\theta} + u) - p^0(u) n^3 \right| \le 312 \sum_{t \ge 1} \frac{e^{-unt/4}}{t} = \varepsilon_5(un). \tag{L.3}$$

In particular every twist ratio to $\theta = 0$ has logarithm at most $2\varepsilon_5(un)$. Averaging the positive determinants over $[-\pi, \pi]^3$ preserves the upper and lower bounds in (L.3). Hence

$$|\log \rho_n^{\mathrm{free}} - p^0(\bar{m}\eta) n^3| \le \varepsilon_5(\bar{m}R). \tag{L.4}$$

L.2. **The coarse twist ratio and its completed determinant.** The terminal coarse ratio is not the native free ratio. We retain its completed determinant explicitly. Let $\delta = L^{-j_*}$, $\mathfrak{N} = \delta^{-1}$ and $b = b_{j_*} \in [b_-, b_+]$, with fixed $0 < b_- \le b_+$. A real constant potential $c_\theta = \theta/(gl)$ has block average Q_c, and write

$$T_c = D_c + m^{\mathrm{ref}} + bQ_c^\sharp Q_c, \qquad D(c) = b - b^2 Q_c T_c^{-1} Q_c^\sharp .$$

Woodbury's identity gives

$$D(c)^{-1} = b^{-1} + Q_c (D_c + m^{\mathrm{ref}})^{-1} Q_c^\sharp, \qquad \|D(c)^{-1}\| \le b_-^{-1} + (m^{\mathrm{ref}})^{-1}. \tag{L.5}$$

The norm follows from the positive real part of the fine Wilson operator. The block average is coisometric at a real potential.

For completeness the cancellation that makes T_c insensitive to a twist can be checked on each alias block. Choose p in the coarse twisted antiperiodic momentum grid, put $q_m = p + 2\pi m$, with m in the centered $\mathfrak{N}^3$ alias cube, and set

$$\begin{aligned} R_m &= m^{\mathrm{ref}} + \delta^{-1}\sum_\nu (1-\cos\delta q_{m,\nu}) + i\delta^{-1}\sum_\nu \gamma_\nu \sin\delta q_{m,\nu}, \\ w(q_m) &= \left(m^{\mathrm{ref}} + \delta^{-1}\sum_\nu (1-\cos\delta q_{m,\nu})\right)^2 + \delta^{-2}\sum_\nu \sin^2\delta q_{m,\nu}, \\ u_1(t) &= \mathfrak{N}^{-1}\sum_{|a|\le(\mathfrak{N}-1)/2} e^{-ita/\mathfrak{N}}, \qquad u_m = \prod_{\nu=1}^{3} u_1(q_{m,\nu}), \\ C(p) &= b^{-1} + \sum_m u_m^2 R_m^{-1}. \end{aligned}$$

On the real axis $C(p) = \alpha - i\gamma\cdot\beta$, with real β and $\alpha \ge b^{-1}$. Thus $\det C(p) > 0$ and $D(c)^{-1}$ is the direct sum of these four-spin matrices. The determinant lemma applied to $\mathrm{diag}(R_m) + b|u\rangle\langle u| \otimes I_4$ gives

$$\det T(p) = b^4 \det C(p) \prod_m w(q_m)^2 > 0.$$

The fine rescaling determinants cancel in ratios. We obtain the exact identity

$$R_\theta := \frac{\det D(c_\theta)}{\det D(0)} = \frac{\det(D_{H_\theta} + \bar{m}\eta)}{\det(D_0 + \bar{m}\eta)} \frac{\det T_0}{\det T_{c_\theta}}. \tag{L.6}$$

We next prove a fixed-strip estimate for the last ratio, uniform in the number of aliases and in $0 < m^{\mathrm{ref}} \le 1$. This is needed because a bound on the full fine trace would lose the cutoff. Fix the other two momentum components real, and allow the first component z to satisfy $|\Re z| \le \pi + 1/2$, $|\Im z| \le 1/4$. Separating the zero alias gives

$$\begin{aligned} R_{\mathrm{reg}} &= b^{-1} + \sum_{m\neq 0} u_m^2 R_m^{-1}, \quad N = u_0^2 + R_0 R_{\mathrm{reg}}, \\ \Phi &:= \log\det T(p) = \log\det N + \sum_{m\neq 0} 2\log w(q_m) + 4\log b. \end{aligned}$$

The apparent zero-alias singularity is absent from N.

Here are uniform bounds sufficient to construct its logarithm. For nonzero aliases the one-coordinate factorization used for (L.3), with complex momentum, implies

$$|w(q_m)| \ge c_1^2 |\Re q_m|_2^2, \quad |R_m^{-1}| \le 5c_1^{-2}|\Re q_m|_2^{-1}, \quad |R_0| \le 22,$$

for an absolute $c_1 > 0$. To see the first bound directly, in the factor $1 - \exp(-E + i\delta z)$ use $|1 - re^{i\phi}|^2 = (1-r)^2 + 4r\sin^2(\phi/2)$. If $E \ge 2\delta|\Im z|$, the first term controls E and the second controls the real momentum; if $E < 2\delta|\Im z|$, the nonzero alias has $|\Re q_m|_2 \ge \pi - 1/2$, so the second term supplies the same bound. The elementary inequalities $\sin t \ge t/2$ for $0 \le t \le 1.66$ and $1 - e^{-t} \ge 0.45t$ for $0 \le t \le 1.85$ give, for example, any sufficiently reduced absolute c_1. The geometric sum for u_1 is

$$u_1(z) = \frac{\sin(z/2)}{\mathfrak{N}\sin(z/(2\mathfrak{N}))}.$$

It gives $|u_1(t + i\kappa)| \le e^{|\kappa|/2}$ and, when $t \neq 0$, $|u_1(t+i\kappa)| \le 4\cosh(\kappa/2)/|t|$. Summing the three alias coordinates separately yields $\sum_m |u_m|^2 \le 400$. Consequently

$$|R_{\mathrm{reg}}| \le B_R := b_-^{-1} + 800c_1^{-2}, \qquad |N| \le B_N := 2 + 22B_R.$$

On the real axis $u_0^2 \geq u_- = 0.056$. If $|R_0|B_R \leq u_-/2$, the Neumann series for N/u_0^2 gives $|N^{-1}| \leq 2/u_-$. Otherwise R_0 is normal and $|R_0^{-1}| \leq 2B_R/u_-$. Since $\Re C \geq b^{-1}$ and $N = R_0 C$, this gives

$$|N^{-1}| \leq B'_N := \frac{2}{u_-}\max(1, b_+ B_R).$$

Cauchy's estimate on a radius-1/8 disk gives $|\partial_z N| \leq 8B_N$. With $\gamma_1 = (32 B_N B'_N)^{-1}$ the relative perturbation of N at distance $2\gamma_1$ from the real axis has norm at most $1/2$. The logarithm of its positive real determinant therefore continues holomorphically to that strip and obeys

$$|\log\det N| \leq B_\Lambda := 4\max(\log B_N, \log B'_N) + 4\log 2.$$

The remaining aliases require a derivative bound, rather than a bound on their total logarithm. Four derivatives of the one-coordinate logarithmic factor, first computed where its series converges and then continued analytically, give

$$|\partial_z^4(2\log w(q_m))| \leq 29 c_1^{-4} |\Re q_m|_2^{-4}.$$

Indeed the derivative series sums to $r(1+4r+r^2)/(1-r)^4$, with r complex replaced by its modulus; the preceding factor lower bound cancels its four powers of δ. Since there are at most $26t^2$ aliases with $|m| = t$ and $|\Re q_m|_2 \geq (\pi - 1/2)(2t-1)$, its sum is bounded by $18c_1^{-4}$. Cauchy's estimate for $\log\det N$ then gives

$$|\partial_z^4 \Phi| \leq C_4 := 24 B_\Lambda \gamma_1^{-4} + 18 c_1^{-4} \quad (|\Im z| \leq \gamma_1).$$

The real function Φ is 2π-periodic; the local logarithms agree on the overlapping real segments and hence throughout the overlaps. It thus has a periodic holomorphic continuation. Four integrations by parts in a largest Fourier coordinate, followed by a contour shift of height γ_1, give

$$|\widehat{\Phi}(x)| \leq C_4 |x|^{-4} e^{-\gamma_1 |x|} \quad (x \neq 0).$$

The zero Fourier coefficient cancels in a twist ratio. Summing the remaining frequencies lw proves

$$\left|\log \frac{\det T_{c_\theta}}{\det T_0}\right| \leq C_\Sigma l^{-1} e^{-\gamma_1 l}, \qquad C_\Sigma = 86 C_4. \tag{L.7}$$

Every constant here depends only on b_-, b_+. In particular (L.6) has not discarded a source-dependent fine Gaussian determinant.

L.3. **The terminal Berezin perturbation.** Let A be a real mean-zero terminal potential with $\|dA\|_\infty \leq p_*$. A periodic real gauge, obtained by integrating the curvature along coordinate paths and subtracting its periods, gives $\|A - d\lambda\|_\infty \leq 2l^2 p_*$. By terminal gauge invariance we use this representative. Its fine minimizer A^0 then has all three norms at most $2c_{\mathrm{row}} l^2 p_* < g^{-\varepsilon}/2$. Put $r_g = g^{-3/4+4\varepsilon} \geq 8$. For $|s| < 2r_g$, $sA^0 + c_\theta$ lies in the full potential domain of Theorem 9.1; the arbitrary real harmonic constant is admitted by a local gauge. Define $D_\theta = D_*(c_\theta)$ and $C_\theta = D_\theta^{-1}$. By (L.5) its covariance norm is at most $b_-^{-1} + (m^{\mathrm{ref}})^{-1}$.

The exact density of the normalized step can now be written

$$Y_*(A + c_\theta) = e^{-\epsilon_* l^3} R_\theta \int d\mu_{C_\theta}(\Psi)\, e^{V_\theta(A,\Psi)}, \tag{L.8}$$

where

$$V_\theta = -\bar\Psi[D_*(A^0 + c_\theta) - D_\theta]\Psi - m_*\langle \bar H_* \bar\Psi, H_*\Psi\rangle + \sum_X E_*(X, A^0 + c_\theta, H_*\Psi).$$

The completed scalar determinant correction is already included in the physical density in (L.8); it is not a further perturbation or a change of photon expectation. Use the unit-lattice Grassmann weights $a_1 = g^{-1/4}$ and $a_2 = c_{\mathrm{row}}^{-1} g^{-1/4+\varepsilon}$. The complete coarse kinetic row bound gives $\|\bar\Psi D_*(sA^0 + c_\theta)\Psi\|_{a_1} \leq c_{\mathrm{row}} g^{-1/2} l^3$. Cauchy's integral on the above s disk therefore gives

$$\|\bar\Psi[D_*(A^0 + c_\theta) - D_\theta]\Psi\|_{a_1} \leq 2 c_{\mathrm{row}} g^{1/4-4\varepsilon} l^3.$$

The fine legs have bounded unit-block rows with their literal δ^3 and δ^6 measures. The mass term is consequently at most $4g^{1/4-8\varepsilon}l^3$ in the a_2 norm. Summing the rooted activity bound over the l^3 unit cubes gives at most $c_\kappa g^{1/4-8\varepsilon}l^3$. Thus

$$\|V_\theta\|_{a_2} \le (2c_{\mathrm{row}} + c_\kappa + 4)g^{1/4-8\varepsilon}l^3.$$

Its covariance has absolute row sum at most $4l^3[b_-^{-1} + (m^{\mathrm{ref}})^{-1}]$. The inequalities (L.1) and (E.7) make this row sum less than a_2 and make the last displayed norm at most $g^{1/4-9\varepsilon}/4$. Wick's determinant inequality in the coefficient norm gives

$$\left|\int d\mu_{C_\theta}(e^{V_\theta} - 1)\right| \le e^{\|V_\theta\|_{a_2}} - 1 \le 2\|V_\theta\|_{a_2} \le \tfrac{1}{2}g^{1/4-9\varepsilon} =: \epsilon_1.$$

All estimates are uniform in θ. The original positive determinant representation also makes the left side of (L.8) nonnegative on the real domain. Since $R_\theta > 0$ and $\epsilon_1 < 1$, we obtain the two-sided real comparison

$$e^{-\epsilon_* l^3}R_\theta(1-\epsilon_1) \le Y_*(A + c_\theta) \le e^{-\epsilon_* l^3}R_\theta(1+\epsilon_1). \tag{L.9}$$

L.4. **The bad-set bound uses the same normalized law.** Denote by $\mathfrak{G}$ the event that every new fluctuation coordinate is in its prescribed box and every terminal plaquette satisfies $|dA_*| \le p_*$. In the original curvature Gaussian coordinates of Appendix G.6, a new fluctuation coordinate has norm one and a terminal plaquette has norm at most one. There are at most $2l^3L^{3(j_*-j)}$ new fluctuation coordinates at level j. The domination theorem 4.1 in the actual averaged probability therefore supplies their one-dimensional tails without a determinant/free expectation transfer.

Here is why the union bound is independent of the number of ultraviolet levels. Put $t = j_* - j$; then $v_j = v_* + t\log\sqrt{L}$ and $p_{0,j} = v_j^{p_0}$. For v_* large and $p_0 > 2$,

$$v_j^{2p_0}/(2c^2) \ge v_*^{p_0}/2 + 4t\log L.$$

The Gaussian tail at level j is at most $2\exp[-p_{0,j}^2/(2c^2)]$, where the fixed c includes the box convention. Multiplication by L^{3t} and summation over $t \ge 0$ bounds the new fluctuation contribution by a fixed multiple of $l^3e^{-p_{0,*}/2}$. There are $3l^3$ terminal plaquettes; their threshold is p_*/c_{row}. By increasing the fixed threshold constants once, these estimates can be written

$$\overline{\mu}(\mathfrak{G}^c) \le \epsilon_3 := 6l^3\left(e^{-p_{0,*}/2} + e^{-p_*^2/(2c_{\mathrm{row}}^2)}\right), \qquad \epsilon_3 \le g/2. \tag{L.10}$$

The terminal free Gaussian law m_* has the same plaquette variance bound, so $\mathsf{m}_*(\chi_* = 0) \le \epsilon_4 \le \epsilon_3$. These conclusions use the four-spin positive law at the actual selected real mass; a positive selected native mass is unnecessary.

We give the normalization step explicitly. Let ν_θ have density

$$e^{-\epsilon_{0,N}n^3}\frac{\det(D_{A(F)+H_\theta} + \mu_N)}{\det(D_0 + \bar{m}\eta)}$$

against the original curvature Gaussian, and let $\overline{\nu}$ be its harmonic average. Its normalization is finite and positive, and $\overline{\mu} = \overline{\nu}/\overline{\nu}(1)$. The exact Gaussian coordinate identity and the iterated normalized map, including every box probability, imply

$$\nu_\theta(\mathfrak{G}) = \int d\mathsf{m}_*(A)\,\chi_*(A)Y_*(A + c_\theta). \tag{L.11}$$

Indeed the downward recursion reconstructs the original potential up to a real exact gauge; adding c_θ reconstructs H_θ up to the same type of gauge. Orthogonal Gaussian coordinates give Fubini's identity, with the original indicator equal to the product of the terminal indicator and all new fluctuation boxes. This is the physical identity of the step, not an identity of separately weakened factors.

Let $I_R = (2\pi)^{-3}\int_{[-\pi,\pi]^3} R_\theta\, d\theta$. Equations (L.9) and (L.11) give

$$e^{-\epsilon_* l^3} I_R(1-\epsilon_1)(1-\epsilon_4) \le \overline{\nu}(\mathfrak{G}) \le e^{-\epsilon_* l^3} I_R(1+\epsilon_1).$$

From the actual normalized estimate (L.10),

$$\overline{\nu}(\mathfrak{G}) \le \overline{\nu}(1) \le \overline{\nu}(\mathfrak{G})/(1-\epsilon_3).$$

There is no circular normalization estimate here: Gaussian domination is a statement about the already defined finite probability, and finiteness and positivity were proved before the renormalization group.

By (L.6) and (L.7), with $\epsilon_2 = C_\Sigma l^{-1} e^{-\gamma_1 l} \le g$, the number I_R lies between $e^{-\epsilon_2}$ and e^{ϵ_2} times $\rho_n^{\mathrm{free}}/\det(D_0+\bar{m}\eta)$. Also $\rho_n = e^{\epsilon_{0,N} n^3}\det(D_0+\bar{m}\eta)\overline{\nu}(1)$. Taking logarithms in the preceding inequalities, using $|\log(1\pm x)| \le 2x$ for $0 \le x \le 1/2$, proves

$$\left|\log\rho_n - \log\rho_n^{\mathrm{free}} - \epsilon_{0,N} n^3 + \epsilon_* l^3\right| \le 2\epsilon_1 + \epsilon_2 + 2\epsilon_3 + 2\epsilon_4 \le 2g^{1/4-9\varepsilon}.$$

Finally (9.20) and (E.7) give $|\epsilon_*| l^3 \le g^{1/4-9\varepsilon}$. Combining this with (L.4) proves (9.21). On R^* and LR^* the same selected pair gives the same p_F; their native site counts differ by exactly L^3. The extensive pressure cancels, which proves (9.22) and completes the proof of Theorem 9.5.

Appendix M. Compatible compact probes

Throughout these appendices the four-dimensional matrices satisfy $\{\gamma_\mu,\gamma_\nu\} = 2\delta_{\mu\nu}$, are Hermitian, and have the paired representation $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$. All continuum norms in the free calculation are Euclidean. Distances in the terminal polymer estimates are sup distances. The change of these norms affects fixed constants only.

The finite source must lie in the transverse cochain space. Sampling $d^*\beta$ alone would not give the exact discrete integration by parts needed in the source identity. Instead average β first. Let

$$J_\eta g(x) = \eta^{-3}\int_{[-\eta/2,\eta/2]^3} g(x+y)\,dy, \qquad \delta_{\eta,\mu} g(x) = \frac{g(x+\eta e_\mu/2) - g(x-\eta e_\mu/2)}{\eta}.$$

Periodize a fixed lattice translate of β on the reference torus and define

$$u_{\mu\nu}(x) = J_\eta \beta_{\mu\nu}(x+\eta(e_\mu+e_\nu)/2), \quad a_\eta = d_\eta^* u, \quad \widehat{a} = \eta^{1/2} a_\eta, \quad v_\eta = \|d\widehat{a}\|_{\ell^2}^2.$$

The adjoint uses $\eta^3\sum$ in every degree; hence $d_\eta^* = \eta^{-1} d^*$, and $\widehat{a} = d^*(\eta^{-1/2}u) \in (\ker d)^\perp$. Introduce the unsmoothed central-difference one-form

$$a'_{\eta,\nu} = -\sum_{\mu\ne\nu} \delta_{\eta,\mu}\beta_{\mu\nu}, \qquad \check{a}_{\eta,\nu} = J_\eta a'_{\eta,\nu}, \qquad f_{\mu\nu}^{(\eta)} = \delta_{\eta,\mu} a'_{\eta,\nu} - \delta_{\eta,\nu} a'_{\eta,\mu}.$$

The test map averages $f^{(\eta)}$ once at the plaquette center. Since J_η commutes with every central difference, summation by parts gives

$$a_{\eta,\nu}(x) = \check{a}_{\eta,\nu}(x+\eta e_\nu/2), \qquad (f^{(\eta)})_\eta = d_\eta a_\eta, \qquad \langle d\widehat{a}, F^u\rangle = X(f^{(\eta)}).$$

The last equality uses $F = \eta^{-3/2} F^u$: both sides equal $\eta^{3/2}\sum (d_\eta a_\eta) F^u$. A reflected copy has the same counting variance and lies in the transverse space.

Lemma M.1 (Probe error and support)**.** *For $r \ge 0$,*

$$\|f^{(\eta)} - f\|_{H^r} \le \eta^2 \|\beta\|_{H^{r+4}}, \qquad \|\check{a}_\eta - a\|_{C^i} \le 3\eta^2 |\beta|_{C^{i+3}}, \qquad |D^i \check{a}_\eta|_\infty \le 2|\beta|_{C^{i+1}}.$$

The smooth functions are supported in $\mathrm{supp}\,\beta + [-\eta,\eta]^3$*. Every nonzero probe link has both endpoints in* $\mathrm{supp}\,\beta + [-3\eta/2, 3\eta/2]^3$*. If that set lies strictly inside a mirror cell, the curvature supports of different mirror copies are disjoint. Moreover*

$$v_\eta^{1/2} \le \|f\|_2 + \|\beta\|_{H^4}, \qquad |v_\eta - \|f\|_2^2| \le C_\beta \eta.$$

Proof. The multiplier of J_η is the product of the three sinc functions; that of $\delta_{\eta,\mu}$ is $2i\eta^{-1}\sin(\eta p_\mu/2)$. The inequalities $|\sin t - t| \le |t|^3/6$ and $|1 - \sin t/t| \le t^2/6$ give the Sobolev estimate for the two unsmoothed central differences in $f^{(\eta)}$, and the smoothed one-form estimates. Taylor's formula with symmetric averaging gives the C^i estimates. The operators read only an η-neighborhood; the indicated supports and link endpoint margin follow. The test-map averaging is an L^2 contraction, so $v_\eta^{1/2} = \|(f^{(\eta)})_\eta\|_\eta \le \|f^{(\eta)}\|_2$. On the fixed support cube the first derivatives of $|J_\eta f^{(\eta)}|^2$ are uniformly bounded. Comparing each sampled value to its cell integral gives a total error at most $C_\beta\eta$; smoothing and central differences have $O(\eta^2)$ error in L^2. This proves the variance bound. Reflections preserve the counting inner product and commute with d, proving the mirror assertions. □

Appendix N. The free loop and its lattice contacts

We first record the analytic estimates that justify the traces and the comparison with the lattice determinant. The explicit kernel is

$$Y_m(x) = \frac{e^{-m|x|}}{4\pi|x|}, \qquad S_m(x) = (m - \gamma\cdot\partial)Y_m(x), \qquad |S_m(p)| = (m^2 + |p|^2)^{-1/2}.$$

In position space $|S_m(x)| \le Ce^{-m|x|}(m|x|^{-1} + |x|^{-2})$, and $\|S_m\|_{L^1} \le 3/m$. Constants may depend on a fixed lower bound for m. If v is a smooth compact multiplication matrix, Plancherel gives

$$\|S_m v S_m\|_{\mathrm{HS}}^2 \le C_m \int |\widehat{v}(k)|^2 (1+|k|)^2\, dk < \infty.$$

Indeed, in the momentum kernel the two inverse factors have product bounded by $C_m(1+|k|)(m+|p|)^{-2}$; its squared p-integral is finite in three dimensions. Thus $(S_m v)^4$ is a product of two Hilbert–Schmidt operators. Charge conjugation, or taking the adjoint and conjugating by the matrix interchanging the Pauli sectors, makes its trace real for $v = i\gamma\cdot a$.

N.1. **Uniform momentum sums.** For $R \ge 1$ and any translate $\mathcal{P}$ of $(2\pi/R)\mathbb{Z}^3$,

$$R^{-3}\sum_{p\in\mathcal{P}} (m+|p|)^{-q} \le C_{m,q} \quad (q > 3).$$

Partition momentum space into cubes of side $2\pi/R$ and compare the value at their centers with the integral on each cube. The comparison factor is bounded by $(1 + 2\pi\sqrt{3}/m)^q$. The same argument bounds the tail beyond A by CA^{3-q}. Smooth compact probes have $|\widehat{a}(k)| \le C_j(1+|k|)^{-j}$, uniformly after the compatible discretization. Their sampled coefficients are the full alias sums

$$\widehat{a}_\eta(k) = \sum_{r\in(2\pi/\eta)\mathbb{Z}^3} \widehat{\breve{a}_\eta}(k+r)\exp[i\eta(k_\nu + r_\nu)/2]$$

in each link component, with its midpoint phase. In particular the alias sum, not just its central term, has the same polynomial bound. This follows by splitting the sum into the closest representative and the remaining translates and choosing $j > 3$.

N.2. **The exact Wilson expansion.** Let $K_\eta(t) = \eta^{-1}D[t\eta a_\eta] + \bar{m}$, $S_\eta = K_\eta(0)^{-1}$, and expand $K_\eta(t) = K_\eta(0) + \sum_{r\ge1} t^r V_r$. With $P_\nu^\pm = (1 \pm \gamma_\nu)/2$ and the forward shift τ_ν, the literal hopping expansion gives

$$V_r = -\frac{\eta^{r-1} i^r}{r!}\sum_\nu \{P_\nu^- a_{\eta,\nu}^r \tau_\nu + (-1)^r P_\nu^+ \tau_\nu^* a_{\eta,\nu}^r\}.$$

The placement of the backward multiplication follows the reversed link; it is not a symmetric continuum vertex. Put $T_r = S_\eta V_r$. Differentiating the finite-dimensional logarithm yields

$$b_{\eta,R} = 24\,\mathrm{Tr}\{T_4 - T_1T_3 - \tfrac{1}{2}T_2^2 + T_1^2T_2\} - 6\,\mathrm{Tr}\,T_1^4. \tag{N.1}$$

This formula retains every Wilson seagull contribution.

The free symbol obeys

$$|S_\eta(p)| \le C(\bar m + |p|)^{-1}, \qquad p \in [-\pi/\eta, \pi/\eta]^3,$$

using principal Brillouin representatives. To see the estimate, the real Wilson term is $\eta^{-1}\sum_\nu(1-\cos\eta p_\nu)$, while the skew term has squared norm $\eta^{-2}\sum\sin^2\eta p_\nu$. On $|\eta p|\le 1$ the latter controls $|p|$; outside this ball the Wilson term controls η^{-1}, and hence $|p|$. The contact vertices satisfy

$$|V_r(p,p')| \le C_r\eta^{r-1}(1+|p-p'|)^{-j}$$

with the full alias distance in the Brillouin cube. The momentum-sum bound then gives

$$|\operatorname{Tr} T_4| + |\operatorname{Tr} T_1T_3| + |\operatorname{Tr} T_2^2| \le C\eta, \qquad |\operatorname{Tr} T_1^2T_2| \le C\eta\log(2/\eta).$$

For clarity, after the rapidly decreasing transfer momenta have been summed, the four expressions respectively have factors $\eta^3\sum(\bar m+|p|)^{-1}$, $\eta^2\sum(\bar m+|p|)^{-2}$, $\eta^2\sum(\bar m+|p|)^{-2}$, and $\eta\sum(\bar m+|p|)^{-3}$, where each sum includes its R^{-3} measure and is cut off at C/η. Radial integration gives the displayed bounds.

For the four linear vertices, write the trace as a loop in the four momenta p_i, with transfers $k_i = p_i - p_{i+1}$ and $\sum k_i = 0$. The Peetre bound

$$(\bar m + |p+k|)^t \le C_{m,t}(\bar m+|p|)^t(1+|k|)^{|t|}$$

allows three transfer sums to be absorbed by the smooth probe bounds. The tails with one momentum outside the principal cube contribute $O(\eta)$. On the central cube the exact symbol and the linear vertex differ from their continuum counterparts by

$$|S_\eta(p) - S^R(p)| \le C\min\{\eta, (\bar m+|p|)^{-1}\},$$

and

$$|V_1(p,p') - i\gamma\cdot\widehat a(p-p')| \le C\min\{\eta(|p|+|p'|+1),1\}(1+|p-p'|)^{-j} + C\eta^2(1+|p-p'|)^{-j}.$$

The elementary inequality $\min(u,1)\le u^{1/2}$, followed by telescoping the eight loop factors, bounds the difference by

$$C\eta^{1/2}(\bar m+|p_1|)^{-7/2}\prod_{i=1}^3(1+|k_i|)^{-7}.$$

The normalized sums are bounded independently of R. Thus $|\operatorname{Tr} T_1^4 - \operatorname{Tr}(S^R i\gamma\cdot a)^4| \le C\eta^{1/2}$. Together with (N.1) this proves the first comparison in Theorem 10.1.

N.3. **Removal of the torus.** The antiperiodic kernel is $S^R(x-y) = \sum_{n\in\mathbb{Z}^3}(-1)^{n_1+n_2+n_3}S_{\bar m}(x-y+Rn)$. On the fixed support cube, every nonzero image has distance at least $R - D_\beta$. The exponential kernel bound and its convolution give an exponentially small difference between S^RVS^R and $S_{\bar m}VS_{\bar m}$ in Hilbert–Schmidt norm, with the fixed compact multiplications on the exterior sides. Expanding the difference of the two fourth powers into terms containing one such difference, and using $|\operatorname{Tr} AB| \le \|A\|_{\mathrm{HS}}\|B\|_{\mathrm{HS}}$, proves $|b_R - b| \le Ce^{-\bar m(R-D_\beta)}$.

Appendix O. The large-mass sign

We give the local expansion carefully because its sign is the probe detecting non-Gaussianity. A pointwise expansion of the propagator alone would be singular on the diagonal. Taking one mass derivative produces a five-propagator loop whose diagonal is continuous, and permits a local Taylor expansion after dilation.

O.1. **Diagonal kernels and the mass derivative.** For $s(x) = C|x|^{-2}e^{-c|x|}$, its convolution powers satisfy

$$s * s(x) \le C|x|^{-1}e^{-c'|x|}, \qquad s^{*3}(x) \le C(1 + |\log|x||)e^{-c'|x|}, \qquad s^{*4}, s^{*5} \in C_b(\mathbb{R}^3).$$

For the first bound split the integration into balls of radius $|x|/2$ around 0 and x, and their complement; scaling the remaining integral gives $|x|^{-1}$. Convolving once more gives the logarithm by radial integration. The logarithm is locally integrable against $|x|^{-2}$; the fourth convolution is bounded. Translation continuity follows by splitting the singular balls from their complement and then using dominated convergence. Polynomial weights in the integration variables can be absorbed by a smaller exponential rate. These bounds also apply when four multiplication fields grow at most linearly.

Let $V = i\gamma \cdot a$, with a either smooth compact or a real linear constant-strength field. The continuous diagonal kernel

$$r_4(x; m, a) = \operatorname{tr}_4[(S_m V)^4 S_m](x, x)$$

is therefore given by the absolutely convergent five-kernel integral. For compact a, Hilbert–Schmidt differentiation and $\partial_m S_m = -S_m^2$ give

$$\frac{d}{dm} q_{4,m}(a) = \int r_4(x; m, a)\, dx, \qquad q_{4,m}(a) = -\tfrac{1}{4} \operatorname{Tr}(S_m V)^4.$$

Every differentiated term is the same trace by cyclicity. The same majorants prove continuity in $m > 0$, and show $q_{4,m} \to 0$ as $m \to \infty$.

O.2. **Local gauge invariance and dilation.** On the Schwartz space introduce the formal inverse

$$(m + \gamma \cdot \partial + uV)^{-1} = \sum_{n \ge 0} (-u)^n (S_m V)^n S_m.$$

No convergence in u is needed. For compact gauge λ, the formal operator is conjugated by $e^{-iu\lambda}$ when a is replaced by $a + d\lambda$. At fourth order its off-diagonal kernel differs by lower-order kernels multiplied by powers of $\lambda(x) - \lambda(y)$. More explicitly, the coefficient with $n \le 3$ insertions has local singularity bounded respectively by $|x-y|^{-2}, |x-y|^{-1}, 1 + |\log|x-y||, 1$, and is multiplied by $(\lambda(x) - \lambda(y))^{4-n}$. Since a smooth gauge has $|\lambda(x) - \lambda(y)| \le C|x - y|$, each product tends to zero. The coefficient with four insertions is continuous by the five-kernel bound. Hence $r_4(x; m, a + d\lambda) = r_4(x; m, a)$. The identity is local, so a gauge whose prescribed jet is cut off outside a neighborhood of x suffices.

Since $S_m(z) = m^2 S_1(mz)$, changing four integration variables to $w_i = m(z_i - x)$ gives a factor m^{-2}. Taylor-expand each $V(x + w_i/m)$ to degree five. If $V_j(x; w) = i\gamma \cdot D^j a(x)[w^j]/j!$, write $\rho_j(x)$ for the sum of the four-factor integrals whose total Taylor degree is j. The polynomially weighted convolution bounds give, uniformly on a fixed neighborhood of the support,

$$r_4(x; m, a) = m^{-2} \sum_{j=0}^{5} m^{-j} \rho_j(x) + O_a(m^{-8}).$$

Outside that neighborhood the compact multiplications force an exponentially long propagation; its integral is $O_a(e^{-cm})$. Thus the displayed remainder is integrable with total $O_a(m^{-8})$.

A radial gauge removes the constant value and symmetric first jet of a at x. The remaining first jet is $A_{F(x),\mu}(w) = \frac{1}{2} \sum_\nu w_\nu F_{\nu\mu}(x)$. Gauge invariance makes $\rho_0, \dots, \rho_3$ vanish and makes $\rho_4(x)$ equal to the constant-strength value $\Psi(F(x))$. Odd total Taylor degrees vanish: conjugation by the matrix interchanging the two Pauli sectors changes the sign of every γ_μ, and changing all integration variables w_i to $-w_i$ changes the sign according to the Taylor degree. Their combination gives $\rho_j = (-1)^j \rho_j$. Consequently

$$\frac{d}{dm} q_{4,m}(a) = m^{-6} \int \Psi(F(x))\, dx + O_a(m^{-8}). \tag{O.1}$$

It remains to evaluate Ψ, including the four-spin trace.

Lemma O.1 (Schwartz action and continuous diagonal)**.** *For $m > 0$, convolution by S_m maps the spinor Schwartz space continuously into itself. Multiplication by a smooth compact field or a linear field does the same. Thus every formal coefficient $(S_m V)^n S_m$ is an operator on that space. For $n = 4$ its five-propagator kernel is continuous, including its diagonal, and has at most polynomial growth for a linear field.*

Proof. In momentum space $S_m(p) = (m + i\gamma \cdot p)^{-1}$ is smooth; each derivative has polynomial bounds, because its denominator is $m^2 + |p|^2$ and $m > 0$. Multiplying a Schwartz function by this symbol preserves every Schwartz seminorm. Smooth compact and linear multiplication fields have polynomially bounded derivatives, so multiplication also preserves these seminorms. For the kernel claim use the five-kernel majorant in Appendix O.1; its polynomially weighted convolution is locally integrable and continuous. Splitting small singular neighborhoods from their complement gives continuity uniformly on compact sets; a linear field contributes at most a fourth-degree exterior polynomial. Equality of two such continuous kernels as operators on Schwartz tests implies their pointwise equality by testing against approximations to delta functions. □

O.3. **Constant-strength calculation.** For this calculation put $\mathscr{S} = \mathcal{S}(\mathbb{R}^3; \mathbb{C}^4)$. Formal coefficients act on this space; equality of such operators determines their continuous kernels. The following heat-kernel computation is coefficientwise, so no complex-field semigroup assumption is made.

Lemma O.2. *Let F be a real antisymmetric 3×3 matrix, and let $B \in \mathbb{R}^3$ be its vector:*

$$B_{\mathfrak{l}} = \frac{1}{2} \sum_{\nu,\mu=1}^{3} \varepsilon_{\mathfrak{l}\nu\mu} F_{\nu\mu} \qquad (\mathfrak{l} = 1, 2, 3),$$

where $\varepsilon_{\mathfrak{l}\nu\mu}$ is the sign of the permutation $(\mathfrak{l}, \nu, \mu)$ of $(1, 2, 3)$ when the three indices are distinct, and $\varepsilon_{\mathfrak{l}\nu\mu} = 0$ otherwise. Then the number $\Psi(F)$ of the local expansion above is given by

$$\Psi(F) = -\frac{4}{45} (4\pi)^{-\frac{3}{2}} \Gamma\left(\frac{7}{2}\right) |B|^4.$$

Proof. Throughout, $\mathsf{M} = 1$. By the definitions of the local expansion above, $\Psi(F) = \rho_4(0; A_F) = r_4(0; 1, A_F)$ for the field A_F of constant strength, $A_{F,\mu}(z) = \frac{1}{2} \sum_\nu z_\nu F_{\nu\mu}$. As $V_1(0; w) = i\gamma \cdot A_F(w)$ and $V_j(0; \cdot) = 0$ for $j \neq 1$, the only term of $\rho_4(0; A_F)$ that survives is the one with $(j_1, j_2, j_3, j_4) = (1, 1, 1, 1)$:

$$\Psi(F) = \operatorname{tr} \int \mathrm{d}w_1\, \mathrm{d}w_2\, \mathrm{d}w_3\, \mathrm{d}w_4\, S_1(-w_1)\, V_1(0; w_1)\, S_1(w_1 - w_2)\, V_1(0; w_2) \\ S_1(w_2 - w_3)\, V_1(0; w_3)\, S_1(w_3 - w_4)\, V_1(0; w_4)\, S_1(w_4).$$

The integral converges absolutely. Since every $|w_i|$ is at most the sum of the five successive edge lengths of this closed loop, the four linear multiplication factors can be absorbed into a polynomial weight on those edges. With $\mathsf{s}(z) = C(1 + |z|)^4 |z|^{-2} e^{-|z|/2}$ the absolute integral is at most $c_{A_F}^4 \mathsf{s}^{*5}(0)$. This is finite by the polynomially weighted convolution bound above.

(1) Reduction to $B = (0, 0, \mathsf{b})$. Let $\mathrm{Rt} \in \mathrm{SO}(3)$. Since $\mathrm{SU}(2)$ is a covering group of $\mathrm{SO}(3)$, there is $U \in \mathrm{SU}(2)$ with $U\,(\sigma \cdot w)\,U^{-1} = \sigma \cdot (\mathrm{Rt}\, w)$ for all $w \in \mathbb{R}^3$, where $\sigma \cdot w := \sum_\mu w_\mu \sigma_\mu$. With $\hat{U} := \operatorname{diag}(U, U)$ and $\gamma \cdot w = \operatorname{diag}(\sigma \cdot w, -\sigma \cdot w)$ (the paired representation fixed above) it follows that

$$\hat{U}\,(\gamma \cdot w)\,\hat{U}^{-1} = \gamma \cdot (\mathrm{Rt}\, w) \qquad (w \in \mathbb{R}^3).$$

By the explicit formula for S_1, $S_1(z) = Y_1(z) - \left(\gamma \cdot \frac{z}{|z|}\right) Y_1'(|z|)$ for $z \neq 0$, where $Y_1(z)$ and $Y_1'(|z|)$ depend on $|z|$ only; as $|\mathrm{Rt}\, z| = |z|$, it follows that $S_1(\mathrm{Rt}\, z) = \hat{U}\, S_1(z)\, \hat{U}^{-1}$ for $z \neq 0$. Further, $A_F(w) = -\frac{1}{2} F w$; so the matrix $F' := \mathrm{Rt} \cdot F \cdot \mathrm{Rt}^T$ satisfies $A_{F'}(\mathrm{Rt}\, w) = \mathrm{Rt}\, A_F(w)$, and therefore

$\gamma\cdot A_{F'}(\mathrm{Rt}\,w)=\hat U\,(\gamma\cdot A_F(w))\,\hat U^{-1}$. After the change of variables $(w_1,\dots,w_4)\mapsto(\mathrm{Rt}\,w_1,\dots,\mathrm{Rt}\,w_4)$, which preserves Lebesgue measure, the integrand of the formula for $\Psi(F')$ becomes $\hat U$ times the integrand of the formula for $\Psi(F)$ times $\hat U^{-1}$; as the trace is invariant under conjugation, $\Psi(F')=\Psi(F)$. The vector of F' is $\mathrm{Rt}\,B$ (indeed $Fw=w\times B$ for $w\in\mathbb{R}^3$, and rotations preserve the vector product). Choosing Rt with $\mathrm{Rt}\,B=(0,0,|B|)$, one may therefore assume $B=(0,0,\mathsf{b})$ with $\mathsf{b}:=|B|\geq 0$. Then $F_{12}=\mathsf{b}$, $F_{13}=F_{23}=0$, and the field $A:=A_F$ is

$$A(z)=\left(-\frac{\mathsf{b}\cdot z_2}{2},\ \frac{\mathsf{b}\cdot z_1}{2},\ 0\right),\qquad \partial\cdot A=0.$$

(2) The square of $\gamma\cdot D_u$. Let E be the ring of all $\mathbb{C}$-linear maps of $\mathscr{S}$ into itself, and let $\mathsf{E}[[u]]$ be the ring of formal power series in u with coefficients in E, as in the proof of the formal gauge calculation above. Let $D_u:=\partial+iuA$, that is, $D_{u,\mu}=\partial_\mu+iuA_\mu$ for $\mu=1,2,3$. The relations $\gamma_\mu\gamma_\nu+\gamma_\nu\gamma_\mu=2\delta_{\mu\nu}$ of the paired representation fixed above give

$$(\gamma\cdot D_u)^2=\sum_\mu D_{u,\mu}^2+\sum_{\mu<\nu}\gamma_\mu\gamma_\nu\,[D_{u,\mu},D_{u,\nu}]\,.$$

One has $[D_{u,1},D_{u,2}]=iu\,(\partial_1A_2-\partial_2A_1)=iu\cdot\mathsf{b}$, the two other commutators vanish, $\gamma_1\gamma_2=i\,\mathrm{diag}(\sigma_3,\sigma_3)$, and $\sum_\mu D_{u,\mu}^2=\Delta+2iu\,A\cdot\partial-u^2|A|^2$ since $\partial\cdot A=0$. Hence, in $\mathsf{E}[[u]]$,

$$(1-\gamma\cdot D_u)(1+\gamma\cdot D_u)=(1+\gamma\cdot D_u)(1-\gamma\cdot D_u)=\mathrm{diag}\,(\mathsf{h}_u,\mathsf{h}_u)\,,$$

$$\mathsf{h}_u:=1-\Delta+uW_1+u^2W_2,\qquad W_1:=-2i\,A\cdot\partial+\mathsf{b}\cdot\sigma_3,\qquad W_2:=|A|^2=\frac{\mathsf{b}^2}{4}\left(z_1^2+z_2^2\right).$$

Here $\mathsf{h}_u\in\mathsf{E}_2[[u]]$, where E_2 is the ring of all $\mathbb{C}$-linear maps of $\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2)$ into itself, and $\mathrm{diag}(\mathsf{h}_u,\mathsf{h}_u)$ acts on $\mathscr{S}=\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2)\oplus\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2)$, according to the splitting $\mathbb{C}^4=\mathbb{C}^2\oplus\mathbb{C}^2$ of the paired representation fixed above. In E_2 the operator $1-\Delta$ has an inverse, the Fourier multiplier $(1+|p|^2)^{-1}$; as $1-\Delta$ is the constant term of the power series h_u, the latter is invertible in $\mathsf{E}_2[[u]]$.

By Lemma O.1, S_1 and $V_A:=i\gamma\cdot A$ map $\mathscr{S}$ into itself. Let $\mathsf{D}_u(A):=1+\gamma\cdot\partial+u\,i\gamma\cdot A=1+\gamma\cdot D_u\in\mathsf{E}[[u]]$, formed as $\mathsf{D}_u(\tilde a)$ in the proof of the formal gauge calculation above, with $\mathsf{M}=1$ and with the field A in place of $\tilde a$. Its constant term is $1+\gamma\cdot\partial$, which is invertible in E with inverse S_1; so $\mathsf{D}_u(A)=(1+\gamma\cdot\partial)(1+u\,S_1V_A)$ is invertible in $\mathsf{E}[[u]]$, and its inverse is

$$\sum_{n\geq0}(-u)^n\,(S_1V_A)^n\,S_1=\sum_{n\geq0}u^n\,\mathfrak{R}_A^{(n)},\qquad \mathfrak{R}_A^{(n)}=(-1)^n\,(S_1V_A)^n\,S_1.$$

By the identity above, $(1-\gamma\cdot D_u)\,\mathrm{diag}(\mathsf{h}_u^{-1},\mathsf{h}_u^{-1})$ is a right inverse of $1+\gamma\cdot D_u$, and inverses in a ring are unique. Therefore

$$\sum_{n\geq0}u^n\,\mathfrak{R}_A^{(n)}=(1+\gamma\cdot D_u)^{-1}=(1-\gamma\cdot D_u)\,\mathrm{diag}\left(\mathsf{h}_u^{-1},\mathsf{h}_u^{-1}\right).$$

(3) Mehler's kernel. For $t>0$ and $z\in\mathbb{R}^3$ let

$$g_t(z):=(4\pi t)^{-\frac{3}{2}}\exp\left(-\frac{|z|^2}{4t}\right)$$

be the heat kernel. For $x,y\in\mathbb{R}^3$ put $x_p:=(x_1,x_2)$ and $x\wedge y:=x_1y_2-x_2y_1$; note that $x\wedge y=(x-y)\wedge y$. For $t>0$ and complex u with $|u|\cdot\mathsf{b}\cdot t<\pi$ let $\boldsymbol{\tau}:=u\cdot\mathsf{b}\cdot t$, and for $x,y\in\mathbb{R}^3$ let $k_u(t;x,y)$ be the complex 2×2 matrix

$$k_u(t;x,y):=g_t(x-y)\,\frac{\boldsymbol{\tau}}{\sinh\boldsymbol{\tau}}\,\exp\left\{-\frac{(\boldsymbol{\tau}\coth\boldsymbol{\tau}-1)\,|x_p-y_p|^2}{4t}+\frac{iu\cdot\mathsf{b}}{2}\,x\wedge y\right\}\exp\left(-\boldsymbol{\tau}\cdot\sigma_3\right).$$

The functions $\boldsymbol{\tau} \mapsto \boldsymbol{\tau}/\sinh\boldsymbol{\tau}$ and $\boldsymbol{\tau} \mapsto \boldsymbol{\tau}\coth\boldsymbol{\tau}$, with the value 1 at $\boldsymbol{\tau} = 0$, are analytic in the disc $|\boldsymbol{\tau}| < \pi$; hence $k_u(t;x,y)$ is an analytic function of u in the disc $|u| \cdot \mathsf{b} \cdot t < \pi$. For all such t and u,

$$(\partial_t + \mathsf{h}_u - 1)\, k_u = 0,$$

where h_u acts on the variable x. For the proof let $b' := u \cdot \mathsf{b}$. Since $A \cdot \partial = \frac{\mathsf{b}}{2}\,(x_1\partial_2 - x_2\partial_1)$, step (2) gives

$$\mathsf{h}_u - 1 = -\partial_3^2 + \check{h} + b'\sigma_3, \qquad \check{h} := -\partial_1^2 - \partial_2^2 - ib'\,(x_1\partial_2 - x_2\partial_1) + \frac{b'^2}{4}\,|x_p|^2\,,$$

while k_u is the product

$$k_u(t;x,y) = (4\pi t)^{-\frac{1}{2}} \exp\left(-\frac{(x_3-y_3)^2}{4t}\right) \cdot \check{k} \cdot \exp\left(-\boldsymbol{\tau}\cdot\sigma_3\right)$$

with

$$\check{k} := \frac{b'}{4\pi\sinh(b't)}\,\exp(\check{\phi}), \qquad \check{\phi} := -\check{\alpha}\cdot|x_p - y_p|^2 + \frac{ib'}{2}\,x\wedge y, \qquad \check{\alpha} := \frac{b'}{4}\coth(b't).$$

(The functions of b' that occur here and below are analytic in the disc $|b'|\,t < \pi$, their values at $b' = 0$ being the limits; for instance $b'/\sinh(b't)$ and $b'\coth(b't)$ take the value $1/t$ at $b' = 0$.) The first factor satisfies the heat equation in the variables (t, x_3), and $\partial_t \exp(-\boldsymbol{\tau}\cdot\sigma_3) = -b'\sigma_3\exp(-\boldsymbol{\tau}\cdot\sigma_3)$, which compensates the term $b'\sigma_3$ of $\mathsf{h}_u - 1$. It remains to prove that $\partial_t\check{k} = -\check{h}\,\check{k}$. One computes

$$\begin{aligned}
\partial_1\check{\phi} &= -2\check{\alpha}\cdot(x_1 - y_1) + \frac{ib'y_2}{2},\\
\partial_2\check{\phi} &= -2\check{\alpha}\cdot(x_2 - y_2) - \frac{ib'y_1}{2}, \qquad \left(\partial_1^2 + \partial_2^2\right)\check{\phi} = -4\check{\alpha},\\
\left(\partial_1\check{\phi}\right)^2 + \left(\partial_2\check{\phi}\right)^2 &= 4\check{\alpha}^2\cdot|x_p - y_p|^2 - \frac{b'^2}{4}\,|y_p|^2 - 2i\,\check{\alpha}\cdot b'\,x\wedge y,\\
(x_1\partial_2 - x_2\partial_1)\,\check{\phi} &= 2\check{\alpha}\cdot x\wedge y - \frac{ib'}{2}\,x_p\cdot y_p.
\end{aligned}$$

Since $\partial_j^2\exp(\check{\phi}) = \left(\partial_j^2\check{\phi} + \left(\partial_j\check{\phi}\right)^2\right)\exp(\check{\phi})$, these identities give

$$\begin{aligned}
\frac{\check{h}\,\check{k}}{\check{k}} &= 4\check{\alpha} - 4\check{\alpha}^2\cdot|x_p - y_p|^2 + \frac{b'^2}{4}\left(|y_p|^2 - 2\,x_p\cdot y_p + |x_p|^2\right)\\
&= 4\check{\alpha} - \left(4\check{\alpha}^2 - \frac{b'^2}{4}\right)|x_p - y_p|^2\,.
\end{aligned}$$

On the other hand

$$\frac{\partial_t\check{k}}{\check{k}} = -b'\coth(b't) - \check{\alpha}'\cdot|x_p - y_p|^2\,, \qquad \check{\alpha}' := \partial_t\check{\alpha} = -\frac{b'^2}{4\sinh^2(b't)} = -\left(4\check{\alpha}^2 - \frac{b'^2}{4}\right),$$

and $b'\coth(b't) = 4\check{\alpha}$. So $\partial_t\check{k} = -\check{h}\,\check{k}$, and the product rule gives $(\partial_t + \mathsf{h}_u - 1)\,k_u = 0$.

(4) Expansion in powers of u. The function $\boldsymbol{\tau}/\sinh\boldsymbol{\tau}$ is a power series in $\boldsymbol{\tau}^2 = u^2\cdot\mathsf{b}^2\cdot t^2$, and $(\boldsymbol{\tau}\coth\boldsymbol{\tau} - 1)/(4t)$ is a power series in u^2 in which the coefficient of u^{2j}, $j \geq 1$, is a real number times $\mathsf{b}^{2j}\cdot t^{2j-1}$. Hence $k_u = \sum_{n\geq 0} u^n k^{(n)}$ with

$$k^{(n)}(t;x,y) = g_t(x-y)\,\check{P}_n(t,x,y),$$

where $\check{P}_n$ is a finite sum of monomials $t^j(x-y)^{\boldsymbol{\alpha}}y^{\boldsymbol{\beta}}$ ($j \geq 0$ an integer, $\boldsymbol{\alpha},\boldsymbol{\beta} \in \{0,1,2,\dots\}^3$ multi-indices) with coefficients in the linear span of 1 and σ_3, each monomial satisfying $2j + |\boldsymbol{\alpha}| \geq n$. To see this, assign to t the weight 2 and to each component of $x - y$ the weight 1. Every occurrence of u in $\boldsymbol{\tau}/\sinh\boldsymbol{\tau}$ and in $\exp(-\boldsymbol{\tau}\cdot\sigma_3)$ comes with a factor t, so that u^k comes with the weight $2k$

there; a term $u^{2j}t^{2j-1}\,|x_p-y_p|^2$ of the exponent has the weight $4j$; and $u\,x\wedge y = u\,(x-y)\wedge y$ has the weight 1. Thus in every product of these terms the weight is at least the power of u. Moreover $\check{P}_0 = 1$. Comparing the coefficients of u^n in the identity of step (3), one finds, for all $t>0$, $x,y\in\mathbb{R}^3$ and $n\geq 0$, with $k^{(-1)} = k^{(-2)} := 0$,

$$\partial_t k^{(n)} - \Delta_x k^{(n)} + W_{1,x} k^{(n-1)} + W_2(x)\,k^{(n-2)} = 0.$$

(5) The inverse of h_u. For an integer $j\geq 0$, a multi-index $\boldsymbol{\alpha}$ and $z\in\mathbb{R}^3$ let

$$\check{m}_{j,\boldsymbol{\alpha}}(z) := \int_0^\infty \mathrm{d}t\,\exp(-t)\,t^j z^{\boldsymbol{\alpha}} g_t(z)$$

whenever the integral converges absolutely. This is the case for every $z\neq 0$, since $\exp\left(-\frac{|z|^2}{4t}\right)$ tends to 0 faster than every power of t as $t\to 0$. At $z=0$ the integrand vanishes if $\boldsymbol{\alpha}\neq 0$, and it equals $(4\pi)^{-\frac{3}{2}}\exp(-t)\,t^{j-\frac{3}{2}}$ if $\boldsymbol{\alpha}=0$, which is integrable over $(0,\infty)$ if $j\geq 1$. Thus $\check{m}_{j,\boldsymbol{\alpha}}$ is defined on $\mathbb{R}^3\setminus\{0\}$, and on all of $\mathbb{R}^3$ if $2j+|\boldsymbol{\alpha}|\geq 1$. The Fourier transform of $\check{m}_{j,\boldsymbol{\alpha}}$ equals $(i\partial_p)^{\boldsymbol{\alpha}}\left[j!\,(1+|p|^2)^{-j-1}\right]$; this function of $p\in\mathbb{R}^3$ is smooth, and it and all its derivatives are bounded. Convolution with $\check{m}_{j,\boldsymbol{\alpha}}$ therefore maps $\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2)$ into itself. Put

$$\check{K}^{(n)}(x,y) := \int_0^\infty \mathrm{d}t\,\exp(-t)\,k^{(n)}(t;x,y) = \sum c_{j,\boldsymbol{\alpha},\boldsymbol{\beta}}\,\check{m}_{j,\boldsymbol{\alpha}}(x-y)\,y^{\boldsymbol{\beta}},$$

where the sum and the coefficients $c_{j,\boldsymbol{\alpha},\boldsymbol{\beta}}\in\mathrm{span}\{1,\sigma_3\}$ are those of $\check{P}_n$. The integral converges absolutely and equals the sum for all $x,y\in\mathbb{R}^3$ with $x\neq y$, and also for $x=y$ if $n\geq 1$, because every monomial of $\check{P}_n$ then has $2j+|\boldsymbol{\alpha}|\geq 1$. Further let

$$\left(\mathfrak{K}^{(n)}\,\boldsymbol{\phi}\right)(x) := \int \mathrm{d}y\,\check{K}^{(n)}(x,y)\,\boldsymbol{\phi}(y) \qquad \left(\boldsymbol{\phi}\in\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2),\ x\in\mathbb{R}^3\right).$$

Then $\mathfrak{K}^{(n)}\in\mathsf{E}_2$; put $\mathfrak{K}^{(-1)} = \mathfrak{K}^{(-2)} := 0$. For $\boldsymbol{\phi}\in\mathcal{S}(\mathbb{R}^3;\mathbb{C}^2)$ let $\boldsymbol{\Phi}_n(t,x) := \int \mathrm{d}y\,k^{(n)}(t;x,y)\,\boldsymbol{\phi}(y)$ for $t>0$ and $x\in\mathbb{R}^3$, and $\boldsymbol{\Phi}_{-1} = \boldsymbol{\Phi}_{-2} := 0$. By step (4),

$$\partial_t\boldsymbol{\Phi}_n = \Delta\boldsymbol{\Phi}_n - W_1\boldsymbol{\Phi}_{n-1} - W_2\boldsymbol{\Phi}_{n-2}.$$

Multiplication by $\exp(-t)$ and integration over (ϵ,∞), $\epsilon>0$, give

$$\begin{aligned}-\exp(-\epsilon)\,\boldsymbol{\Phi}_n(\epsilon,\cdot) + (1-\Delta)\int_\epsilon^\infty \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_n(t,\cdot)\\ + W_1\int_\epsilon^\infty \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_{n-1}(t,\cdot) + W_2\int_\epsilon^\infty \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_{n-2}(t,\cdot) = 0.\end{aligned}$$

Now let $\epsilon\to 0$. As $\check{P}_0 = 1$ and $(g_t)_{t>0}$ is an approximate identity, $\boldsymbol{\Phi}_0(\epsilon,\cdot)\to\boldsymbol{\phi}$ uniformly. For each monomial $c\,t^j(x-y)^{\boldsymbol{\alpha}}y^{\boldsymbol{\beta}}$ of $\check{P}_m$, $m\geq 0$, one has

$$\left|\int \mathrm{d}y\,g_t(x-y)\,c\,t^j(x-y)^{\boldsymbol{\alpha}}y^{\boldsymbol{\beta}}\boldsymbol{\phi}(y)\right| \leq |c|\cdot\sup_{y\in\mathbb{R}^3}\left|y^{\boldsymbol{\beta}}\boldsymbol{\phi}(y)\right|\cdot t^j\int \mathrm{d}z\ |z^{\boldsymbol{\alpha}}|\,g_t(z) = c'\,t^{j+\frac{|\boldsymbol{\alpha}|}{2}},$$

with a number c' independent of t and x, because $\int \mathrm{d}z\,|z^{\boldsymbol{\alpha}}|\,g_t(z)$ is a constant multiple of $t^{\frac{|\boldsymbol{\alpha}|}{2}}$ and $y^{\boldsymbol{\beta}}\boldsymbol{\phi}$ is bounded. Hence $|\boldsymbol{\Phi}_m(t,x)|$ is bounded uniformly in $0<t\leq 1$ and $x\in\mathbb{R}^3$, for every $m\geq 0$; and for $n\geq 1$, where $2j+|\boldsymbol{\alpha}|\geq 1$ for every monomial of $\check{P}_n$, $\sup_x|\boldsymbol{\Phi}_n(\epsilon,x)|\leq c''\cdot\epsilon^{\frac{1}{2}}\to 0$ for $\epsilon\leq 1$, with c'' independent of ϵ. So $\int_0^\epsilon \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_m(t,x)\to 0$ uniformly in x, and since $\int_0^\infty \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_m(t,x) = \left(\mathfrak{K}^{(m)}\,\boldsymbol{\phi}\right)(x)$ by Fubini's theorem, $\int_\epsilon^\infty \mathrm{d}t\,\exp(-t)\,\boldsymbol{\Phi}_m(t,\cdot)\to\mathfrak{K}^{(m)}\,\boldsymbol{\phi}$ uniformly. Uniform convergence implies convergence in $\mathcal{S}'(\mathbb{R}^3;\mathbb{C}^2)$, on which differential operators with polynomial coefficients act continuously. It follows that

$$(1-\Delta)\,\mathfrak{K}^{(n)}\,\boldsymbol{\phi} + W_1\,\mathfrak{K}^{(n-1)}\,\boldsymbol{\phi} + W_2\,\mathfrak{K}^{(n-2)}\,\boldsymbol{\phi} = \delta_{n0}\,\boldsymbol{\phi} \qquad (n\geq 0),$$

that is, $\mathsf{h}_u \sum_{n\geq 0} u^n\, \mathfrak{K}^{(n)} = 1$ in $\mathsf{E}_2[[u]]$. Being a right inverse of the invertible element h_u of this ring, the series $\sum_{n\geq 0} u^n\, \mathfrak{K}^{(n)}$ is its inverse: $\mathsf{h}_u^{-1} = \sum_{n\geq 0} u^n\, \mathfrak{K}^{(n)}$.

(6) The value at the origin. By steps (2) and (5),

$$\sum_{n\geq 0} u^n\, \mathfrak{R}_A^{(n)} = (1 - \gamma\cdot\partial - u\, i\gamma\cdot A) \times \sum_{n\geq 0} u^n \operatorname{diag}\left(\mathfrak{K}^{(n)}, \mathfrak{K}^{(n)}\right).$$

The coefficient of u^4 is

$$\mathfrak{R}_A^{(4)} = (1-\gamma\cdot\partial)\operatorname{diag}\left(\mathfrak{K}^{(4)}, \mathfrak{K}^{(4)}\right) - i\gamma\cdot A \operatorname{diag}\left(\mathfrak{K}^{(3)}, \mathfrak{K}^{(3)}\right).$$

For $j \geq 0$ and $s \geq 0$ let

$$\check{\mu}_j(s) := \int_0^\infty \mathrm{d}t\, \exp(-t)\, t^j\, (4\pi t)^{-\frac{3}{2}} \exp\left(-\frac{s^2}{4t}\right)$$

whenever the integral converges; as in step (5), this is the case for every $s > 0$, and for $s = 0$ if $j \geq 1$. Then $\check{m}_{j,\boldsymbol{\alpha}}(z) = z^{\boldsymbol{\alpha}}\cdot \check{\mu}_j(|z|)$ for $z \neq 0$, and also for $z = 0$ if $j \geq 1$, while $\check{m}_{0,\boldsymbol{\alpha}}(0) = 0$ if $\boldsymbol{\alpha} \neq 0$. One has $\check{\mu}_0(|z|) = Y_1(z) = \exp(-|z|)/(4\pi|z|)$ for $z \neq 0$. For $j \geq 1$ the function $\check{\mu}_j$ is bounded and continuous on $[0,\infty)$, by dominated convergence, the integrand being at most $(4\pi)^{-\frac{3}{2}} \exp(-t)\, t^{j-\frac{3}{2}}$; and $\check{\mu}_j'(s) = -\frac{s}{2}\, \check{\mu}_{j-1}(s)$ for $s > 0$. Hence the function $\check{m}_{j,\boldsymbol{\alpha}}$, defined on all of $\mathbb{R}^3$ when $2j + |\boldsymbol{\alpha}| \geq 1$, is continuous on $\mathbb{R}^3$ if $2j + |\boldsymbol{\alpha}| \geq 2$, and continuously differentiable on $\mathbb{R}^3$ if $2j + |\boldsymbol{\alpha}| \geq 3$. On $\mathbb{R}^3 \setminus \{0\}$ both properties follow from the formulas for Y_1 and for $\check{\mu}_j'$. At 0, for $j = 0$ use $\check{m}_{0,\boldsymbol{\alpha}}(0) = 0$, $|z^{\boldsymbol{\alpha}}| \leq |z|^{|\boldsymbol{\alpha}|}$, $Y_1(z) \leq (4\pi|z|)^{-1}$ and the fact that the gradient of Y_1 at $z \neq 0$ has the length $\left(1 + |z|^{-1}\right) Y_1(z)$; for $j \geq 1$ use that the gradient of $z \mapsto \check{\mu}_j(|z|)$ at $z \neq 0$ is $-\frac{1}{2}\, \check{\mu}_{j-1}(|z|)\, z$, which is bounded if $j = 1$ and tends to 0 as $z \to 0$ if $j \geq 2$. In this way one finds that, if $2j + |\boldsymbol{\alpha}| \geq 3$, the gradient of $\check{m}_{j,\boldsymbol{\alpha}}$ on $\mathbb{R}^3 \setminus \{0\}$ has a limit at 0; and a continuous function on $\mathbb{R}^3$ that is continuously differentiable on $\mathbb{R}^3 \setminus \{0\}$, with a gradient that has a limit at 0, is continuously differentiable on $\mathbb{R}^3$. Since, for $n \geq 1$, $\check{K}^{(n)}(x,y)$ is the finite sum of step (5) for all $x, y \in \mathbb{R}^3$, and every monomial of $\check{P}_n$ has $2j + |\boldsymbol{\alpha}| \geq n$, it follows that $\check{K}^{(3)}$, $\check{K}^{(4)}$ and $\partial_x \check{K}^{(4)}$ are continuous on $\mathbb{R}^3 \times \mathbb{R}^3$ and grow at most polynomially; thus the operator on the right-hand side of the last formula for $\mathfrak{R}_A^{(4)}$ has the continuous kernel

$$(1-\gamma\cdot\partial_x)\operatorname{diag}\left(\check{K}^{(4)}, \check{K}^{(4)}\right)(x,y) - i\gamma\cdot A(x) \operatorname{diag}\left(\check{K}^{(3)}, \check{K}^{(3)}\right)(x,y).$$

On the other hand, $\mathfrak{R}_A^{(4)} = (S_1 V_A)^4 S_1$ has the continuous kernel $\mathrm{Rk}_A^{(4)}$ by the continuous-diagonal bounds above, where $\mathrm{Rk}_A^{(4)}$ is the kernel $\mathrm{Rk}^{(4)}$ of the diagonal-kernel calculation above formed with $\mathsf{M} = 1$ and the field A. A continuous kernel is determined by the operator that it defines on $\mathscr{S}$; so the two kernels coincide. At $x = y = 0$ one has $A(0) = 0$, and $\operatorname{tr}\left[\gamma_\mu \operatorname{diag}(\check{X}, \check{X})\right] = \operatorname{tr}(\sigma_\mu \check{X}) - \operatorname{tr}(\sigma_\mu \check{X}) = 0$ for every complex 2×2 matrix $\check{X}$ and $\mu = 1,2,3$. Therefore, $\check{K}^{(4)}(0,0)$ being given by the integral over t of step (5) at $x = y = 0$,

$$\Psi(F) = \operatorname{tr} \mathrm{Rk}_A^{(4)}(0,0) = 2 \operatorname{tr} \check{K}^{(4)}(0,0) = 2\int_0^\infty \mathrm{d}t\, \exp(-t) \operatorname{tr} k^{(4)}(t;0,0).$$

Finally, $\operatorname{tr}\exp(-\boldsymbol{\tau}\cdot\sigma_3) = 2\cosh\boldsymbol{\tau}$, so that

$$\operatorname{tr} k_u(t;0,0) = (4\pi t)^{-\frac{3}{2}}\, \frac{\boldsymbol{\tau}}{\sinh\boldsymbol{\tau}} \cdot 2\cosh\boldsymbol{\tau} = 2\,(4\pi t)^{-\frac{3}{2}}\, \boldsymbol{\tau}\coth\boldsymbol{\tau},$$

and by $s\coth s = 1 + \frac{s^2}{3} - \frac{s^4}{45} + O(s^6)$ the coefficient of u^4 in this function is $-\frac{2}{45}\,(4\pi t)^{-\frac{3}{2}}\,\mathsf{b}^4 \cdot t^4$. Since $\int_0^\infty \mathrm{d}t\, \exp(-t)\, t^{\frac{5}{2}} = \Gamma\left(\frac{7}{2}\right)$, it follows that

$$\Psi(F) = -\frac{4}{45}\,(4\pi)^{-\frac{3}{2}}\,\Gamma\left(\frac{7}{2}\right)\cdot \mathsf{b}^4, \qquad \mathsf{b} = |B|.$$

□

O.3.1. *Large mass and dilated probes.* Integrating (O.1) from m to infinity now proves

$$q_{4,m}(a) = \frac{1}{240\pi m^5}\int |\operatorname{curl} a|^4 + O_a(m^{-7}).$$

Indeed $-\Psi/5 = (240\pi)^{-1}|B|^4$, since $\Gamma(7/2) = 15\sqrt{\pi}/8$. For the dilation $a_l(x) = la(x/l)$, the unitary map $U_l\psi(x) = l^{-3/2}\psi(x/l)$ gives $U_l^{-1}S_{\bar m}i\gamma\cdot a_l U_l = l^2 S_{\bar m l} i\gamma\cdot a$. Thus $q_{4,\bar m}(a_l) = l^8 q_{4,\bar m l}(a)$, proving the positive-probe assertion in Theorem 10.1.

Appendix P. Normalized reference jets

Write a_*, e_*, l_* for the terminal physical spacing, charge and side in terminal units, and $\eta_* = \eta/a_*$ for its fine mesh. The terminal data from Theorem 9.5 satisfy

$$a_* \in \{h'/L, h'/L^2\}, \quad e_* = e\sqrt{a_*}, \quad l_* = Mr_*, \quad v_* = -\log e_*, \quad v_*^r \le r_* < Lv_*^r, \tag{P.1}$$

where $r \ge 2$, $k_* \ge 1$, and the reference mass $\mathfrak{m}_* = \bar m a_*$ lies in a fixed positive interval. In particular the terminal geometry is independent of the fine cutoff.

Place the compact probe in the central three quarters of a mirror cell. Let $B_c = e^{-1}\widehat a_c^{[*]}$, and set $B(t) = \sum_c t_c B_c$. The native cochain rescaling is $A^{[j]}(b) = L^{j/2}A(L^j b)$, on bonds in level coordinates. In physical coordinates this gives the exact source formula

$$\widehat a_\nu^{[*]}(y) = (a_*/\eta)^{1/2}\widehat a_\nu(a_* y) = a_*^{1/2}\check a_{\eta,\nu}(a_* y + \eta e_\nu/2).$$

Thus the terminal source is the rescaling of the same native shift used in 3.1; no extra factor $\sqrt{\eta}$ is retained in it. Let P_c contain the terminal unit blocks meeting source-link endpoints and let P_c^+ be its $(M+1)$-neighborhood. Then

$$|P_c^+| \le p_\beta = (D_\beta/a_* + 2M + 4)^3, \qquad d(P_c, P_{c'}) \ge d_* := Mr_*/(8L) - 1. \tag{P.2}$$

The value, derivative and local derivative Hölder seminorms of $\widehat a^{[*]}$ have sum at most $C_\beta = 2\sqrt{h'}(1 + h' + 3h'^2)|\beta|_{C^3}$. Hölder pairs have distance at most one; separation greater than four makes each seminorm of the sum a maximum over cells. Thus

$$t_x = \frac{e_*^{3\epsilon}}{1 + \sqrt{h'}C_\beta}, \qquad r_x = t_x/4$$

is a common source radius. All full connection supports at earlier levels are admitted too, by scaling and Theorem 9.1.

P.1. **Terminal identities.** We state exactly what is used from the four RG interfaces. Locally

$$S(A) = (D_A + \mathfrak{m}_* + b_* Q^T(-A)Q(A))^{-1}, \quad D_*(A) = b_* - b_*^2 Q(A)S(A)Q^T(-A).$$

For real fields $Q^T(-A) = Q(A)^*$. The fine inverse, field map and covariant Hölder map have two-end majorants $C_F e^{-\gamma_F d(y,y')}$ in coarse block variables, on their common open domain $\widetilde G_* \supset \widetilde R_* + \mathfrak{X}_*$. These domains admit real gauge and real constant additions. The relevant correction is $|m_*| \le e_*^{3/4-8\epsilon}/8$; it is separate from the positive reference mass $\mathfrak{m}_*$. The balanced activity satisfies

$$\sum_{n,m} \frac{e_*^{-n/4} e_*^{(-1/4+\epsilon)m}}{n!m!}\|E_{*,nm}(X;\mathscr{A},B)\| \le 4e_*^{1/4-6\epsilon}e^{-\kappa d_M(X)}. \tag{P.3}$$

Its connection support consists of the links meeting X, and the endpoints of every fermion and covariant Hölder argument are in its full support. The scalar determinant correction P_* is the reblocked sum of $E_j^d(X,B) - E_j^d(X,0)$, where each E_j^d is even, real-gauge invariant, analytic on its complete domain and bounded by $CM^3 e^{-(\kappa+1)d_M(X)}$. Exact Gaussian telescoping gives

$$N_0(t) = e^{\sum_X P_*(X,B(t))} \frac{\det D_*(B(t))}{\det D_*(0)}. \tag{P.4}$$

The orthogonal photon coordinates are $(A_*;\zeta_j)$. White coordinates are unit linear curvature reads, at most $2l_*^3 L^{3(k_*-j)}$ at level j; each plaquette of dA_* is a read of norm at most one. On the retained small-field set a real gauge gives $|A_*^g| \le 2l_*^2 p_*$, $\mathscr{A} = HA_*^g \in \widetilde{R}_*/4$, and the segment $u\mathscr{A} + c_\theta$, $0 \le u \le 1$, is legal for real harmonic constants c_θ. With $\mathscr{T} = \mathscr{A} + c_\theta + B$, the exact terminal density is

$$Y(A_* + c_\theta; B) = \frac{e^{-\epsilon_* l_*^3 + \sum_X P_*(X,B)}}{\det D_*(0)} \int D\Psi\, e^{-\bar\Psi D_*(\mathscr{T})\Psi - m_*\langle\bar\psi,\psi\rangle + \sum_X E_*(X;\mathscr{A}+c_\theta,B;\psi^\#)}. \tag{P.5}$$

Here $\psi^\# = H^\#(\mathscr{T})\Psi$, the fine pairing keeps its η_*^3 weight, and integration against the terminal photon measure equals the original small-field integral with its scalar normalizer. These are consequences of Theorems 9.1, 9.3, 9.4 and 9.5; the tuned pair is the one fixed in those theorems.

For real A, an additional all-background bound is

$$\|D_*^{-1}\| \le b_*^{-1} + \mathfrak{m}_*^{-1}, \quad |D_*^{-1}(y,y')| \le b_*^{-1}\mathbf{1}_{y=y'} + \frac{2\exp(1)}{\mathfrak{m}_*} e^{-\mathfrak{m}_* d(y,y')/(12\exp(1))}. \tag{P.6}$$

The Schur identity is $D_*^{-1} = b_*^{-1} + Q(D_A + \mathfrak{m}_*)^{-1}Q^*$. The fine operator's Hermitian part is at least $\mathfrak{m}_*$. Exponential conjugation of slope $\mathfrak{m}_*/(12\exp(1))$ changes the finite-range operator by at most $\mathfrak{m}_*/2$. Its inverse has norm at most $2/\mathfrak{m}_*$; averaging over unit blocks proves the entry bound. This argument uses reference positivity, not positivity of the selected bare mass.

P.2. **Whole Gram bounds.** For a finite Grassmann algebra let $|F|_h$ be the absolute sum of ordered-monomial coefficients with weight $h^{\deg}$. It is independent of ordering and submultiplicative. Normalize $\int D\Psi e^{-\bar\Psi K\Psi} = \det K$, and let μ_G denote the normalized Gaussian integral with covariance G. Wick's rule, also for singular G by polynomial continuation, gives

$$\int \mu_G F(\bar\Psi,\Psi) = \int \mu_I F(\bar\Psi, G\Psi).$$

If all absolute row sums of G are at most $h \ge 1$, then $|\int \mu_G F| \le |F|_h$. Substitute the unbarred generators, use the row sums, and then $|\int \mu_I F| \le |F|_1$. Moreover $|\bar\Psi K\Psi|_h = h^2 \sum |K_{ij}|$. The paired field-row bounds in Theorem 9.1 imply, with $a_2 = C_{\mathrm{jet}}^{-1} e_*^{-1/4+\epsilon}$,

$$|\langle\bar\psi,\psi\rangle|_{a_2} \le 4e_*^{-1/2+2\epsilon} l_*^3, \quad |E_*(X;H^\#\Psi)|_{a_2} \le \|E_*(X)\|_{(e_*^{-1/4}, e_*^{-1/4+\epsilon})}. \tag{P.7}$$

For the first inequality sum the exterior coarse rows on both sides and retain the η_*^3 sum of the intermediate fine site. For the second do this separately for ordinary and covariant Hölder arguments. Short pairs use the paired kernel; long pairs use $|\delta_{\alpha,A}\psi(x,x')| \le \exp(1)|\psi(x')| + |\psi(x)|$. Choose C_{jet} to absorb the fixed row factors. No fine-rank factor appears after this substitution.

For the connected estimate write an even polynomial without scalar part in antisymmetric kernels $F = \sum_{m\ge2}(m!)^{-1}\sum K_m \Psi^m$. For a symmetric weight w put

$$N_h(F;w) = \sum_{m\ge2} \frac{h^m}{(m-1)!} \sum |K_m| w,$$

$$\rho_h(F;w) = \sum_{m\ge2} \frac{h^m}{(m-1)!} \max_j \sup_{\xi_j} \sum_{\xi_i,\, i\ne j} |K_m| w.$$

Distances ignore spin and bar/unbar signs. Use

$$w_\theta(\xi_1,\dots,\xi_m)=e^{\theta\max_{a,b}d(\xi_a,\xi_b)},\qquad w_P(\xi_1,\dots,\xi_m)=e^{\theta\max_a d(\xi_a,P)}.$$

Suppose the covariance has weighted row and column sums at most A_G, and a Gram representation with vector norms at most g. Put $h_G=1+g$. Polar decomposition gives the Gram representation; (P.6) gives the weighted sums for a sufficiently small fixed θ.

Lemma P.1 (Connected Gaussian estimate)**.** *For even inputs F_i, $n\ge 2$,*

$$|\langle F_1;\dots;F_n\rangle_G^T|\le (n-2)!A_G^{n-1}N_{h_G}(F_1;w_\theta)\prod_{i=2}^n\rho_{h_G}(F_i;w_\theta).$$

If $A_G\rho_{h_G}(V;w_\theta)\le 1/2$, the logarithm $W(V)=\log\int\mu_G e^V$, continued from zero, is analytic and

$$\begin{aligned}|\partial_P\partial_{P'}W|&\le 2N_{h_G}(V_{PP'};1)\\&\quad+2A_GN_{h_G}(V_P;w_P)\rho_{h_G}(V_{P'};w_{P'})e^{-\theta d(P,P')}.\end{aligned}$$

Proof. Give each input independent generators and interpolate cross covariances. For precision, let Δ_{ij} contract a generator at input i with one at input j, with its covariance and the Grassmann differentiation sign; both bar orientations are included. Let Δ_{ii} be the contractions within one input. The connected contraction identity is

$$\langle F_1;\dots;F_n\rangle_G^T=\sum_{T\ \text{tree}}\int_{[0,1]^T}ds\left[\prod_{\{i,j\}\in T}\Delta_{ij}\right]\exp\left\{\sum_i\Delta_{ii}+\sum_{i<j}s_{ij}^T\Delta_{ij}\right\}\prod_i F_i\Big|_{\Psi=0},$$

where s_{ij}^T is the minimum of the parameters on the unique tree path, and its diagonal is one. This is a finite polynomial identity. One derivation applies the fundamental theorem of calculus when joining two current components of a partition, records the joining edge, and repeats within the joined components. On each resulting ordered forest the allowed edge parameters lie below the last parameter at which its endpoint components separate; their successive upper limits are exactly the path minima. Summing the orders partitions the parameter cubes into these ordered regions. The forest terms with more than one component factor into their component integrals and cancel under the defining partition subtraction of the truncated expectation; the remaining forests are the trees in the display. Thus the identity also proves that all disconnected terms have been removed before the bound.

On a tree the interpolation matrix has entries equal to the minimum of the parameters along the path; it is the integral of the positive indicator matrices of its level-set partitions. It is therefore a Gram matrix of unit vectors. Tensor the original covariance vectors by them. After the tree contractions, the remaining Wick determinant is one Gram determinant, bounded by g to the number of remaining generators.

If vertex i has input degree m_i and tree degree k_i, its derivatives give $m_i!/(m_i-k_i)!$. Orient the tree toward vertex 1 and sum leaves successively, using an absolute weighted row or column on each edge. For the unmarked bound the weights are at least one. For two marks follow the unique tree path between their vertices: the triangle inequality pays its covariance-edge lengths, the diameter at each intermediate vertex, and the maximal distances to the marks at its two ends. Its total is at least $d(P,P')$. There remain one unrestricted root norm and $n-1$ anchored norms. The number of trees of degrees (k_i) is $(n-2)!/\prod_i(k_i-1)!$, and summing the degree choices uses

$$\sum_{k=1}^m\frac{g^{m-k}}{(m-k)!(k-1)!}=\frac{(1+g)^{m-1}}{(m-1)!}.$$

This proves the truncated estimate. The forest partition identity gives the connected series for the logarithm. Its normal convergence follows when $A_G\rho(V)<1$. Two marked derivatives either

strike one input, or separate inputs joined by a path. The path pays $e^{-\theta d(P,P')}$; the remaining unmarked-input sums are bounded by two when $A_G\rho(V) \le 1/2$. This proves the last bound without splitting the residual Gram determinant. □

P.3. **Source derivatives and their supports.** For a direction B supported near P, put $\epsilon_B = e_*|B|_\infty$, $E(y,z) = e^{-\gamma_F d(y,z)}$. The exact resolvent derivatives are

$$S_B = -SK_BS, \qquad S_{BB'} = SK_BSK_{B'}S + SK_{B'}SK_BS - SK_{BB'}S.$$

Wilson derivatives read their link endpoints; block derivatives read the full marked path. The fixed path lengths and two-end bounds give

$$|\partial_B H^{\#}|(\xi; y') \le C\epsilon_B \sum_{w\in \mathrm{pos}(\xi)} \sum_{z\in P^+} E(w,z)E(z,y'),$$

and

$$|\partial_B\partial_{B'} H^{\#}|(\xi; y') \le C\epsilon_B\epsilon_{B'} \sum_{w,z,z'} [E(w,z)E(z,z')E(z',y') + E(w,z')E(z',z)E(z,y')],$$

where $z \in P^+$, $z' \in P'^+$. Contacts obey the same bound, and separated local contacts vanish. The covariant Hölder endpoint phases are differentiated too. Short pairs use their phase derivative and paired kernel; long pairs use both endpoint phases and their unit modulus. Differentiation of $D_* = b_* - b_*^2 QSQ^T$ likewise yields

$$\|D_{*,B}\|_{\gamma_F/2} \le C\epsilon_B, \qquad \sum_{y,y'} |D_{*,B}(y,y')| \le C\epsilon_B|P|,$$

and a two-mark bound containing the connecting kernel $E(z,z')$. Both row orientations are used.

Keep $\mathscr{A} = HA_*^g$ for the photon minimizer and put $\mathscr{A}_\theta = \mathscr{A} + c_\theta$, $\mathscr{T}(t) = \mathscr{A}_\theta + B(t)$. Set

$$V(t) = -\bar\Psi[D_*(\mathscr{T}(t)) - D_*(\mathscr{A}_\theta)]\Psi - m_*\langle\bar\psi(t), \psi(t)\rangle + \sum_X E_*(X; \mathscr{A}_\theta, B(t); \psi^{\#}(t)) = V_0 + V_+.$$

Here V_0 is the scalar part. Distribute the mass over unit blocks so each input has a polymer owner. With $q_x = 2/t_x$ and $0 < \theta \le \min(\gamma_F/8, 1/M)$, the preceding identities, (P.3) and Cauchy circles of radius $t_x/2$ give the bounds below. We spell out the degree and spatial weights used here. The root norm has the extra degree factor $m/m! = 1/(m-1)!$. If $|F|_{2h;w}$ is the absolute coefficient norm with factorial $m!$, then

$$N_h(F; w) \le |F|_{2h;w}, \qquad m2^{-m} \le 1 \quad (m \ge 2).$$

The same inequality holds degree by degree with one fixed kernel argument in place of the full sum. Every physical leg has a weighted row and transposed-row bound C_θ at this fixed rate. Choose the charge so that $2C_\theta h_G \le \min(h_1, h_2)$. For a coefficient with n fine and m Hölder legs, the extra root factor is then paid by

$$(n+m)(C_\theta h_G)^{n+m} \le h_1^n h_2^m.$$

This uses the unused field-weight margin, rather than changing the coefficient activity norm. The quadratic precision and mass inputs have degree two and obey the same comparison directly.

For an unmarked owner X, the generator diameter is bounded by $M[d_M(X) + 2]$ plus the two longest distances of its substituted legs. For an owner meeting P, every original generator is at distance at most $M[d_M(X) + 2] + 2M$ from P; substitution adds the longest leg distance. Bounding a maximum of leg distances by their sum distributes its exponential among the leg rows. The factor $e^{\theta M[d_M(X)+4]}$ costs at most a fixed constant and one unit of the reserved polymer decay. Fixing one substituted generator uses a transposed row for that leg and rows for the others. Rooted polymer summation therefore gives the anchored ρ bound with the same field-weight margin. There is no

weight for a tree joining all generators. These comparisons apply as well to the coefficient and leg derivatives on the smaller Cauchy circles. They give

$$\rho_{h_G}(V_+;w_\theta)\le Ce_*^{2\epsilon}+Ce_*^{1/4-7\epsilon}=:\mathfrak{r}, \tag{P.8}$$

$$N_{h_G}(V_{+,c};w_{P_c})\le Cq_xp_\beta,\qquad \rho_{h_G}(V_{+,c};w_{P_c})\le Cq_x, \tag{P.9}$$

$$N_{h_G}(V_{+,cc'};1)+|V_{0,cc'}|\le Cq_x^2p_\beta e^{-\theta(d_*-4M-6)}\quad (c\neq c'). \tag{P.10}$$

For clarity, the precision difference is quadratic and pays $e_*|B(t)|\le e_*^{3\epsilon}$. The mass and activity have the powers in (P.3). A coefficient derivative forces its full connection support to meet the mark; two such derivatives force both marks and give $Md_M(X)\ge d_*-2M$. A leg derivative instead supplies the displayed connecting chain. The remaining polymer sum is bounded by $\sum_{X\supset\square}e^{-\kappa_0 d_M(X)}\le K_0$, using the reserved RG decay to pay the $4M+6$ enlargements. These arguments establish the three bounds uniformly in the real twist and retained real background. For small charge $A_G\mathfrak{r}\le 1/2$, so Lemma P.1 gives

$$|\partial_c\partial_{c'}W(t)|\le Cq_x^2p_\beta e^{-\theta(d_*-4M-6)}. \tag{P.11}$$

The scalar determinant sum has the same separation bound because its full connection support must meet both marks. This controls every nonsingleton source component of the normalized logarithm.

P.4. **The terminal ratio and its zero-free logarithm.** Let $\mathcal{Z}(t)$ be the Berezin integral in (P.5), and compare it with the determinant at the constant background:

$$f_A(t)=\frac{\mathcal{Z}(t)}{\det D_*(c_\theta+B(t))}.$$

The coarse inverse at the denominator is legal on the source disc. Perturbing the real constant precision by $B(t)$ costs $e_*|B(t)|<e_*^{3\epsilon}$, and $\mathfrak{m}_*$ has a fixed positive lower bound. The resolvent identity and (P.6) give a fixed operator bound, hence absolute row sums at most $4g_1l_*^3$. Choose the charge so that $4g_1l_*^3\le a_2$. With this covariance,

$$f_A(t)=\int\mu_{D_*(c_\theta+B(t))^{-1}}e^{\mathcal{V}(t)},$$

where

$$\mathcal{V}=-\bar\Psi[D_*(\not{A}+c_\theta+B)-D_*(c_\theta+B)]\Psi-m_*\langle\bar\psi,\psi\rangle+\sum_X E_*(X;\psi^\#).$$

The derivative bound for full connection dependence, integrated on the legal real segment, and $|\not{A}|<e_*^{-\epsilon}/2$, give three contributions:

$$|\mathcal{V}_{\text{precision}}|_{a_2}\le 2C_Kl_*^3e_*^{1/2+\epsilon},$$

$$|\mathcal{V}_{\text{mass}}|_{a_2}\le \tfrac12 l_*^3e_*^{1/4-6\epsilon},$$

$$|\mathcal{V}_{\text{activity}}|_{a_2}\le 4K_0(l_*/M)^3e_*^{1/4-6\epsilon}.$$

The second is (P.7) times the selected mass bound. For the third, every polymer contains an M-cube, so $\sum_X e^{-\kappa d_M(X)}\le K_0(l_*/M)^3$. Since l_* is polynomial in v_*, the three terms together are at most $y/4$, where $y=e_*^{1/4-9\epsilon}$. The whole Gaussian coefficient estimate therefore yields

$$|f_A(t)-1|\le e^{|\mathcal{V}|_{a_2}}-1\le y/2\le 1/16. \tag{P.12}$$

The finite Berezin integral and the inverse entries are analytic on the common disc, so f_A is analytic there. This proves nonvanishing of the terminal integral with its actual interaction.

The scalar $e^{-\epsilon_* l_*^3}/\det D_*(0)$ cancels in the ratio, but the scalar source correction does not. The exact ratio and its normalized logarithm are

$$\mathcal{R}(t;\theta,A_*)=\frac{Y(A_*+c_\theta;B(t))}{Y(A_*+c_\theta;0)}, \tag{P.13}$$

$$L_A(t)=\sum_X P_*(X,B(t))+\log\frac{\det D_*(c_\theta+B(t))}{\det D_*(c_\theta)}+\log f_A(t)-\log f_A(0). \tag{P.14}$$

All logarithms are continued from $t=0$, where $L_A=0$. This choice is unambiguous on the source polydisc: integrate $\sum_c t_c\partial_c\mathcal{R}(ut)/\mathcal{R}(ut)$ over $0\le u\le 1$. Real source ratios are positive, by the paired determinant and (P.12); hence this logarithm is real on real sources. It is independent of the chosen real gauge of A_*, because the exact density is gauge invariant. The connected estimate computes derivatives of this very logarithm, including the scalar determinant correction, and proves (P.11) for it.

P.5. **The free scalar determinant sum on a source disc.** Write $\ell_0(t)=\sum_X P_*(X,B_ct)+\log[\det D_*(B_ct)/\det D_*(0)]$. By (P.4), $e^{\ell_0(t)}=N_0(t)$. Its bound must be uniform in the number of earlier scales. We supply the local gauge argument that makes this possible.

At level j, a source of physical wavelength one has

$$e_j|\partial\chi_j|\le C_\beta a_j^2 a_*^{-3/2},\qquad \chi_j=e^{-1}\widehat{a}^{[j]}.$$

Consider a polymer X of diameter s_X whose doubled connection support is liftable. A real linear gauge removes the constant part of χ_j at a point in this support. The remainder there has absolute size at most $Cs_X|\partial\chi_j|$. In the two complex parameters (t,u), the expression $E_j^d(X,(t\chi_j-u\,d\lambda)\mathbf{1}_X)$ is analytic on the convex potential domain containing the legal real gauge segment. For real u, gauge invariance makes it independent of u. The identity theorem first in u and then in t proves the same identity at $u=t$, whenever that endpoint belongs to the domain. This does not require scaling a large real gauge to an arbitrary complex potential outside its domain.

Since E_j^d is even, its zero-subtracted value on $|t|\le t_x$ is bounded by its domain supremum times the square of the source radius divided by the legal curvature-gauged circle radius. Using the determinant bound, absorbing s_X^2 in its reserved decay, and summing the polymers that meet the source cube gives

$$\sum_{X\ \text{liftable}}|E_j^d(X,t\chi_j)-E_j^d(X,0)|\le C_\beta e_*^{-2\epsilon}|t|^2(a_j/a_*)^{1-\epsilon}.$$

Indeed there are $O(a_j^{-3})$ source cubes, whereas the squared derivative gain has a_j^4; their product is a_j. The complex domain contributes $e_j^{-2\epsilon}$, and $e_j/e_*=(a_j/a_*)^{1/2}$ leaves the displayed power $1-\epsilon$. The earlier-scale sum is still geometric, with ratio $L^{-(1-\epsilon)}<1$. If s_X is too large to admit the lift, its diameter is at least half the torus side. Its reserved decay gives

$$\sum_{j,X\ \text{nonliftable}}|E_j^d(X,t\chi_j)-E_j^d(X,0)|\le Cp_\beta e^{-r_*}.$$

The same estimates follow when the improved curvature circle is smaller than $|t|$, by using the absolute domain bound; thus there is no missing branch in this Cauchy estimate.

The coarse determinant at the end has a bounded logarithm on the source disc. Integrating its exact derivative $\mathrm{Tr}[D_*^{-1}D_{*,B_c}]$ gives a bound $Cp_\beta e_*^{2\epsilon}$, after using the source radius and the analytic perturbation of the real constant inverse. Therefore

$$|\ell_0(t)|\le C_\beta e_*^{-2\epsilon}|t|^2+Cp_\beta e^{-r_*}+Cp_\beta e_*^{2\epsilon}=:B_0(t). \tag{P.15}$$

On $|t|\le t_x$, $B_0\to 0$. Make it at most $1/4$. The one-variable jet ϕ_0 of N_0 then obeys

$$\|\phi_0\|_{r_x}\le 2,\quad |c_2|\le Cr_x^{-2},\quad |c_4|\le Cr_x^{-4},\quad c_4-3c_2^2=b_{\eta,R^*}.$$

The coefficients are real and odd coefficients vanish by charge conjugation. In particular $|c_2|+|c_4|^{1/2} \le Ce_*^{-6\epsilon}$.

P.6. **Harmonic tubes and one-cell comparison.** For the constant-background determinant let

$$g(t;\theta) = \log \frac{\det D_*(c_\theta + B_c t)}{\det D_*(c_\theta)}.$$

The phase c_θ is the harmonic connection corresponding to the three boundary angles θ. Shifting an angle by 2π is an exact real gauge transformation; the normalized logarithm is periodic because its value at $t = 0$ is zero. A complex boundary angle of size τ changes the hopping precision by $O(\tau/l_*)$. Reference accretivity therefore supplies a tube $|\operatorname{Im}\theta_\nu| \le \tau_* = c_1 l_*$, uniformly in the real angles. The source remains in the same tube with a fixed margin. The trace derivative and the support count give $|g| \le B_1$, $B_1 = Cp_\beta e_*^{2\epsilon}$, there.

For a 2π-periodic analytic function on this three-dimensional tube its Fourier coefficients satisfy $|g_k| \le B_1 e^{-\tau_*|k|_1}$. Shift the integration contour by the sign of each Fourier index. Summing the nonzero indices gives

$$|g(t;\theta) - g(t;0)| \le 52 B_1 e^{-\tau_*} \quad (\theta \in \mathbb{R}^3),$$

once $\tau_* \ge 1$. This is exponentially small in the terminal side, even though the harmonic connection is not pinned. Combining it with (P.12), (P.14) and (P.4), the one-cell correction $w_c(t) = L_A(te_c) - \ell_0(t)$ satisfies

$$w_c(0) = 0, \qquad \sup_{|t|\le t_x} |w_c(t)| \le 3e_*^{1/4-9\epsilon}.$$

Choose the charge so $52B_1 e^{-c_1 l_*} \le e_*^{1/4-9\epsilon}$. This is uniform in the fine cutoff, because $l_* \asymp v_*^r$.

P.7. **Cell components of the logarithm.** For an analytic function $F(t)$ with $F(0) = 0$, let t^U mean that coordinates outside U are set to zero and put

$$F_S(t) = \sum_{U \subset S} (-1)^{|S|-|U|} F(t^U).$$

The binomial identity gives $F = \sum_{S \ne \varnothing} F_S$. Every nonzero Taylor monomial of F_S uses all variables in S, and no others. For $|S| \ge 2$, choose distinct $c, c' \in S$, pair the four terms in those two coordinates and integrate twice. On $|t_c| \le t_x/2$,

$$|F_S(t)| \le 2^{|S|-2} (t_x/2)^2 \sup |\partial_c \partial_{c'} F|.$$

Applying this to L_A with (P.11) and the scalar determinant bound gives

$$\sup |L_{A,S}| \le \varpi_{\text{sep}} := C 2^{n_c} p_\beta e^{-\theta(d_* - 4M - 6)}.$$

Since $d_* \ge Mr_*/(8L) - 1$, $r_* \ge v_*^2$, and n_c is fixed, reducing the charge makes

$$2\varpi_{\text{sep}} \le \bar\kappa^{n_c}, \qquad \bar\kappa = 6e_*^{1/4-9\epsilon}, \qquad n_c \bar\kappa^{1-s} \le 1/2.$$

The separation estimate pays all nonsingleton components; it is not limited to components of size two.

P.8. **The actual-law large-field complement.** We now bound the complement in the original probability law. Let $\mathcal{G}$ require the orthogonal white coordinates to satisfy their thresholds $p_{0,j} = v_j^{p_0}$, and the terminal curvature coordinates to be at most $p_* = v_*^p$, where $p_0 > 3r + 2$, $p > 2p_0$, and $v_j = v_* + (k_* - j)\log L/2$. The measure is the actual harmonic-averaged law, denoted here by μ_{av}. The coordinates are linear curvature reads of norm at most one. The domination theorem 4.1 gives for each such read Z, $\mu_{\text{av}}(|Z| > u) \le 2e^{-u^2/2}$. After shifts of norm at most one, using the weaker bound $(p_{0,j} - 1)^2 \ge p_{0,j}$, the union bound gives

$$\mu_{\text{av}}(\mathcal{G}^c + G) \le \varpi_0 := 4l_*^3 e^{-p_{0,*}/2} \sum_{i \ge 0} L^{-4i} + 6l_*^3 e^{-(p_*-1)^2/2}, \qquad \|G\|_2 \le 1. \tag{P.16}$$

To obtain L^{-4i}, choose the charge so that $p_0 v_*^{p_0-1} \geq 28$. Convexity then gives $p_{0,j} \geq p_{0,*} + 14i \log L$, and its exponential pays the L^{3i} coordinate count with room to spare. The constant in the coarse norm, if its read norm is written as C_{curv}, replaces the last exponent by $-(p_*-1)^2/(2C_{\mathrm{curv}}^2)$; this fixed change has no effect below.

Let $\Phi(t)$ be the normalized source function in 3.2. Because the source supports are disjoint,

$$\Phi(t) = E^{\mathrm{av}} \exp\left\{ e^{-1} \sum_c t_c X_c - \frac{v_\eta}{2e^2} \sum_c t_c^2 \right\}.$$

Split its translated determinant integral into $\mathcal{G}$ and its complement, obtaining $\Phi = \Phi_S + \Phi_L$. For real $t = ex$, the exact source translation changes the event to $\mathcal{G}^c + \sum_c x_c d\widehat{a}_c$. Choose $t_2 = ez_2$, with fixed $z_2 > 0$ small enough that this shift has norm at most one when $|x_c| \leq z_2$. Cauchy–Schwarz and the actual Gaussian bound then give

$$0 \leq \Phi_L(t) \leq \varpi_L := \sqrt{\varpi_0}\, e^{D/2} \quad (t \text{ real},\ |t_c| \leq t_2),$$

where D is a fixed upper bound for $v_\eta \sum_c x_c^2$. The normalization is the original $Z(0)$, not the small-field mass. The exact terminal identity expresses

$$\Phi_S(t) = \int d\varphi(\theta, A_*)\, e^{L_A(t)}, \quad d\varphi = Z(0)^{-1} \chi_*(A_*) Y(A_* + c_\theta; 0)(2\pi)^{-3} d\theta\, dm_*(A_*).$$

This measure is positive: the real paired determinant is nonnegative, and the terminal integral is nonzero with the same positive sign at zero source by (P.12). Its total mass is exactly $\mu_{\mathrm{av}}(\mathcal{G}) \in [1-\varpi_0, 1]$.

The entire source identity and domination imply

$$|\Phi(t)| \leq \exp\left\{ \frac{v_\eta}{2e^2} \sum_c (\operatorname{Im} t_c)^2 \right\}.$$

On $|t_c| \leq t_2$, the analytic scalar determinant and terminal bounds give $|e^{L_A(t)}| \leq 2$, after reducing z_2 and imposing $8 n_c t_2 \leq t_x$. Therefore $|\Phi_L(t)| \leq M_L := 2 + e^{D/2}$ on this complex polydisc. This is an analytic bound on the actual complement, not a Gaussian expectation for it.

Lemma P.2 (A two-constants disc estimate)**.** *Let F be analytic near the unit polydisc in n variables, $|F| \leq M$ there and $|F| \leq a \leq M$ on its real cube. On the polydisc of radius $1/(2n)$, $|F| \leq a^{1/3} M^{2/3}$.*

Proof. In one variable use either half-disc. The harmonic measure of its semicircle, at a point of modulus at most q, is at most $(4/\pi)\arctan q$. The subharmonic function $\log|F|$, regularized at zeros, is bounded by the corresponding convex combination of $\log a$ and $\log M$. Apply this successively in the n variables, leaving the other variables real at each stage. The exponent of a is at least $(1 - (4/\pi)\arctan(1/(2n)))^n \geq 1 - 2/\pi > 1/3$. Since $a \leq M$, the stated weaker exponent follows. □

Applying the lemma at radius t_2 yields $|\Phi_L| \leq \varpi_L^{1/3} M_L^{2/3}$ on radius $t_2/(2n_c)$. The degree-four jet norm below is consequently at most $\lambda_L^{n_c}$, with

$$\lambda_L = 5 \left(\frac{2 n_c r_x}{t_2} \right)^4 [\varpi_L^{1/3} M_L^{2/3}]^{1/n_c}. \tag{P.17}$$

Indeed each of the five retained powers in each variable obeys its Cauchy bound and $r_x \geq t_2/(2n_c)$. Now

$$\log \lambda_L \leq -\frac{v_*^{p_0}}{13 n_c} + 4 v_* + C.$$

Here $l_*^3 \leq C v_*^{3r}$, while $t_2 \asymp e$ and $r_x \leq 1$. Since $p_0 > 3r + 2$, the negative term dominates every fixed linear or logarithmic loss. In particular $\lambda_L \leq \bar{\kappa}^s$ for sufficiently small charge, uniformly in the fine cutoff.

P.9. **Finite jets and the covering cancellation.** The ring $\mathcal{J} = \mathbb{C}[t_c : c]/(t_c^5 : c)$ has a unique representative $F = \sum_{0\le i_c\le 4} F_i t^i$. Its norm $\|F\|_{r_x} = \sum_i |F_i| r_x^{|i|}$ is submultiplicative; terms discarded in multiplication only reduce the norm. Taking the jet of an analytic function is a ring homomorphism, so exponential factorization remains exact in this finite ring.

The Möbius decomposition of L_A gives

$$L_A(t) = \sum_c [\ell_0(t_c) + w_c(t_c)] + \sum_{|S|\ge 2} L_{A,S}(t).$$

Let $k_{\{c\}} = e^{w_c} - 1$ and $k_S = e^{L_{A,S}} - 1$. On the jet radius $r_x = t_x/4$, Cauchy's estimates and the vanishing of each variable on its coordinate hyperplane give

$$\|k_{\{c\}}\|_{r_x} \le \bar\kappa, \qquad \|k_S\|_{r_x} \le \bar\kappa^{n_c} \quad (|S| \ge 2).$$

For the Cauchy norm, a component containing every variable in S has a sum of geometric coefficients at radius half the analytic radius at most one per variable. Thus its norm is bounded by its supremum, not by a factor exponential in the degree. The exponential uses $e^u - 1 \le 2u$ for $0 \le u \le 1/2$.

Taking jets coefficientwise under the positive finite measure gives the exact representation

$$\Phi(t) = \prod_c \phi_0(t_c) \int d\varphi \prod_{\varnothing \ne S \subset \mathrm{Cel}} (1 + k_S(t)) + \mathcal{L}(t), \tag{P.18}$$

where $\|\mathcal{L}\|_{r_x} \le \lambda_L^{n_c}$. Fubini is justified by the bound two for the small-field analytic integrand on a source torus. All coefficients are real. For every monomial of k_S, the variables appearing are exactly those in S; this follows from its Möbius construction, even after taking its exponential.

For $m = 2, 4$ apply the tensor functional $\prod_c (\partial_{t_c}^m - c_m)|_{t=0}$. An uncovered factor gives $(\partial^m - c_m)\phi_0(0) = 0$. Thus only families $\mathcal{F}$ covering all cells contribute. Write $\kappa_S = \bar\kappa$ for a singleton and $\kappa_S = \bar\kappa^{n_c}$ otherwise. For $0 < s < 1$ put

$$\theta_s = \max_S \kappa_S^{s/|S|} \le \bar\kappa^s, \qquad \Sigma_s = \max_c \sum_{S \ni c} \kappa_S^{1-s}.$$

The number of subsets is fixed, and reducing the charge gives $\Sigma_s \le 1$. A covering family obeys $\prod_{S\in\mathcal{F}} \kappa_S^s \le \theta_s^{n_c}$. Summing the remaining factors over all families, including those not covering, gives

$$\sum_{\mathcal{F}\ \text{covering}} \prod_{S\in\mathcal{F}} \kappa_S \le \theta_s^{n_c} \prod_S (1 + \kappa_S^{1-s}) \le (\theta_s e^{\Sigma_s})^{n_c}.$$

This is the covering bound; no connectedness assumption on $\mathcal{F}$ is made or needed.

The one-variable functional has norm at most $m! r_x^{-m} + |c_m| \le C m! r_x^{-m}$. Using $\|\phi_0\|_{r_x} \le 2$, the mass of φ at most one, and (P.18), we obtain

$$\Theta_m^0 \le m! r_x^{-m} (2\exp(1)\bar\kappa^s + \lambda_L)$$

after a fixed harmless enlargement of constants. Since $r_x^{-4} \le C e_*^{-12\epsilon}$, $\lambda_L \le \bar\kappa^s$, and $1 + |c_2| \le C e_*^{-6\epsilon}$,

$$\Theta_4^0 \le C e_*^{s(1/4-9\epsilon)-12\epsilon}, \qquad (1 + |c_2|)\Theta_2^0 \le C e_*^{s(1/4-9\epsilon)-12\epsilon}.$$

The exact source/Hermite identity 3.3 identifies this tensor functional with the actual centered all-cell expectation in Theorem 11.1. Its nonnegativity is the finite chessboard positivity in A.3.

P.10. **Choice of the charge threshold.** All preceding smallness conditions are simultaneous. They are of the forms $C v_*^A e^{-b v_*} \le 1$, $b > 0$, $C e^{-c v_*^2} \le e^{-b v_*}$, or $C e^{-v_*^{p_0}/(13 n_c) + 4 v_*} \le e^{-b v_*}$. There are finitely many such conditions. Their constants depend only on the scheme, $\bar m$, and the fixed probe seminorms; they do not depend on the fine cutoff. Taking a maximum of their thresholds and using $v_* \ge -\log(e\sqrt{h'})$ selects a positive e_β. The same choice makes the fixed probe fit the central cell, since $R^*/(2L) \to \infty$. It also makes $6 n_c e^2 R^* \log e^{-1} \le \pi^2$, because R^* grows only polynomially in $\log e^{-1}$. This proves Theorem 11.1 with one threshold uniform over all sufficiently fine cutoffs.

Appendix Q. The normalized pressure defect

The pressure used in the transfer estimate is an unscaled partition function of the native lattice. This section keeps the gauge and harmonic volumes as well as the transverse Gaussian determinant. Their separate powers are essential: after subtraction on two tori the answer contains $q^2n = e^2R^*$, rather than q^2.

For a cubic native torus of side n, let Z_n denote the all-flux partition function and $\bar{Z}_n = \omega_n^{(0)}(1)$ the zero-flux partition function. Write

$$r_q = 2\pi/q, \quad \Gamma_n = r_q\{m \in C^1(n;\mathbb{Z}) : dm = 0\}, \quad Z_n^{\mathrm{ph}} = \int_{C^1(n)/\Gamma_n} e^{-\|dA\|^2/2}\, dA.$$

The measure is the quotient of the counting-coordinate Lebesgue measure. The exact zero-flux identification following Corollary A.4 unfolds the compact link box to this quotient. Let

$$\rho_n = (2\pi)^{-3} \int_{[-\pi,\pi]^3} d\theta \int_{V_{\mathrm{t}}} d\widetilde{\gamma}_n(A) \det(D_{A+H_\theta} + \mu_N), \qquad P_n(0) = \bar{Z}_n/Z_n.$$

Here $\widetilde{\gamma}_n$ is the normalized transverse Maxwell Gaussian and H_θ is the constant potential with holonomy θ; thus $q(H_\theta)_\nu = \theta_\nu/n$. The determinant is the actual determinant, with the selected native mass μ_N, in this definition of ρ_n.

Q.1. **Gauge, harmonic and transverse factors.** Put

$$\mathrm{Ld}(n_1,n_2,n_3) = \sum_{k\neq 0} \log\left[\sum_{\nu=1}^{3}\left(2 - 2\cos\frac{2\pi k_\nu}{n_\nu}\right)\right], \qquad \mathrm{Ld}(n) = \mathrm{Ld}(n,n,n).$$

The sum is over the discrete momentum torus with its zero point removed. It is the logarithm of the nonzero scalar Laplacian determinant.

Lemma Q.1 (The photon volume)**.** *For $n \geq 2$,*

$$Z_n^{\mathrm{ph}} = (2\pi/q)^{n^3+2}(2\pi)^{n^3-1}e^{-\mathrm{Ld}(n)/2}, \qquad \bar{Z}_n = Z_n^{\mathrm{ph}}\rho_n, \quad Z_n = Z_n^{\mathrm{ph}}\rho_n/P_n(0). \tag{Q.1}$$

Moreover

$$1 - \varsigma_n \leq P_n(0) \leq 1, \qquad \varsigma_n = \sum_{w\in\mathbb{Z}^3\setminus\{0\}} \exp[-2\pi^2|w|_2^2/(q^2n)]. \tag{Q.2}$$

If $q^2n \leq 1$, then $\varsigma_n \leq 7\exp[-2\pi^2/(q^2n)] < 1/8$.

Proof. The Hodge decomposition is $C^1 = V_{\mathrm{g}} \oplus V_{\mathrm{h}} \oplus V_{\mathrm{t}}$, with dimensions $n^3-1, 3, 2n^3-2$. The lattice Γ_n is contained in the first two summands. If $d\lambda$ is integer, connectedness says that $\lambda - \lambda(0)$ is integer. Consequently $\Gamma_n \cap V_{\mathrm{g}} = r_q dC^0(n;\mathbb{Z})$. The projection of $\mathbb{Z}^{n^3}$ on the mean-zero subspace has covolume $n^{-3/2}$. For example the projections of the first n^3-1 coordinate vectors have Gram matrix $I - n^{-3}\mathbf{1}\mathbf{1}^T$, of determinant n^{-3}. On that subspace d has absolute determinant $e^{\mathrm{Ld}(n)/2}$, so

$$\mathrm{covol}(\Gamma_n \cap V_{\mathrm{g}}) = r_q^{n^3-1}e^{\mathrm{Ld}(n)/2}n^{-3/2}.$$

The unit constant cochain in direction ν has value $n^{-3/2}$. The period of a closed integer cochain along a line in that direction is the same integer on every parallel line. Its harmonic coordinate is therefore $n^{1/2}k_\nu$. All triples of periods occur, by taking integer sheets on three coordinate cuts. Thus $\mathrm{covol}(\pi_{\mathrm{h}}\Gamma_n) = r_q^3 n^{3/2}$. Choosing first a basis for the gauge lattice and then lifts of a harmonic basis gives a block triangular volume matrix, whence

$$\mathrm{covol}(\Gamma_n) = r_q^{n^3+2}e^{\mathrm{Ld}(n)/2}.$$

The one-form Laplacian is the scalar Laplacian in each component. Its nonzero eigenvalue product is $e^{3\mathrm{Ld}(n)}$. The gauge summand contributes $e^{\mathrm{Ld}(n)}$, the harmonic summand contributes no nonzero

eigenvalue, and hence $\det(d^*d|_{V_t}) = e^{2\mathrm{Ld}(n)}$. Integrating the $2n^3-2$ transverse coordinates now proves the first identity in (Q.1).

The determinant is gauge invariant, and its harmonic period is precisely $\pi_{\mathrm{h}}\Gamma_n$. A fundamental harmonic box is parametrized by H_θ, with its normalized measure $(2\pi)^{-3}d\theta$. Fubini on the gauge–harmonic quotient, followed by dividing the transverse Gaussian by its integral, therefore proves $\bar{Z}_n = Z_n^{\mathrm{ph}}\rho_n$. The all-flux identity is its definition through $P_n(0)$.

Corollary A.4 gives $Z_n^{-1}\omega_n^{(w)}(1) \le e^{-2\pi^2|w|_2^2/(q^2n)}$. Sum over nonzero fluxes to obtain (Q.2). If $x = e^{-2\pi^2/(q^2n)} \le e^{-2\pi^2}$, then $\sum_{i\in\mathbb{Z}} x^{i^2} \le 1 + 2x/(1-x) \le 1 + 2.001x$. The cube of this bound minus one is at most $7x$, which proves the last assertion. □

Q.2. **The finite-size Laplacian cancellation.**

Lemma Q.2 (A cubic Laplacian difference)**.** *For integers $n, j \ge 2$,*

$$\frac{1}{2}[\mathrm{Ld}(jn) - j^3\mathrm{Ld}(n)] \le -(j^3-1)\log n + K_j, \tag{Q.3}$$

where

$$G_j = (1 + j/3.2)\, 2j^2/2.56, \qquad K_j = (1 + j + j^2)(\log j + jG_j).$$

In particular the constant is independent of the ultraviolet cutoff.

Proof. For $p \in [-\pi,\pi]^2$, define $\omega_p = \operatorname{arcosh}(3 - \cos p_1 - \cos p_2)$. Factoring the roots of unity, or multiplying $(e^\omega - e^{2\pi ik/n_1})(e^\omega - e^{-2\pi ik/n_1})e^{-\omega}$, gives

$$\prod_{k_1=0}^{n_1-1} (2\cosh\omega - 2\cos(2\pi k_1/n_1)) = 4\sinh^2(n_1\omega/2).$$

At $\omega = 0$ the zero factor is removed and the remaining product is n_1^2, by the same identity after division by $2\cosh\omega - 2$ and passage to zero. Consequently

$$\mathrm{Ld}(n_1,n_2,n_3) = 2\log n_1 + 2\sum_{k_s\neq 0} \log[2\sinh(n_1\omega_{p(k_s)}/2)], \tag{Q.4}$$

with $p(k_s) = (2\pi k_2/n_2, 2\pi k_3/n_3)$ taken in the principal square.

Let $g(x) = -\log(1 - e^{-2x})$, $x > 0$. Since $\log(2\sinh x) = x - g(x)$,

$$\log(2\sinh jx) - j\log(2\sinh x) \le jg(x).$$

Thus lengthening the first side in (Q.4) costs at most

$$2\log j + 2(1-j)\log n_1 + 2j\sum_{k_s\neq 0} g(n_1\omega_{p(k_s)}/2).$$

The elementary bounds $4\pi^{-2}|p|^2 \le \sum_i(2 - 2\cos p_i) \le 8$ and $\operatorname{arcsinh} y \ge .81y$ for $0 \le y \le \sqrt{2}$ show that $\omega_p \ge .51|p|$. If $n_1 = n$ and $n_2, n_3 \in \{n, jn\}$, it follows that $n\omega_{p(k_s)}/2 \ge c_j|k_s|_2$, $c_j = 1.6/j$. Also $g(x) \le e^{-2x}(1 + (2x)^{-1})$. Extending the sum to $\mathbb{Z}^2 \setminus \{0\}$ and using $|k_s|_2 \ge (|k_2| + |k_3|)/\sqrt{2}$ gives

$$\sum_{k_s\neq 0} g(n\omega_{p(k_s)}/2) \le (1 + (2c_j)^{-1})[\coth^2(c_j/\sqrt{2}) - 1] \le (1 + (2c_j)^{-1})\, 2/c_j^2 = G_j.$$

Finally lengthen the three sides in succession. The difference $\mathrm{Ld}(jn,jn,jn) - j^3\mathrm{Ld}(n,n,n)$ is the sum of the corresponding one-side differences with coefficients $1, j, j^2$. After permuting the lengthened side into the first position, each has the bound just proved. Since $(1 + j + j^2)(1 - j) = 1 - j^3$, division by two yields (Q.3). □

Q.3. **The matter and flux contributions.** Let $n = R^*/\eta$, $Z_1 = Z_n$, $Z_L = Z_{Ln}$, and $\bar{Z}_L = \bar{Z}_{Ln}$. The two defects that will be used are

$$D_L = L^3 \log Z_1 - \log Z_L, \qquad \bar{D}_L = L^3 \log Z_1 - \log \bar{Z}_L. \tag{Q.5}$$

They are not the same number: the latter retains the zero-sector normalizer at the larger reference torus. Put

$$D_L^{\mathrm{mat}} = L^3 \log \rho_n - \log \rho_{Ln}, \qquad D_L^{\mathrm{ph}} = (L^3 - 1) \log \frac{2\pi}{q^2} + \frac{1}{2}[\mathrm{Ld}(Ln) - L^3 \mathrm{Ld}(n)].$$

Substitution of (Q.1) is exact and gives

$$D_L = D_L^{\mathrm{ph}} + D_L^{\mathrm{mat}} - L^3 \log P_n(0) + \log P_{Ln}(0), \tag{Q.6}$$

$$\bar{D}_L = D_L^{\mathrm{ph}} + D_L^{\mathrm{mat}} - L^3 \log P_n(0). \tag{Q.7}$$

The native-to-physical identity $q^2 n = e^2 R^*$ and Lemma Q.2 imply

$$D_L^{\mathrm{ph}} \le (L^3 - 1) \log \frac{2\pi}{e^2 R^*} + K_L.$$

The reference agreement and tuned pair give

$$|D_L^{\mathrm{mat}}| \le (L^3 + 1)[\epsilon_5(\bar{m} R^*) + 3 e_K^{1/4 - 9\epsilon}]. \tag{Q.8}$$

This is Theorem 9.5 applied to the same ρ_n as above, not to a determinant with its harmonic variable fixed. For small charge its right side is bounded by a scheme constant. The threshold $6 n_c e^2 R^* \log e^{-1} \le \pi^2$, $e \le 1/3$, implies

$$\frac{2\pi^2}{e^2 R^*} \ge 12 n_c \log e^{-1}, \qquad \varsigma_n \le 7 e^{12 n_c}, \qquad -L^3 \log(1 - \varsigma_n) \le 1.$$

For the last inequality one may use $-\log(1 - x) \le x/(1 - x)$, $\varsigma_n \le 1/8$, and $L^3 = n_c/8$. Equations (Q.6) and (Q.7) therefore show that both defects are at most

$$B_L = (L^3 - 1) \log \frac{2\pi}{e^2 R^*} + K_L + D_L^{\mathrm{mat}} + 1. \tag{Q.9}$$

Their nonnegativity will follow from transfer positivity below. No pressure limit is needed: these are two finite-volume differences, with every scalar volume retained.

Appendix R. Chessboard transfer on the actual finite laws

The transfer uses the same native mass on every torus. None of the arguments in this section needs positive native mass: the finite model, positive determinant and transfer factorization require only $\mu_N > -1/2$. The positive reference mass enters earlier in the RG and free-loop estimates.

R.1. **Half-integer cells and their slab operators.** Let $\lambda = m_{\mathrm{p}} + 1/2$, with integer $m_{\mathrm{p}} \ge 2$. On a native torus with sides $n_\nu = \lambda N_\nu$, each N_ν even, define the closed cells

$$\mathfrak{c}_c = \prod_{\nu=0}^{2} [\lambda c_\nu, \lambda(c_\nu + 1)].$$

The boundary planes alternate between site and link planes. Use only plaquettes with every vertex in the cell and contained in no boundary plane; call this set $P^+(\mathfrak{c}_c)$. These sets are disjoint. The lattice isometry g_c carrying the base cell to $\mathfrak{c}_c$ acts coordinatewise by

$$x_\nu \longmapsto \begin{cases} \lambda c_\nu + x_\nu, & c_\nu \text{ even}, \\ \lambda(c_\nu + 1) - x_\nu, & c_\nu \text{ odd}. \end{cases}$$

The translation constants are integers. For a function O of the curvature on $P^+(\mathfrak{c}_0)$, put

$$O^{[c]} = O \circ g_c^* \quad (|c|_1 \text{ even}), \qquad O^{[c]} = \overline{O \circ g_c^*} \quad (|c|_1 \text{ odd}), \qquad O^\sharp = \prod_c O^{[c]}.$$

The pull-back includes the sign of each oriented two-form component. The reflections generating these copies act simply transitively on the cells: in one direction they are the dihedral group generated by two neighboring boundary reflections; the three groups commute. Consequently taking a mirror copy from one cell to another is consistent with these formulas.

For the reference geometry $\lambda = R^*/(2L\eta)$, $2m_{\mathrm{p}} + 1 = R^*/(L\eta)$ is an odd power of L. The reference has $N_\nu = 2L$ and $n_c = 8L^3$ cells; a target with side kR^* has $N_\nu = 2kL$. The source supports in Appendix M.1 lie strictly inside these cells. They are a subset of P^+, so their statistics belong to the cell algebra just defined.

Take direction zero as time. Slab i contains the integer slices from $t_i^- = \lceil \lambda i \rceil$ to $t_i^+ = \lfloor \lambda(i+1) \rfloor = t_i^- + m_{\mathrm{p}}$. A cell product P_i in that slab reads its electric variables and magnetic variables on these slices. At a site boundary it reads no magnetic plaquette. At the opposite boundary it may read an end magnetic variable. Lemma A.5 expresses the latter using the interior variables and gives the adjoint mirror identity for $C_{m_{\mathrm{p}},\vartheta}[P_i]$.

Use the Bloch space and X_ϑ of Proposition A.3. Let Π be the identity or the orthogonal projection onto one spatial-flux summand. Define

$$Y_i[P] = \begin{cases} X_\vartheta \Pi C_{m_{\mathrm{p}},\vartheta}[P], & i \text{ even}, \\ C_{m_{\mathrm{p}},\vartheta}[P] \Pi X_\vartheta^*, & i \text{ odd}. \end{cases} \tag{R.1}$$

All these operators are Hilbert–Schmidt. The projection commutes with the slab operators and $T_\vartheta = X_\vartheta^* X_\vartheta$. Multiplying an odd operator by the preceding even operator gives

$$Y_{i+1}[P_{i+1}] Y_i[P_i] = \Pi C_{m_{\mathrm{p}},\vartheta}[P_{i+1}] C_{1,\vartheta}[1] C_{m_{\mathrm{p}},\vartheta}[P_i].$$

This pays the unoccupied step across the intervening link plane. Pairs of cells have total thickness $2m_{\mathrm{p}} + 1$, as required. The exact cyclic trace, including its Gaussian scalar, is therefore

$$\omega_n^{\vartheta,\Pi}\Big(\prod_i P_i\Big) = \kappa^{n_0} \operatorname{Tr}\big(Y_{N_0-1}[P_{N_0-1}] \cdots Y_0[P_0]\big), \tag{R.2}$$

where $\omega_n^{\vartheta,\Pi}$ is the Bloch-weighted flux sum, with the spatial flux restricted if $\Pi \neq I$. The factor κ, space and every operator on the right depend only on the cross-section and not on n_0.

Reflection at the neighboring boundary gives $Y_i[P]^* = Y_{i\pm1}[P^\theta]$. Indeed the electric arguments are reversed and negated and the magnetic arguments reversed, as in Lemma A.5; the excluded end variable is exchanged with the other end variable. Alternating P and its mirror in (R.2) consequently yields

$$\omega_n^{\vartheta,\Pi}\Big(\prod_{i'} P^{(i')}\Big) = \kappa^{n_0} \|Y_i[P]\|_{N_0}^{N_0} = \kappa^{n_0} \operatorname{Tr}\big(Y_i[P]^* Y_i[P]\big)^{N_0/2} \geq 0. \tag{R.3}$$

No half-step scalar has been deleted in this identity. Even if the native side is odd, the number N_0 of cells is even.

R.2. **The chessboard inequality.**

Proposition R.1 (Finite cell products)**.** *For $\omega = \omega_n$ or $\omega_n^{(0)}$,*

$$\omega(O^\sharp) \geq 0, \qquad \Big|\omega\Big(\prod_c O_c^{[c]}\Big)\Big| \leq \prod_c \omega(O_c^\sharp)^{1/N_n}, \quad N_n = N_0 N_1 N_2. \tag{R.4}$$

The functions may be any polynomials in the finitely many cell curvature variables. More generally the statement holds whenever the products used below are absolutely integrable.

Proof. First take bounded functions G_c in their respective cell algebras and put $P_i = \prod_{c:c_0=i} G_c$. Schatten Hölder applied to (R.2) with $\vartheta = 0, \Pi = I$ gives

$$|\omega_n(\prod_c G_c)| \le \prod_i [\kappa^{n_0} \|Y_i[P_i]\|_{N_0}^{N_0}]^{1/N_0}.$$

By (R.3), a factor in brackets is the integral of the family obtained by replacing every slab by the mirror of slab i. These integrals are nonnegative. For zero flux use Π at spatial flux zero, integrate the same trace over $[0,1]^2$, and apply ordinary Hölder with exponents N_0 to this integral. Absolute convergence of the Bloch Fourier series permits the integration and identifies its result with $\omega_n^{(0)}$. Thus the same slab-replacement bound holds for both functionals.

Apply it first in direction zero, then in direction one to each resulting family, and then in direction two. Reflections in distinct directions commute. After the three replacements, the family indexed by the original cell c consists of the mirror copies of G_c in all cells. All intermediate factors are nonnegative, so the three successive roots multiply to $1/N_n$. Taking $G_c = O_c^{[c]}$ proves the inequality. If all G_c are copies of one O, the first slab identity (R.3), before any Hölder inequality, already gives positivity of $\omega(O^\sharp)$.

Finally truncate each O by $O\mathbf{1}_{|O|\le M}$. Truncation commutes with mirrors, and its product is dominated by the product of the absolute untruncated functions. Such products of curvature polynomials are integrable by Gaussian domination; the flux-series absolute values are bounded by the full positive integral of their absolute product. Dominated convergence proves (R.4) for the stated class. □

R.3. **Three Fourier projections and zero flux.** The zero-flux comparison needs more than positivity at zero Bloch angle. The positive integrand for every angle in (R.3) supplies the missing information.

Lemma R.2 (Zero-flux domination for a mirror product)**.** *On every torus with the cell structure,*

$$0 \le \omega_n^{(0)}(O^\sharp) \le \omega_n(O^\sharp). \tag{R.5}$$

Proof. It suffices first to use a bounded O. Write $a(w_{12}, w_{02}, w_{01}) = \omega_n^{(w)}(O^\sharp)$. Its series is absolutely summable. With w_{12} fixed, (R.3) gives

$$\sum_{w_{01},w_{02}} e^{2\pi i(\vartheta_1 w_{01} + \vartheta_2 w_{02})} a(w) \ge 0 \quad (\vartheta \in \mathbb{R}^2).$$

Exchanging directions zero and one, and then zero and two, gives the analogous positive series with respectively w_{02} and w_{01} fixed. The signs from oriented pull-back are absorbed by changing signs of the corresponding angles; positivity holds for every angle, so these changes do not affect it.

At zero angle the first family says that $\sum_{w_{01},w_{02}} a(w_{12}, w_{02}, w_{01}) \ge 0$ for every w_{12}. Hence

$$A_1 := \sum_{w_{01},w_{02}} a(0, w_{02}, w_{01}) \le \sum_w a(w) = \omega_n(O^\sharp).$$

In the second family set the angle paired with w_{01} to zero and integrate the angle paired with w_{12}. This gives $\sum_{w_{01}} a(0, w_{02}, w_{01}) \ge 0$ for each w_{02}, and consequently

$$A_2 := \sum_{w_{01}} a(0, 0, w_{01}) \le A_1.$$

Integrate both angles in the third family. Then $a(0,0,w_{01}) \ge 0$ for every w_{01}, so $0 \le a(0,0,0) \le A_2$. These three projections prove (R.5). Uniform absolute convergence justifies every Fourier integral. The truncation argument in Proposition R.1 removes the boundedness restriction. □

R.4. **Partition convexity and lengthening.** For native sides (na, nb, nc), with fixed $n \geq 2$, put $\phi(a,b,c) = \log Z_{(na,nb,nc)}$ and $\psi(a,b,c) = \log \bar{Z}_{(na,nb,nc)}$. The last symbol c in these arguments is an integer multiplier, not a cell index. The following elementary consequences of the positive transfer operator will be used for arbitrary integer target multipliers, not only powers of L.

Lemma R.3 (Convexity and power bounds)**.** *Both ϕ and ψ are separately convex on $\mathbb{N}^3_{\geq 1}$, $\psi \leq \phi$, and*

$$\phi(ka,b,c) \leq k\phi(a,b,c) \quad (k \geq 1), \tag{R.6}$$

with the analogous inequalities in the other coordinates. If $D_j = j^3\phi(1,1,1) - \phi(j,j,j)$ and $\bar{D}_j = j^3\phi(1,1,1) - \psi(j,j,j)$, then $0 \leq D_j \leq \bar{D}_j$, and for $k \geq j \geq 2$,

$$\phi(k,k,k) \geq k^3\phi(1,1,1) - \left(\frac{k-1}{j-1}\right)^3 D_j, \tag{R.7}$$

$$\psi(k,k,k) \geq k^3\phi(1,1,1) - \left(\frac{k-1}{j-1}\right)^3 \bar{D}_j. \tag{R.8}$$

Proof. At fixed cross-section the all-flux partition is $\kappa^{n_0} \operatorname{Tr} T_0^{n_0}$. The zero-flux partition is

$$\bar{Z}_{(n_0,n_1,n_2)} = \int_{[0,1]^2} \kappa^{n_0} \operatorname{Tr}_{\mathscr{H}_\vartheta^{(0)}} T_\vartheta^{n_0}\, d\vartheta.$$

This follows from A.3 by restricting the spatial flux and integrating the two electric Fourier variables. The positive eigenvalues give $\operatorname{Tr} T^{ua} \leq (\operatorname{Tr} T^{u(a-1)} \operatorname{Tr} T^{u(a+1)})^{1/2}$, for $a \geq 2$. Cauchy–Schwarz for the angle integral proves the same inequality for $\bar{Z}$; the scalar powers κ^{ua} split exactly between the two sides. Axis exchange proves separate convexity in all directions. For a positive summable sequence, $\sum_i x_i^k \leq (\sum_i x_i)^k$. Apply this to the eigenvalues of T_0^{na} and keep the scalar power to obtain (R.6). Positivity of each sector proves $\bar{Z} \leq Z$, and hence $\psi \leq \phi$.

Iterating (R.6) gives $\phi(a,b,c) \leq abc\phi(1,1,1)$, regardless of the sign of $\phi(1,1,1)$. In particular $D_j \geq 0$ and $\bar{D}_j \geq D_j$. Write $x = (k-1)/(j-1) \geq 1$. Discrete convexity, equivalently monotonicity of secant slopes, gives $v(k) \geq xv(j) - (x-1)v(1)$ for any such one-variable restriction of ϕ or ψ. For ψ, apply this in the three coordinates and bound the negatively signed endpoint values using $\psi \leq \phi$ and (R.6). The result is

$$\psi(k,k,k) \geq x^3\psi(j,j,j) - (x-1)(k^2 + xjk + x^2j^2)\phi(1,1,1).$$

Insert $\psi(j,j,j) = j^3\phi(1,1,1) - \bar{D}_j$. The coefficient of $\phi(1,1,1)$ is k^3, because $xj - (x-1) = k$; successive cancellation gives $x^3j^3 - x^2(x-1)j^2 - x(x-1)jk - (x-1)k^2 = k^3$. This proves (R.8). The identical argument with ϕ proves (R.7). □

For a full mirror product, lengthening one side by k leaves the cross-section, the base insertion and its Y unchanged. In (R.3) it only replaces N_0 by kN_0. The positive trace power inequality therefore gives

$$0 \leq \omega_{n^{(\nu,k)}}(O^\sharp) \leq \omega_n(O^\sharp)^k. \tag{R.9}$$

The scalar on the longer torus is $(\kappa^{n_0})^k$, so there is no additional normalization in this bound. The even cell count also preserves the alternating mirror pattern for odd k. Lengthening all three directions yields

$$\omega_{kn}(O^\sharp) \leq \omega_n(O^\sharp)^{k^3}. \tag{R.10}$$

R.5. **Normalized products in every target volume.** Define the two reference cell norms by

$$\|O\|_* = \left(\omega_n(O^\sharp)/Z_n\right)^{1/n_c}, \qquad \|O\|_{0*} = \left(\omega_n^{(0)}(O^\sharp)/\bar{Z}_n\right)^{1/n_c}.$$

These nonnegative quantities are only used on the cell algebra; no claim about a norm on arbitrary global functions is needed. For a subset J of target cells, fill the remaining cells by the function

one in (R.4) and divide by the actual target partition. Each unoccupied cell contributes its partition-function root, which cancels in this division. Thus

$$|E^{\mathrm{af}}_{kn}\prod_{c\in J}O^{[c]}_c|\le\prod_{c\in J}[\omega_{kn}(O^\sharp_c)/Z_{kn}]^{1/(k^3n_c)},$$

and the identical statement with $\omega^{(0)},\bar Z$ holds for E^{av}. Equations (R.7) and (R.10) give, for $k\ge L$,

$$|E^{\mathrm{af}}_{kn}\prod_{c\in J}O^{[c]}_c|\le\prod_{c\in J}e^{D_L/((L-1)^3n_c)}\|O_c\|_*. \tag{R.11}$$

The same inequality holds at $k=1$, since $D_L\ge0$. For the averaged law use the zero-flux comparison (R.5) on the target, then the full-law lengthening (R.10), and finally (R.8). This gives

$$|E^{\mathrm{av}}_{kn}\prod_{c\in J}O^{[c]}_c|\le\prod_{c\in J}e^{\bar D_L/((L-1)^3n_c)}\|O_c\|_*\quad(k\ge L). \tag{R.12}$$

At $k=1$ chessboard gives $\prod_{c\in J}\|O_c\|_{0*}$. For the centered Hermite statistics below the latter is $(\Theta^0_m)^{|J|}$, so a common upper constant also covers this case. These are same-law normalized estimates, with the full partition in (R.11) and the zero partition in (R.12).

R.6. **The reference flux tail and the Hermite statistics.** For an even integer $p\ge2$, Gaussian domination gives

$$\|X\|_{L^p(E^{\mathrm{af}})}\le\sqrt{\exp(1)\,p\,v_\eta}. \tag{R.13}$$

Indeed $\cosh(tX)\ge(tX)^p/p!$, its expectation is at most $e^{v_\eta t^2/2}$, and optimizing at $t^2=p/v_\eta$, with $p!\le p^p$, gives the assertion; $v_\eta=0$ is immediate. Minkowski's inequality consequently gives, for every integer $M\ge1$,

$$\|H_4(X;v_\eta)\|_{L^{2M}}\le(64\exp(2)+24\exp(1)+3)M^2v_\eta^2\le600M^2v_\eta^2,\quad\|H_2(X;v_\eta)\|_{L^{2M}}\le12Mv_\eta.$$

The estimates hold in any surrounding volume with the same local probe. At $M=n_c$, define

$$\dot M_4=600n_c^2v_\eta^2e^{-4}+|c_4|,\qquad\dot M_2=12n_cv_\eta e^{-2}+|c_2|.$$

For $O_m=e^{-m}H_m(X;v_\eta)-c_m$, $\|O_m\|_{L^{2n_c}(E^{\mathrm{af}}_n)}\le\dot M_m$. The exact source identity gives $\|O_m\|_{0*}=\Theta^0_m$. Splitting the full reference law into zero and nonzero flux, and using positivity of each sector, gives

$$E^{\mathrm{af}}_nO^\sharp_m\le(\Theta^0_m)^{n_c}+\varsigma_n^{1/2}(E^{\mathrm{af}}_n|O^\sharp_m|^2)^{1/2}.$$

Hölder for the n_c copies and their common distribution bounds the last moment by $\dot M_m^{n_c}$. Taking the n_c-th root, using positivity of the left side and $(u+v)^{1/n_c}\le u^{1/n_c}+v^{1/n_c}$, proves

$$\|O_m\|_*\le\Theta^0_m+\varsigma_n^{1/(2n_c)}\dot M_m\le\Theta^0_m+2e^6\dot M_m. \tag{R.14}$$

The last inequality is exactly the reference tail payment: $\varsigma_n\le7\exp[-2\pi^2/(e^2R^*)]$ and $6n_ce^2R^*\log e^{-1}\le\pi^2$.

Combining (R.11), (R.12), and (R.14) proves Theorem 12.1. A convenient common constant is

$$\Gamma_*=exp[B_L/((L-1)^3n_c)],\qquad\Theta_m=\Theta^0_m+2e^6\dot M_m. \tag{R.15}$$

Both defects are nonnegative by Lemma R.3, so $B_L\ge0$ and $\Gamma_*\ge1$; the reference averaged case is also covered. For small charge,

$$\Gamma_*\le C_Le^{-2a_L/n_c},\qquad a_L=(L^3-1)/(L-1)^3. \tag{R.16}$$

Here the e on the right is the charge. The bound follows from (Q.9), boundedness of (Q.8) and $R^*\ge1$; it has no target-volume or cutoff dependence. Translating the entire cell grid by an integer native vector leaves every estimate unchanged. Ordinary copies in even cells are separated by R^*/L in physical units; these are the translates used in the component argument.

Appendix S. Cumulants and invariant conditional statistics

S.1. **The positive exponent and the weighted second statistic.** Fix the scheme before the charge. Set

$$\alpha_{\mathrm{jet}} = s(1/4 - 9\epsilon) - 12\epsilon, \qquad \sigma = \alpha_{\mathrm{jet}} - 2a_L/n_c > 0. \tag{S.1}$$

For $s = 1/2$, $\epsilon \le 10^{-4}$, and odd $L \ge 5$, $\alpha_{\mathrm{jet}} \ge .12335$ and $2a_L/n_c \le .003875$. The latter expression decreases for $L \ge 5$. Thus this is a nonempty window, preserved when the RG proof further increases L or decreases ϵ. The letter α_{jet} is distinct from the Hölder exponent in the primitive domains.

Use the common constants Γ_*, Θ_m from (R.15) and write $u_m = \Gamma_* \Theta_m$, $m = 2, 4$. By Theorem 11.1, $|c_2| + |c_4|^{1/2} \le Ce^{-6\epsilon}$. The probe variance is uniformly bounded by Lemma M.1. Consequently the tail terms obey

$$2e^6 \dot{M}_4 \le C(e^2 + e^{6-12\epsilon}), \qquad (1 + |c_2|)2e^6 \dot{M}_2 \le C(e^{4-6\epsilon} + e^{6-12\epsilon}).$$

All the exponents on the right exceed α_{jet} in the selected parameter window. Multiplication by the transfer loss (R.16) therefore gives

$$u_4 \le Ce^{\sigma}, \qquad (1 + |c_2|)u_2 \le Ce^{\sigma}. \tag{S.2}$$

This is the reason for retaining the weight $1 + |c_2|$ at the reference calculation. An unweighted second-jet bound would lose another power when converting the fourth Hermite statistic to the fourth cumulant. The reference flux tail in (R.14) preserves this same weight.

S.2. **Finite cumulants.** Let $X_\eta = X(f^{(\eta)})$ at the chosen placement on any target torus. Charge conjugation makes this actual finite law centered. The one-cell case of Theorem 12.1 gives real numbers δ_2, δ_4 with

$$e^{-2}(EX_\eta^2 - v_\eta) = c_2 + \delta_2, \quad e^{-4}EH_4(X_\eta; v_\eta) = c_4 + \delta_4, \qquad |\delta_m| \le u_m. \tag{S.3}$$

For any centered real variable X,

$$EH_4(X; v) - 3(EH_2(X; v))^2 = EX^4 - 3(EX^2)^2 = \kappa_4(X).$$

The subtraction cancels the variance v exactly. Thus

$$e^{-4}\kappa_4(X_\eta) = c_4 - 3c_2^2 + \delta_4 - 6c_2\delta_2 - 3\delta_2^2. \tag{S.4}$$

Use $c_4 - 3c_2^2 = b_{\eta,R^*}$ and (S.2). For small enough charge,

$$|\kappa_4(X_\eta) - e^4 b_{\eta,R^*}| \le e^4[u_4 + 6|c_2|u_2 + 3u_2^2] \le Ce^{4+\sigma}, \qquad |EX_\eta^2 - v_\eta| \le Ce^{2-6\epsilon}.$$

The free comparison in Theorem 10.1 pays $Ce^4\eta^{1/2}$. Its torus-image error is $Ce^4 e^{-\bar{m}(R^* - D_\beta)}$. The reference construction has $R^* = h' l_*$, $l_* = Mr_*$, and $r_* \asymp (\log e^{-1})^r$, $r \ge 2$. For any fixed $\sigma > 0$, decreasing the charge ceiling therefore makes $e^{-\bar{m}(R^* - D_\beta)} \le Ce^{\sigma}$.

The compatible source uses $f^{(\eta)}$, whereas the theorem uses the fixed continuum test f. This change costs $O(\eta^2)$ in every fixed moment through order four, uniformly over the finite laws. To see this directly, expand a difference of products by replacing one factor at a time. Each replaced factor is a curvature readout of $f^{(\eta)} - f$; Gaussian domination bounds its L^p norm by a fixed test seminorm times η^2, using Lemma M.1. The other factors have uniform fixed moments by the same bound. Hölder gives the claimed error. The variance comparison $v_\eta - \|f\|_2^2 = O(\eta)$ is separately paid in the second cumulant. Hence

$$\begin{aligned} |\kappa_4(X(f)) - e^4 b(f)| &\le Ce^{4+\sigma} + C(\eta^2 + e^4\eta^{1/2}), \\ |\kappa_2(X(f)) - \|f\|_2^2| &\le Ce^{2-6\epsilon} + C\eta. \end{aligned} \tag{S.5}$$

The constants have no target-volume or cutoff dependence. The counterterm assignment has been selected by the RG construction before the probe is chosen; only the charge ceiling needed for this fixed probe is reduced here.

S.3. **Passing every fixed product to a limit.** Let μ be any cofinal limit point of one of the two finite families at a fixed small charge. Theorem 5.1 gives convergence of all fixed moments. In particular (S.5) passes to μ. The component conclusion needs the stronger all-order product assertion; its passage is recorded explicitly next.

Along a subsequence realizing μ, the bounded numbers $c_2(N), c_4(N), u_2(N), u_4(N)$ have a further convergent subsequence. This changes neither the limiting law nor its mass assignment. Denote the limits by $c_2^\infty, c_4^\infty, u_2^\infty, u_4^\infty$. Their inequalities survive:

$$u_4^\infty \le Ce^\sigma, \quad (1+|c_2^\infty|)u_2^\infty \le Ce^\sigma, \quad |c_2^\infty| \le Ce^{-6\epsilon}, \quad c_4^\infty - 3(c_2^\infty)^2 = b_{R^*}(f). \tag{S.6}$$

The last equality is the lattice free comparison at the fixed reference side, followed by $\eta \to 0$.

Let $\ell = R^*/L$, and let $y_1, \dots, y_J$ be any pairwise distinct points of $\ell\mathbb{Z}^3$. For all sufficiently large target tori these are ordinary, unreflected copies in distinct even cells, after translating the whole grid by an integer native vector if necessary. A fixed compact set of translations lies within the target period when its side tends to infinity. Thus Theorem 12.1 bounds their product by u_m^J. Define on the limiting field

$$G_m(F) = e^{-m} H_m(F(f); \|f\|_2^2) - c_m^\infty, \qquad m = 2, 4.$$

For each fixed J, replace the compatible probes by the fixed translates of f, and v_η by $\|f\|_2^2$, in the finite product. Each replacement tends to zero: the first is $O(\eta^2)$ and the second is $O(\eta)$, with constants allowed to depend on this fixed J and charge. Expanding one factor at a time and applying Hölder with the uniform moments justifies this statement even for high J. Moment convergence from 5.1 now yields

$$\left| \int \prod_{i=1}^{J} U(y_i) G_m \, d\mu \right| \le (u_m^\infty)^J \tag{S.7}$$

for every fixed J and every such distinct collection. This is a countable family of tests on $\ell\mathbb{Z}^3$, so the same subsequence suffices. We have not interchanged a growing product order with the cutoff limit.

S.4. **From lattice-invariant to translation-invariant statistics.** The variables G_m have every finite moment by Gaussian domination of μ. Apply Lemma 7.1 to the action of $\ell\mathbb{Z}^3$ and (S.7). If $\mathcal{I}_\ell$ is its invariant sigma algebra, the result is

$$|E_\mu[G_m \mid \mathcal{I}_\ell]| \le u_m^\infty \quad \mu\text{-almost surely.}$$

The invariant sigma algebra $\mathcal{I}$ of the full translation action is contained in $\mathcal{I}_\ell$. The tower property and positivity of conditional expectation give

$$|E_\mu[G_m \mid \mathcal{I}]| = |E_\mu[E_\mu[G_m \mid \mathcal{I}_\ell] \mid \mathcal{I}]| \le u_m^\infty. \tag{S.8}$$

This step does not identify the two invariant sigma algebras.

Let $\mu = \int \mu_\xi \, d\nu(\xi)$ be the translation-ergodic decomposition. The conditional expectations in (S.8) equal the component expectations for ν-almost every ξ. Theorem 6.1 gives centeredness and the required moments of these components. For this fixed probe put

$$\delta_2(\xi) = e^{-2}(\mu_\xi(F(f)^2) - \|f\|_2^2) - c_2^\infty, \quad \delta_4(\xi) = e^{-4}\mu_\xi(H_4(F(f); \|f\|_2^2)) - c_4^\infty.$$

They satisfy $|\delta_m(\xi)| \le u_m^\infty$ on the intersection of two conull sets. The exact centered cumulant algebra gives

$$e^{-4}\kappa_4^{\mu_\xi}(f) = c_4^\infty - 3(c_2^\infty)^2 + \delta_4(\xi) - 6c_2^\infty\delta_2(\xi) - 3\delta_2(\xi)^2. \tag{S.9}$$

Using (S.6), its weighted second bound, and the torus-image estimate already paid above, proves

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le Ce^{4+\sigma} \quad \nu\text{-almost surely.}$$

If $b(f) > 0$, choose the final charge ceiling so that $Ce^\sigma \le b(f)/2$. Then the fourth cumulant is positive in almost every component. The same expansion for the full law was already obtained from (S.5), so it too is non-Gaussian. The exceptional set may depend on the fixed probe and charge, as

the theorem allows. The conclusion asserts neither a universal limiting hierarchy nor uniqueness of a cutoff limit point.

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